



NCERT EXEMPLAR

PROBLEMS-SOLUTIONS

SCIENCE **Class X**

**Detailed Explanation to all
Objective & Subjective Problems**

Highly Useful for
**School Examinations & to Build
Foundation for Entrance Exams**

NCERT EXEMPLAR PROBLEMS-SOLUTIONS SCIENCE Class X

**Detailed Explanations to all
Objective & Subjective Problems**

*Rajesh Singh • Indu Gupta
Sikha Sharma*



ARIHANT PRAKASHAN



(School Division Series)



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PREFACE

The Department of Education in Science & Mathematics (DESM) & National Council of Educational Research & Training (NCERT) developed **Exemplar Problems** in Science and Mathematics for Secondary and Senior Secondary Classes with the objective to provide the students a large number of quality problems in various forms and format viz. Multiple Choice Questions, Short Answer Questions, Long Answer Questions etc., with varying levels of difficulty.

NCERT Exemplar Problems are very important for both; School & Board Examinations as well as competitive examinations like NTSE, Olympiad etc. The questions given in exemplar book are mainly of higher difficulty order by practicing these problems, you will be able to manage with the margin between a good score and a very good or an excellent score.

Approx 20% problems asked in any Board Examination or Competitive Examinations are of higher difficulty order, exemplar problems will make you ready to solve these difficult problems.

This book **NCERT Exemplar Problems-Solutions Science X** contains Explanatory & Accurate Solutions to all the questions given in NCERT Exemplar Science book.

For the overall benefit of the students' we have made unique this book in such a way that it presents not only hints and solutions but also detailed and authentic explanations. Through these detailed explanations, students can learn the concepts which will enhance their thinking and learning abilities.

We have introduced some additional features with the solutions which are as follows

- **Thinking Process** Along with the solutions to questions we have given  thinking process that tell how to approach to solve a problem. Here, we have tried to cover all the loopholes which may lead to confusion. All formulae and hints are discussed in detail.
- **Note** We have provided notes also to solutions in which special points are mentioned which are of great value for the students.

For the completion of this book, we would like to thank Priyanshi Garg who helped us at project management level.

*With the hope that this book will be of great help to the students,
we wish great success to our readers.*

Authors

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Chemical Reactions and Equations

Multiple Choice Questions (MCQs)

Q. 1 Which of the following is not a physical change?

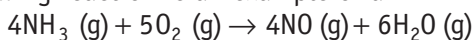
- (a) Boiling of water to give water vapour
- (b) Melting of ice to give water
- (c) Dissolution of salt in water
- (d) Combustion of Liquefied Petroleum Gas (LPG)

💡 Thinking Process

- (i) Chemical change takes place on the molecular level and it produces a new substance.
- (ii) Physical change involves change in physical properties and it does not produce a new substance.

Ans. (d) Combustion of Liquefied Petroleum Gas (LPG) is a chemical change. Because it is an irreversible reaction and new products, carbon dioxide and water vapour are formed and lot of heat is also produced during the reaction.
Rest three are physical changes as no new products are formed.

Q. 2 The following reaction is an example of a



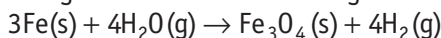
- 1. displacement reaction
 - 2. combination reaction
 - 3. redox reaction
 - 4. neutralisation reaction
- (a) 1 and 4 (b) 2 and 3 (c) 1 and 3 (d) 3 and 4

💡 Thinking Process

In this reaction, NH_3 is changing into NO . Removal of hydrogen from a substance is called oxidation. So, NH_3 is undergoing oxidation. O_2 is changing into H_2O . The addition of hydrogen to a substance is called reduction. So, O_2 is undergoing reduction.

Ans. (d) It is a redox reaction because oxidation and reduction both take place simultaneously in this reaction. Also, it is a displacement reaction because H of NH_3 has been displaced by oxygen.

Q. 3 Which of the following statements about the given reaction are correct?



1. Iron metal is getting oxidised.
2. Water is getting reduced.
3. Water is acting as reducing agent.
4. Water is acting as oxidising agent.

(a) 1, 2 and 3

(b) 3 and 4

(c) 1, 2 and 4

(d) 2 and 4

Ans. (c) Fe is gaining oxygen to give Fe_3O_4 . H_2O is losing oxygen to give H_2 . The substance which oxidises the other substance in a chemical reaction is known as an oxidising agent. So, water is acting as oxidising agent.

Q. 4 Which of the following are exothermic processes?

1. Reaction of water with quick lime
2. Dilution of an acid
3. Evaporation of water
4. Sublimation of camphor (crystals)

(a) 1 and 2

(b) 2 and 3

(c) 1 and 4

(d) 3 and 4

💡 Thinking Process

Exothermic processes are those in which heat is released along with the formation of products.

Ans. (a) When quick lime reacts with water, a large amount of heat is released along with the formation of calcium hydroxide. Similarly, the process of dissolving an acid or base in water is a highly exothermic reaction. Evaporation of water and sublimation of camphor are endothermic reactions.

Q. 5 Three beakers labelled as A, B and C each containing 25 mL of water were taken. A small amount of NaOH, anhydrous CuSO_4 and NaCl were added to the beakers A, B and C respectively. It was observed that there was an increase in the temperature of the solutions contained in beakers A and B, whereas in case of beaker C, the temperature of the solution falls. Which one of the following statement(s) is(are) correct?

1. In beakers A and B, exothermic process has occurred.
2. In beakers A and B, endothermic process has occurred.
3. In beaker C, exothermic process has occurred.
4. In beaker C, endothermic process has occurred.

(a) Only 1

(b) Only 2

(c) 1 and 4

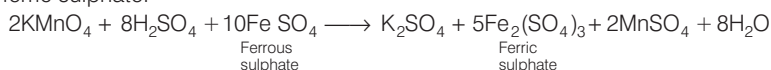
(d) 2 and 3

Ans. (c) As in case of beakers A and B, heat is given out, so temperature became high, hence it is an exothermic reaction while in beaker C, heat is absorbed from water, so temperature falls, hence it is an endothermic process.

Q. 6 A dilute ferrous sulphate solution was gradually added to the beaker containing acidified permanganate solution. The light purple colour of the solution fades and finally disappears. Which of the following is the correct explanation for the observation?

- (a) KMnO_4 is an oxidising agent, it oxidises FeSO_4
- (b) FeSO_4 acts as an oxidising agent and oxidises KMnO_4
- (c) The colour disappears due to dilution, no reaction is involved
- (d) KMnO_4 is an unstable compound and decomposes in the presence of FeSO_4 to a colourless compound

Ans. (a) Potassium permanganate (KMnO_4) in presence of dil H_2SO_4 , i.e., in acidic medium acts as strong oxidising agent. In acidic medium, KMnO_4 oxidises ferrous sulphate to ferric sulphate.



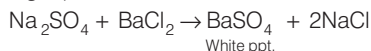
Q. 7 Which among the following is (are) double displacement reaction(s)?

- (i) $\text{Pb} + \text{CuCl}_2 \longrightarrow \text{PbCl}_2 + \text{Cu}$
- (ii) $\text{Na}_2\text{SO}_4 + \text{BaCl}_2 \longrightarrow \text{BaSO}_4 + 2\text{NaCl}$
- (iii) $\text{C} + \text{O}_2 \longrightarrow \text{CO}_2$
- (iv) $\text{CH}_4 + 2\text{O}_2 \longrightarrow \text{CO}_2 + 2\text{H}_2\text{O}$

- (a) (i) and (iv) (b) Only (ii) (c) (i) and (ii) (d) (iii) and (iv)

Ans. (b) In double displacement reaction, two compounds exchange their ions to form two new compounds.

Only in reaction (ii), exchange of ions is taking place (cations and anions of both reactants are exchanged).



Q. 8 Which among the following statement(s) is /are true? Exposure of silver chloride to sunlight for a long duration turns grey due to

- (i) the formation of silver by decomposition of silver chloride.
- (ii) sublimation of silver chloride.
- (iii) decomposition of chlorine gas from silver chloride.
- (iv) oxidation of silver chloride.

- (a) (i) Only (b) (i) and (iii) (c) (ii) and (iii) (d) Only (iv)

Ans. (a) Decomposition of silver chloride takes place in the presence of sunlight.



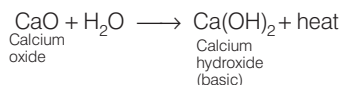
In this reaction, the white colour of silver chloride changes greyish white due to the formation of silver metal.

Q. 9 Solid calcium oxide reacts vigorously with water to form calcium hydroxide accompanied by liberation of heat. This process is called slaking of lime. Calcium hydroxide dissolves in water to form its solution called lime water. Which among the following is true about slaking of lime and the solution formed?

- (i) It is an endothermic reaction.
- (ii) It is exothermic reaction.
- (iii) The pH of the resulting solution will be more than seven.
- (iv) The pH of the resulting solution will be less than seven.

- (a) (i) and (ii)
- (b) (ii) and (iii)
- (c) (i) and (iv)
- (d) (iii) and (iv)

Ans. (b) It is an exothermic reaction because heat is given out and the resulting compound is Ca(OH)_2 which is basic in nature so, the pH of the resulting solution will be more than seven.



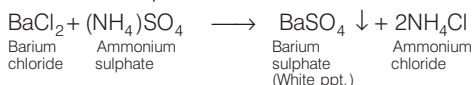
Ca(OH)_2 turns red litmus solution to blue. So, its pH value is greater than seven.

Q. 10 Barium chloride on reacting with ammonium sulphate forms barium sulphate and ammonium chloride. Which of the following correctly represents the type of the reaction involved?

- (i) Displacement reaction
- (ii) Precipitation reaction
- (iii) Combination reaction
- (iv) Double displacement reaction

- (a) Only (i)
- (b) Only (ii)
- (c) Only (iv)
- (d) (ii) and (iv)

Ans. (d) The reaction is a double displacement reaction as follows

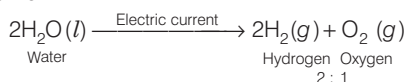


It is also called precipitation reaction due to the formation of white precipitate of barium sulphate.

Q. 11 Electrolysis of water is a decomposition reaction. The mole ratio of hydrogen and oxygen gases liberated during electrolysis of water is

- (a) 1 : 1
- (b) 2 : 1
- (c) 4 : 1
- (d) 1 : 2

Ans. (b) The water decomposes during electrolysis to form hydrogen and oxygen gases in the ratio 2 : 1 by volume.



Q. 12 Which of the following is(are) an endothermic process(es)?

- (i) Dilution of sulphuric acid
- (ii) Sublimation of dry ice
- (iii) Condensation of water vapours
- (iv) Evaporation of water

- (a) Both (i) and (iii)
- (b) Only (ii)
- (c) Only (iii)
- (d) Both (ii) and (iv)

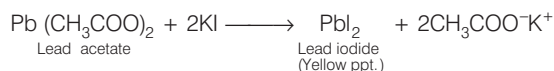
Ans. (d) Sublimation of dry ice and evaporation of water both are endothermic process and take heat (energy) from their surrounding to get sublime or evaporate. Hence, both these processes produce cooling. This is the reason for using dry ice as refrigerant. Dilution of sulphuric acid and condensation of water vapours evolve heat. Hence, these are exothermic processes.

Q. 13 In the double displacement reaction between aqueous potassium iodide and aqueous lead nitrate, a yellow precipitate of lead iodide is formed. While performing the activity if lead nitrate is not available, which of the following can be used in place of lead nitrate?

- (a) Lead sulphate (insoluble)
- (b) Lead acetate
- (c) Ammonium nitrate
- (d) Potassium sulphate

Ans. (b) Lead acetate can be used in place of lead nitrate because like lead nitrate, it is also a soluble salt in water.

The reaction is



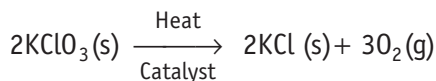
Lead sulphate is insoluble in water, so it cannot be used.

Q. 14 Which of the following gases can be used for storage of fresh sample of an oil for a long time?

- (a) Carbon dioxide or oxygen
- (b) Nitrogen or oxygen
- (c) Carbon dioxide or helium
- (d) Helium or nitrogen

Ans. (d) Helium and nitrogen both the gases provide inert atmosphere. When the packed food is surrounded by unreactive gas (nitrogen or helium), there is no oxygen (or air) to cause its oxidation and make it rancid.

Q. 15 The following reaction is used for the preparation of oxygen gas in the laboratory



Which of the following statement(s) is/are correct about the reaction?

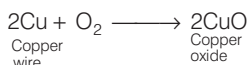
- (a) It is a decomposition reaction and endothermic in nature
- (b) It is a combination reaction
- (c) It is a decomposition reaction and accompanied by release of heat
- (d) It is a photochemical decomposition reaction and exothermic in nature

Ans. (a) It is a decomposition reaction and endothermic in nature because decomposition of KClO_3 requires heat for products formation.

Q. 16 Which one of the following processes involve chemical reactions?

- (a) Storing of oxygen gas under pressure in a gas cylinder
- (b) Liquefaction of air
- (c) Keeping petrol in a China dish in the open
- (d) Heating copper wire in the presence of air at high temperature

Ans. (d) A chemical reaction is a change in which one or more substance (s) or reactant (s) react to form new substance (s) with entirely different properties.



Q. 17 In which of the following chemical equations, the abbreviations represent the correct states of the reactants and products involved at reaction temperature?

- (a) $2\text{H}_2(l) + \text{O}_2(l) \longrightarrow 2\text{H}_2\text{O}(g)$
- (b) $2\text{H}_2(g) + \text{O}_2(l) \longrightarrow 2\text{H}_2\text{O}(l)$
- (c) $2\text{H}_2(g) + \text{O}_2(g) \longrightarrow 2\text{H}_2\text{O}(l)$
- (d) $2\text{H}_2(g) + \text{O}_2(g) \longrightarrow 2\text{H}_2\text{O}(g)$

Ans. (c) It is because, the standard state for hydrogen and oxygen is gas and for water is liquid at reaction temperature.

Q. 18 Which of the following are combination reactions?

- (i) $2\text{KClO}_3 \xrightarrow{\text{Heat}} 2\text{KCl} + 3\text{O}_2$
- (ii) $\text{MgO} + \text{H}_2\text{O} \longrightarrow \text{Mg}(\text{OH})_2$
- (iii) $4\text{Al} + 3\text{O}_2 \longrightarrow 2\text{Al}_2\text{O}_3$
- (iv) $\text{Zn} + \text{FeSO}_4 \longrightarrow \text{ZnSO}_4 + \text{Fe}$

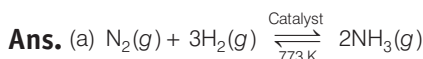
- (a) Both (i) and (iii)
- (b) Both (iii) and (iv)
- (c) Both (ii) and (iv)
- (d) Both (ii) and (iii)

Ans. (d) A reaction in which two or more reactants react together to form a single product, is called a combination reaction. In reaction (ii), MgO and H_2O combine to form $\text{Mg}(\text{OH})_2$ and in reaction (iii), Al and O_2 combine to form Al_2O_3 . Reaction (i) is decomposition reaction and reaction (iv) is displacement reaction.

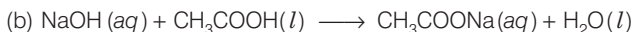
Short Answer Type Questions

Q. 19 Write the balanced chemical equations for the following reactions and identify the type of reaction in each case.

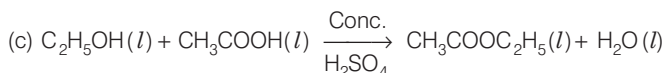
- (a) Nitrogen gas is treated with hydrogen gas in the presence of a catalyst at 773 K to form ammonia gas.
- (b) Sodium hydroxide solution is treated with acetic acid to form sodium acetate and water.
- (c) Ethanol is warmed with ethanoic acid to form ethyl acetate in the presence of concentrated H_2SO_4 .
- (d) Ethane is burnt in the presence of oxygen to form carbon dioxide, water and releases heat and light.



This is an example of combination reaction.



This is an example of neutralisation reaction as well as double displacement reaction.



This is an example of double displacement reaction.



This is an example of combustion reaction.

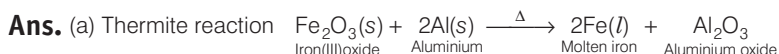
Q. 20 Write the balanced chemical equation for the following reactions and identify the type of reaction in each case.

(a) In Thermite reaction, iron (III) oxide reacts with aluminium and gives molten iron and aluminium oxide.

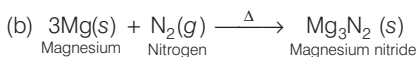
(b) Magnesium ribbon is burnt in an atmosphere of nitrogen gas to form solid magnesium nitride.

(c) Chlorine gas is passed in an aqueous potassium iodide solution to form potassium chloride solution and solid iodine.

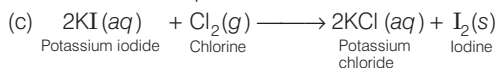
(d) Ethanol is burnt in air to form carbon dioxide, water and releases heat.



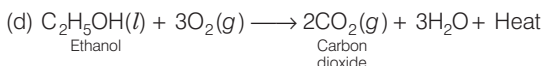
This is an example of displacement reaction.



This is an example of combination reaction.

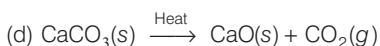
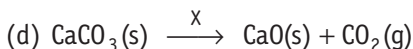
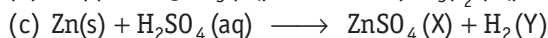
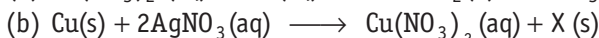
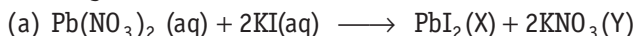


This is an example of displacement reaction.



This is an example of combustion reaction.

Q. 21 Complete the missing components/variables given as **X** and **Y** in the following.



Q. 22 Which among the following changes are exothermic or endothermic in nature?

- (a) Decomposition of ferrous sulphate
- (b) Dilution of sulphuric acid
- (c) Dissolution of sodium hydroxide in water
- (d) Dissolution of ammonium chloride in water

Ans. (a) Decomposition of ferrous sulphate is an endothermic reaction because heat is absorbed during the reaction.
 (b) The process of dissolving sulphuric acid in water (i.e., dilution of sulphuric acid) is a highly exothermic reaction.
 (c) The process of dissolving sodium hydroxide (a base) in water is again a highly exothermic reaction.
 (d) Dissolution of ammonium chloride in water is an endothermic reaction as heat is absorbed during the reaction.

Q. 23 Identify the reducing agent in the following reactions.

- (a) $4\text{NH}_3 + 5\text{O}_2 \longrightarrow 4\text{NO} + 6\text{H}_2\text{O}$
- (b) $\text{H}_2\text{O} + \text{F}_2 \longrightarrow \text{HF} + \text{HOF}$
- (c) $\text{Fe}_2\text{O}_3 + 3\text{CO} \longrightarrow 2\text{Fe} + 3\text{CO}_2$
- (d) $2\text{H}_2 + \text{O}_2 \longrightarrow 2\text{H}_2\text{O}$

Thinking Process

A substance that helps in the reduction of another substance is called reducing agent. It either adds oxygen, gives hydrogen or donates electrons to the substance that is reduced.

Ans. (a) NH_3 is the reducing agent because it gives hydrogen and itself gets oxidised to NO (Removal of H and addition of O). O_2 has been reduced to H_2O (Addition of hydrogen).
 (b) H_2O is the reducing agent because of the addition of electronegative F to get oxidised to HOF. It reduces F_2 to HF (by addition of hydrogen).
 (c) Fe_2O_3 has been reduced to Fe (Removal of oxygen). Also, CO has been oxidised to CO_2 (Addition of oxygen) and hence, CO is a reducing agent.
 (d) H_2 is the reducing agent which itself gets oxidised to H_2O (Addition of oxygen). O_2 has been reduced to H_2O (Addition of hydrogen).

Q. 24 Identify the oxidising agent (oxidant) in the following reactions.

- (a) $\text{Pb}_3\text{O}_4 + 8\text{HCl} \longrightarrow 3\text{PbCl}_2 + \text{Cl}_2 + 4\text{H}_2\text{O}$
- (b) $2\text{Mg} + \text{O}_2 \longrightarrow 2\text{MgO}$
- (c) $\text{CuSO}_4 + \text{Zn} \longrightarrow \text{Cu} + \text{ZnSO}_4$
- (d) $\text{V}_2\text{O}_5 + 5\text{Ca} \longrightarrow 2\text{V} + 5\text{CaO}$
- (e) $3\text{Fe} + 4\text{H}_2\text{O} \longrightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$
- (f) $\text{CuO} + \text{H}_2 \longrightarrow \text{Cu} + \text{H}_2\text{O}$

Thinking Process

Oxidising agent is the substance that itself gets reduced in the reactions. It either gives oxygen, adds hydrogen or accepts electrons from the substance to be oxidised.

Ans. (a) In this reaction, HCl has been oxidised to Cl_2 (Removal of H) and Pb_3O_4 has been reduced to PbCl_2 (Removal of O). Hence, Pb_3O_4 is the oxidising agent (oxidant).
 (b) Mg has been oxidised to MgO (Addition of O). Hence, O_2 is the oxidant.
 (c) This reaction occurs as

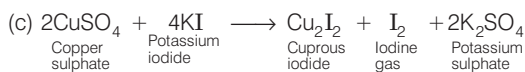
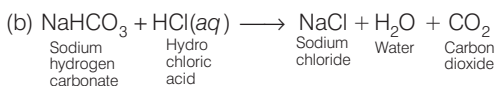
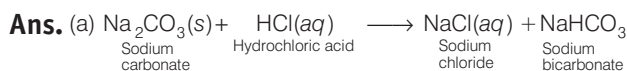


Here Cu^{2+} has gained two electrons to form Cu. Hence, Cu^{2+} has been reduced to Cu. But Zn has donated two electrons to form Zn^{2+} . Hence, Zn has been oxidised to Zn^{2+} . Thus, CuSO_4 is the oxidising agent.

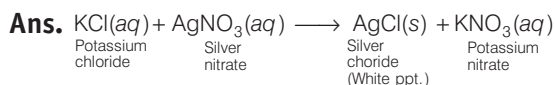
- (d) Ca has been oxidised to CaO (Addition of oxygen), whereas V_2O_5 has been reduced to V. Hence, V_2O_5 is the oxidant.
- (e) Fe has been oxidised to Fe_3O_4 whereas H_2O has been reduced to H_2 . Hence, H_2O is the oxidant.
- (f) CuO has been reduced to Cu whereas H_2 has been oxidised to H_2O . Hence, CuO is the oxidant.

Q. 25 Write the balanced chemical equations for the following reactions.

- (a) Sodium carbonate on reaction with hydrochloric acid in equal molar concentrations gives sodium chloride and sodium hydrogen carbonate.
- (b) Sodium hydrogen carbonate on reaction with hydrochloric acid gives sodium chloride, water and liberates carbon dioxide.
- (c) Copper sulphate on treatment with potassium iodide precipitates cuprous iodide (Cu_2I_2), liberates iodine gas and also forms potassium sulphate.

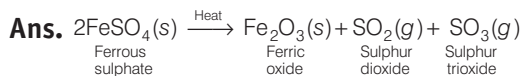


Q. 26 A solution of potassium chloride when mixed with silver nitrate solution, an insoluble white substance is formed. Write the chemical reaction involved and also mention the type of the chemical reaction.



This is an example of double displacement reaction.

Q. 27 Ferrous sulphate decomposes with the evolution of a gas having a characteristic odour of burning sulphur. Write the chemical reaction involved and identify the type of reaction.



This is an example of decomposition reaction (Thermal decomposition).

Q. 28 Why do fire flies glow at night?

Ans. Fire flies have a special kind of substance (Protein) i.e., luciferin that undergoes oxidation (by atmospheric oxygen) in the presence of an enzyme (luciferase). This reaction is accompanied by emission of light. Therefore, fire flies glow at night.

Q. 29 Grapes hanging on the plant do not ferment but after being plucked from the plant can be fermented. Under what conditions do these grapes ferment? Is it a chemical or a physical change?

Ans. When grapes are attached to the plant, aerobic respiration takes place in the cells as the oxygen reaches there. But when plucked, no oxygen reaches the cell, therefore fermentation starts. It is obvious that fermentation takes place only in the absence of oxygen (anaerobic condition). Fermentation is a chemical change in which sugar is converted to alcohol.

Q. 30 Which among the following are physical or chemical changes?

- (a) Evaporation of petrol
- (b) Burning of Liquefied Petroleum Gas (LPG)
- (c) Heating of an iron rod to red hot
- (d) Curdling of milk
- (e) Sublimation of solid ammonium chloride

Ans. (a) Evaporation of petrol is a physical change because no new chemical substance is formed.
 (b) Burning of LPG is a chemical change because on burning, LPG forms CO_2 and H_2O .
 (c) Heating of an iron rod to red hot is a physical change.
 (d) Curdling of milk is a chemical change because in the conversion of milk to curd, new chemical substances are formed.
 (e) It is a physical change, because solid NH_4Cl vapours and no new chemical substance is formed.

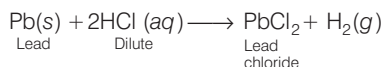
Q. 31 During the reaction of some metals with dilute hydrochloric acid, following observations were made.

- (a) Silver metal does not show any change
 - (b) The temperature of the reaction mixture rises when aluminium (Al) is added
 - (c) The reaction of sodium metal is found to be highly explosive
 - (d) Some bubbles of a gas are seen when lead (Pb) is reacted with the acid
- Explain these observations giving suitable reasons.

Ans. (a) This is because silver is below hydrogen in the reactivity series so, cannot displace hydrogen when reacted with acid.
 (b) The reaction of Al with dilute HCl is exothermic *i.e.*, heat is produced in the reaction, hence the temperature of the reaction mixture rises.

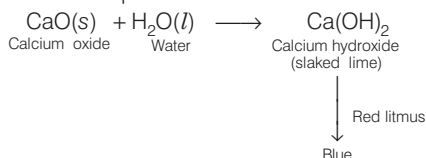
$$2\text{Al} + 6\text{HCl} \longrightarrow 2\text{AlCl}_3 + 3\text{H}_2 + \text{Heat}$$

 (c) Sodium is a very reactive metal. It reacts explosively (extremely rapidly) with hydrochloric acid to form sodium chloride and hydrogen with the evolution of heat too. H_2 gas produced catches fire immediately.
 (d) Lead is present just above the hydrogen in the activity series of metals. Hence, it is slightly more reactive and displace hydrogen from acid very slowly and to a small extent. Therefore, only bubbles of H_2 are seen to evolve.



Q. 32 A substance X, which is an oxide of a group 2 element, is used intensively in the cement industry. This element is present in bones also. On treatment with water it forms a solution which turns red litmus blue. Identify X and also write the chemical reactions involved.

Ans. X is calcium oxide (CaO), also called quick lime. It reacts with water to form calcium hydroxide which is a basic compound.



Q. 33 Write a balanced chemical equation for each of the following reactions and also classify them.

- Lead acetate solution is treated with dilute hydrochloric acid to form lead chloride and acetic acid solution.
- A piece of sodium metal is added to absolute ethanol to form sodium ethoxide and hydrogen gas.
- Iron (III) oxide on heating with carbon monoxide gas reacts to form solid iron and liberates carbon dioxide gas.
- Hydrogen sulphide gas reacts with oxygen gas to form solid sulphur and liquid water.

Ans. (a) $\text{Pb}(\text{CH}_3\text{COO})_2(\text{aq}) + 2\text{HCl}(\text{dil}) \longrightarrow \text{PbCl}_2(\text{s}) \downarrow + 2\text{CH}_3\text{COOH}(\text{aq})$

This is a double displacement reaction. It is also a precipitation reaction.

(b) $2\text{C}_2\text{H}_5\text{OH}(\text{l}) + 2\text{Na}(\text{s}) \longrightarrow 2\text{C}_2\text{H}_5\text{O}^-\text{Na}^+ + \text{H}_2 \uparrow$

This is displacement reaction.

(c) $\text{Fe}_2\text{O}_3(\text{s}) + 3\text{CO}(\text{g}) \longrightarrow 2\text{Fe}(\text{s}) + 3\text{CO}_2(\text{g}) \uparrow$

This is a redox reaction.

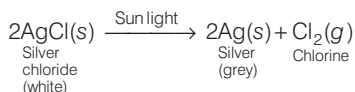
(d) $2\text{H}_2\text{S}(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{S}(\text{s}) + 2\text{H}_2\text{O}(\text{l})$

This is a redox reaction.

Q. 34 Why do we store silver chloride in dark coloured bottles?

Ans. Dark coloured bottles interrupt the path of light such that light cannot reach silver chloride in the bottles and its decomposition is prevented. It is known that silver chloride decomposes to silver and chlorine in the presence of light.

This is shown in the reaction given below



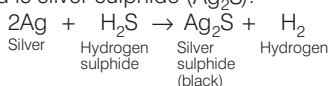
Hence, it is stored in dark coloured bottles.

Q. 38 A silver article generally turns black when kept in the open for a few days. The article when rubbed with toothpaste again starts shining.

- (a) Why do silver articles turn black when kept in the open for a few days? Name the phenomenon involved.
 (b) Name the black substance formed and give its chemical formula.

Ans. (a) Silver article turns black when kept in the air because silver article reacts with sulphur compounds such as hydrogen sulphide (H_2S) present in air. The phenomenon is called corrosion. For silver particularly, it is called tarnishing of silver.

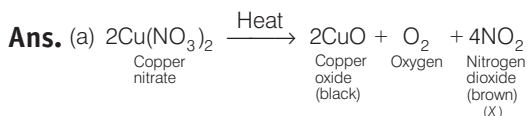
(b) The black substance formed is silver sulphide (Ag_2S).



Long Answer Type Questions

Q. 39 On heating blue coloured powder of copper (II) nitrate in a boiling tube, copper oxide (black), oxygen gas and a brown gas X is formed.

- (a) Write a balanced chemical equation of the reaction.
 (b) Identify the brown gas X evolved.
 (c) Identify the type of reaction.
 (d) What could be the pH range of aqueous solution of the gas X?

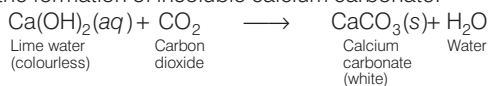


- (b) X is nitrogen dioxide gas (NO_2).
 (c) This is a thermal decomposition reaction.
 (d) The aqueous solution of the gas is acidic because it is an oxide of a non-metal, so its pH range would be in between 1 to 7.

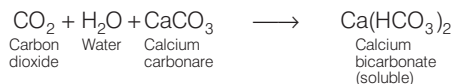
Q. 40 Give the characteristic tests for the following gases.

- (a) CO_2 (b) SO_2
 (c) O_2 (d) H_2

Ans. (a) **Test for CO_2 gas** (lime water test) When CO_2 gas is passed through lime water, it turns milky due to the formation of insoluble calcium carbonate.

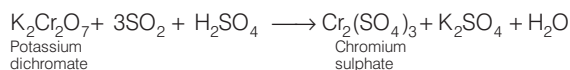


If the gas is in excess, milky appearance disappears due to the formation of soluble calcium bicarbonate.

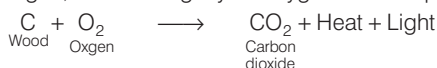


(b) Test for SO₂ gas

- (i) Sulphur dioxide gas turns moist litmus paper from blue to red because SO₂ is acidic in nature.
- (ii) Sulphur dioxide gas turns acidified potassium dichromate (vi) solution from orange to green.



- (c) **Test for O₂ gas** When a wooden splinter is brought near the mouth of the gas jar containing oxygen gas, it burns brightly as oxygen is the supporter of combustion.



- (d) **Test for H₂ gas** Hydrogen gas burns in the presence of air or oxygen with a pop sound when a burning candle is brought near it.

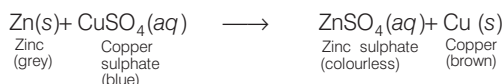
Q. 41 What happens when a piece of

- (a) zinc metal is added to copper sulphate solution?
 (b) aluminium metal is added to dilute hydrochloric acid?
 (c) silver metal is added to copper sulphate solution?

Also, write the balanced chemical equation, if reaction occurs.

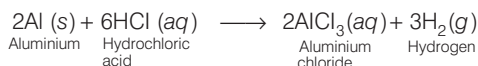
Ans. (a) Blue colour of CuSO₄ solution disappears.

Reason Zinc reacts with copper sulphate to form colourless zinc sulphate solution and solid copper in deposited form as Zn is more reactive than Cu, so, displaces Cu from CuSO₄ solution.



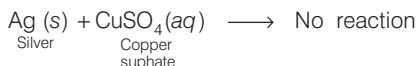
- (b) Hydrogen gas is evolved.

Reason Aluminium reacts with dilute hydrochloric acid to form aluminium chloride and hydrogen gas.



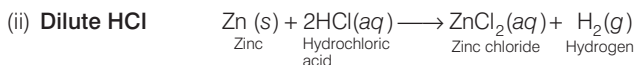
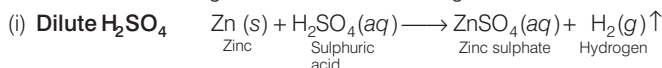
- (c) No reaction occurs.

Reason Silver metal is less reactive than copper, therefore, it cannot displace copper from copper sulphate solution.

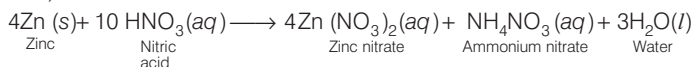


Q. 42 What happens when zinc granules are treated with dilute solution of H₂SO₄, HCl, HNO₃, NaCl and NaOH, also write the chemical equations, if reaction occurs ?

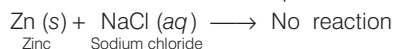
Ans. The reactions of zinc granules with various reagents are as follows



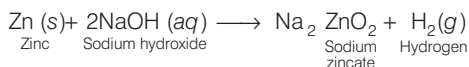
- (iii) **Dilute HNO_3** Dilute HNO_3 is an oxidising acid therefore, its reaction with a metal (e.g., Zn) is different than other dilute acids.



- (iv) **NaCl solution** Reaction does not take place



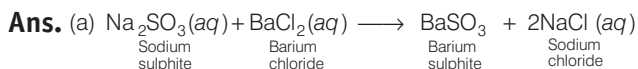
- (v) **NaOH solution**



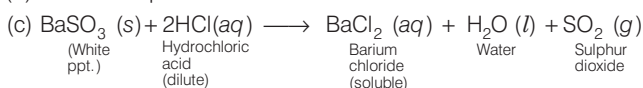
This hydrogen gas is evolved with H_2SO_4 , HCl and NaOH , nitrous oxide gas is evolved with dilute HNO_3 and no reaction takes place with NaCl solution.

Q. 43 On adding a drop of barium chloride solution to an aqueous solution of sodium sulphite, white precipitate is obtained.

- Write a balanced chemical equation of the reaction involved.
- What other name can be given to this precipitation reaction?
- On adding dilute hydrochloric acid to the reaction mixture, white precipitate disappears. Why?



- (b) Double displacement reaction



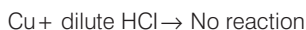
As the reaction product barium chloride is water soluble, the white precipitate disappears.

Q. 44 You are provided with two containers made up of copper and aluminium. You are also provided with solutions of dilute HCl , dilute HNO_3 , ZnCl_2 and H_2O . In which of the containers these solutions can be kept?

Ans. The container made up of copper or aluminium is suitable for storing the given solutions, can be decided by studying their reactions.

- (i) **Reactions of copper with**

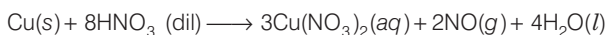
- (a) **Dilute HCl**



So, it can be stored in Cu container.

- (b) **Dilute HNO_3**

Being a strong oxidising agent, dilute HNO_3 reacts with copper, so it cannot be stored in copper container.



- (c) **ZnCl_2**

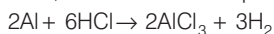
Copper is less reactive than zinc so it does not react with ZnCl_2 solution. Therefore, it can be stored in copper container.

(d) **H₂O**

Copper does not react with water. So, its container can store H₂O in it. Thus, dil HCl, ZnCl₂ solution and H₂O can be kept in copper vessel but dil. HNO₃ cannot be kept in copper vessel.

(ii) **Reactions of aluminium with**(a) **Dilute HCl**

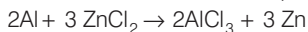
Al reacts with dilute HCl, so it cannot be kept in aluminium container.

(b) **Dilute HNO₃**

When dil. HNO₃ is kept in Al container, it forms a protective layer of aluminium oxide on it which makes it passive towards action with acid, therefore, it can be kept in Al container.

(c) **ZnCl₂**

Al is more reactive than zinc, so it cannot keep ZnCl₂ solution in it.

(d) **H₂O**

Al does not react with H₂O. So, its container can store H₂O in it. Thus, dil. HNO₃ and H₂O(l) can be kept in aluminium vessel but dil HCl and ZnCl₂ solution cannot be kept in aluminium vessel.

2

Acids, Bases and Salts

Multiple Choice Questions (MCQs)

Q. 1 What happens when a solution of an acid is mixed with a solution of a base in a test tube?

- (i) The temperature of the solution increases.
- (ii) The temperature of the solution decreases.
- (iii) The temperature of the solution remains the same.
- (iv) Salt formation takes place.

(a) Only (i) (b) (i) and (iii) (c) (ii) and (iii) (d) (i) and (iv)

💡 Thinking Process

- (i) Neutralisation is defined as the process in which an acid reacts with a base to form salt and water.
- (ii) In exothermic process, heat is evolved.

Ans. (d) When an acid reacts with a base, neutralisation reaction takes place to form salt and water and it is an exothermic reaction. So, the temperature of the solution increases.

Q. 2 An aqueous solution turns red litmus solution blue. Excess addition of which of the following solution would reverse the change?

- (a) Baking powder (b) Lime
(c) Ammonium hydroxide solution (d) Hydrochloric acid

Ans. (d) Since, the aqueous solution turns red litmus solution blue. So, it is a basic compound. An acid solution (HCl) would reverse the change. That is HCl would turn blue litmus solution to red.

Other options baking powder, lime and ammonium hydroxide solution are basic compounds. So, the blue litmus solution would not change the colour by adding these compounds.

Q. 3 During the preparation of hydrogen chloride gas on a humid day, the gas is usually passed through the guard tube containing calcium chloride. The role of calcium chloride taken in the guard tube is to

- (a) absorb the evolved gas (b) moisten the gas
(c) absorb moisture from the gas (d) absorb Cl^- ions from the evolved gas

Ans. (c) The role of calcium chloride (CaCl_2) is to absorb moisture from the gas. Because calcium chloride is a good dehydrating agent.

Q. 4 Which of the following salts does not contain water of crystallisation?

- (a) Blue vitriol (b) Baking soda (c) Washing soda (d) Gypsum

💡 Thinking Process

The water molecules which form part of the structure of a crystal (of a salt) are called water of crystallisation. The salts which contain water of crystallisation are called hydrated salts.

Ans. (b) Chemical formula of baking soda is NaHCO_3 (sodium hydrogen carbonate). Chemical formulae of blue vitriol is $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, washing soda is $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ and gypsum is $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$.

So, baking powder does not contain water of crystallisation.

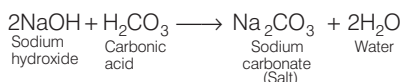
Q. 5 Sodium carbonate is a basic salt because it is a salt of

- (a) strong acid and strong base (b) weak acid and weak base
(c) strong acid and weak base (d) weak acid and strong base

💡 Thinking Process

The salts of weak acids and strong bases give basic solutions (or alkaline solutions) having pH more than 7.

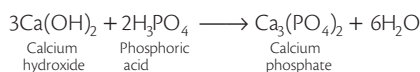
Ans. (d) Sodium carbonate (Na_2CO_3) is the salt of weak acid carbonic acid (H_2CO_3) and a strong base sodium hydroxide (NaOH).



Q. 6 Calcium phosphate is present in tooth enamel. Its nature is

- (a) basic (b) acidic (c) neutral (d) amphoteric

💡 Thinking Process



Ans. (a) Calcium phosphate $\text{Ca}_3(\text{PO}_4)_2$ is basic salt, as it is a salt of weak acid (phosphoric acid) and slightly stronger base (calcium hydroxide) (though both are weak). Also when pH of our mouth falls below 5.5 due to eating of sweets etc., i.e., mouth is acidic, the dissolution of enamel (calcium phosphate) starts which shows that calcium phosphate is basic in nature.

Q. 7 A sample of soil is mixed with water and allowed to settle. The clear supernatant solution turns the pH paper yellowish-orange. Which of the following would change the colour of this pH paper to greenish-blue?

- (a) Lemon juice (b) Vinegar (c) Common salt (d) An antacid

Ans. (d) As pH paper turns greenish blue for weakly basic compound and antacids contain weak base like Mg(OH)_2 . So, an antacid would change the colour of this pH paper to greenish-blue. Other options (a) and (b) contain acids and option (c) is a neutral salt.

Q. 8 Which of the following gives the correct increasing order of acid strength?

- (a) Water < acetic acid < hydrochloric acid
- (b) Water < hydrochloric acid < acetic acid
- (c) Acetic acid < water < hydrochloric acid
- (d) Hydrochloric acid < water < acetic acid

Ans. (a) Hydrochloric acid is a mineral acid and ionises completely in water, that's why it is a strong acid. Acetic acid is an organic acid and ionises only partially in water, hence, it is a weak acid. Water has some what neutral nature. Thus, the order of acidity is water < acetic acid < hydrochloric acid.

Q. 9 If a few drops of a concentrated acid accidentally spills over the hand of a student, what should be done?

- (a) Wash the hand with saline solution
- (b) Wash the hand immediately with plenty of water and apply a paste of sodium hydrogen carbonate
- (c) After washing with plenty of water apply solution of sodium hydroxide on the hand
- (d) Neutralise the acid with a strong alkali

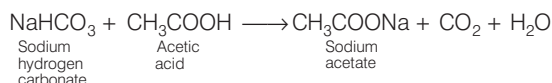
Ans. (b) Wash the hand immediately with plenty of water to wash away most of the acid and then apply a paste of baking soda (NaHCO_3) to neutralise the little acid left. Here a strong base cannot be used to neutralise the acid due to its **corrosive nature**.

Q. 10 Sodium hydrogen carbonate when added to acetic acid evolves a gas. Which of the following statements are true about the gas evolved?

- (i) It turns lime water milky.
- (ii) It extinguishes a burning splinter.
- (iii) It dissolves in a solution of sodium hydroxide.
- (iv) It has a pungent odour.

- (a) (i) and (ii) (b) (i), (ii) and (iii) (c) (ii), (iii) and (iv) (d) (i) and (iv)

Ans. (b) When sodium hydrogen carbonate is added to acetic acid then carbon dioxide (CO_2) gas is evolved.



CO_2 turns lime water milky, it is a non-supporter of combustion and is absorbed by strong alkalies like NaOH .

Q. 11 Common salt besides being used in kitchen can also be used as the raw material for making

- (i) washing soda
- (ii) bleaching powder
- (iii) baking soda
- (iv) slaked lime

- (a) (i) and (ii) (b) (i), (ii) and (iv) (c) (i), (ii) and (iii) (d) (i), (iii) and (iv)

Ans. (c) Common salt (sodium chloride) is used as a raw material for making a large number of chemicals in industry such as sodium hydroxide, washing soda, baking soda, hydrochloric acid, hydrogen, chlorine and sodium metal. Chlorine gas obtained is used for making bleaching powder.

Q. 12 One of the constituents of baking powder is sodium hydrogen carbonate, the other constituent is

- | | |
|-----------------------|--------------------|
| (a) hydrochloric acid | (b) tartaric acid |
| (c) acetic acid | (d) sulphuric acid |

Ans. (b) Baking powder is a mixture of baking soda NaHCO_3 , (sodium hydrogen carbonate) and a mild edible acid such as tartaric acid. When baking powder mixes with water (for making cake or bread), sodium hydrogen carbonate reacts with tartaric acid to evolve carbon dioxide gas.

Q. 13 To protect tooth decay we are advised to brush our teeth regularly. The nature of the tooth paste commonly used is

- | | |
|------------|---------------|
| (a) acidic | (b) neutral |
| (c) basic | (d) corrosive |

💡 Thinking Process

When pH of our mouth falls below 5.5 due to eating of sweets etc., (i.e., our mouth is moderately acidic) then the acid becomes strong enough to attack the enamel which is made up of calcium phosphate of our teeth and corrodes it. This sets in tooth decay.

Ans. (c) The tooth paste commonly used is basic so that the extra acid formed during tooth decay is neutralised and prevent tooth decay.

Q. 14 Which of the following statements is correct about an aqueous solution of an acid and a base?

- | | |
|---------------------------------------|--------------------|
| (i) Higher the pH, stronger the acid | |
| (ii) Higher the pH, weaker the acid | |
| (iii) Lower the pH, stronger the base | |
| (iv) Lower the pH, weaker the base | |
| (a) (i) and (iii) | (b) (ii) and (iii) |
| (c) (i) and (iv) | (d) (ii) and (iv) |

💡 Thinking Process

The common pH scale having pH value from 0 to 14. At pH 7, a solution is neutral. As the pH of solution decreases from 7 to 0, the hydrogen ion concentration in the solution goes on increasing and hence the strength of acid goes on increasing. As the pH of solution increases from 7 to 14, the hydroxide ion concentration in the solution goes on increasing, due to which the strength of base also goes on increasing.

Ans. (d) It depends on the solution i.e., higher the pH, weaker the acid and lower the pH weaker the base.

Q. 15 The pH of the gastric juices released during digestion is

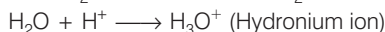
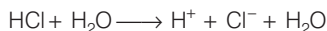
- | | |
|-----------------|-----------------|
| (a) less than 7 | (b) more than 7 |
| (c) equal to 7 | (d) equal to 0 |

Ans. (a) Our stomach produces hydrochloric acid (of pH about 1.4). This dilute hydrochloric acid helps in digesting our food.

Q. 16 Which of the following phenomena occur, when a small amount of acid is added to water?

- (i) Ionisation (ii) Neutralisation
(iii) Dilution (iv) Formation
(a) (i) and (ii) (b) (i) and (iii) (c) (ii) and (iii) (d) (ii) and (iv)

Ans. (b) (i) When water is added to an acid, their molecules dissociate to form ions.



(iii) Mixing of an acid with water is called dilution, it results in the decrease in the concentration of ions, (H_3O^+) per unit volume.

Q. 17 Which one of the following can be used as an acid-base indicator by a visually impaired student?

- (a) Litmus (b) Turmeric
(c) Vanilla essence (d) Petunia leaves

Ans. (c) Vanilla essence is an olfactory indicator. So, its smell is different in acid and basic media which can be detected easily by a visually impaired student. Vanilla extract has a characteristic pleasant smell.

If a basic solution like sodium hydroxide solution is added to vanilla extract then we cannot detect the characteristic smell of vanilla extract. An acidic solution like hydrochloric acid, however, does not destroy the smell of vanilla extract.

Q. 18 Which of the following substances will not give carbon dioxide on treatment with dilute acid?

- (a) Marble (b) Limestone
(c) Baking soda (d) Lime

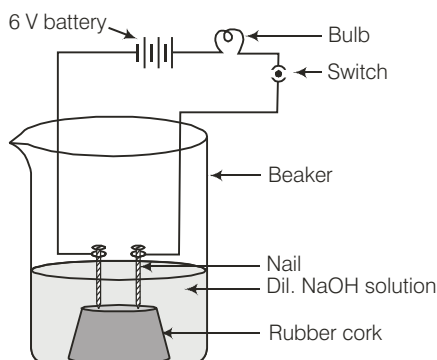
Ans. (d) Lime, CaO (calcium oxide) does not evolve CO_2 when reacted with dilute acid. Other given compounds are carbonates and hydrogen carbonates, so evolve CO_2 with dilute acid.

Q. 19 Which of the following is acidic in nature?

- (a) Lime juice (b) Human blood
(c) Lime water (d) Antacid

Ans. (a) Lime juice is acidic in nature because it contains citric acid. Human blood is slightly basic (*i.e.*, having pH 7.8). Lime water and antacid are basic in nature as they contain hydroxide (OH^-) ion.

Q. 20 In an attempt to demonstrate electrical conductivity through an electrolyte, the following apparatus (figure) was set up.



Which among the following statement(s) is/are correct?

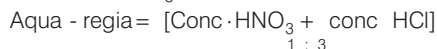
- (i) Bulb will not glow because electrolyte is not acidic.
 - (ii) Bulb will glow because NaOH is a strong base and furnishes ions for conduction.
 - (iii) Bulb will not glow because circuit is incomplete.
 - (iv) Bulb will not glow because it depends upon the type of electrolytic solution.
- (a) (i) and (iii) (b) (ii) and (iv) (c) Only (ii) (d) Only (iv)

Ans. (c) Bulb will glow because NaOH being a strong base furnishes OH^- and Na^+ ions (which are responsible for electrical conductivity).

Q. 21 Which of the following is used for dissolution of gold?

- (a) Hydrochloric acid (b) Sulphuric acid
- (c) Nitric acid (d) Aqua-regia

Ans. (d) Aqua-regia is used for the dissolution of gold. Gold dissolves only in aqua-regia. Aqua-regia is a mixture of conc. HNO_3 and conc. HCl in the ratio 1 : 3.



Q. 22 Which of the following is not a mineral acid?

- (a) Hydrochloric acid (b) Citric acid
- (c) Sulphuric acid (d) Nitric acid

💡 Thinking Process

Mineral acids or inorganic acids are generally prepared from the minerals present in the earth's crust. e.g, HCl (hydrochloric acid), H_2SO_4 (sulphuric acid) and HNO_3 (nitric acid) are mineral acids.

Edible acids or organic acids are produced by plants or animals. e.g, acetic acid, citric acid, lactic acid etc., are edible acids.

Ans. (b) Citric acid is an example of organic acid or edible acid.

Q. 23 Which among the following is not a base?

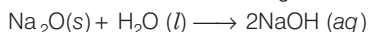
- (a) NaOH (b) KOH (c) NH_4OH (d) $\text{C}_2\text{H}_5\text{OH}$

Ans. (d) $\text{C}_2\text{H}_5\text{OH}$ is not a base. It is an example of an organic compound known as alcohol (ethyl alcohol) which has somewhat acidic nature. And also $\text{C}_2\text{H}_5\text{OH}$ does not give OH^- ions in the solution, so it is not a base.

Q. 24 Which of the following statements is not correct?

- (a) All metal carbonates react with acid to give a salt, water and carbon dioxide
 (b) All metal oxides react with water to give salt and acid
 (c) Some metals react with acids to give salt and hydrogen
 (d) Some non-metal oxides react with water to form an acid

Ans. (b) Most metal oxides are insoluble in water but some of these (e.g., Na_2O , CaO) dissolve in water to form alkalies not salt and acid e.g.,



Q. 25 Match the chemical substances given in Column I with their appropriate application given in Column II.

Column I	Column II
A. Bleaching powder	1. Preparation of glass
B. Baking soda	2. Production of H_2 and Cl_2
C. Washing soda	3. Decolourisation
D. Sodium chloride	4. Antacid

Codes

- A B C D
 (a) 2 1 4 3
 (c) 3 4 1 2

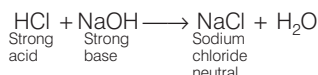
- A B C D
 (b) 3 2 4 1
 (d) 2 4 1 3

Ans. (c) Bleaching powder bleaches the clothes and other coloured substances, baking soda is a constituent of antacid, washing soda is used in the preparation of glass and sodium chloride when subjected to electrolyses gives H_2 and Cl_2 gases.

Q. 26 Equal volumes of hydrochloric acid and sodium hydroxide solutions of same concentration are mixed and the pH of the resulting solution is checked with a pH paper. What would be the colour obtained? (You may use colour guide given in figure of NCERT Book (Science Class X) on page 26).

- (a) Red (b) Yellow
 (c) Yellowish green (d) Blue

Ans. (c) Because the resulting solution is obtained as a result of neutralisation reaction.



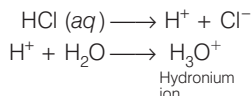
The colour of the neutral solution (with $\text{pH} = 7$) obtained is yellowish green.

Q. 27 Which of the following is/are true when HCl(g) is passed through water?

- (i) It does not ionise in the solution as it is a covalent compound.
- (ii) It ionises in the solution.
- (iii) It gives both hydrogen and hydroxyl ion in the solution.
- (iv) It forms hydronium ion in the solution due to the combination of hydrogen ion with water molecule.

- (a) Only (i) (b) Only (iii) (c) (ii) and (iv) (d) (iii) and (iv)

Ans. (c) When HCl is passed through water then HCl being a polar covalent compound, ionises in water as



Q. 28 Which of the following statement is true for acids?

- (a) Bitter and change red litmus to blue
- (b) Sour and change red litmus to blue
- (c) Sour and change blue litmus to red
- (d) Bitter and change blue litmus to red

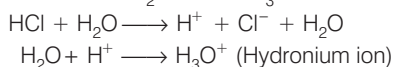
Ans. (c) Acids are those chemical substances which have a sour taste and turn blue litmus solution to red. On the other hand, bases are bitter in taste and soapy to touch and turn red litmus solution to blue.

Q. 29 Which of the following are present in a dilute aqueous solution of hydrochloric acid?

- (a) $\text{H}_3\text{O}^+ + \text{Cl}^-$ (b) $\text{H}_3\text{O}^+ + \text{OH}^-$ (c) $\text{Cl}^- + \text{OH}^-$ (d) Unionised HCl

Ans. (a) When acid is mixed with water, their molecules dissociate to form ions.

The H^+ ions combine with H_2O to form H_3O^+ ions.



Q. 30 Identify the correct representation of reaction occurring during chloralkali process.

- (a) $2\text{NaCl (l)} + 2\text{H}_2\text{O(l)} \longrightarrow 2\text{NaOH (l)} + \text{Cl}_2\text{(g)} + \text{H}_2\text{(g)}$
- (b) $2\text{NaCl (aq)} + 2\text{H}_2\text{O(aq)} \longrightarrow 2\text{NaOH (aq)} + \text{Cl}_2\text{(l)} + \text{H}_2\text{(aq)}$
- (c) $2\text{NaCl (aq)} + 2\text{H}_2\text{O(l)} \longrightarrow 2\text{NaOH (aq)} + \text{Cl}_2\text{(aq)} + \text{H}_2\text{(aq)}$
- (d) $2\text{NaCl (aq)} + 2\text{H}_2\text{O(l)} \longrightarrow 2\text{NaOH (aq)} + \text{Cl}_2\text{(g)} + \text{H}_2\text{(g)}$

Ans. (d) $2\text{NaCl (aq)} + 2\text{H}_2\text{O (l)} \longrightarrow 2\text{NaOH (aq)} + \text{Cl}_2\text{(g)} + \text{H}_2\text{(g)}$ (because state of Cl_2 and H_2 is gaseous, H_2O is liquid and that of NaCl and NaOH is aqueous).

Short Answer Type Questions

Q. 31 Match the acids given in Column I with their correct source given in Column II

Column I	Column II
A. Lactic acid	1. Tomato
B. Acetic acid	2. Lemon
C. Citric acid	3. Vinegar
D. Oxalic acid	4. Curd

Ans.

Column I	Column II
A. Lactic acid	Curd
B. Acetic acid	Vinegar
C. Citric acid	Lemon
D. Oxalic acid	Tomato

Q. 32 Match the important chemicals given in Column I with the chemical formulae given in Column II.

Column I	Column II
A. Plaster of Paris	1. Ca(OH)_2
B. Gypsum	2. $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$
C. Bleaching powder	3. $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
D. Slaked lime	4. CaOCl_2

Ans.

Column I	Column II
A. Plaster of Paris	$\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$
B. Gypsum	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
C. Bleaching powder	CaOCl_2
D. Slaked lime	Ca(OH)_2

Q. 33 What will be the action of the following substances on litmus paper?

Dry HCl gas, moistened NH_3 gas, lemon juice, carbonated soft drink, curd, soap solution.

Ans. (i) **Dry HCl gas** No change on litmus paper

(ii) **Moistened NH_3 gas** (basic) Red litmus will turn blue.

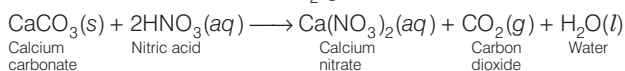
- (iii) **Lemon Juice** (contains citric acid) Blue litmus will turn red.
- (iv) **Carbonated soft drinks** (contains carbonic acid) Blue litmus will turn red.
- (v) **Curd** (contains lactic acid) Blue litmus will turn red.
- (vi) **Soap solution** (basic) Red litmus will turn blue.

Q. 34 Name the acid present in ant sting and give its chemical formula. Also give the common method to get relief from the discomfort caused by the ant sting.

Ans. Formic acid (or methanoic acid) is present in the ant sting. The chemical formula is HCOOH . By applying some wet baking soda on the affected area, it gives relief.

Q. 35 What happens when nitric acid is added to egg shell?

Ans. Egg shells contain calcium carbonate (CaCO_3). When nitric acid is added to it, brisk effervescence due to the formation of CO_2 gas is observed. The reaction is



Q. 36 A student prepared solutions of (i) an acid and (ii) a base in two separate beakers. She forgot to label the solutions and litmus paper is not available in the laboratory. Since, both the solutions are colourless, how will she distinguish between the two?

Ans. In the absence of litmus, any other indicator like methyl orange, phenolphthalein, etc., can be used. Otherwise a natural indicator like turmeric can also be used.

Some common indicators with characteristic colours are tabulated below

S.No.	Indicator	Colour in acidic solution	Colour in neutral solution	Colour in basic solution
1.	Litmus	Red	Purple	Blue
2.	Phenolphthalein	Colourless	Colourless	Pink
3.	Methyl orange	Red/Pink	Orange	Yellow
4.	Turmeric juice	Yellow	Yellow	Reddish brown

Q. 37 How would you distinguish between baking powder and washing soda by heating?

Ans. On heating NaHCO_3 (baking soda), CO_2 (carbon dioxide) gas is given out that turns lime water milky.



While on heating $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ (washing soda) water of crystallisation is given out and the salt becomes anhydrous. The presence of water of crystallisation given as product can be tested by treating it with anhydrous CuSO_4 (white) which becomes blue in colour in its contact.



Q. 38 Salt A commonly used in bakery products on heating gets converted into another salt B which itself is used for removal of hardness of water and a gas C is evolved. The gas C when passed through lime water, turns it milky. Identify A, B and C.

Ans. Salt A is sodium bicarbonate NaHCO_3 (as it is used in bakery products and gives Na_2CO_3 on heating). Salt B is sodium carbonate Na_2CO_3 (it is used for removal of hardness of water) Gas C is carbon dioxide CO_2 (as it turns lime water milky).

This can be shown as follows

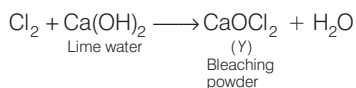
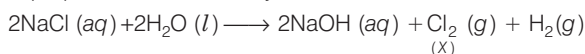


Q. 39 In one of the industrial processes for manufacture of sodium hydroxide, a gas X is formed as by-product. The gas X reacts with lime water to give a compound Y which is used as a bleaching agent in chemical industry. Identify X and Y giving the chemical equation of the reactions involved.

Ans. In the manufacture of sodium hydroxide, hydrogen gas and chlorine gas (X) are formed as by-product. Chlorine gas reacts with lime water to give bleaching powder, a bleaching agent. Thus, X is chlorine gas (Cl_2 gas).

Y is calcium oxychloride or bleaching powder (CaOCl_2).

The equation for the preparation of sodium hydroxide is



Q. 40 Fill in the missing data in the given table.

Name of the salt		Formula	Salt obtained from	
			Base	Acid
(i)	Ammonium chloride	NH_4Cl	NH_4OH	—
(ii)	Copper sulphate	—	—	H_2SO_4
(iii)	Sodium chloride	NaCl	NaOH	—
(iv)	Magnesium nitrate	$\text{Mg(NO}_3)_2$	—	HNO_3
(v)	Potassium sulphate	K_2SO_4	—	—
(vi)	Calcium nitrate	$\text{Ca(NO}_3)_2$	Ca(OH)_2	—

Ans. (i) Acid : HCl $[\because \text{NH}_4\text{OH} + \text{HCl} \longrightarrow \text{NH}_4\text{Cl} + \text{H}_2\text{O}]$

(ii) Formula : CuSO_4 $[\because \text{Cu(OH)}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{CuSO}_4 + 2\text{H}_2\text{O}]$
 Base : Cu(OH)_2

(iii) Acid : HCl $[\because \text{NaOH} + \text{HCl} \longrightarrow \text{NaCl} + \text{H}_2\text{O}]$

(iv) Base : Mg(OH)_2 $[\because \text{Mg(OH)}_2 + 2\text{HNO}_3 \longrightarrow \text{Mg(NO}_3)_2 + 2\text{H}_2\text{O}]$

(v) Base : KOH $[\because 2\text{KOH} + \text{H}_2\text{SO}_4 \longrightarrow \text{K}_2\text{SO}_4 + 2\text{H}_2\text{O}]$
 Acid : H_2SO_4

(vi) Acid : HNO_3 $[\because \text{Ca(OH)}_2 + 2\text{HNO}_3 \longrightarrow \text{Ca(NO}_3)_2 + 2\text{H}_2\text{O}]$

Q. 41 What are strong and weak acids? In the following list of acids, separate strong acids from weak acids. Hydrochloric acid, citric acid, acetic acid, nitric acid, formic acid, sulphuric acid.

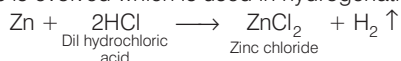
Ans. Strong acid The acid that ionises completely in aqueous solution, thus producing a high concentration of H_3O^+ ions, is called a strong acid, e.g., HCl , H_2SO_4 , HNO_3 etc.

Weak acid Weak acid ionises only partially in aqueous solution and thus it produces ions as well as molecules, e.g., acetic acid, carbonic acid.

Strong acid	Weak acid
Hydrochloric acid	Citric acid
Nitric acid	Acetic acid
Sulphuric acid	Formic acid

Q. 42 When zinc metal is treated with a dilute solution of a strong acid, a gas is evolved, which is utilised in the hydrogenation of oil. Name the gas evolved. Write the chemical equation of the reaction involved and also write a test to detect the gas formed.

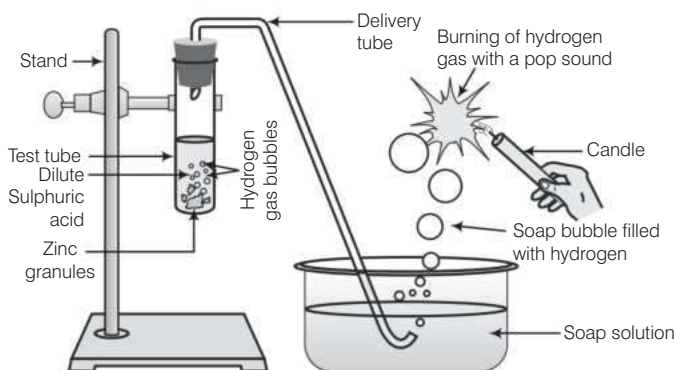
Ans. When zinc reacts with dilute solution of strong acid (like hydrochloric acid HCl), it forms salt and hydrogen gas is evolved which is used in hydrogenation of oil.



To test the presence of H_2 gas when a burning splinter is brought near the mouth of the test tube, the gas burns with a pop sound.

Long Answer Type Questions

Q. 43 In the following schematic diagram for the preparation of hydrogen gas as shown in the figure, what would happen if the following changes are made?



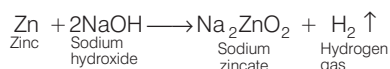
- In place of zinc granules, same amount of zinc dust is taken in the test tube.
- Instead of dilute sulphuric acid, dilute hydrochloric acid is taken.
- In place of zinc, copper turnings are taken.
- Sodium hydroxide is taken in place of dilute sulphuric acid and the tube is heated.

Ans. (a) If same amount of zinc dust is taken in the test tube then the reaction will be comparatively faster and hydrogen gas will evolve with greater speed. It is because zinc dust has larger surface area than zinc granules.

(b) With dilute hydrochloric acid, almost same amount of gas is evolved.

(c) With copper turnings, hydrogen gas will not evolve because copper is less reactive, so, it does not react with dil. H_2SO_4 or dil. HCl . Hence, no reaction will take place.

(d) Zinc also reacts with NaOH . So, if sodium hydroxide is taken, then hydrogen gas will be evolved.



Q. 44 For making cake, baking powder is taken. If at home your mother uses baking soda instead of baking powder in cake.

(a) How will it affect the taste of the cake and why?

(b) How can baking soda be converted into baking powder?

(c) What is the role of tartaric acid added to baking soda?

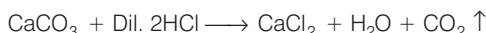
Ans. (a) The advantage of using baking powder is that tartaric acid present in baking powder reacts with sodium carbonate (Na_2CO_3) produced during decomposition of NaHCO_3 and neutralises it. If only sodium hydrogen carbonate (baking soda) is used in making cake, then sodium carbonate formed from it by the action of heat (during baking) will give a bitter taste to cake.

(b) By adding tartaric acid to baking soda we can form baking powder.

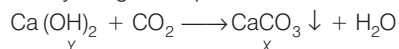
(c) Tartaric acid neutralises the sodium carbonate formed during decomposition of NaHCO_3 hence, making the cake tasty and not bitter in taste.

Q. 45 A metal carbonate X on reacting with an acid gives a gas which when passed through a solution Y gives the carbonate back. On the other hand, a gas G that is obtained at anode during electrolysis of brine is passed on dry Y, it gives a compound Z, used for disinfecting drinking water. Identify X, Y, G and Z.

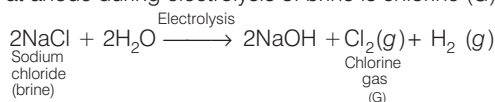
Ans. X is calcium carbonate and the gas evolved is carbon dioxide, when calcium carbonate reacts with acid.



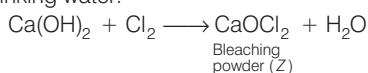
Solution Y is lime water $\text{Ca}(\text{OH})_2$ because, when CO_2 is passed through it, it gives the carbonate back as shown by the given equation.



The gas evolved at anode during electrolysis of brine is chlorine (G).



When chlorine gas is passed through dry $\text{Ca}(\text{OH})_2$ (Y), it produces bleaching powder (Z), used for disinfecting drinking water.



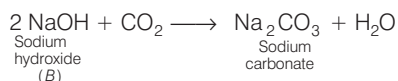
Hence, Z is calcium oxy-chloride (CaOCl_2) or bleaching powder.

Q. 46 A dry pellet of a common base B, when kept in open absorbs moisture and turns sticky. The compound is also a by-product of chlor-alkali process. Identify B, what type of reaction occurs when B is treated with an acidic oxide? Write a balanced chemical equation for one such solution.

Ans. Sodium hydroxide (NaOH) is commonly used base and is hygroscopic, that is, it absorbs moisture from the atmosphere and becomes sticky.

Thus, base B is NaOH (sodium hydroxide). It is also a by-product of chlor-alkali process. The acidic oxide reacts with base to give salt and water.

If CO₂ is the acidic oxide taken, then the following reaction takes place with 'B'.

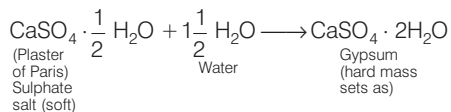


Such reaction is called neutralisation reaction.

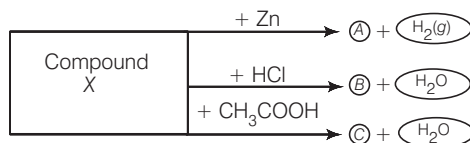
Q. 47 A sulphate salt of group 2 element of the periodic table is a white, soft substance, which can be moulded into different shapes by making its dough. When this compound is left in open for some time, it becomes a solid mass and cannot be used for moulding purposes. Identify the sulphate salt and why does it show such a behaviour? Give the reaction involved.

Ans. The sulphate salt which is used for making different shapes is plaster of Paris. Its chemical name is calcium sulphate hemihydrate ($\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$). The two formula units of CaSO₄ share one molecule of water. As a result, it is soft. When it is left open for some time, it absorbs moisture from the atmosphere and forms gypsum which is a hard solid mass.

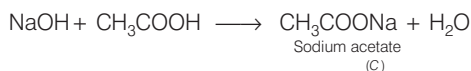
So, gypsum sets as a hard solid mass and cannot be used for moulding purposes.



Q. 48 Identify the compound X on the basis of the reactions given below. Also, write the name and chemical formulae of A, B and C.



Ans. Compound X is NaOH (sodium hydroxide).



Metals and Non-metals

Multiple Choice Questions (MCQs)

Q. 1 Which of the following property is generally not shown by metals?

- (a) Electrical conduction
- (b) Sonorous in nature
- (c) Dullness
- (d) Ductility

💡 Thinking Process

Metals are not dull, they have a shining surface. That is metals are lustrous. The property of a metal of having a shining surface is called 'metallic lustre'.

Ans. (c) A metal has a shining surface only when it is fresh. The metals lose their shine or brightness on keeping in air for a long time and acquire a dull appearance due to the formation of a thin layer of oxide, carbonate or sulphide on their surface.

If we rub the dull surface with a sand paper, the metal objects becomes shiny and bright once again.

Q. 2 The ability of metals to be drawn into thin wire is known as

- (a) ductility
- (b) malleability
- (c) sonorosity
- (d) conductivity

Ans. (a) These are all physical properties of metals. The ability of metals to be drawn into thin wire is known as ductility. Malleability is the property which allows the metals to be hammered into thin sheets.

Sonority or sonorosity is that property of metals due to which metals make sound when hit with an object. Conductivity allows heat to pass through metals easily.

Q. 3 Aluminium is used for making cooking utensils. Which of the following properties of aluminium are responsible for the same?

- (i) Good thermal conductivity
- (ii) Good electrical conductivity
- (iii) Ductility
- (iv) High melting point

- (a) (i) and (ii)
- (b) (i) and (iii)
- (c) (ii) and (iii)
- (d) (i) and (iv)

Ans. (d) Good thermal conductivity, malleability, light weight and high melting point are the properties of aluminium due to which it is used for making cooking utensils.

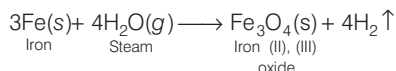
Q. 4 Which one of the following metals do not react with cold as well as hot water?

- (a) Na (b) Ca
(c) Mg (d) Fe

💡 Thinking Process

The intensity of reaction of a metal with water depends on its chemical reactivity.

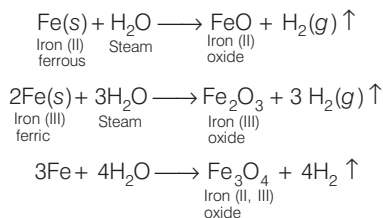
Ans. (d) Some metals like (Na, K and Ca) react with cold water. Mg metal reacts with hot water but Zn and Fe react with steam. Metals like lead, copper, silver and gold do not react with water (or even steam). Red hot iron reacts with steam to form iron (II, III) oxide and hydrogen.



Q. 5 Which of the following oxide(s) of iron would be obtained on prolonged reaction of iron with steam?

- (a) FeO (b) Fe₂O₃
(c) Fe₃O₄ (d) Fe₂O₃ and Fe₃O₄

Ans. (c) Reactions of iron metal with water

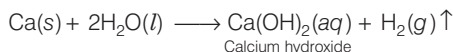


Q. 6 What happens when calcium is treated with water?

- (i) It does not react with water.
(ii) It reacts violently with water.
(iii) It reacts less violently with water.
(iv) Bubbles of hydrogen gas formed stick to the surface of calcium.

- (a) (i) and (iv) (b) (ii) and (iii)
(c) (i) and (ii) (d) (iii) and (iv)

Ans. (d) Calcium reacts less violently with water and the bubbles of hydrogen gas produced stick to the surface of calcium. Due to which it floats over water surface.



Much less heat is produced in this reaction due to which hydrogen gas formed does not catch fire.

Q. 7 Generally metals react with acids to give salt and hydrogen gas. Which of the following acids does not give hydrogen gas on reacting with metals (except Mn and Mg)?

- (a) H_2SO_4 (b) HCl (c) HNO_3 (d) All of these

💡 Thinking Process

Very dilute nitric acid, however, reacts with magnesium and manganese metals to evolve hydrogen gas. This is because, very dilute nitric acid is a weak oxidising agent which is not able to oxidise hydrogen to water.

Ans. (c) Nitric acid (HNO_3) does not give hydrogen gas on reacting with metals (except Mn and Mg) because nitric acid is a strong oxidising agent so, as soon as hydrogen gas is formed in the reaction between metal and dil. HNO_3 , the HNO_3 oxidises this hydrogen to water.

Q. 8 The composition of aqua-regia is

- (a) Dil. HCl : Conc. HNO_3 3 : 1 (b) Conc. HCl : Dil. HNO_3 3 : 1
(c) Conc. HCl : Conc. HNO_3 3 : 1 (d) Dil. HCl : Dil. HNO_3 3 : 1

💡 Thinking Process

Aqua-regia is a freshly prepared mixture of two conc. acids.

Ans. (c) Conc. HCl and conc. HNO_3 in 3 : 1 ratio form aqua-regia.

Aqua-regia is a highly corrosive, fuming liquid. It can dissolve all metals even gold and platinum also.

Q. 9 Which of the following are not ionic compounds?

- (i) KCl (ii) HCl (iii) CCl_4 (iv) NaCl
(a) (i) and (ii) (b) (ii) and (iii) (c) (iii) and (iv) (d) (i) and (iii)

💡 Thinking Process

Ionic compounds are the result of electron transfer.

Ans. (b) HCl is a polar covalent compound while CCl_4 is a non-polar covalent compound because both are formed by sharing of electrons.

KCl and NaCl are formed by the transfer of electrons and are ionic compounds.

Q. 10 Which one of the following properties is not generally exhibited by ionic compounds?

- (a) Solubility in water (b) Electrical conductivity in solid state
(c) High melting and boiling points (d) Electrical conductivity in molten state

💡 Thinking Process

Concern the ionic properties of metals.

Ans. (b) Ionic compounds can conduct electricity in molten or aqueous state only. In solid state, these are non-conductor of electricity because of the absence of free ions. In the solid ionic compounds, the ions are held together in fixed position by strong electrostatic force and cannot move freely.

So, electrical conductivity in solid state is not generally exhibited by ionic compounds.

Q. 11 Which of the following metals exist in their native state in nature?

- (i) Cu (ii) Au
(iii) Zn (iv) Ag
(a) (i) and (ii) (b) (ii) and (iii) (c) (ii) and (iv) (d) (iii) and (iv)

Ans. (c) Gold and silver (Au and Ag) are less reactive metals, so they are usually found in free state (native state) in nature.

Q. 12 Metals are refined by using different methods. Which of the following metals are refined by electrolytic refining?

- (i) Au (ii) Cu
(iii) Na (iv) K
(a) (i) and (ii) (b) (i) and (iii) (c) (ii) and (iii) (d) (ii) and (iv)

💡 Thinking Process

The process of purifying impure metal is called refining of metals.

Ans. (a) Electrolytic refining is used for metals like Cu, Zn, Ag, Au etc.

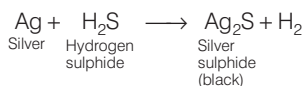
The method to be used for refining an impure metal depends on the nature of the metal as well as on the nature of impurities present in it.

So, metals Au (gold) and Cu (copper) are refined by electrolytic refining.

Q. 13 Silver articles become black on prolonged exposure to air. This is due to the formation of

- (a) Ag_3N (b) Ag_3O
(c) Ag_2S (d) Ag_2S and Ag_3N

Ans. (c) Silver article become black because silver reacts with H_2S gas present in air to form black coating of Ag_2S . The reaction is



Q. 14 Galvanisation is a method of protecting iron from rusting by coating it with a thin layer of

- (a) gallium (b) aluminium (c) zinc (d) silver

Ans. (c) Galvanisation is a method of protecting iron from rusting by coating it with a thin layer of zinc (Zn) metal.

Q. 15 Stainless steel is very useful material for our life. In stainless steel, iron is mixed with

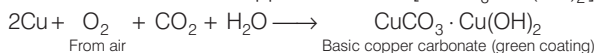
- (a) Ni and Cr (b) Cu and Cr
(c) Ni and Cu (d) Cu and Au

Ans. (a) In stainless steel iron is mixed with nickel (Ni), chromium (Cr) and carbon (C).
Stainless steel contains iron ($\approx 74\%$), chromium ($\approx 18\%$) and nickel ($\approx 8\%$).

Q. 16 If copper is kept open in air, it slowly loses its shining brown surface and gains a green coating. It is due to the formation of

- (a) CuSO_4 (b) CuCO_3 (c) $\text{Cu(NO}_3)_2$ (d) CuO

Ans. (b) Copper reacts with CO_2 present in air and forms a green coating on its surface due to the formation of basic copper carbonate $[\text{CuCO}_3 \cdot \text{Cu(OH)}_2]$ as



Q. 17 Generally, metals are solid in nature. Which one of the following metals is found in liquid state at room temperature?

- (a) Na (b) Fe (c) Cr (d) Hg

Ans. (d) Mercury (Hg) is the only metal which is found in liquid state at room temperature.

Q. 18 Which of the following metals are obtained by electrolysis of their chlorides in molten state?

- (i) Na (ii) Ca
(iii) Fe (iv) Cu
(a) (i) and (iv) (b) (iii) and (iv) (c) (i) and (iii) (d) (i) and (ii)

Ans. (d) Sodium (Na) and calcium (Ca) being highly reactive are obtained by electrolysis of their chlorides in molten state.

Q. 19 Generally, non-metals are not lustrous. Which of the following non-metals is lustrous?

- (a) Sulphur (b) Oxygen (c) Nitrogen (d) Iodine

Ans. (d) Iodine is a non-metal having lustrous appearance. It has a shining surface (like that of metals).

Q. 20 Which one of the following four metals would be displaced from the solution of its salts by other three metals?

- (a) Mg (b) Ag (c) Zn (d) Cu

Ans. (b) Silver (Ag) metal would be displaced from the solution of its salts by other three metals because Ag is less reactive than Mg, Zn and Cu.

Q. 21 2 mL each of conc. HCl , HNO_3 and a mixture of conc. HCl and conc. HNO_3 in the ratio of 3 : 1 were taken in test tubes labelled as A, B and C. A small piece of metal was put in each test tube. No change occurred in test tubes A and B but the metal got dissolved in test tube C. The metal could be

- (a) Al (b) Au (c) Cu (d) Pt

Ans. (b, d)

A mixture of conc. HCl and conc. HNO_3 in the ratio of 3 : 1 is known as aqua-regia. Gold (Au) and platinum (Pt) dissolve only in aqua-regia.

Q. 22 An alloy is

- (a) an element (b) a compound
(c) a homogeneous mixture (d) a heterogeneous mixture

Ans. (c) An alloy is a homogeneous mixture of different metals or a metal and a non-metal.

Q. 23 An electrolytic cell consists of

- (i) positively charged cathode (ii) negatively charged anode
(iii) positively charged anode (iv) negatively charged cathode
(a) (i) and (ii) (b) (iii) and (iv) (c) (i) and (iii) (d) (ii) and (iv)

Ans. (b) An electrolytic cell consists of positively charged anode and negatively charged cathode.

Q. 24 During electrolytic refining of zinc, it gets

- (a) deposited on cathode (b) deposited on anode
(c) deposited on cathode as well as anode (d) remains in the solution

Ans. (a) During electrolytic refining of zinc (Zn), the pure zinc metal is deposited on cathode. It is connected to the negative terminal of the battery. Zn^{2+} ions from the solution move towards the negatively charged cathode. Hence, Zn is deposited on the cathode.

Q. 25 An element A is soft and can be cut with a knife. This is very reactive to air and cannot be kept open in air. It reacts vigorously with water. Identify the element from the following.

- (a) Mg (b) Na (c) P (d) Ca

Ans. (b) From the properties of the element described in the question, it can be said surely that the element is sodium (Na).

Q. 26 Alloys are homogeneous mixtures of a metal with a metal or non-metal. Which among the following alloys contain non-metal as one of its constituents?

- (a) Brass (b) Bronze (c) Amalgam (d) Steel

Ans. (d) Steel is an alloy which contains iron (a metal) and carbon (a non-metal). If we consider other options, then—Brass is an alloy of two metals copper and zinc. Bronze is an alloy of copper and tin. Amalgam is also formed by mixing mercury with other metal.

Q. 27 Which among the following statements is incorrect for magnesium metal?

- (a) It burns in oxygen with a dazzling white flame
(b) It reacts with cold water to form magnesium oxide and evolves hydrogen gas
(c) It reacts with hot water to form magnesium hydroxide and evolves hydrogen gas
(d) It reacts with steam to form magnesium hydroxide and evolves hydrogen gas

Ans. (b) Magnesium metal does not react with cold water.

Mg reacts with hot water and steam both, to give magnesium hydroxide and hydrogen.



Q. 28 Which among the following alloys contain mercury as one of its constituents?

- (a) Stainless steel (b) Alnico (c) Solder (d) Zinc amalgam

💡 Thinking Process

An alloy of mercury metal with one or more other metals is known as an amalgam.

Ans. (d) When mercury reacts with a zinc metal, it forms the zinc amalgam.

Q. 29 Reaction between X and Y, forms compound Z. X loses electron and Y gains electron. Which of the following properties is not shown by Z?

- (a) Has high melting point (b) Has low melting point
(c) Conducts electricity in molten state (d) Occurs as solid

💡 Thinking Process

Ionic compounds are formed by transfer of electrons.

Ans. (b) During reaction between X and Y, the compound Z is formed, which is an ionic compound. Ionic compounds have high melting point, conduct electricity in molten state and usually occur as solids.

Q. 30 The electronic configurations of three elements X, Y and Z are X — 2, 8; Y — 2, 8, 7 and Z — 2, 8, 2. Which of the following is correct?

- (a) X is a metal (b) Y is a metal
(c) Z is a non-metal (d) Y is a non-metal and Z is a metal

Ans. (d) Y is a non-metal because it requires one electron to complete its octate. Z is a metal because it loses two electrons to complete its octate. X is an inert gas as it has complete octate.

Q. 31 Although metals form basic oxides, which of the following metals form an amphoteric oxide?

- (a) Na (b) Ca (c) Al (d) Cu

💡 Thinking Process

Though most of the metal oxides are basic in nature but some of the metal oxides show basic as well as acidic behaviour. These metal oxides are known as amphoteric oxides. Aluminium metal and zinc metal form amphoteric oxides.

Ans. (c) Aluminium (Al) metal forms amphoteric oxide (Al_2O_3) which shows basic as well as acidic behaviour.

Q. 32 Generally, non-metals are not conductors of electricity. Which of the following is a good conductor of electricity?

- (a) Diamond (b) Graphite (c) Sulphur (d) Fullerene

💡 Thinking Process

Carbon is a non-metal and it has two allotropes—Diamond and graphite. A graphite crystal consists of layers of carbon atoms or sheets of carbon atoms. Each carbon atom in a graphite layer is joined to three other carbon atoms by strong covalent bonds to form flat hexagonal rings. The fourth valence electron of each carbon is 'free' to move. Due to the presence of 'free electrons' in a graphite crystal, it conducts electricity.

Ans. (b) Carbon (in the form of graphite) is a non-metal which conducts electricity.

Q. 33 Electrical wires have a coating of an insulating material. The material, generally used is

- (a) sulphur (b) graphite (c) PVC (d) All can be used

Ans. (c) PVC is an insulating substance. It does not allow electric current to pass through it. The electrical wires have a covering of PVC around them.

Q. 34 Which of the following non-metals is a liquid?

- (a) Carbon (b) Bromine (c) Phosphorus (d) Sulphur

Ans. (b) Bromine is a liquid non-metal.

Q. 35 Which of the following can undergo a chemical reaction?

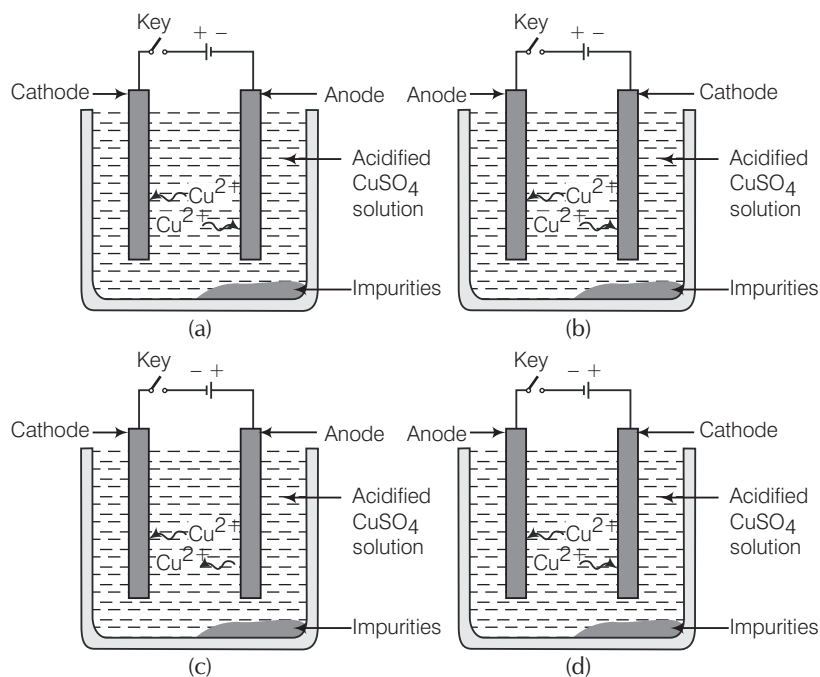
- (a) $\text{MgSO}_4 + \text{Fe}$ (b) $\text{ZnSO}_4 + \text{Fe}$ (c) $\text{MgSO}_4 + \text{Pb}$ (d) $\text{CuSO}_4 + \text{Fe}$

Ans. (d) In options (a) and (b), Fe is less reactive than Mg and Zn, so it is not able to replace these, hence, no reaction takes place.

In option (c), Pb is less reactive than Mg, so it is not able to replace it, hence no reaction takes place.

In option (d), Fe is more reactive than Cu, so it replaces Cu from CuSO_4 , hence reaction takes place.

Q. 36 Which one of the following figures correctly describes the process of electrolytic refining?



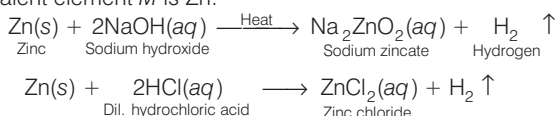
Ans. (c) For the process of electrolytic refining, impure metal is made anode and is connected to the positive terminal of battery. Pure metal is made cathode and is connected to the negative terminal of the battery.

Cu^{2+} ions from the solution are deposited on the cathode while Cu from impure anode dissolves into the solution and the impurities settle down below anode as anode mud.

Short Answer Type Questions

Q. 37 Iqbal treated a lustrous, divalent element *M* with sodium hydroxide. He observed the formation of bubbles in reaction mixture. He made the same observations when this element was treated with hydrochloric acid. Suggest how can he identify the produced gas. Write chemical equations for both the reactions.

Ans. The divalent element *M* is Zn.



The produced gas can be identified by bringing a burning splinter near the gas (mouth of the reaction vessel). The gas burns with a pop sound, hence it is hydrogen.

Q. 38 During extraction of metals, electrolytic refining is used to obtain pure metals. (a) Which material will be used as anode and cathode for refining of silver metal by this process? (b) Suggest a suitable electrolyte also. (c) In this electrolytic cell, where do we get pure silver after passing electric current?

Ans. (a) Thin strip of pure silver is made the cathode (It is connected to the negative terminal of the battery). And impure, silver block is made anode (It is connected to the positive terminal of the battery).

(b) The electrolyte is any salt solution of silver like AgNO_3 but sodium argentocyanide $\text{Na}[\text{Ag}(\text{CN})_2]$ is preferred.

(c) On passing electric current, impure metal dissolves from the anode and goes into the electrolyte solution and pure metal from the electrolyte deposits on the cathode.

Q. 39 Why should the metal sulphides and carbonates be converted to metal oxides in the process of extraction of metal from them?

Ans. This is because it is easier to obtain metals from their oxides (by reduction) than from carbonate or sulphide ores. So, before reduction the ore must be converted into metal oxide which can then be reduced.

Calcination or roasting is done depending on the nature of ore, to convert the ore into oxide. A carbonate ore is converted into oxide by calcination whereas a sulphide ore is converted into oxide by roasting.

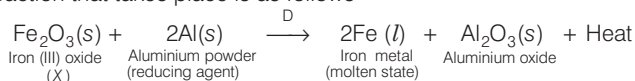
Q. 40 Generally, when metals are treated with mineral acids, hydrogen gas is liberated but when metals (except Mn and Mg) are treated with HNO_3 , hydrogen is not liberated, why?

Ans. Nitric acid (HNO_3) is a strong oxidising agent. So, as soon as hydrogen gas is formed in the reaction between a metal and dilute nitric acid, the nitric acid oxidises this hydrogen to water and itself HNO_3 gets reduced to NO_2 or NO or N_2O .

So, in the reaction of metals (except Mn and Mg), with dilute nitric acid, no hydrogen gas is evolved.

Q. 41 Compound X and aluminium are used to join railway tracks. (a) Identify the compound X (b) Name the reaction (c) Write down its reaction.

- Ans.** (a) Compound X is iron (III) oxide, Fe_2O_3 .
 (b) The reaction is known as thermite reaction or 'alumino thermy'
 (c) The reaction that takes place is as follows



Q. 42 When a metal X is treated with cold water, it gives a basic salt Y with molecular formula XOH (molecular mass = 40) and liberates a gas Z which easily catches fire. Identify X, Y and Z and also write the reaction involved.

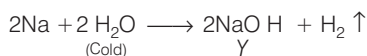
Ans. Molecular formula of $\text{Y} = \text{XOH}$

Let the atomic weight of metal X is a.

Then, molecular mass of $\text{XOH} = a + 16 + 1 = 40$ (Given)

$$a = 40 - 17 = 23$$

Thus, metal X is sodium (Na) because 23 is the atomic weight of sodium metal. Sodium reacts with water as



Sodium (Na) gives hydrogen gas (Z) on reaction with cold water.

So, Y is NaOH (sodium hydroxide).

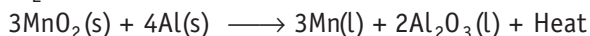
Z is H_2 (hydrogen gas).

Q. 43 A non-metal X exists in two different forms Y and Z. Y is hardest natural substance, whereas Z is a good conductor of electricity. Identify X, Y and Z.

Ans. A non-metal X is carbon (C). Carbon exists in two different forms called the allotropes of carbon. These allotropes are diamond and graphite.

Y is diamond because diamond is the hardest natural substance. Z is graphite which is good conductor of electricity.

Q. 44 The following reaction takes place when aluminium powder is heated with MnO_2



(a) Is aluminium getting reduced?

(b) Is MnO_2 getting oxidised?

💡 Thinking Process

(i) The addition of oxygen to a substance or removal of hydrogen from a substance is called oxidation.

(ii) The addition of hydrogen to a substance or removal of oxygen from a substance is called reduction.

Ans. (a) No, aluminium is getting oxidised because there is addition of oxygen.

(b) No, MnO_2 is getting reduced because there is removal of oxygen.

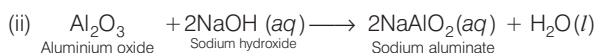
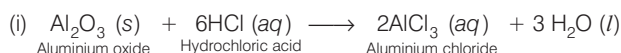
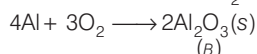
Q. 45 What are the constituents of solder alloy? Which property of solder makes it suitable for welding electrical wires?

Ans. Solder is an alloy of lead (Pb) and tin (Sn). It contains 50% lead and 50% tin. Solder has a low melting point. So, it is used for soldering (or welding) electrical wires together.

Q. 46 A metal A, which is used in thermite process, when heated with oxygen gives an oxide B, which is amphoteric in nature. Identify A and B. Write down the reactions of oxide B with HCl and NaOH.

Ans. Metal A is aluminium (Al) which is used in thermite process.

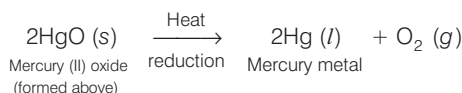
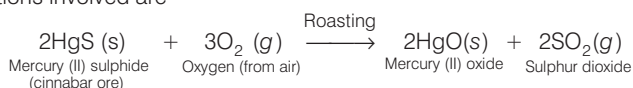
Al reacts with oxygen to form aluminium oxide Al_2O_3 (B) which is amphoteric in nature.



Q. 47 A metal that exists as a liquid at room temperature is obtained by heating its sulphide in the presence of air. Identify the metal and its ore and give the reactions involved.

Ans. The metal is mercury (Hg) because it is the only metal which exists in liquid state. Mercury is obtained from the sulphide ore called cinnabar ore (HgS).

The reactions involved are



Q. 48 Give the formulae of the stable binary compounds that would be formed by the combination of following pairs of elements.

(a) Mg and N_2

(b) Li and O_2

(c) Al and Cl_2

(d) K and O_2

Ans. (a) Mg_3N_2 Magnesium nitride

(b) Li_2O Lithium oxide

(c) AlCl_3 Aluminium chloride

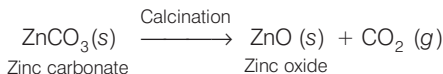
(d) K_2O Potassium oxide

Q. 49 What happens when

(a) ZnCO_3 is heated in the absence of oxygen?

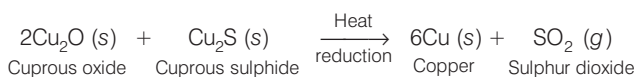
(b) a mixture of Cu_2O and Cu_2S is heated?

Ans. (a) Zinc oxide and carbon dioxide are produced when ZnCO_3 is heated in the absence of oxygen. This process is called calcination.



Moisture and volatile impurities are also expelled out during the process of calcination.

(b) Copper metal and sulphur dioxide gas are produced when a mixture of Cu_2O and Cu_2S is heated.



Q. 50 A non-metal A is an important constituent of our food and forms two oxides B and C. Oxide B is toxic whereas C causes global warming.

(a) Identify A, B and C.

(b) To which group of periodic table does A belong?

Ans. (a) Non-metal A is carbon. Its two oxides are carbon monoxide (CO) and carbon dioxide. From these, B is carbon monoxide (CO) as it is toxic while C is carbon dioxide (CO_2), because it is responsible for global warming.

(b) Electronic configuration of C is 2, 4 (atomic weight of C = 6). It is present in 14th group (10 + valence electrons), i.e., IV A of the periodic table.

Q. 51 Give two examples each of the metals that are good conductors and comparatively poor conductors of heat respectively.

Ans. **Good conductors** Silver and copper.

Poor conductors Lead and mercury.

Q. 52 Name one metal and one non-metal that exist in liquid state at room temperature. Also name two metals having melting point less than 310 K (37°C).

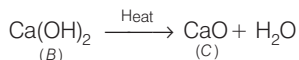
Ans. Metal which exists in liquid state is mercury (Hg) and non-metal which exists in liquid state is bromine (Br).

Two metals having melting point less than 310 K (37°C) are caesium (Cs) and gallium (Ga) which have melting point 28.5°C and 30°C respectively.

Q. 53 An element A reacts with water to form a compound B which is used in white washing. The compound B on heating forms an oxide C which on treatment with water gives back B. Identify A, B and C and give the reactions involved.

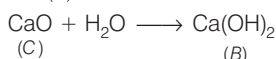
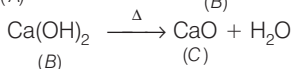
Ans. Element A is calcium (Ca). When it reacts with water, it forms calcium hydroxide. Thus, compound B is calcium hydroxide $[\text{Ca}(\text{OH})_2]$, which is used in white washing.

Compound *B* on heating gives *CaO*.



Thus, *C* is calcium oxide (*CaO*).

The reactions involved are $\underset{(A)}{\text{Ca}} + 2\text{H}_2\text{O} \longrightarrow \underset{(B)}{\text{Ca(OH)}_2} + \text{H}_2 \uparrow$



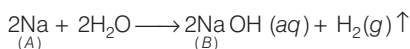
Q. 54 An alkali metal *A* gives a compound *B* (molecular mass = 40) on reacting with water. The compound *B* gives a soluble compound *C* on treatment with aluminium oxide. Identify *A*, *B* and *C* and give the reactions involved.

Ans. Let, the atomic weight of alkali metal *A* is *x*. When it reacts with water it forms a compound *B* having molecular mass 40. Let, the reaction is $2A + 2\text{H}_2\text{O} \longrightarrow 2\text{AOH} + \text{H}_2 \uparrow$
According to question, molecular mass of compound

$$B = x + 16 + 1 = 40 \quad (\text{Given})$$

$$\therefore x = 40 - 17 = 23$$

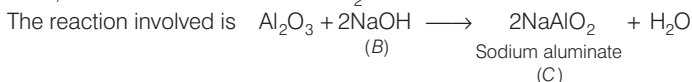
It is the atomic weight of Na (sodium), Therefore, the alkali metal (*A*) is Na and the reaction is



So, compound *B* is sodium hydroxide (*NaOH*)

Sodium hydroxide reacts with aluminium oxide (*Al₂O₃*) to give sodium aluminate (*NaAlO₂*)

Thus, *C* is sodium aluminate *NaAlO₂*

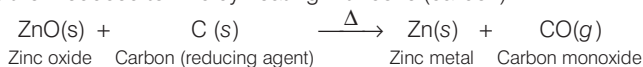


Q. 55 Give the reaction involved during extraction of zinc from its ore by
(a) roasting of zinc ore. (b) calcination of zinc ore.

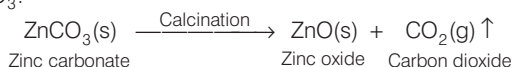
Ans. (a) **Roasting of zinc ore** Roasting is done for the sulphide ore. Sulphide ore of zinc is *ZnS* (zinc blende)



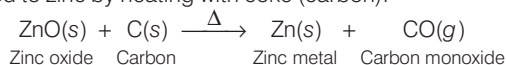
ZnO is then reduced to zinc by heating with coke (carbon)



(b) **Calcination of zinc ore** Calcination is done for the carbonate ore. Carbonate ore of Zn is calamine *ZnCO₃*.



ZnO is then reduced to zinc by heating with coke (carbon).

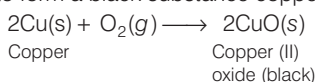


Q. 56 A metal M does not liberate hydrogen from acids but reacts with oxygen to give a black colour product. Identify M and black coloured product and also explain the reaction of M with oxygen.

Ans. Metal M is Cu as Cu is less reactive than hydrogen. So, it does not react with acid to liberate hydrogen.



Copper metal does not burn in air even on strong heating. Copper reacts with the oxygen of air on prolonged heating to form a black substance copper (II) oxide.

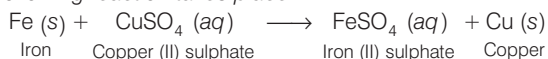


Q. 57 An element forms an oxide A_2O_3 which is acidic in nature. Identify A as a metal or non-metal.

Ans. Oxides of non-metals are acidic in nature while those of metals are basic. As the element A forms an acidic oxide, therefore, A must be a non-metal. Also it should have + 3 charge, i.e., it should have 3 valence electrons. Therefore, A is boron, (electronic configuration is 2, 3) and its oxide is B_2O_3 .

Q. 58 A solution of CuSO_4 was kept in an iron pot. After few days the iron pot was found to have a number of holes in it. Explain the reason in terms of reactivity. Write the equation of the reaction involved.

Ans. As iron is more reactive metal than copper, it displaces copper from copper sulphate solution. *The following reaction takes place*



Since, iron takes part in this reaction, therefore holes are produced at places where iron metal has reacted to form iron (II) sulphate.

Long Answer Type Questions

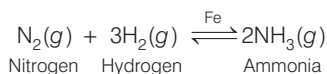
Q. 59 A non-metal A which is the largest constituent of air, when heated with H_2 in 1 : 3 ratio in the presence of catalyst (Fe) gives a gas B. On heating with O_2 it gives an oxide C. If this oxide is passed into water in the presence of air, it gives an acid D which acts as a strong oxidising agent.

(a) Identify A, B, C and D.

(b) To which group of the periodic table does this non-metal belongs?

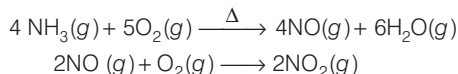
Ans. (a) Non-metal A is nitrogen gas N_2 since, the percentage of N_2 in air is largest, i.e., 78%.

When N_2 is heated with H_2 in 1:3 ratio in presence of Fe as catalyst, it forms ammonia (NH_3).



Thus, *B* is ammonia (NH_3) gas.

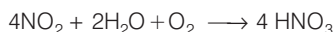
Gas *B*, when heated with O_2 , it forms nitric oxide (NO). NO further gets oxidised to NO_2 by O_2 of the air. Thus, *C* is NO_2 (nitrogen dioxide).



When this oxide is passed into water in the presence of air (O_2), it gives nitric acid (HNO_3) which acts as a strong oxidising agent.

Thus,

D is nitric acid (HNO_3)

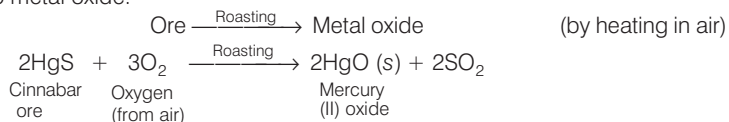


(b) The non-metal belongs to group 15 or VA because N has 5 valence electrons (electronic configuration is 2, 5).

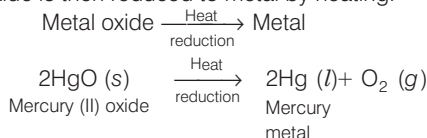
Q. 60 Give the steps involved in the extraction of metals of low and medium reactivity from their respective sulphide ores.

Ans. For metals, low in the reactivity series like mercury (Hg), the following steps are involved

(i) **Roasting** (heating the ore strongly in the presence of excess of air) Metal sulphide is converted into metal oxide.



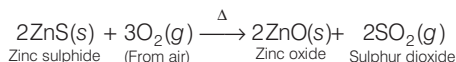
(ii) **Reduction** Metal oxide is then reduced to metal by heating.



(iii) **Refining of metal.**

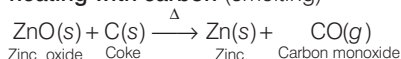
For metals, medium in reactivity series like zinc (Zn), the following steps are involved

(i) **Roasting** (for sulphide ores) The sulphide ore is converted into metal oxide by heating the sulphide ore strongly in the presence of air.

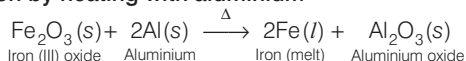


(ii) **Reduction** Metal oxides are reduced to metals by using a suitable reducing agent. This can be done by either of two.

(a) **Reduction by heating with carbon** (smelting)



(b) **Reduction by heating with aluminium**



Q. 61 Explain the following

- Reactivity of Al decreases if it is dipped in HNO_3 .
- Carbon cannot reduce the oxides of Na or Mg.
- NaCl is not a conductor of electricity in solid state whereas it does conduct electricity in aqueous solution as well as in molten state.
- Iron articles are galvanised.
- Metals like Na, K, Ca and Mg are never found in their free state in nature.

Ans. (a) HNO_3 is an oxidising agent.

When aluminium is dipped in HNO_3 , an oxide layer of aluminium is formed on the surface of the metal, which prevents it from further reaction. Thus, reactivity of Al decreases.

- (b) Na, Mg, etc., metals are quite reactive and present towards the top of the reactivity series. Hence, their affinity with oxygen is higher than carbon.

Therefore, their oxides are stable. To reduce them with carbon, very high temperature is required and at that temperature they will form their corresponding carbides. Hence, their oxides cannot be reduced by carbon.

- (c) Ions of NaCl cannot move to carry the charge in dry or solid state as they are fixed. But they are free in molten state and in aqueous solution to carry the charge and hence, NaCl conducts electricity in molten state.

- (d) In galvanisation a thin layer of zinc is formed over the iron articles by dipping them in molten zinc, which prevents iron from corrosion as zinc is more reactive than iron.

- (e) These metals are quite reactive so they cannot exist in free state and hence, they are found in nature in the form of their compounds.

Q. 62 (a) Given below are the steps for extraction of copper from its ore. Write the reaction involved.

(i) Roasting of copper (I) sulphide.

(ii) Reduction of copper (I) oxide with copper (I) sulphide.

(iii) Electrolytic refining

- (b) Draw a neat and well labelled diagram for electrolytic refining of copper.

Ans. (a) (i) $2\text{Cu}_2\text{S}(\text{s}) + 3\text{O}_2(\text{g}) \xrightarrow{\text{Heat}} 2\text{Cu}_2\text{O}(\text{s}) + 2\text{SO}_2(\text{g})$

(ii) $2\text{Cu}_2\text{O}(\text{s}) + \text{Cu}_2\text{S}(\text{s}) \xrightarrow{\text{Heat}} 6\text{Cu}(\text{s}) + \text{SO}_2(\text{g})$

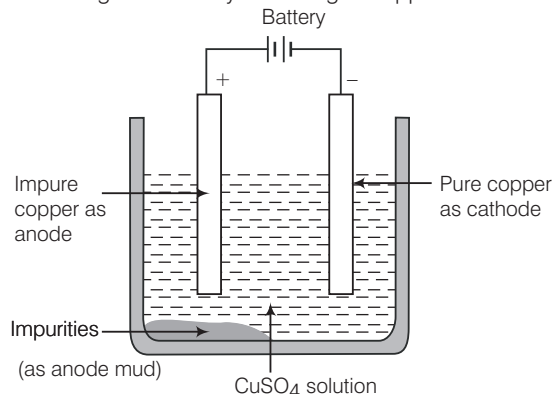
(iii) **Electrolytic refining**

- A thick block of the impure metal is made anode (+ ve).
- A thin strip of pure metal is made cathode (– ve).
- CuSO_4 solution is taken as electrolyte.

At cathode $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \longrightarrow \text{Cu}(\text{s})$

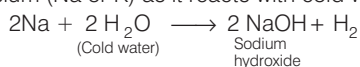
At anode $\text{Cu}(\text{s}) \longrightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$

(b) The diagram showing the electrolytic refining of copper.

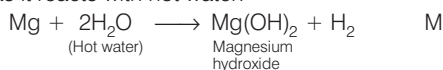


Q. 63 Of the three metals X, Y and Z, X reacts with cold water, Y with hot water and Z with steam only. Identify X, Y and Z and also arrange them in order of increasing reactivity.

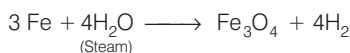
Ans. X is sodium or potassium (Na or K) as it reacts with cold water.



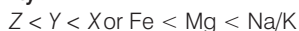
Y is magnesium (Mg) as it reacts with hot water.



Z is iron (Fe).



Increasing order of reactivity

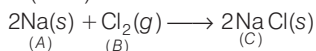


Q. 64 An element A burns with golden flame in air. It reacts with another element B, atomic number 17 to give a product C. An aqueous solution of product C on electrolysis gives a compound D and liberates hydrogen. Identify A, B, C and D. Also write down the equations for the reactions involved.

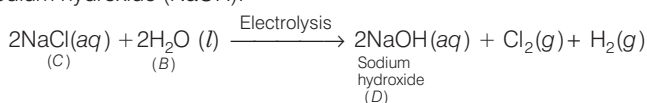
Ans. A is sodium (Na) because it burns with golden flame in air.

B is chlorine (Cl) as its atomic number is 17. When Na reacts with Cl_2 , NaCl is formed.

Hence, C is sodium chloride (NaCl).



Aqueous solution of NaCl, on electrolysis gives sodium hydroxide. Thus, D is sodium hydroxide (NaOH).

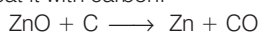


Q. 65 Two ores A and B were taken. On heating, ore A gives CO_2 whereas, ore B gives SO_2 . What steps will you take to convert them into metals?

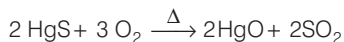
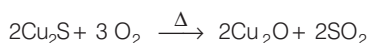
Ans. Since ore A gives CO_2 on heating, it must be a carbonate of moderately reactive metal like Zn. Hence, it will be first subjected to calcination (to convert carbonate ore into metal oxide) followed by reduction (to convert metal oxide to metal).
e.g., ZnCO_3 (calamine).



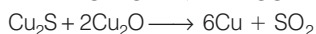
To convert it into metal treat it with carbon.



Since, B gives SO_2 , B must be a sulphide ore, e.g., Cu_2S , ZnS , HgS , etc. Hence it will be first subjected to roasting (to convert sulphide ore to metal oxide) followed by reduction (to convert metal oxide to metal).



From these oxides metal is obtained either by using reducing agent like C or by autoreduction.



Carbon and its Compounds

Multiple Choice Questions (MCQs)

Q. 1 Carbon exists in the atmosphere in the form of

- (a) Only carbon monoxide
- (b) carbon monoxide in traces and carbon dioxide
- (c) Only carbon dioxide
- (d) coal

Ans. (c) Carbon exists in the atmosphere in the form of carbon dioxide gas (CO_2) in air (only 0.03%). Carbon also occurs in the earth's crust in the form of minerals like carbonates. It also occurs in the form of fossil fuels, organic compounds, wood, cotton and wool, etc.

Q. 2 Which of the following statements are usually correct for carbon compounds? These

- (i) are good conductors of electricity.
 - (ii) are poor conductors of electricity.
 - (iii) have strong forces of attraction between their molecules.
 - (iv) do not have strong forces of attraction between their molecules.
- (a) (i) and (iii) (b) (ii) and (iii) (c) (i) and (iv) (d) (ii) and (iv)

💡 Thinking Process

Atomic number of carbon is 6. So, its electronic configuration is 2, 4. It can achieve the inert gas electron arrangement only by the sharing of electrons.

Ans. (d) Carbon mainly forms covalent compounds, which due to the absence of free ions, are generally poor conductors of electricity. Also the forces of attraction in between their molecules are weak.

Q. 3 A molecule of ammonia (NH_3) has

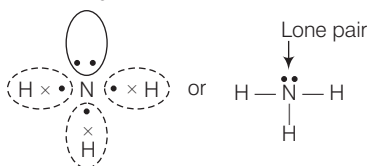
- (a) Only single bonds
- (b) Only double bonds
- (c) Only triple bonds
- (d) two double bonds and one single bond

💡 Thinking Process

Atomic number of N is 7.

Its electronic configuration is 2, 5 so it needs 3 more electrons to complete its octet. It gains 3 electrons each from 3 hydrogen atoms. Also N has one lone pair of electrons.

Ans. (a) A molecule of ammonia (NH_3) has only single bonds and these are covalent bonds.



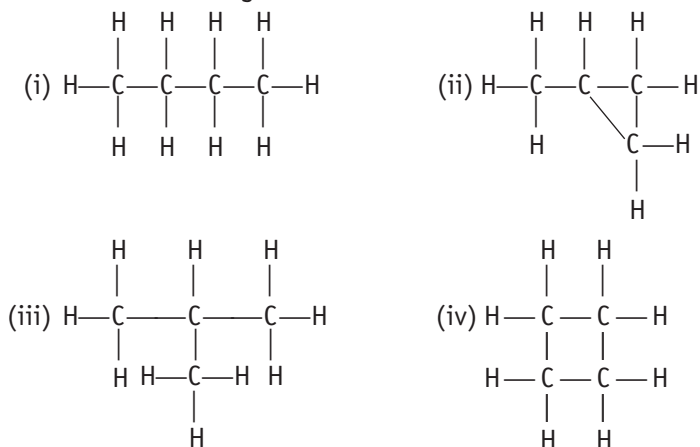
Note The pair of electrons on the nitrogen atom in NH_3 molecule which is not involved in bond formation is called lone pair.

Q. 4 Buckminsterfullerene is an allotropic form of

- (a) phosphorus (b) sulphur (c) carbon (d) tin

Ans. (c) Buckminsterfullerene is an allotrope of carbon containing clusters of 60 carbon atoms joined together to form spherical molecules. Its formula is C_{60} (C-Sixty). It is a dark solid at room temperature and as compared to another allotropic form of carbon (diamond and graphite), it is neither very hard nor soft.

Q. 5 Which of the following are correct structural isomers of butane?



- (a) (i) and (iii) (b) (ii) and (iv) (c) (i) and (ii) (d) (iii) and (iv)

Thinking Process

*Isomers are the compounds having the same molecular formula but different structures.
Molecular formula of butane is C_4H_{10} .*

Ans. (a) Structure (i) is *n*-butane

Structure (iii) is *iso*-butane

Since, molecular formula is same, only structures are different. So, (i) and (iii) are isomers while structure (ii) and (iv) have molecular formula C_5H_{12} .

Q. 6 $\text{CH}_3 - \text{CH}_2 - \text{OH} \xrightarrow{\text{Alkaline KMnO}_4 + \text{Heat}} \text{CH}_3 - \text{COOH}$

In the above given reaction, alkaline KMnO_4 acts as

- (a) reducing agent (b) oxidising agent
(c) catalyst (d) dehydrating agent

💡 Thinking Process

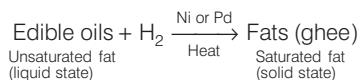
Oxidising agent is a substance which gives oxygen to the compound for oxidation (i.e., oxygen is removed from it) or which removes hydrogen from the compound. In other words we can say that $\text{CH}_3\text{CH}_2\text{OH}$ is oxidised to CH_3COOH in the presence of oxidising agent like alk KMnO_4 .

Ans. (b) KMnO_4 acts as oxidising agent because it removes hydrogen from $\text{CH}_3\text{CH}_2\text{OH}$ and adds one oxygen to it.

Q. 7 Oils on treating with hydrogen in the presence of palladium or nickel catalyst form fats. This is an example of

- (a) addition reaction (b) substitution reaction
(c) displacement reaction (d) oxidation reaction

Ans. (a) Oils are unsaturated compounds containing double bonds. Addition reactions are characteristic property of unsaturated hydrocarbons. The given reaction is an example of addition reaction.



Q. 8 In which of the following compounds $-\text{OH}$ is the functional group?

- (a) Butanone (b) Butanol (c) Butanoic (d) Butanal

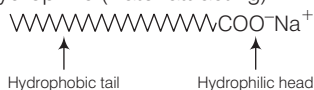
Ans. (b) Butanol, $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{OH}$

The general formula of alcohols is $\text{C}_n\text{H}_{2n+1}-\text{OH}$. For butanol, $n = 4$ So, formula is $\text{C}_4\text{H}_9-\text{OH}$ or $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{OH}$

Q. 9 The soap molecule has a

- (a) hydrophilic head and a hydrophobic tail
(b) hydrophobic head and a hydrophilic tail
(c) hydrophobic head and a hydrophobic tail
(d) hydrophilic head and a hydrophilic tail

Ans. (a) A soap molecule is made up of two parts: A long hydrocarbon part and a short ionic part $-\text{COO}^-\text{Na}^+$ group. The long hydrocarbon chain is hydrophobic (water repelling) and ionic portion is hydrophilic (water attracting).



Q. 10 Which of the following is the correct representation of electron dot structure of nitrogen?

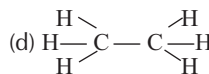
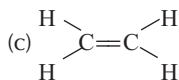
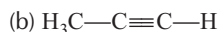
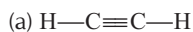
- (a) $\cdot\ddot{\text{N}}:\ddot{\text{N}}\cdot$ (b) $\cdot\ddot{\text{N}}::\ddot{\text{N}}\cdot$ (c) $\cdot\ddot{\text{N}}:\ddot{\text{N}}\cdot$ (d) $\cdot\text{N}::\text{N}\cdot$

Ans. (d) Electronic configuration of N (atomic number 7) is $\begin{smallmatrix} K & L \\ 2 & 5 \end{smallmatrix}$

Therefore, it needs three more electrons to complete its octet. Each nitrogen atom shares three electrons to form a molecule of N_2 as



Q. 11 Structural formula of ethyne is



Thinking Process

General formula of alkyne is $\text{C}_n\text{H}_{2n-2}$

Ans. (a) 'eth' shows presence of two carbon atoms and 'yne' shows presence of a triple bond. Thus, ethyne has the structure, $\text{H}-\text{C}\equiv\text{C}-\text{H}$. It is also known as acetylene.

Q. 12 Identify the unsaturated compounds from the following.

(i) Propane (ii) Propene (iii) Propyne (iv) Chloropropane

(a) (i) and (ii) (b) (ii) and (iv) (c) (iii) and (iv) (d) (ii) and (iii)

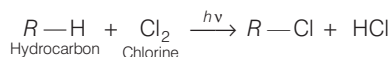
Ans. (d) A hydrocarbon in which the two carbon atoms are connected by a 'double bond' or a 'triple bond' is called an unsaturated hydrocarbon.

(ii) Propene, $\text{CH}_3\text{CH}=\text{CH}_2$ and (iii) propyne $\text{CH}_3-\text{C}\equiv\text{CH}$ both have double and triple bonds, respectively hence unsaturated. Propane and chloropropane are saturated hydrocarbons which contain only single bonds.

Q. 13 Chlorine reacts with saturated hydrocarbons at room temperature in the

(a) absence of sunlight (b) presence of sunlight
(c) presence of water (d) presence of hydrochloric acid

Ans. (b) Chlorine reacts with saturated hydrocarbon at room temperature in the presence of sunlight.

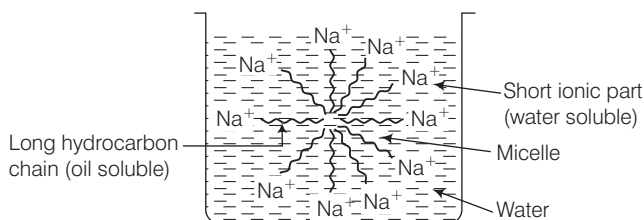


(—R is any alkyl group i.e., $-\text{CH}_3$, $-\text{C}_2\text{H}_5$, etc.)

Q. 14 In the soap micelles

(a) the ionic end of soap is on the surface of the cluster while the carbon chain is in the interior of the cluster
(b) ionic end of soap is in the interior of the cluster and the carbon chain is out of the cluster
(c) both ionic end and carbon chain are in the interior of the cluster
(d) both ionic end and carbon chain are on the exterior of the cluster

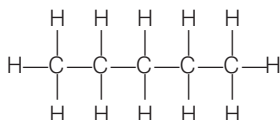
Ans. (a) A 'spherical aggregate of soap molecules' in the soap solution in water is called a 'micelle'. In a soap micelle, the soap molecules are arranged readily with hydrocarbon ends directed towards the centre and ionic ends directed outwards.



Q. 15 Pentane has the molecular formula C_5H_{12} . It has

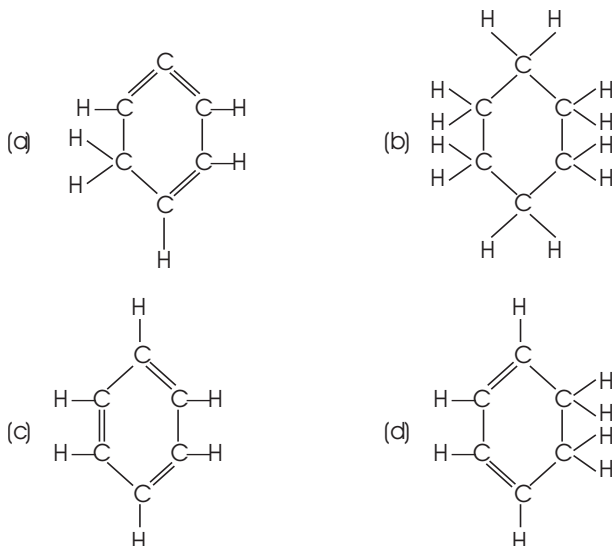
- (a) 5 covalent bonds (b) 12 covalent bonds
(c) 16 covalent bonds (d) 17 covalent bonds

Ans. (c) The structural formula of pentane C_5H_{12} is



It contains 16 covalent bonds.

Q. 16 Structural formula of benzene is



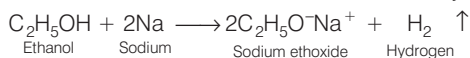
Ans. (c) Benzene molecule contains alternate single and double bonds. Its formula is C_6H_6 in structure (b) formula is C_6H_8 .

In structure (c) double bond is not at alternate position. In (d) formula is C_6H_{12} . Therefore, option (c) is correct.

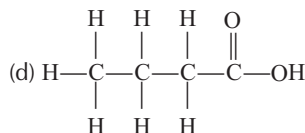
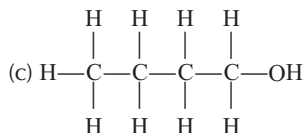
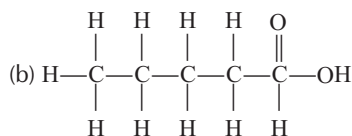
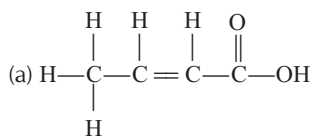
Q. 17 Ethanol reacts with sodium and forms two products. These are

- (a) sodium ethanoate and hydrogen
(b) sodium ethanoate and oxygen
(c) sodium ethoxide and hydrogen
(d) sodium ethoxide and oxygen

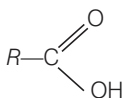
Ans. (c) Ethanol reacts with sodium to form sodium ethoxide and hydrogen gas.



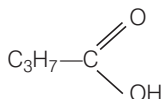
Q. 18 The correct structural formula of butanoic acid is



Ans. (d) The general formula of carboxylic acid is $R-\text{COOH}$ where R is an alkyl group. Its representation is



So, its formula is



Because 'butane' shows presence of 4 single bonded carbon atoms and 'oic acid'

shows presence of $\begin{array}{c} \text{O} \\ || \\ -\text{C}-\text{OH} \end{array}$ group.

Q. 19 Vinegar is a solution of

- (a) 50%-60% acetic acid in alcohol
- (b) 5%-8% acetic acid in alcohol
- (c) 5%-8% acetic acid in water
- (d) 50%-60% acetic acid in water

Ans. (c) A 5%-8% solution of acetic acid in water is called vinegar.

Q. 20 Mineral acids are stronger acids than carboxylic acids because

- (i) mineral acids are completely ionised.
- (ii) carboxylic acids are completely ionised.
- (iii) mineral acids are partially ionised.
- (iv) carboxylic acids are partially ionised.

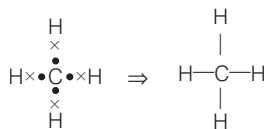
- (a) (i) and (iv)
- (b) (ii) and (iii)
- (c) (i) and (ii)
- (d) (iii) and (iv)

Ans. (a) Mineral acids are strong acids which ionise almost completely and carboxylic acids are weak acids which ionise only partially.

Q. 21 Carbon forms four covalent bonds by sharing its four valence electrons with four univalent atoms, e.g., hydrogen. After the formation of four bonds, carbon attains the electronic configuration of

- (a) helium
- (b) neon
- (c) argon
- (d) krypton

Ans. (b) Electronic configuration of carbon (C) = 2, 4 when it forms four covalent bonds by sharing its four valence electrons with hydrogen, it forms CH_4 molecule like this



Now, electronic configuration of C in $\text{CH}_4 = 2, 8$

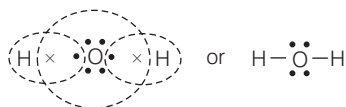
Atomic number of Ne is 10. Its electronic configuration is $\overset{K}{2}, \overset{L}{8}$

Therefore, after the formation of four bonds, carbon attains the electronic configuration of neon.

Q. 22 The correct electron dot structure of a water molecule is

- (a) $\text{H} \cdot \ddot{\text{O}} \cdot \text{H}$ (b) $\text{H} : \ddot{\text{O}} : \text{H}$ (c) $\text{H} : \ddot{\text{O}} : \text{H}$ (d) $\text{H} \cdot \cdot \text{O} \cdot \cdot \text{H}$

Ans. (c) Electronic configuration of ${}_8\text{O} = 2, 6$ therefore, it has 6 valence electrons, while H has 1. Hence, water molecule has 2 bond pairs and 2 lone pairs of electrons.



Q. 23 Which of the following is not a straight chain hydrocarbon?

- (a) $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_3$ (b) $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_3$

- (c) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H}_2\text{C}-\text{H}_2\text{C}-\text{H}_2\text{C}-\text{CH}_2 \\ | \\ \text{CH}_3 \end{array}$ (d) $\begin{array}{c} \text{H}_3\text{C} \\ \diagup \\ \text{CH}-\text{CH}_2-\text{CH}_2-\text{CH}_3 \\ \diagdown \\ \text{H}_3\text{C} \end{array}$

Ans. (d) $\begin{array}{c} \text{H}_3\text{C}^1 \\ \diagup \\ \text{CH}^2-\text{CH}_2^3-\text{CH}_2^4-\text{CH}_3^5 \\ \diagdown \\ \text{H}_3\text{C} \end{array}$

or $\begin{array}{c} \text{H}_3\text{C}^1 \\ | \\ \text{CH}^2-\text{CH}_2^3-\text{CH}_2^4-\text{CH}_3^5 \\ | \\ \text{CH}_3 \end{array}$ is a branched chain hydrocarbon not straight

chain hydrocarbon. Rest three are straight chain hydrocarbons.

Q. 24 Which among the following are unsaturated hydrocarbons?

- (i) $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_3$ (ii) $\text{H}_3\text{C}-\text{C} \equiv \text{C}-\text{CH}_3$
 (iii) $\begin{array}{c} \text{H}_3\text{C}-\text{CH}-\text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$ (iv) $\begin{array}{c} \text{H}_3\text{C}-\text{C}=\text{CH}_2 \\ | \\ \text{CH}_3 \end{array}$

(a) (i) and (iii)

(b) (ii) and (iii)

(c) (ii) and (iv)

(d) (iii) and (iv)

Ans. (c) Unsaturated hydrocarbons have double or triple bond in the structure. Both (ii) and (iv) structures have triple and double carbon-carbon bonds respectively.

Q. 25 Which of the following does not belong to the same homologous series?

(a) CH_4

(b) C_2H_6

(c) C_3H_8

(d) C_4H_8

Thinking Process

A homologous series is a group of organic compounds having similar structures and similar chemical properties in which the successive compounds differ by CH_2 group.

Ans. (d) Because successive members of a homologous series differ by —CH_2 unit.

$$\begin{array}{r} \text{C}_3\text{H}_8 \\ + \text{CH}_2 \\ \hline \text{C}_4\text{H}_{10} \end{array}$$

Thus, C_4H_{10} is the next member of this series.

So, homologous series of alkanes is

Methane CH_4 ,

Ethane C_2H_6 ,

Propane C_3H_8 ,

Butane C_4H_{10}

So, C_4H_8 does not belong to the homologous series.

Q. 26 The name of the compound $\text{CH}_3\text{—CH}_2\text{—CHO}$ is

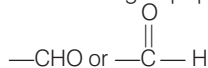
(a) propanal

(b) propanone

(c) ethanol

(d) ethanal

Ans. (a) Functional group present in $\text{CH}_3\text{—CH}_2\text{—CHO}$ is aldehyde. Aldehyde group is



It is propanal containing 3 carbon atoms in which 'propane' represents 3 carbon atoms while 'al' represents aldehyde group. So, propane + al = propanal.

Q. 27 The heteroatoms present in



(i) oxygen

(ii) carbon

(iii) hydrogen

(iv) chlorine

(a) (i) and (ii)

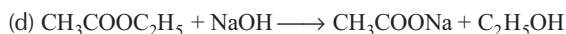
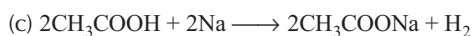
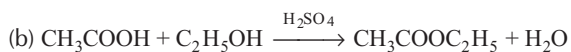
(b) (ii) and (iii)

(c) (iii) and (iv)

(d) (i) and (iv)

Ans. (d) Atoms other than C and H, if present in organic compound, are called heteroatoms.

Q. 28 Which of the following represents saponification reaction?



Ans. (d) In saponification reaction, when an ester is heated with sodium hydroxide solution, ester gets hydrolysed (breaks down) to form the parent alcohol and sodium salt of carboxylic acid.

Q. 29 The first member of alkyne homologous series is

- (a) ethyne (b) ethene (c) propyne (d) methane

Ans. (a) The general formula of alkyne is C_nH_{2n-2} where, n is the number of carbon atoms. The first member of alkyne homologous series is ethyne C_2H_2 .

Short Answer Type Questions

Q. 30 Draw the electron dot structure of ethyne and also draw its structural formula.

Ans. Molecular formula of ethyne is C_2H_2 . Electronic configuration of C = 2, 4.

(valence electrons = 4)

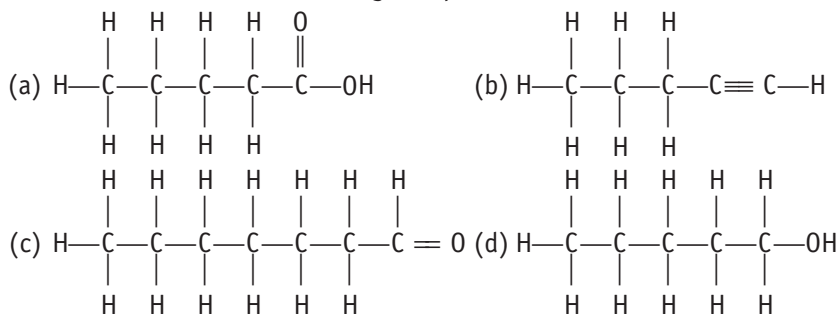
Electronic configuration H = 1 (valence electron = 1)

Electron dot structure is



Structural formula is $H-C \equiv C-H$
Ethyne

Q. 31 Write the names of the following compounds.



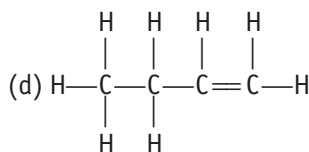
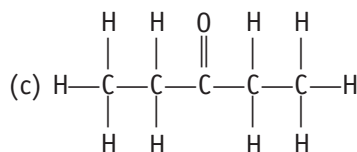
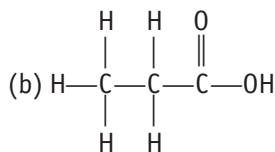
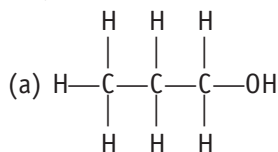
Ans. (a) Pentane – e + oic acid = Pentanoic acid (because it contains five C atoms and one $-COOH$ group).

(b) Pentane – e + yne = Pentyne (due to the presence of five carbon atoms along with a triple bond).

(c) Heptane – e + al = Heptanal (as it contains 7 carbon atoms and a $-CHO$ group)

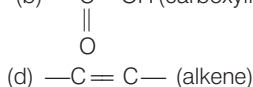
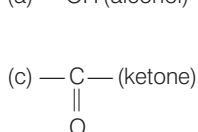
(d) Pentane – e + ol = Pentanol (as it contains 5 carbon atoms and one $-OH$ group)

Q. 32 Identify and name the functional groups present in the following compounds.



Ans. (a) —OH (alcohol)

(b) —C—OH (carboxylic acid)

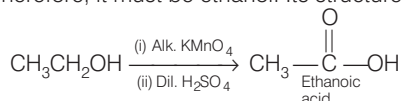


Q. 33 A compound X is formed by the reaction of a carboxylic acid $\text{C}_2\text{H}_4\text{O}_2$ and an alcohol in the presence of a few drops of H_2SO_4 . The alcohol on oxidation with alkaline KMnO_4 followed by acidification gives the same carboxylic acid as used in this reaction. Give the names and structures of (a) carboxylic acid, (b) alcohol and (c) the compound X. Also write the reaction.

Ans. (a) Carboxylic acid having molecular formula $\text{C}_2\text{H}_4\text{O}_2$ is (acetic acid or ethanoic acid). Its structure is $\text{CH}_3-\text{C}-\text{OH}$



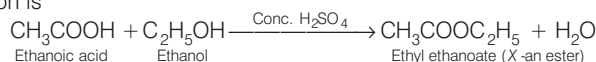
(b) Since, an alcohol which on oxidation with alkaline KMnO_4 followed by acidification gives ethanoic acid. Therefore, it must be ethanol. Its structure is $\text{CH}_3\text{CH}_2-\text{OH}$.



(c) Since compound X is formed by the reaction of ethanoic acid with ethanol in presence of a few drops of conc. H_2SO_4 . Therefore, compound X must be an ester, i.e., ethyl ethanoate. Its structure is $\text{CH}_3-\text{C}-\text{O}-\text{C}_2\text{H}_5$.



The reaction is



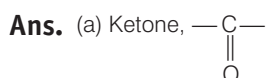
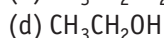
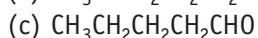
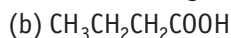
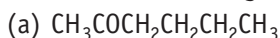
Q. 34 Why detergents are better cleansing agents than soaps? Explain.

Ans. Detergents are better cleansing agents than soaps because they can be used even with hard water. The charged ends of detergents do not form insoluble precipitates with calcium and magnesium ions in hard water.

When soap is used for washing clothes with hard water, it reacts with the calcium and magnesium ions of hard water to form an insoluble precipitate called scum. Scum sticks to the cloth to be washed and it makes the cleaning of clothes difficult.

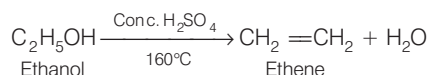
Detergents have a stronger cleansing action than soaps and are more soluble in water than soaps.

Q. 35 Name the functional groups present in the following compounds.



Q. 36 How is ethene prepared from ethanol? Give the reaction involved in it.

Ans. Ethene is prepared by the dehydration of ethyl alcohol in the presence of conc. H_2SO_4 at 160°C .

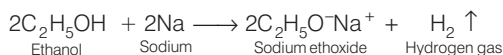


Q. 37 Intake of small quantity of methanol can be lethal. Comment.

Ans. Methanol is very poisonous because it is oxidised to methanal in the liver. Methanal reacts rapidly with the components of cells and causes the protoplasm to get coagulated (In the same way, as an egg gets coagulated on boiling). Methanol also affects the optic nerve causing blindness.

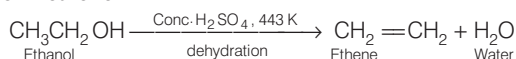
Q. 38 A gas is evolved when ethanol reacts with sodium. Name the gas evolved and also write the balanced chemical equation of the reaction involved.

Ans. The gas evolved is hydrogen. *The reaction is as follows*



Q. 39 Ethene is formed when ethanol at 443 K is heated with excess of concentrated sulphuric acid. What is the role of sulphuric acid in this reaction? Write the balanced chemical equation of this reaction.

Ans. When ethanol is heated with excess of concentrated sulphuric acid at 443 K, it gets dehydrated to form ethene.



In this reaction, concentrated sulphuric acid acts as a dehydrating agent which removes water molecule from the ethanol molecule.

Q. 40 Carbon, group (14) element in the periodic table, is known to form compounds with many elements.

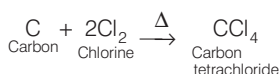
Write an example of a compound formed with

- (a) chlorine (group 17 of periodic table)
- (b) oxygen (group 16 of periodic table)

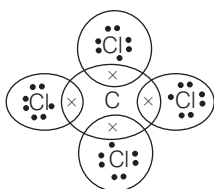
Ans. (a) Carbon (C) electronic configuration of carbon, C (6), is $\begin{smallmatrix} K & L \\ 2 & 4 \end{smallmatrix}$

Electronic configuration of Cl (17) $\begin{smallmatrix} K & L & M \\ 2 & 8 & 7 \end{smallmatrix}$

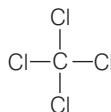
With chlorine, carbon forms carbon tetrachloride



Electron dot structure and structural formula of CCl_4 is as follows.



(Carbon tetrachloride)



(b) Electronic configuration of O (8) is $\begin{smallmatrix} K & L \\ 2 & 6 \end{smallmatrix}$

Its electron dot structure and structural formula are as follows



or $\text{O}=\text{C}=\text{O}$

With O_2 , C forms carbon dioxide (CO_2).

Q. 41 In electron dot structure, the valence shell electrons are represented by crosses or dots.

- (a) The atomic number of chlorine is 17. Write its electronic configuration.
- (b) Draw the electron dot structure of chlorine molecule.

Ans. (a) Electronic configuration of Cl (17) $\begin{smallmatrix} K & L & M \\ 2 & 8 & 7 \end{smallmatrix}$

(b) Electron dot structure of chlorine molecule is



$\text{Cl}-\text{Cl}$ or Cl_2 (chlorine molecule)

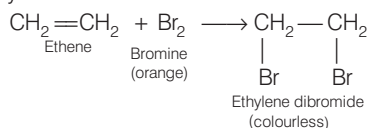
Q. 42 Catenation is the ability of an atom to form bonds with other atoms of the same element. It is exhibited by both carbon and silicon. Compare the ability of catenation of the two elements. Give reasons.

Ans. Both carbon (C) and silicon (Si) have similar valence shell electronic configuration *i.e.*, each has four electrons in the valence shell and hence show the phenomenon of catenation.

But carbon exhibits catenation much more than silicon or any other element due to its smaller size which makes the C—C bonds strong while the Si—Si bonds are comparatively weaker due to its large size. Thus, due to greater strength of C—C over Si—Si bonds, carbon shows catenation to a greater extent than silicon.

Q. 43 Unsaturated hydrocarbons contain multiple bonds between the two C-atoms and show addition reaction. Give the test to distinguish ethane from ethene.

Ans. Bromine solution in CCl_4 has an orange colour. When it is added dropwise to ethene, the orange colour disappears due to the formation of the colourless ethylene dibromide. This reaction is not shown by ethane as it is addition reaction.



The two can be distinguished also by subjecting them to the flame. Saturated hydrocarbon (ethane) generally gives a clear flame while unsaturated hydrocarbon (ethene) gives a yellow flame with lots of black smoke.

Q. 44 Match the reactions given in Column I with the names given in Column II.

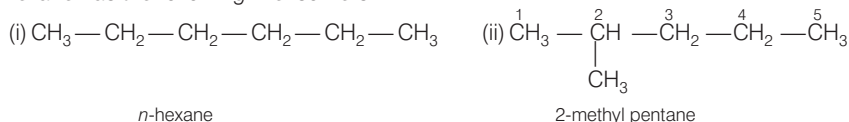
Column I	Column II
A. $\text{CH}_3\text{OH} + \text{CH}_3\text{COOH} \xrightarrow{\text{H}^+} \text{CH}_3\text{COOCH}_3 + \text{H}_2\text{O}$	1. Addition reaction
B. $\text{CH}_2 = \text{CH}_2 + \text{H}_2 \xrightarrow{\text{Ni}} \text{CH}_3 - \text{CH}_3$	2. Substitution reaction
C. $\text{CH}_4 + \text{Cl}_2 \xrightarrow{\text{Sunlight}} \text{CH}_3\text{Cl} + \text{HCl}$	3. Neutralisation reaction
D. $\text{CH}_3\text{COOH} + \text{NaOH} \longrightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$	4. Esterification reaction

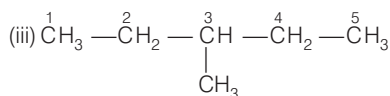
Ans. The matching of Column I and Column II is as given

Column I	Column II
A. $\text{CH}_3\text{OH} + \text{CH}_3\text{COOH} \xrightarrow{\text{H}^+} \text{CH}_3\text{COOCH}_3 + \text{H}_2\text{O}$	Esterification reaction
B. $\text{CH}_2 = \text{CH}_2 + \text{H}_2 \xrightarrow{\text{Ni}} \text{CH}_3 - \text{CH}_3$	Addition reaction
C. $\text{CH}_4 + \text{Cl}_2 \xrightarrow{\text{Sunlight}} \text{CH}_3\text{Cl} + \text{HCl}$	Substitution reaction
D. $\text{CH}_3\text{COOH} + \text{NaOH} \longrightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$	Neutralisation reaction

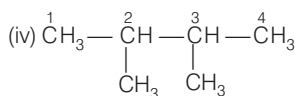
Q. 45 Write the structural formulae of all the isomers of hexane.

Ans. Hexane has the following five isomers

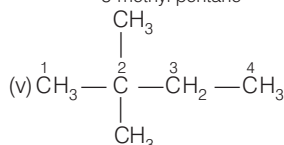




3-methyl pentane

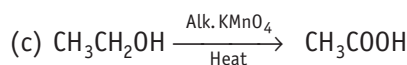
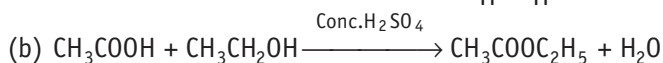
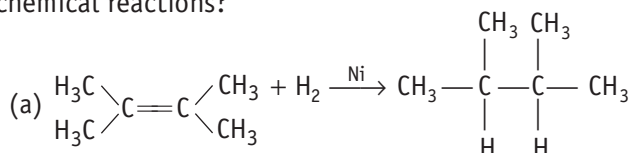


2, 3- dimethyl butane



2, 2-dimethyl butane

Q. 46 What is the role of metal or reagents written on arrows in the given chemical reactions?



Ans. (a) Nickel (Ni) acts as the catalyst during the reaction. It first adsorbs the hydrogen molecule on its surface as hydrogen atoms and then the alkene molecule side by side. The two hydrogen atoms then add across the double bond of the alkene to form the addition product *i.e.*, 2, 3- dimethyl butane.

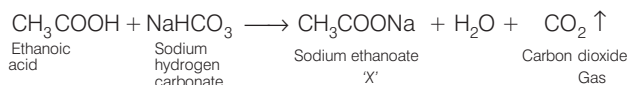
(b) Conc. H_2SO_4 increases the rate of the forward reaction by removing H_2O formed during the reaction. In other words, conc. H_2SO_4 acts as a dehydrating agent.

(c) Alkaline KMnO_4 acts as an oxidising agent and oxidises ethanol to ethanoic acid.

Long Answer Type Questions

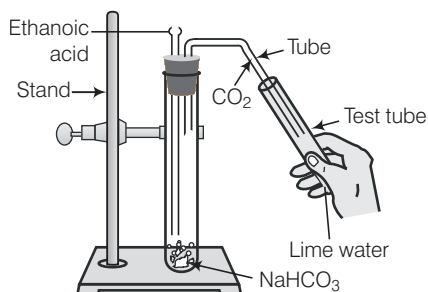
Q. 47 A salt **X** is formed and a gas is evolved when ethanoic acid reacts with sodium hydrogen carbonate. Name the salt **X** and the gas evolved. Describe an activity and draw the diagram of the apparatus to prove that the evolved gas is the one which you have named. Also, write the chemical equation of the reaction involved.

Ans. When ethanoic acid reacts with sodium hydrogen carbonate, sodium ethanoate is formed. Thus, **X** is sodium ethanoate and the gas evolved is carbon dioxide (CO_2).

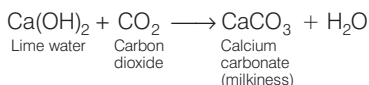


Activity

(i) Set up the apparatus as shown.



- (ii) Take NaHCO_3 in a test tube and add 2 mL ethanoic acid.
 (iii) CO_2 gas is evolved with brisk effervescence.
 (iv) Pass the gas through freshly prepared lime water. It will turn milk due to the formation of insoluble calcium carbonate.



- Q. 48** (a) What are hydrocarbons? Give examples.
 (b) Give the structural differences between saturated and unsaturated hydrocarbons with two examples each.
 (c) What is functional group? Give examples of four different functional groups.

Ans. (a) The compounds that are made up of carbon and hydrogen atoms are called hydrocarbons. *e.g.*, methane (CH_4), ethene ($\text{CH}_2 = \text{CH}_2$), Ethyne (C_2H_2), cyclohexane (C_6H_{12}), benzene (C_6H_6) etc.

(b) In saturated hydrocarbons, all the four valencies of carbon are satisfied by a single covalent bond while in unsaturated hydrocarbons, double or triple bonds are required to satisfy the valencies. *e.g.*,

(i) Saturated hydrocarbons

Methane (CH_4), Ethane ($\text{CH}_3 - \text{CH}_3$)

(ii) Unsaturated hydrocarbons

Ethene ($\text{H}_2\text{C} = \text{CH}_2$), Ethyne ($\text{HC} \equiv \text{CH}$)

(c) A functional group is an atom or group of atoms that defines the structure (or the properties) of organic compounds. The four examples are

(i) $-\text{OH}$ Alcohol

(ii) $-\text{COOH}$ Carboxylic acid

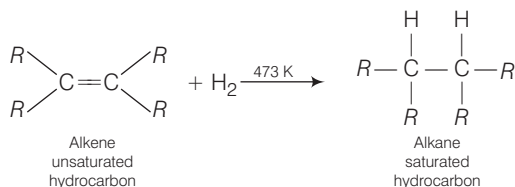
(iii) $-\text{CHO}$ Aldehyde

(iv) $-\text{X}$ Halogens

Q. 49 Name the reaction which is commonly used in the conversion of vegetable oils to fats. Explain the reaction involved in detail.

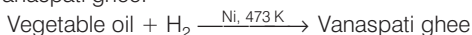
Ans. The reaction is known as hydrogenation reaction. When vegetable oils are treated with hydrogen and passed over finely divided nickel or palladium at 200°C, the hydrogen molecules are added to the unsaturated carbon-carbon bonds and hence, saturated vegetable fats are obtained.

e.g.,



where, R is any alkyl group i.e., $-\text{CH}_3$, $-\text{C}_2\text{H}_5$ etc.

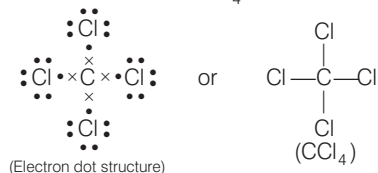
Industrial Application The process of hydrogenation is used in industry to convert vegetable oils to vanaspati ghee.



Q. 50 (a) Write the formula and draw electron dot structure of carbon tetrachloride.

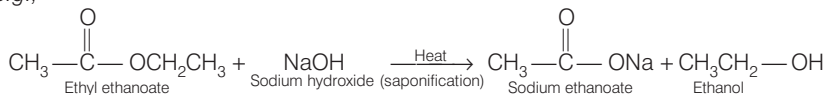
(b) What is saponification? Write the reaction involved in this process.

Ans. (a) The formula of carbon tetrachloride is CCl_4 . Its electron dot structure is as follows

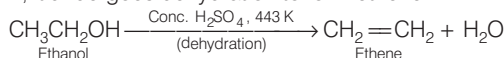


(b) **Saponification** Alkaline hydrolysis of an ester to give the salt of the corresponding acid and the alcohol is called saponification. It is reverse of esterification reaction.

e.g.,



Dehydration means removal of a molecule of water. When ethanol is heated with conc. H_2SO_4 at 443 K, it undergoes dehydration to form ethene.

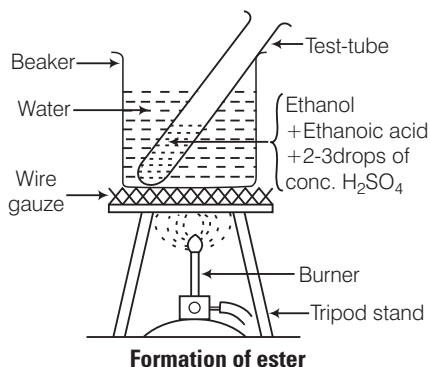


Q. 51 Esters are sweet-smelling substances and are used in making perfumes. Suggest some activity and the reaction involved for the preparation of an ester with well labelled diagram.

Ans. Activity

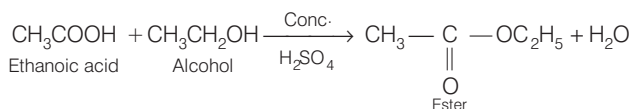
- Take 1 mL ethanol and 1 mL glacial acetic acid along with a few drops of concentrated sulphuric acid in a test tube.
- Warm the contents in a water bath for atleast five minutes.

- (iii) Pour the content into a beaker containing 20-50 mL of water and smell the resulting mixture.



- (iv) Sweet smell would be observed.

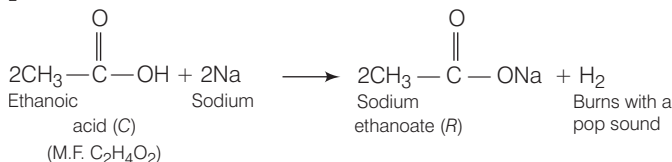
Reaction



Q. 52 A compound C (molecular formula, $\text{C}_2\text{H}_4\text{O}_2$) reacts with Na metal to form a compound R and evolves a gas which burns with a pop sound. Compound C on treatment with an alcohol A in the presence of an acid forms a sweet smelling compound S (molecular formula, $\text{C}_3\text{H}_6\text{O}_2$). On addition of NaOH to C, it also gives R and water. S on treatment with NaOH solution gives back R and A.

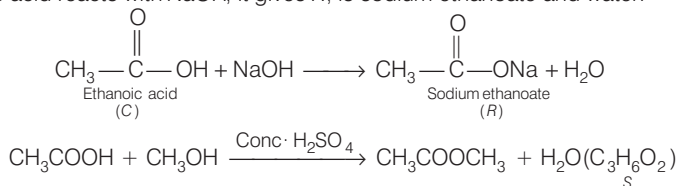
Identify **C**, **R**, **A**, **S** and write down the reactions involved.

- Ans.** (i) Since, compound C (M.F. $\text{C}_2\text{H}_4\text{O}_2$) contains two oxygen atoms, therefore most probably it may be a carboxylic acid, *i.e.*, ethanoic acid (CH_3COOH).
- (ii) Since, an acid, *i.e.*, ethanoic acid reacts with a base, *i.e.*, Na metal to evolve a gas which burns with a pop sound along with the formation of compound (R), therefore, R must be a salt, *i.e.*, sodium ethanoate and the gas which burns with a pop sound must be H_2 gas.



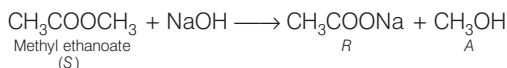
H_2 gas burns with 'pop' sound.

Compound R is sodium ethanoate. This is also supported by the observation that when ethanoic acid reacts with NaOH, it gives R, is sodium ethanoate and water.

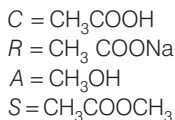


Since, compound C on treatment with an alcohol A in presence of acid forms a sweet smelling compound S (M.F. $C_3H_6O_2$), therefore, S is methyl ethanoate (ester).

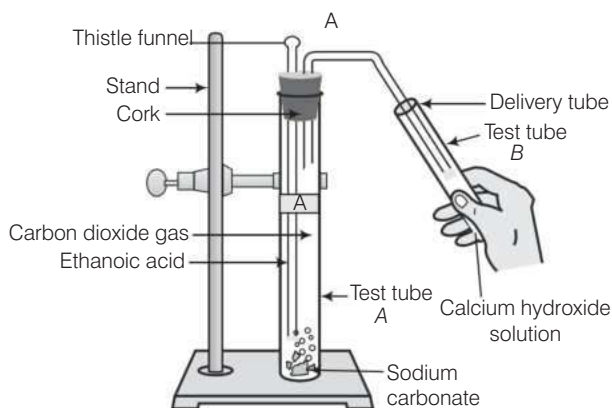
Since, ester S has three carbon atoms and the acid C has two carbon atoms, therefore alcohol A must contain one C atom, i.e., A is methanol.



Thus,



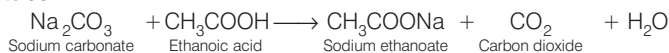
Q. 53 Look at the figure and answer the following questions.



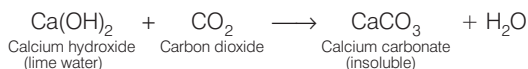
- What change would you observe in the calcium hydroxide solution taken in tube B?
- Write the reaction involved in test tubes A and B respectively.
- If ethanol is given instead of ethanoic acid, would you expect the same change?
- How can a solution of lime water be prepared in the laboratory?

Ans. (a) When CO_2 is passed through calcium hydroxide solution (taken in test tube B), i.e., lime water, it turns milky due to the formation of insoluble calcium carbonate ($CaCO_3$).

(b) **Test tube A**



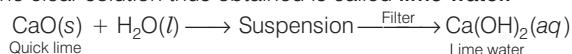
Test tube B



- (c) If ethanol is taken instead of ethanoic acid, no change will occur because ethanol is a very weak acid and hence cannot decompose Na_2CO_3 to give CO_2 gas.



- (d) When quick lime (CaO) is added to water in a test tube. Some of it will dissolve to form calcium hydroxide (lime water) while majority of it remains suspended. Filter the solution. The clear solution thus obtained is called **lime water**.



5

Periodic Classification of Elements

Multiple Choice Questions (MCQs)

Q. 1 Upto which element, the law of octaves was found to be applicable?

- (a) Oxygen (b) Calcium (c) Cobalt (d) Potassium

Ans. (b) Newlands' law of octaves was applicable only to lighter elements having atomic masses upto 40 u *i.e.*, upto calcium. After calcium, every eighth element did not possess properties similar to that of the first element.

Q. 2 According to Mendeleev's periodic law, the elements were arranged in the periodic table in the order of

- (a) increasing atomic number (b) decreasing atomic number
(c) increasing atomic masses (d) decreasing atomic masses

💡 Thinking Process

Mendeleev's periodic law states that, the physical and chemical properties of elements are a periodic function of their atomic masses.

Ans. (c) According to Mendeleev's periodic law, the elements were arranged in the periodic table in the order of increasing atomic masses.

Q. 3 In Mendeleev's periodic table, gaps were left for the elements to be discovered later. Which of the following elements found a place in the periodic table later?

- (a) Germanium (b) Chlorine (c) Oxygen (d) Silicon

Ans. (a) Mendeleev's left some gaps in the periodic table for those elements which were not known at that time.

Germanium element found a place in the periodic table later and Mendeleev's predictions were found to be remarkably correct.

Ans. (c) Electronic configuration of A (atomic number 9) is 2, 7.
 Electronic configuration of B (atomic number 11) is 2, 8, 1.
 Electronic configuration of C (atomic number 17) is 2, 8, 7
 Electronic configuration of D (atomic number 12) is 2, 8, 2.
 Electronic configuration of E (atomic number 13) is 2, 8, 3.
 Elements which differ in atomic number by 8 *i.e.*, 9 (A fluorine) and 17 (Cl, chlorine) lie in the same group *i.e.*, group 17 (halogens).

Q. 8 Where would you locate the element with electronic configuration 2, 8 in the modern periodic table?

- (a) Group 8 (b) Group 2 (c) Group 18 (d) Group 10

💡 Thinking Process

Group 18 elements are known as noble gases or inert gases. They are helium, neon, argon, krypton etc. All the noble gases have 8 electrons in their outermost shell (except helium which has only 2 electrons) in the K valence shell. Thus, the valence shells of all the noble gases are completely filled with electrons.

Ans. (c) Element with electronic configuration 2, 8, is an inert gas *i.e.*, neon and hence, belongs to group 18.

Q. 9 An element which is an essential constituent of all organic compounds belongs to

- (a) group 1 (b) group 14 (c) group 15 (d) group 16

Ans. (b) Constituent of all organic compounds is carbon. It belongs to group 14.

Q. 10 Which of the following is the outermost shell for elements of period 2?

- (a) K shell (b) L shell (c) M shell (d) N shell

Ans. (b) The elements of 2nd period involve the filling of 2nd shell *i.e.*, L shell. Because in period 2, there are two shell, K and L.

Q. 11 Which one of the following elements exhibit maximum number of valence electrons?

- (a) Na (b) Al (c) Si (d) P

💡 Thinking Process

Amongst representative elements, the number of valence electrons increases as the group number increases from left to right.

Ans. (d) Thus, Na (group 1) has one, Al (group 13) has three (13-10), Si (group 14) has four (14-10) and P (group 15) has five (15-10) valence electrons. Therefore, P has maximum number of valence electrons *i.e.*, 5 (maximum among the given).

Q. 12 Which of the following gives the correct increasing order of the atomic radii of O, F and N?

- (a) O, F, N (b) N, F, O (c) O, N, F (d) F, O, N

💡 Thinking Process

Atomic radii is the distance between the centre of the nucleus and the outermost shell which contains electrons in an isolated system.

Ans. (d) As we move from left to right in a period, the atomic number of each succeeding element increases by 1. So, the electrons are attracted close to the nucleus and hence, the atomic size decreases, along a period from left to right.

Therefore, the increasing order of radii is F, O, N as atomic number of F, O and N are 9, 8 and 7 respectively. Due to which F belongs to 17th group. O belongs to 16th group and N belongs to 15th group

Q. 13 Which among the following elements has the largest atomic radii?

(a) Na

(b) Mg

(c) K

(d) Ca

💡 Thinking Process

Atomic radii increases on moving down the group as new shell of electrons is added at each succeeding element.

Ans. (c) Na and K are in the same group and K is below than Na, so K will have higher atomic radius *i.e.*, $K > Na$. In a period on moving from left to right, atomic radius decreases. Since, K and Ca are in the same period and K is in 1 group and Ca is in 2 group, so atomic radius of K will be more than Ca, *i.e.*, $K > Ca$.

Also Na and Mg are in the same period, but Na belongs to 1 group and Mg belongs to 2 group, so atomic radius of Na is more than Mg, *i.e.*, $Na > Mg$. Thus, if we take all these together we get $K > Na > Mg$ and $K > Ca > Mg$. Hence, we can say the atomic radius of K is largest.

Q. 14 Which of the following elements would lose an electron easily?

(a) Mg

(b) Na

(c) K

(d) Ca

Ans. (c) Electronic configuration of Mg (atomic number 12) 2, 8, 2.

Electronic configuration of Na (atomic number 11) 2, 8, 1.

Electronic configuration of K (atomic number 19) 2, 8, 8, 1.

Electronic configuration of Ca (atomic number 20) 2, 8, 8, 2.

From the above electronic configurations, it is clear that K and Na will lose electron easily to achieve stable configuration. But out of K and Na, K will lose electron more easily because the force of attraction on valence electron of K is least among the given.

In other words, larger the atomic radius of an element more easily it can lose its valence electron. Among all these, K has the largest atomic radius, therefore it can lose its valence electron more easily.

For atomic size refer to Q. 13.

Q. 15 Which of the following elements does not lose an electron easily?

(a) Na

(b) F

(c) Mg

(d) Al

💡 Thinking Process

Elements of 2nd period have smaller size than those of the corresponding elements of the 3rd period. Further in a period, halogens have the smallest size. F belongs to 2nd period and rest three to 3rd period.

Ans. (b) Smaller the atomic radius of an element, more difficult is to lose electron. Among Na, F, Mg and Al, F has the smallest size. Also it contains 7 electrons in its valence shell and it is most electronegative element. Hence, it does not lose an electron easily rather it gains an electron easily.

Q. 16 Which of the following are the characteristics of isotopes of an element?

- (i) Isotopes of an element have same atomic masses.
 - (ii) Isotopes of an element have same atomic number.
 - (iii) Isotopes of an element show same physical properties.
 - (iv) Isotopes of an element show same chemical properties.
- (a) (i), (iii) and (iv) (b) (ii), (iii) and (iv) (c) (ii) and (iii) (d) (ii) and (iv)

Thinking Process

All the isotopes of an element have the same atomic number. Therefore, they can be placed at one place in the same group of the periodic table. Chemical properties depend on the electronic configuration of the element, i.e., atomic number of the element.

Ans. (d) Isotopes are the atoms of the same element having same atomic number and hence, similar chemical properties but different atomic masses.

Q. 17 Arrange the following elements in the order of their decreasing metallic character Na, Si, Cl, Mg, Al.

- (a) $\text{Cl} > \text{Si} > \text{Al} > \text{Mg} > \text{Na}$ (b) $\text{Na} > \text{Mg} > \text{Al} > \text{Si} > \text{Cl}$
 (c) $\text{Na} > \text{Al} > \text{Mg} > \text{Cl} > \text{Si}$ (d) $\text{Al} > \text{Na} > \text{Si} > \text{Ca} > \text{Mg}$

Ans. (b) Metals lie on the extreme left side of the periodic table. Metallic character decreases from left to right in a period.

Na, Si, Cl, Mg, Al belong to same period in the order Na, Mg, Al, Si, Cl. On moving in a period from left to right, the metallic character decreases.

Thus, the order of decreasing metallic character is



Q. 18 Arrange the following elements in the order of their increasing non-metallic character Li, O, C, Be, F

- (a) $\text{F} < \text{O} < \text{C} < \text{Be} < \text{Li}$ (b) $\text{Li} < \text{Be} < \text{C} < \text{O} < \text{F}$
 (c) $\text{F} < \text{O} < \text{C} < \text{Be} < \text{Li}$ (d) $\text{F} < \text{O} < \text{Be} < \text{C} < \text{Li}$

Ans. (b) Li, O, C, Be, F belong to same period i.e., 2nd period. In a period on moving from left to right, non-metallic character increases. Thus, the order of increasing non-metallic character is $\text{Li} < \text{Be} < \text{C} < \text{O} < \text{F}$.

Q. 19 What type of oxide would Eka-aluminium form?

- (a) EO_3 (b) E_3O_2 (c) E_2O_3 (d) EO

Thinking Process

Eka-aluminium (first element after aluminium) i.e., gallium lies in group III of the Mendeleev's periodic table.

Ans. (c) Gallium has a valency of 3. Hence, it forms an oxide having molecular formula E_2O_3 . In other options, valency of E is not 3.

Q. 20 Three elements B, Si and Ge are

- (a) metals
- (b) non-metals
- (c) metalloids
- (d) metal, non-metal and metalloid respectively

Ans. (c) The elements boron (B), silicon (Si) and germanium (Ge) are metalloids. In the periodic table, metals have been separated from non-metals by some elements called 'metalloids' which are placed diagonally in the periodic table. These metalloids are B, Si, Ge, As (arsenic), Sb (antimony), Te (tellurium) and (Po) polonium.

Q. 21 Which of the following elements will form an acidic oxide?

- (a) An element with atomic number 7
- (b) An element with atomic number 3
- (c) An element with atomic number 12
- (d) An element with atomic number 19

Ans. (a) Non-metals can form acidic oxides in general. Non-metals have 4 to 8 electrons in the outermost shell. The electronic configuration of given elements are (a) *i.e.*, $7 = 2, 5$ (b) *i.e.*, $3 = 2, 1$ (c) *i.e.*, $12 = 2, 8, 2$ (d) *i.e.*, $19 = 2, 8, 8, 1$.

So, element with atomic number 7 (electronic configuration 2, 5) with is non-metal (N) and it will form an acidic oxide. Rest three elements atomic numbers, 3 (Li), 12 (Mg) and 19 (K) are metals and hence form basic oxides.

Q. 22 The element with atomic number 14 is hard and forms acidic oxide and a covalent halide. To which of the following categories does the element belong?

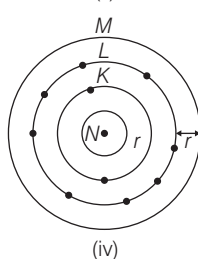
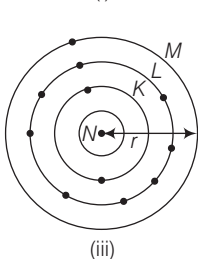
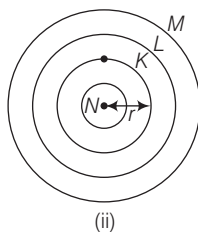
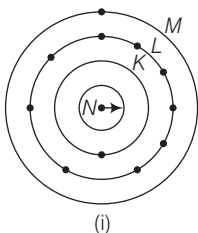
- (a) Metal
- (b) Metalloid
- (c) Non-metal
- (d) Left-hand side element

💡 Thinking Process

Atomic number 14 has electronic configuration 2, 8, 4.

Ans. (c) Its outermost shell has 4 electrons. So, it is a non-metal. Non-metal forms acidic oxide and by sharing of electrons with halogen, it forms covalent halide.
(Also, refer to Q. 21)

Q. 23 Which one of the following depict the correct representation of atomic radius (r) of an atom?



(a) (i) and (ii)

(b) (ii) and (iii)

(c) (iii) and (iv)

(d) (i) and (iv)

Ans. (b) Atomic radius is the distance between the centre of the nucleus and the outermost shell which contains electrons in an isolated atom. Figure (ii) and (iii) are the correct representation of atomic radius from nucleus to the outermost shell.

Q. 24 Which one of the following does not increase while moving down the group of the periodic table?

- (a) Atomic radius
- (b) Metallic character
- (c) Valence
- (d) Number of shells in an element

💡 Thinking Process

On moving down in a particular group of the periodic table, the number of valence electrons in the elements remains the same.

All the remaining three properties increase down the group.

Ans. (c) All the elements of a group of the periodic table have the same number of valence electrons. e.g., all the elements of group 1 have one valence electron each in their atoms.

All the elements of group 18 have 8 valence electrons each in their atoms, except helium which has only 2 valence electrons in its atom. All the remaining three properties increase down the group.

Q. 25 On moving from left to right in a period in the periodic table, the size of the atom

- (a) increases
- (b) decreases
- (c) does not change appreciably
- (d) first decreases and then increases

💡 Thinking Process

As we move from left to right in a period, the atomic number of elements increases which means that the number of protons and electrons in the atoms increases (the extra electrons being added to the same shell).

Ans. (b) Due to large positive charge on the nucleus, the electrons are pulled in more close to the nucleus and the size of the atom decreases. However, the size of atom of an inert gas is bigger than that of the preceding halogen atom. The greater size of the inert gas atom in a period is due to the structural stability of its outermost shell consisting of an octet of electrons.

Q. 26 Which of the following set of elements is written in order of their increasing metallic character?

- (a) Be, Mg, Ca
- (b) Na, Li, K
- (c) Mg, Al, Si
- (d) C, O, N

Ans. (a) Metallic character increases down a group and decreases along a period. Thus, metallic character increases in the order Be, Mg, Ca as they belong to same group.

Short Answer Type Questions

Q. 27 The three elements **A**, **B** and **C** with similar properties have atomic masses **X**, **Y** and **Z** respectively. The mass of **Y** is approximately equal to the average mass of **X** and **Z**. What is such an arrangement of elements called as? Give one example of such a set of elements.

Ans. The arrangement of elements in which the atomic mass of middle element is almost the mean of atomic masses of first and third elements is known as Dobereiner's triads.

e.g.,

Ca (40), Sr (88), Ba (137)

$$\text{Atomic mass of Sr} = \frac{40 + 137}{2} = 88.5$$

Other example is Li (7), Na (23), K (39).

Q. 28 Elements have been arranged in the following sequence on the basis of their increasing atomic masses. F, Na, Mg, Al, Si, P, S, Cl, Ar, K.

(a) Pick two sets of elements which have similar properties.

(b) The given sequence represents which law of classification of elements?

Ans. (a) Here, the elements are arranged in the order of increasing atomic masses, so according to Newlands' law of octaves there is a repetition of every eighth element as compared to the given element. *The two sets of elements which have similar properties are*

Set I $\rightarrow$ F, Cl

Set II $\rightarrow$ Na, K

F and Cl are first and eighth element in the above sequence, therefore, they have similar properties. Although Na and K have similar properties but they are not related as first and eighth element in the above sequence.

(b) *The given sequence is according to Newlands' law of octaves represented as*

F Na Mg Al Si P S Cl Ar K

Q. 29 Can the following groups of elements be classified as Dobereiner's triad?

(a) Na, Si, Cl

(b) Be, Mg, Ca

Atomic mass of Be 9; Na 23; Mg 24; Si 28; Cl 35; Ca 40

Explain by giving reason.

Ans. (a) Na, Si and Cl have different properties, therefore, they do not form a Dobereiner's triad even though the atomic mass of the middle atom (Si) is approximately the average of the atomic masses of Na and Cl i.e.,

Na (23); Si (28); Cl (35)

$$\text{Atomic mass of Si} = \frac{23 + 35}{2} = \frac{58}{2} = 29$$

(b) Be, Mg and Ca have many similar properties and also the atomic mass of the middle element Mg is approximately the average of the atomic masses of Be and Ca i.e.,

Be (9); Mg (24); Ca (40)

$$\text{Atomic mass of Mg} = \frac{9 + 40}{2} = \frac{49}{2} = 24.5$$

Therefore, they form a Dobereiner's triad.

Q. 30 In Mendeleev's periodic table, the elements were arranged in the increasing order of their atomic masses. However, cobalt with atomic mass of 58.93 amu was placed before nickel having an atomic mass of 58.71 amu. Give reason for the same.

Ans. In Mendeleev's periodic table, cobalt (Co) with a higher atomic mass of 58.93 u is placed before nickel (Ni) due to the following reasons

- The properties of cobalt are similar to those of rhodium (Rh) and iridium (Ir) (same group) and
- The properties of nickel are similar to those of palladium (Pd) and platinum (Pt) (same group).

Q. 31 'Hydrogen occupies a unique position in modern periodic table', justify the statement.

Ans. Hydrogen occupies a unique position in the modern periodic table due to the following reasons

- Both hydrogen and alkali metals have similar outer electronic configuration as both have one electron in the valence shell. Therefore, some of the properties of hydrogen are similar to those of alkali metals and hence, it can be placed in group 1 alongwith alkali metals.
- Both hydrogen and halogens have similar outer electronic configuration (both have one electron less than the nearest inert gas configuration). Therefore, some of the properties of hydrogen are similar to those of halogens and hence, it can be placed in group 17 alongwith halogens.
- In some properties, it differs from both hydrogen and halogens. e.g., the oxide of hydrogen, i.e., H_2O is neutral but the oxides of alkali metals (i.e., Na_2O , K_2O etc.) are basic while those of halogens (i.e., Cl_2O_7 , Br_2O_5 , I_2O_5 etc.) are acidic.

Q. 32 Write the formulae of chlorides of Eka-silicon and Eka-aluminium, the elements predicted by Mendeleev.

Ans. Eka-silicon is germanium (Ge). It lies in group 4 of the Mendeleev's periodic table and thus, has a valency of 4.

∴ The formula of its chloride is $GeCl_4$.

Eka-aluminium is gallium (Ga). It lies in group 3 of the Mendeleev's periodic table and thus, has a valency of 3.

∴ The formula of its chloride is $GaCl_3$.

Q. 33 Three elements **A**, **B** and **C** have 3, 4 and 2 electrons respectively in their outermost shell. Give the group number to which they belong in the modern periodic table. Also, give their valencies.

Ans. (i) Element A has 3 valence electrons, therefore, its valency is 3 and thus belongs to group 13 ($3 + 10$). As such it could be any one of the following elements : B, Al, Ga, In or Tl.

(ii) Element B has 4 valence electrons, therefore, its valency is 4 and it belongs to group 14 ($4 + 10$). The element B could be any one of the following : C, Si, Ge, Sn or Pb.

(iii) Element C has two valence electrons, therefore, its valency is 2 and it belongs to group 2. The element C could be any one of the following : Be, Mg, Ca, Sr, Ba or Ra.

Q. 34 If an element **X** is placed in group 14, what will be the formula and the nature of bonding of its chloride?

Ans. Element X is placed in group 14. Therefore, its electronic configuration is 2, 8, 4.

∴ Number of valence electrons is 4.

Further, since it is difficult to lose all the four valence electrons or gain four more electrons. Therefore, it prefers to share these four electrons to acquire the stable electronic configuration of the nearest inert gas.

Thus, the nature of the chloride of element X is covalent. Its chloride will be XCl_4

Q. 35 Compare the radii of two species **X** and **Y**. Give reasons for your answer.

(a) **X** has 12 protons and 12 electrons

(b) **Y** has 12 protons and 10 electrons

Ans. Since, species X has 12 protons and 12 electrons, it is electrically neutral. Since, species Y has 12 protons and 10 electrons, therefore, it has two units positive charge. The electronic configuration of the two species are

Species X			Species Y	
K	L	M	K	L
2	8	2	2	8

Since, species X has three shells while species Y has two shells, therefore, species Y has smaller radius than species X.

Q. 36 Arrange the following elements in increasing order of their atomic radii.

(a) Li, Be, F, N

(b) Cl, At, Br, I

Ans. (a) Atomic radii decrease along a period from left to right due to increasing nuclear charge. Li, Be, F and N belong to same period. Thus, the atomic radii of Li, Be, F and N increase in the order : $\text{F} < \text{N} < \text{Be} < \text{Li}$.

(b) Atomic radii increase in a group from top to bottom due to the corresponding increase in the number of filled electronic shells. Cl, At, Br, I belong to same group. Thus, atomic radii of Cl, At, Br and I increase in the order : $\text{Cl} < \text{Br} < \text{I} < \text{At}$.

Q. 37 Identify and name the metals out of the following elements whose electronic configurations are given below.

(a) 2, 8, 2

(b) 2, 8, 1

(c) 2, 8, 7

(d) 2, 1

💡 Thinking Process

Elements having 1-3 electrons in the valence shell are metals, those having 4 valence electrons may be non-metals, metalloids or may even metals while those, having 5-8 valence electrons are non-metals.

Ans. The following elements whose electronic configurations are given below

Electronic configuration	Number of valence electrons	Metal/Non-metal	Atomic number	Name of the element
(a) 2, 8, 2	2	Metal	$2 + 8 + 2 = 12$	Magnesium
(b) 2, 8, 1	1	Metal	$2 + 8 + 1 = 11$	Sodium
(c) 2, 8, 7	7	Non-metal	$2 + 8 + 7 = 17$	Chlorine
(d) 2, 1	1	Metal	$2 + 1 = 3$	Lithium

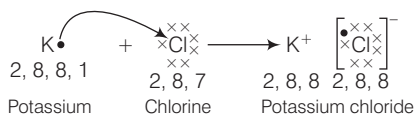
Thus, elements (a), (b) and (d) are metals while (c) is a non-metal.

Q. 38 Write the formula of the product formed when the element **A** (atomic number 19) combines with the element **B** (atomic number 17). Draw its electronic dot structure. What is the nature of the bond formed?

Thinking Process

Now as the metal loses the valence electron and the non-metal accepts the electron lost by metal to acquire the stable electronic configuration of the nearest noble gas. As a result, metal forms positive ion (cation) and non-metal forms negative ion (anion).

Ans. Atomic number of A = 19
 Electronic configuration is 2, 8, 8, 1
 Hence, element A is metal potassium (K) and
 Atomic number of B = 17
 Electronic configuration is 2, 8, 7
 It is a non-metal, chlorine (Cl).
 So, the electron dot structure of KCl is



The bond formed between K^+ and Cl^- is ionic bond and formula of the product formed $\text{K}^+ \text{Cl}^-$ or KCl.

Q. 39 Arrange the following elements in the increasing order of their metallic character Mg, Ca, K, Ge, Ga.

Ans. Arranging the given elements into different groups and periods in order of their increasing atomic numbers. So, we have
 Group number 1 2 13 14
 Third period : — Mg — —
 Fourth period : K Ca Ga Ge
 Since, the metallic character decreases along a period from left to right, therefore, metallic character increases in the order : $\text{Ge} < \text{Ga} < \text{Ca} < \text{K}$ (as all these belong to same period).
 Now we compare the metallic character of Ca and Mg. Since, metallic character increases down the group from top to bottom, therefore, Ca is more metallic than Mg.
 Combining the above two results, the overall metallic character increases in the order $\text{Ge} < \text{Ga} < \text{Mg} < \text{Ca} < \text{K}$.

Q. 40 Identify, the elements with the following property and arrange them in increasing order of their reactivity.

- An element which is a soft and reactive metal.
- The metal which is an important constituent of limestone.
- The metal which exists in liquid state at room temperature.

Ans. (a) An element which is soft and reactive metal is sodium (Na) potassium (K) as alkali metals are soft and reactive.
 (b) Limestone is calcium carbonate CaCO_3 . The important constituent of limestone is calcium (Ca).
 (c) Metal which exists in liquid state at room temperature is mercury (Hg) So, the increasing order of their reactivity

$$\text{Mercury} < \text{Calcium} < \text{Sodium}$$

 or

$$\text{Hg} < \text{Ca} < \text{Na}.$$

Q. 41 Properties of the elements are given below. Where would you locate the following elements in the periodic table?

- A soft metal stored under kerosene.
- An element with variable (more than one) valency stored under water.
- An element which is tetravalent and forms the basis of organic chemistry.
- An element which is an inert gas with atomic number 2.
- An element whose thin oxide layer is used to make other elements corrosion resistant by the process of "anodising".

Ans. (a) Alkali metals are soft and reactive towards air and moisture. Therefore, are stored under kerosene. e.g., sodium and potassium. Sodium metal (Na) is placed in group 1 and period 3 or potassium metal (K) is placed in group 1 and period 4.

(b) Phosphorus shows variable valency of 3 and 5. It is reactive towards air and not towards water. Therefore, it is stored under water. Phosphorus is placed in group 15 and period 3.

(c) The element, carbon (C) is tetravalent and is the basis of organic compounds. It is placed in group 14 and period 2.

(d) The lightest inert gas with atomic number 2 is helium (He). It is placed in group 18 and period 1.

(e) The metal whose thin oxide layer (i.e., Al_2O_3) is used to make other elements corrosion resistant by the process of anodising is aluminium (Al). It is placed in group 13 and period 3.

Long Answer Type Questions

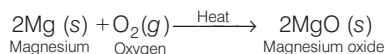
Q. 42 An element is placed in 2nd group and 3rd period of the periodic table, burns in presence of oxygen to form a basic oxide.

- Identify the element.
- Write the electronic configuration.
- Write a balanced equation when it burns in the presence of air.
- Write a balanced equation when this oxide is dissolved in water
- Draw the electron dot structure for the formation of this oxide.

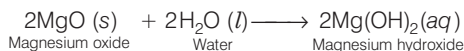
Ans. (a) Since, the element lies in group 2, it must be an alkaline earth metal. Since, it lies in the third period, it must be magnesium (Mg).

- (b) Atomic number of Mg is 12, therefore, its electronic configuration is : $\begin{matrix} K & L & M \\ & 2 & 8 & 2 \end{matrix}$

(c) When Mg burns in the presence of air it forms a basic oxide, MgO.

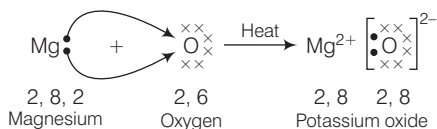


- (d) When MgO is dissolved in water it forms magnesium hydroxide.



- (e) Mg has 2 valence electrons [as electronic configuration of $_{12}\text{Mg} = 2, 8, 2$] oxygen has 6 valence electron [as electronic configuration of $_8\text{O} = 2, 6$].

Electron dot structure for the formation of magnesium oxide.



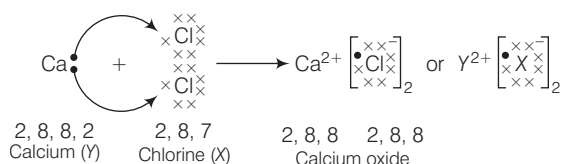
Q. 43 An element **X** (atomic number 17) reacts with an element **Y** (atomic number 20) to form a divalent halide.

- Where in the periodic table are elements **X** and **Y** placed?
- Classify **X** and **Y** as metal (s), non-metal (s) or metalloid (s).
- What will be the nature of oxide of element **Y**? Identify the nature of bonding in the compound formed.
- Draw the electron dot structure of the divalent halide.

Ans. (a) The electronic configuration of element 'X' with atomic number 17 is 2, 8, 7. Since, it has 7 valence electrons. Therefore, it lies in group 17 (10 + 7). Further, since in element X, third shell is being filled, it lies in third period. In other words, X is chlorine.

The electronic configuration of element 'Y' with atomic number 20 is 2, 8, 8, 2. Since, it has 2 valence electrons, it lies in group 2. Further, since in element Y, fourth shell is being filled, it lies in 4th period. In other words, Y is calcium.

- Since, element X (i.e., Cl) has seven electrons in the valence shell and needs one more electron to complete its octet. Therefore it is a **non-metal**. Further, since element Y has two electrons in the valence shell which it can easily lose to achieve the stable electronic configuration of the nearest inert gas, therefore, it is a **metal**.
- Since, element Y (i.e., Ca) is a metal, therefore, its oxide (i.e., CaO) must be basic in nature. Further, since metals and non-metals form ionic compounds, therefore, the nature of bonding in calcium oxide is ionic.
- Electronic configuration of $_{20}\text{Ca} = 2, 8, 8, 2$ [valence electrons = 2] electronic configuration of $_{17}\text{Cl} = 2, 8, 7$ [valence electrons = 7]. The electron dot structure of divalent metal halide, i.e., CaCl_2 is



Q. 44 Atomic number of a few elements are given below 10, 20, 7, 14

- Identify the elements.
- Identify the group number of these elements in the periodic table.
- Identify the periods of these elements in the periodic table.
- What would be the electronic configuration for each of these elements?
- Determine the valency of these elements.

💡 Thinking Process

In the modern periodic table, elements are arranged in order of their increasing atomic number. There are 18 groups and 7 periods in the periodic table. Each period starts with the filling of a new energy shell and the number of valence electrons increases regularly by one as we move from left to right.

Ans.

Atomic number	Electronic configuration	Group number	Period number	Valency	Element
10	2, 8	$(8 + 10) = 18$	2 nd	Zero	Neon
20	2, 8, 8, 2	$= 2$	4 th	2	Calcium
7	2, 5	$(5 + 10) = 15$	2 nd	$(8 - 5) = 3$	Nitrogen
14	2, 8, 4	$(4 + 10) = 14$	3 rd	4	Silicon

Q. 45 Complete the following crossword puzzle (Figure).**Across**

- (1) An element with atomic number 12.
- (3) Metal used in making cans and member of group 14.
- (4) A lustrous non-metal which has 7 electrons in its outermost shell.

Down

- (2) Highly reactive and soft metal which imparts yellow colour when subjected to flame and is kept in kerosene.
- (5) The first element of second period.
- (6) An element which is used in making fluorescent bulbs and is second member of group 18 in the modern periodic table.
- (7) A radioactive element which is the last member of halogen family.
- (8) Metal which is an important constituent of steel and forms rust when exposed to moist air.
- (9) The first metalloid in modern periodic table whose fibres are used in making bullet-proof vests.

	1	7				2					
		3	8				9		5		
						4				6	

Ans. Across

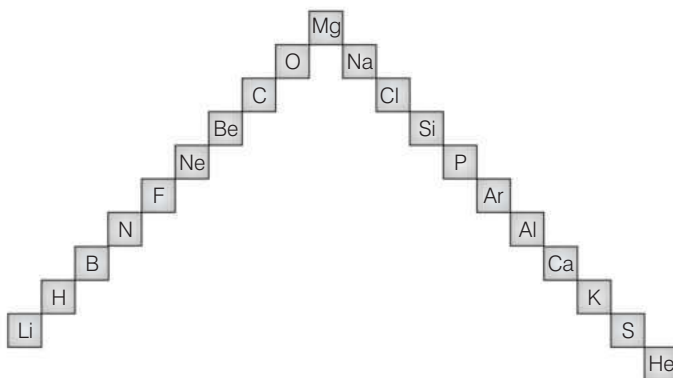
- (1) An element with atomic number 12 is magnesium
- (3) Metal used in making cans and member of group 14 is tin
- (4) A lustrous non-metal which has 7 electrons in its outermost shell is iodine.

Down

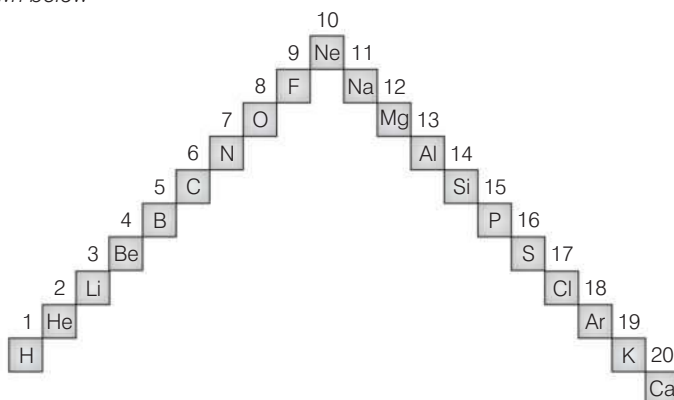
- (2) Highly reactive and soft metal which imparts yellow colour when subjected to flame and is kept in kerosene is sodium.
- (5) The first element of second period is lithium.
- (6) An element which is used in making fluorescent bulbs and is second member of group 18 in the modern periodic table is neon.
- (7) A radioactive element which is the last member of halogen family is astatine.
- (8) Metal which is an important constituent of steel and forms rust when exposed to moist air is iron.
- (9) The first metalloid in modern periodic table whose fibres are used in making bullet-proof vests is boron.

	M ¹	A ⁷	G	N	E	S ²	I	U	M		
		S				O					
		T ³	I ⁸	N		D	B ⁹		L ⁵		
		A	R			I ⁸	O	D	I	N ⁶	E
		T	O			U	R		T	E	
		I	N			M	O		H	O	
		N					N		I	N	
		E							U		
									M		

- Q. 46** (a) In this ladder (Figure) symbols of elements are jumbled up. Rearrange these symbols of elements in the increasing order of their atomic numbers in the periodic table.
- (b) Arrange them in the order of their group also.



Ans. (a) The symbols of elements arranged in increasing order of their atomic numbers are as shown below



Atomic numbers are written on each rectangle

(b) The symbols of elements arranged in increasing order of their group numbers are as follows.

1	2	13	14	15	16	17	18
H							He
Li	Be	B	C	N	O	F	Ne
Na	Mg	Al	Si	P	S	Cl	Ar
K	Ca						

Q. 47 Mendeleev predicted the existence of certain elements not known at that time and named two of them as Eka-silicon and Eka-aluminium.

- Name the elements which have taken the place of these elements.
- Mention the group and the period of these elements in the modern periodic table.
- Classify these elements as metals, non-metals or metalloids.
- How many valence electrons are present in each one of them?

Ans. (a) Germanium (Ge) has taken the place of Eka-silicon while gallium (Ga) has taken the place of Eka-aluminium.

(b) **Germanium** (Ge) group 14, period 4.

Gallium (Ga) group 13, period 4.

(c) Gallium is **other metal** while germanium (Ge) is a **metalloid**.

Note In periodic table, **other metals** are the metallic elements in group 13 to 16 of the periodic table. They are physically weak metals having their location between 'true metals' and 'metalloids'.

(d) Ga lies in group 13

∴ It has $13 - 10 = 3$ valence electrons.

Ge lies in group 14.

∴ It has $14 - 10 = 4$ valence electrons.

- Q. 48** (a) Electropositive nature of the element (s) increases down the group and decreases across the period.
 (b) Electronegativity of the element decreases down the group and increases across the period.
 (c) Atomic size increases down the group and decreases across a period (left to right).
 (d) Metallic character increases down the group and decreases across a period.

On the basis of the above trends of the periodic table, answer the following about the element with atomic numbers 3 to 9.

- (a) Name the most electropositive element among them.
 (b) Name the most electronegative element.
 (c) Name of the element with smallest atomic size.
 (d) Name the element which is a metalloid.
 (e) Name the element which shows maximum valency.

Ans. The names and symbols of elements having atomic numbers 3 — 9 are

Atomic number	3	4	5	6	7	8	9
Name	Lithium	Beryllium	Boron	Carbon	Nitrogen	Oxygen	Fluorine
Symbol	Li	Be	B	C	N	O	F

- (a) Most electropositive element is lithium (Li).
 (b) Most electronegative element is fluorine (F).
 (c) Element with smallest atomic size is fluorine (F).
 (d) Metalloid element is boron (B).
 (e) Carbon (C) shows maximum valency *i.e.*, 4.

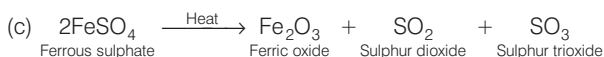
Q. 49 An element **X** which is a yellow solid at room temperature shows catenation and allotropy. **X** forms two oxides which are also formed during the thermal decomposition of ferrous sulphate crystals and are the major air pollutants.

- (a) Identify the element **X**.
 (b) Write the electronic configuration of **X**.
 (c) Write the balanced chemical equation for the thermal decomposition of ferrous sulphate crystals?
 (d) What would be the nature (acidic/basic) of oxides formed?
 (e) Locate the position of the element in the modern periodic table.

Ans. (a) Element **X** which is a yellow solid at room temperature, shows catenation and allotropy is sulphur (S). It forms two oxides SO_2 and SO_3 which are also formed during thermal decomposition of ferrous sulphate crystals and are the major air pollutants.

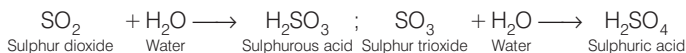
- (b) Atomic number of S = 16

Its electronic configuration is $\begin{matrix} K & L & M \\ 2 & 8 & 6 \end{matrix}$



Both SO_2 and SO_3 are major air pollutants.

- (d) Since, sulphur is a non-metal, therefore, both SO_2 and SO_3 are acidic oxides. (non-metals form acidic oxides). They dissolve in water to form the corresponding acids.

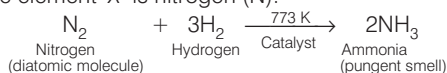


- (e) Since, sulphur contains six valence electrons, therefore, it lies in group $6 + 10 = 16$. Further, since atomic number of sulphur (S) is 16, electronic configuration is $\begin{matrix} K & L & M \\ 2 & 8 & 6 \end{matrix}$. So, it lies in 3rd period.

Q. 50 An element **X** of group 15 exists as diatomic molecule and combines with hydrogen at 773 K in presence of the catalyst to form a compound, ammonia which has a characteristic pungent smell.

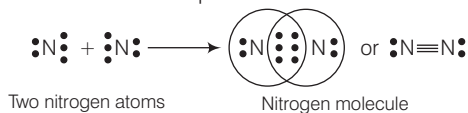
- Identify the element **X**. How many valence electrons does it have?
- Draw the electron dot structure of the diatomic molecule of . What type of bond is formed in it?
- Draw the electron dot structure for ammonia and what type of bond is formed in it?

Ans. Since, the element 'X' of group 15 exists as a diatomic molecule and combines with hydrogen at 773 K in presence of a catalyst to form ammonia which has a characteristic smell, therefore, the element 'X' is nitrogen (N).

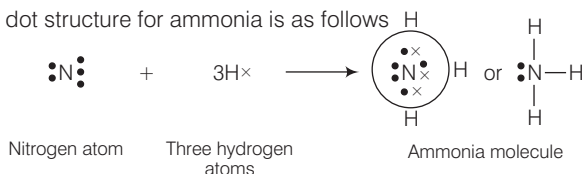


- The atomic number of nitrogen is 7. So, its electronic configuration is 2, 5. Thus, it has five valence electrons.
- Nitrogen has 5 valence electrons. Therefore, it needs 3 more electrons to complete its octet. To do so, it shares three of its electrons with three electrons of the other nitrogen atom to form a diatomic molecule of N_2 gas.

Thus, three **covalent bonds** are formed between two nitrogen atoms and each nitrogen atom is left with one lone pair of electrons.



- (c) Electron dot structure for ammonia is as follows



In NH_3 molecule, there are three N—H single covalent bonds and one lone pair of electrons on the nitrogen atom.

Q. 51 Which group of elements could be placed in mendeleev's periodic table without disturbing the original order? Give reason.

Ans. The noble gases could be placed in a new group without disturbing the existing order because they were very inert and present in extremely low concentrations in our atmosphere.

When Mendeleev gave his periodic table, at that time noble gases such as helium (He), neon (Ne), argon (Ar), krypton (Kr), xenon (Xe) and radon (Rn) were not known. When these gases were discovered much later, these were placed in new group called the zero group.

Q. 52 Give an account of the process adopted by Mendeleev for the classification of elements. How did he arrive at "periodic law"?

Ans. When Mendeleev selected his work, 63 elements were known. He examined the relationship between the atomic masses of the elements and their physical and chemical properties. Among the chemical properties, he concentrated on the compounds formed by elements with hydrogen and oxygen. He selected hydrogen and oxygen as they are very reactive and formed compounds with most elements. The formulae of the hydrides and oxides formed by elements were treated as one of the basic properties of the elements for their classification.

He then took 63 cards and on each card he wrote down the properties of one element. He sorted out the elements with similar properties and paired the cards together on a wall. He observed that most of the elements get a place in a periodic table and were arranged in order of their increasing atomic masses.

It was also observed that there occurs a periodic recurrence of elements with similar physical and chemical properties. On this basis, Mendeleev formulated a periodic law, which states that "the properties of elements are the periodic function of their atomic masses".

Life Processes

Multiple Choice Questions (MCQs)

Q. 1 Which of the following statements about the autotrophs is incorrect?

- (a) They synthesise carbohydrates from carbon dioxide and water in the presence of sunlight and chlorophyll
- (b) They store carbohydrates in the form of starch
- (c) They convert carbon dioxide and water into carbohydrates in the absence of sunlight
- (d) They constitute the first trophic level in food chains

Ans. (c) It is incorrect as, autotrophs can convert carbon dioxide and water into carbohydrates only in the presence of sunlight. The autotrophs contain the green pigment called chlorophyll which is capable of trapping sun energy.

This trapped energy is utilised by them to make food by combining inorganic materials like CO_2 and H_2O by the process of photosynthesis. The food that is not used immediately, is stored in the form of starch. These autotrophs constitute the first trophic level in food chains.

Q. 2 In which of the following groups of organisms, food material is broken down outside the body and absorbed?

- (a) Mushroom, green plants, *Amoeba*
- (b) Yeast, mushroom, bread mould
- (c) *Paramecium*, *Amoeba*, *Cuscuta*
- (d) *Cuscuta*, lice, tapeworm

Ans. (b) **Yeast, mushroom and bread mould** are saprophytes. They break down and convert complex organic molecules present in dead and decaying matter into simpler substances outside their body. These simpler substances are then absorbed by them, (i.e., saprotrophic nutrition).

Green plants are autotrophic (make their own food through photosynthesis).

Amoeba and ***Paramecium*** are holozoic (take solid or fluid organic food) and ***Cuscuta***, lice, tapeworm are parasitic (live on food material obtained from the body of host).

Q. 3 Select the correct statement

- (a) Heterotrophs do not synthesise their own food
- (b) Heterotrophs utilise solar energy for photosynthesis
- (c) Heterotrophs synthesise their own food
- (d) Heterotrophs are capable of converting carbon dioxide and water into carbohydrates

Ans. (a) Heterotrophs are those organisms which cannot make their own food from inorganic substances like CO_2 and water as they do not have chlorophyll to trap solar energy, e.g., all animals, most bacteria and fungi.

They depend on other organisms for their food. Autotrophs synthesise their own food through photosynthesis by utilising solar energy. e.g., green plants.

Q. 4 Which is the correct sequence of parts in human alimentary canal?

- (a) Mouth → Stomach → Small intestine → Oesophagus → Large intestine
- (b) Mouth → Oesophagus → Stomach → Large intestine → Small intestine
- (c) Mouth → Stomach → Oesophagus → Small intestine → Large intestine
- (d) Mouth → Oesophagus → Stomach → Small intestine → Large intestine

Ans. (d) The various organs in sequence are mouth, oesophagus, stomach, small intestine, large intestine and anus. The nutrition in human beings takes place through the digestive system, that consists of alimentary canal and associated glands.

Q. 5 If salivary amylase is lacking in the saliva, which of the following events in the mouth cavity will be affected?

- (a) Proteins breaking down into amino acids
- (b) Starch breaking down into sugars
- (c) Fats breaking down into fatty acids and glycerol
- (d) Absorption of vitamins

Ans. (b) The human saliva contains an enzyme called salivary amylase, which breaks down the starch present in food into sugar. The digestion of starch (carbohydrates) begins in the mouth. In case, saliva is lacking, it will affect the break down of starch.

The protein digestion begins in the stomach by the enzyme pepsin and completes in small intestine by enzyme trypsin, chymotrypsin, carboxypeptidases, aminopeptidases and dipeptidases.

The fats are broken down into small particles by bile in small intestine and then broken down into fatty acids and glycerol by lipases present in pancreatic juice.

Absorption of vitamins takes place in the small intestine.

Q. 6 The inner lining of stomach is protected by one of the following from hydrochloric acid. Choose the correct one

- (a) Pepsin (b) Mucus (c) Salivary amylase (d) Bile

Ans. (b) The mucus is secreted in the gastric juice by the glands present in the stomach wall. It helps to protect the wall of stomach from its own secretions of hydrochloric acid. If mucus is not secreted, HCl will cause the erosion of inner lining of stomach leading to ulcer formation.

Pepsin acts on proteins in the stomach. Salivary amylase is present in saliva, it acts on starch in the mouth.

Bile helps in emulsification of fat in small intestine.

Q. 7 Which part of alimentary canal receives bile from the liver?

- (a) Stomach (b) Small intestine (c) Large intestine (d) Oesophagus

Ans. (b) Small intestine receives bile from the liver, which is normally stored in the gall bladder. It performs two functions, i.e., makes the food alkaline and break the fats present in food into small globules by the process of emulsification.

Q. 8 A few drops of iodine solution were added to rice water. The solution turned blue-black in colour. This indicates that rice water contains

- (a) complex proteins (b) simple proteins
(c) fats (d) starch

💡 Thinking Process

A starch solution on addition of iodine turns blue-black in colour.

Ans. (d) The blue-black colour of rice water confirms the presence of starch.

No colour change will be seen in case of proteins or fats in iodine solution colour, when iodine is added with them.

Q. 9 In which part of the alimentary canal food is finally digested?

- (a) Stomach (b) Mouth cavity (c) Large intestine (d) Small intestine

Ans. (d) The small intestine in human beings is the site of complete digestion of food. It is also the main region for absorption of digested food. However, digestion begins in the mouth cavity, food is further digested in stomach to form a semi-solid paste. The undigested food from small intestine passes into large intestine that absorb most of the water.

Q. 10 Choose the function of the pancreatic juice from the following

- (a) Trypsin digests proteins and lipase carbohydrates
(b) Trypsin digests emulsified fats and lipase proteins
(c) Trypsin and lipase digest fats
(d) Trypsin digests proteins and lipase emulsified fats

Ans. (d) *Pancreas secretes pancreatic juice which contain following digestive enzymes amylases, lipases, trypsin. Amylase breaks down the starch, trypsin digests the proteins and lipase breaks down the emulsified fats.*

Q. 11 When air is blown from mouth into a test-tube containing lime water, the lime water turned milky due to the presence of

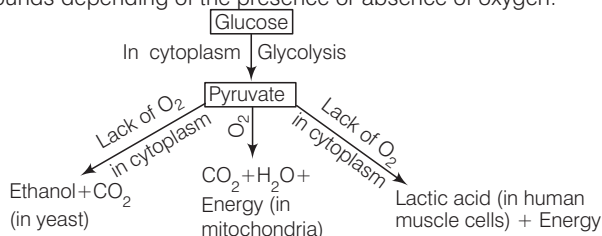
- (a) oxygen (b) carbon dioxide (c) nitrogen (d) water vapour

Ans. (b) Carbon dioxide gas turns lime water milky. When air is blown from mouth into test tube, the lime water turned milky because the air we breathe out has more CO_2 .

Q. 12 The correct sequence of anaerobic reactions in yeast is

- (a) Glucose $\xrightarrow{\text{Cytoplasm}}$ Pyruvate $\xrightarrow{\text{Mitochondria}}$ Ethanol + Carbon dioxide
(b) Glucose $\xrightarrow{\text{Cytoplasm}}$ Pyruvate $\xrightarrow{\text{Cytoplasm}}$ Lactic acid
(c) Glucose $\xrightarrow{\text{Cytoplasm}}$ Pyruvate $\xrightarrow{\text{Mitochondria}}$ Lactic acid
(d) Glucose $\xrightarrow{\text{Cytoplasm}}$ Pyruvate $\xrightarrow{\text{Cytoplasm}}$ Ethanol + Carbon dioxide

Ans. (d) Glucose is converted into pyruvate (glycolysis) in the cytoplasm of the cell. It is the first stage of respiration. After glycolysis, pyruvate gets converted into different compounds depending of the presence or absence of oxygen.



Breakdown of glucose by various pathways

Q. 13 Which of the following is most appropriate for aerobic respiration?

- (a) Glucose $\xrightarrow{\text{Mitochondria}}$ Pyruvate $\xrightarrow{\text{Cytoplasm}}$ $\text{CO}_2 + \text{H}_2\text{O} + \text{Energy}$
 (b) Glucose $\xrightarrow{\text{Cytoplasm}}$ Pyruvate $\xrightarrow{\text{Mitochondria}}$ $\text{CO}_2 + \text{H}_2\text{O} + \text{Energy}$
 (c) Glucose $\xrightarrow{\text{Cytoplasm}}$ Pyruvate + Energy $\xrightarrow{\text{Mitochondria}}$ $\text{CO}_2 + \text{H}_2\text{O}$
 (d) Glucose $\xrightarrow{\text{Cytoplasm}}$ Pyruvate + Energy $\xrightarrow{\text{Mitochondria}}$ $\text{CO}_2 + \text{H}_2\text{O} + \text{Energy}$

Ans. (b) Aerobic respiration uses oxygen for respiration. It takes place in the mitochondria. After glycolysis, pyruvate (from cytoplasm) enters the mitochondria and is oxidised in a series of reactions, to form considerable amount of energy (ATP), CO_2 and H_2O .

Q.14 Which of the following statements (s) is (are) true about respiration?

- (i) During inhalation, ribs move inward and diaphragm is raised
 (ii) In the alveoli, exchange of gases takes place, *i.e.*, oxygen from alveolar air diffuses into blood and carbon dioxide from blood into alveolar air
 (iii) Haemoglobin has greater affinity for carbon dioxide than oxygen
 (iv) Alveoli increases surface area for exchange of gases
 (a) (i) and (iv) (b) (ii) and (iii) (c) (i) and (iii) (d) (ii) and (iv)

Ans. (d) When we breathe in, we lift our ribs and flatten our diaphragm, the chest cavity becomes larger. Alveoli is balloon like structure which provides a surface where the exchange of gases occurs the walls of olveoli consist of network of blood-vessels. The air is sucked into the lungs to fill the expanded alveoli.

The blood brings carbon dioxide from the rest of the body for release into the alveoli. and the oxygen in the alveolar air is taken up by blood in the alveolar blood vessels to be transported to all the cells in the body.

In human beings, the respiratory pigment is haemoglobin which has a very high affinity for oxygen. This pigment is present in the red blood corpuscles.

The human lungs have millions of air sacs called alveoli whose surface area is many times that of the body. These are involved in exchange of gases.

Q.15 Which is the correct sequence of air passage during inhalation?

- (a) Nostrils $\rightarrow$ Larynx $\rightarrow$ Pharynx $\rightarrow$ Trachea $\rightarrow$ Lungs
 (b) Nasal passage $\rightarrow$ Trachea $\rightarrow$ Pharynx $\rightarrow$ Larynx $\rightarrow$ Alveoli
 (c) Larynx $\rightarrow$ Nostrils $\rightarrow$ Pharynx $\rightarrow$ Lungs
 (d) Nostrils $\rightarrow$ Pharynx $\rightarrow$ Larynx $\rightarrow$ Trachea $\rightarrow$ Alveoli

Ans. (d) The air for respiration is drawn into our body through the **nostrils**, it then goes into nasal passage. From their, air enters into **pharynx**, **larynx**, then into the windpipe (or trachea), bronchi, lungs and finally to the **alveoli** where gaseous exchange takes place.

Q. 16 During respiration exchange of gases take place in

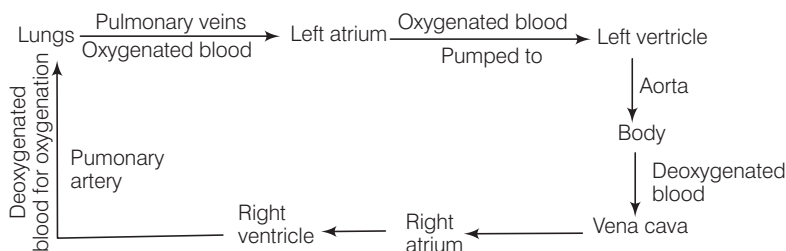
- (a) trachea and larynx (b) alveoli of lungs
 (c) alveoli and throat (d) throat and larynx

Ans. (b) It is in the alveoli in the lungs where gaseous exchange takes place. The walls of the alveoli are very thin and surrounded by network of very thin blood capillaries. The presence of millions of alveoli in the lungs provides a large area for the exchange of gases.

Q. 17 Which of the following statement (s) is (are) true about heart?

- (i) Left atrium receives oxygenated blood from different parts of body while right atrium receives deoxygenated blood from lungs
 - (ii) Left ventricle pumps oxygenated blood to different body parts while right ventricle pumps deoxygenated blood to lungs
 - (iii) Left atrium transfers oxygenated blood to right ventricle which sends it to different body parts
 - (iv) Right atrium receives deoxygenated blood from different parts of the body while left ventricle pumps oxygenated blood to different parts of the body
- (a) (i) (b) (ii) (c) (ii) and (iv) (d) (i) and (iii)

Ans. (c) Route of blood circulation in heart is as follows



Q. 18 What prevents backflow of blood inside the heart during contraction?

- (a) Valves in heart
- (b) Thick muscular walls of ventricles
- (c) Thin walls of atria
- (d) All of these

Ans. (a) Valves ensure that blood does not flow backwards when the atria or ventricles contract. Since ventricles have to pump blood into various organs with high pressure, they have thicker walls than atria. The separation of the right side and left side of heart by valve also helps to keep oxygenated and deoxygenated blood from mixing so as to supply high amount of oxygen to the body.

Q. 19 Single circulation, *i.e.*, blood flows through the heart only once during one cycle of passage through the body, is exhibited by

- (a) Labeo, Chameleon, Salamander
- (b) Hippocampus, Exocoetus, Anabas
- (c) Hyla, Rana, Draco
- (d) whale, dolphin, turtle

Ans. (b) The examples and the relation of animal group and heart are as

Animal Group	Heart	Examples
Fishes	2-chambered heart (one atrium and one ventricle) single circulation	Labeo, Hippocampus, Exocoetus, Anabas, etc. Fishes.
Amphibians and Reptiles	3-chambered heart (two atria and one ventricle) mixing of oxygenated and deoxygenated blood.	Salamander, Hyla, Rana (Amphibians) Chameleon, Draco, Turtle (Reptiles)
Birds and Mammals	4-chambered heart (two atria and two ventricle) No mixing of blood	Pigeon, parrot (birds), whale, dolphins, humans (Mammals)

According to this chart 2 chambered heart is present only in fishes and blood flows only once during one cycle. Group (b) contains fishes so correct answers are in option 'b'.

Q. 20 In which of the following vertebrate group/groups, heart does not pump oxygenated blood to different parts of the body?

- (a) Pisces and amphibians
- (b) Amphibians and reptiles
- (c) Amphibians only
- (d) Pisces only

Ans. (d) In Pisces (fishes), the heart is 2 chambered and pumps deoxygenated blood to the gills, where oxygenation takes place by diffusion. The oxygenated blood from the gills is supplied to the body parts where oxygen is utilised and CO_2 enters it making the blood deoxygenated.

This deoxygenated blood returns to the heart. Thus, it has single circulation, i.e., blood passes through the heart only once during one complete cycle.

Q. 21 Choose the correct statement that describes arteries.

- (a) They have thick elastic walls, blood flows under high pressure; collect blood from different organs and bring it back to the heart
- (b) They have thin walls with valves inside, blood flows under low pressure and carry blood away from the heart to various organs of the body
- (c) They have thick elastic walls, blood flows under low pressure carry blood from the heart to various organs of the body
- (d) They have thick elastic walls without valves inside, blood flows under high pressure and carry blood away from the heart to different parts of the body.

Ans. (d) Arteries are the vessels which carry blood away from the heart to various organs of the body. Since the blood emerges from the heart under high pressure, the arteries have thick, elastic walls.

Veins collect the blood from different organs and bring it back to the heart. They do not need thick walls because the blood is no longer under pressure, instead they have valves that ensure that the blood flows in only one direction.

Q. 22 The filtration units of kidneys are called

- (a) ureter
- (b) urethra
- (c) neurons
- (d) nephrons

Ans. (d) Nephrons are the filtration units of kidneys. It is the structural and functional unit of a kidney and has three functions - filtration, reabsorption and secretion. Urine from the kidney passes through a tube called ureter, that carry urine into the urinary bladder.

The urethra is a canal that carries urine from bladder and expels it out of body. Neurons are the nerve cells and are functional units of nervous system.

Q. 23 Oxygen liberated during photosynthesis comes from

- (a) water
- (b) chlorophyll
- (c) carbon dioxide
- (d) glucose

Ans. (a) Oxygen liberated during photosynthesis comes from water. *The following events occur during photosynthesis*

- (i) Absorption of light energy by chlorophyll.
- (ii) Conversion of light energy to chemical energy and splitting of water molecules into hydrogen and oxygen.
- (iii) Reduction of carbon dioxide to carbohydrates, which is used as food for plants.

Q. 24 The blood leaving the tissues becomes richer in

- (a) carbon dioxide (b) water (c) haemoglobin (d) oxygen

Ans. (a) The blood leaving the tissues becomes richer in CO_2 . The main artery (aorta) carries oxygenated blood to all the organs of the body from the heart. When the oxygenated blood passes through the capillaries of the body organs/tissues, it gives oxygen to the body cells.

At the same time, carbon dioxide produced during respiration enters into blood. Thus, deoxygenated blood (carrying CO_2) leaves the tissues and enters the vena cava, from where they are again passed to the lungs for oxygenation.

Q. 25 Which of the following is an incorrect statement?

- (a) Organisms grow with time
(b) Organisms must repair and maintain their structure
(c) Movement of molecules does not take place among cells
(d) Energy is essential for life processes

Ans. (c) A cell is the basic structural and functional unit of a living being. Each cell is made up of molecules. The molecules exhibit movement during cellular reactions that lead to life activities but their movements of molecules do not take place in the cells.

The growth of a living organism starts with the division of its cells. All living things have an organised structure, which if breaks down leads to various diseases, disorders and even death. Therefore, the living things have to repair and maintain their structures through various processes. These life processes require energy which is obtained from food.

Q. 26 The internal (cellular) energy reserve in autotrophs is

- (a) glycogen (b) protein (c) starch (d) fatty acid

Ans. (c) In autotrophs, the cellular energy reserve is starch. However, carbohydrates serve as a major fuel in the cells to provide energy for life processes.

The sugar (glucose) produced that is not used immediately gets stored in the form of starch in plants. In animals, food is stored in the form of glycogen. These food reserves provide energy as and when required by the organisms.

Q. 27 Which of the following equations is the summary of photosynthesis?

- (a) $6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 + 6\text{H}_2\text{O}$
(b) $6\text{CO}_2 + \text{H}_2\text{O} + \text{Sunlight} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + \text{O}_2 + 6\text{H}_2\text{O}$
(c) $6\text{CO}_2 + 12\text{H}_2\text{O} + \text{Chlorophyll} + \text{Sunlight} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 + 6\text{H}_2\text{O}$
(d) $6\text{CO}_2 + 12\text{H}_2\text{O} + \text{Chlorophyll} + \text{Sunlight} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{CO}_2 + 6\text{H}_2\text{O}$

Ans. (c) The summary equation of photosynthesis is

$6\text{CO}_2 + 12\text{H}_2\text{O} + \text{Chlorophyll} + \text{Sunlight} \longrightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{CO}_2 + 6\text{H}_2\text{O}$ by using

Photosynthesis is a process by which green plants make their own food, e.g., glucose), CO_2 and H_2O in the presence of sunlight energy and chlorophyll in the presence of. Oxygen gas is released during the process of photosynthesis.

Q. 28 Choose the event that does not occur in photosynthesis.

- (a) Absorption of light energy by chlorophyll
(b) Reduction of carbon dioxide to carbohydrates
(c) Oxidation of carbon to carbon dioxide
(d) Conversion of light energy to chemical energy

Ans. (c) Carbon dioxide is used in photosynthesis for the products on of food. The oxidation of carbon compounds occurs in respiration. (*Also, refer to Q. 23*)

Q. 29 The opening and closing of the stomatal pore depends upon

- (a) oxygen
- (b) temperature
- (c) water in guard cells
- (d) concentration of CO_2 in stomata

Ans. (c) Large number of tiny pores called stomata are present on the surface of the leaves of plants. The gaseous exchange as well as transpiration takes place through these pores. The opening and closing of the pore is a function of the guard cells.

The guard cells swell when water flows into them, causing the stomatal pores to open. Similarly the pore closes if the guard cells shrink or losses water in them.

Q. 30 Choose the forms in which most plants absorb nitrogen

- (i) Proteins
 - (ii) Nitrates and Nitrites
 - (iii) Urea
 - (iv) Atmospheric nitrogen
- (a) (i) and (ii) (b) (ii) and (iii) (c) (iii) and (iv) (d) (i) and (iv)

Ans. (b) Nitrogen is an essential element used in the synthesis of proteins and other compounds. This is taken up in the from of inorganic nitrates or nitrites. Sometimes it is taken up as organic compounds *i.e.*, urea, which have been prepared by bacteria from atmospheric nitrogen.

Q. 31 Which is the first enzyme to mix with food in the digestive tract?

- (a) Pepsin
- (b) Cellulase
- (c) Amylase
- (d) Trypsin

Ans. (c) Digestion process starts from the mouth where salivary glands are present. Salivary glands produces amyloses. So **amylase** is the first enzyme to mix with food. The saliva secreted by the salivary glands in mouth contains this enzyme that breaks down starch.

Pepsin is a protein digesting enzyme secreted by the gastric glands present in the stomach wall.

The pancreas secretes pancreatic juice in the small intestine. It contains enzymes like trypsin for digesting proteins.

Cellulase is an enzyme complex which breaks down cellulose. Most animals, including humans (except ruminants) do not produce cellulase and are therefore unable to use most energy contained in plant material.

Q. 32 Which of the following statements (s) is (are) correct?

- (i) Pyruvate can be converted into ethanol and carbon dioxide by yeast
 - (ii) Fermentation takes place in aerobic bacteria
 - (iii) Fermentation takes place in mitochondria
 - (iv) Fermentation is a form of anaerobic respiration
- (a) (i) and (iii) (b) (ii) and (iv)
(c) (i) and (iv) (d) (ii) and (iii)

Ans. (c) Fermentation is a form of anaerobic respiration. Pyruvate can be converted into ethanol and CO_2 in yeast during fermentation. Since this process takes place in the absence of air (oxygen), it is called anaerobic respiration. It takes place in cytoplasm. Break down of pyruvate using oxygen, *i.e.*, aerobic respiration takes place in the mitochondria.

Q. 33 Lack of oxygen in muscles often leads to cramps among cricketers. This results due to

- (a) conversion of pyruvate to ethanol
- (b) conversion of pyruvate to glucose
- (c) non-conversion of glucose to pyruvate
- (d) conversion of pyruvate to lactic acid

Ans. (d) Lactic acid is produced in the muscle from the metabolism of carbohydrate in the cells. When muscles work hard (as during the time of exercise), the amount of O_2 needed is used during energy production. Thus O_2 lacks in the muscles.

This results to another process called anaerobic metabolism. The product of this process of obtaining energy by anaerobically is lactic acid which is build up in muscles and caused pain. However, lactic acid produced helps in healing process actually.

Q. 34 Choose the correct path of urine in our body

- (a) Kidney → Ureter → Urethra → Urinary bladder
- (b) Kidney → Urinary bladder → Urethra → Ureter
- (c) Kidney → Ureters → Urinary bladder → Urethra
- (d) Urinary → Bladder → Kidney → Ureter → Urethra

Ans. (c) The correct path of urine is kidney → Ureters → Urinary bladder → Urethra.

The urine formed in each kidney eventually enters a long tube, the ureter, which connects the kidneys with the urinary bladder. Urine is stored in the urinary bladder until the pressure of the expanded bladder leads to the urge, to pass it out through the urethra.

Q. 35 During deficiency of oxygen in tissues of human beings, pyruvic acid is converted into lactic acid in the

- (a) cytoplasm (b) chloroplast (c) mitochondria (d) Golgi body

Ans. (a) Pyruvic acid is converted into lactic acid in the cytoplasm of muscle cells during deficiency of oxygen in human beings. (Also, refer to Q. 12)

Short Answer Type Questions

Q. 36 Name the following

- (a) The process in plants that links light energy with chemical energy.
- (b) Organisms that can prepare their own food.
- (c) The cell organelle where photosynthesis occurs.
- (d) Cells that surround a stomatal pore.
- (e) Organisms that cannot prepare their own food
- (f) An enzyme secreted from gastric glands in stomach that acts on proteins.

Ans. (a) Photosynthesis It is a process by which green plants having chlorophyll synthesise simple sugar (glucose) from water and CO_2 using the energy of sunlight. The light energy is used in splitting of water molecules into hydrogen and oxygen (i.e., chemical energy).

- (b) **Autotrophs** Those organisms which can make their own food from inorganic substances present in the environment, e.g., green plants, autotrophic bacteria etc. are called autotrophs.
- (c) **Chloroplasts** Chlorophyll is the green pigment capable of trapping light energy, required for photosynthesis. This pigment is present in chloroplast present in the leaves.
- (d) **Guard cells** Each stomatal pore is surrounded by a pair of guard cells, that control their opening and closing by the inflow and outflow of water.
- (e) **Heterotrophs** Those organisms which cannot make their own food from inorganic substances and depend on other organisms for their food. e.g., all animals, yeast, etc.
- (f) **Pepsin** It is a protein digesting enzyme secreted from gastric glands present in the wall of the stomach.

Q. 37 "All plants give out oxygen during day and carbon dioxide during night". Do you agree with this statement? Give reason.

Ans. Yes, the statement is correct.

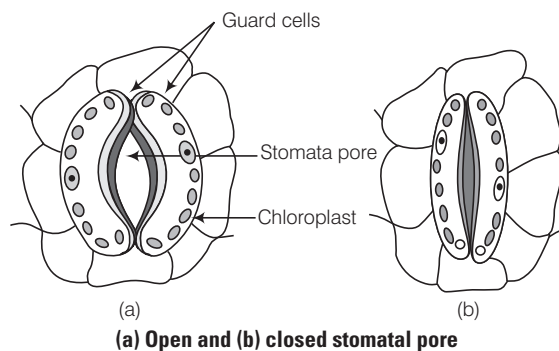
Plants respire throughout the day, while photosynthesis takes place only in the presence of sunlight. In daytime, CO_2 produced during respiration is used by the plants in photosynthesis. So, CO_2 is not released into the environment. The oxygen produced during photosynthesis, is released into air through stomata.

At night, when there is no photosynthesis, no oxygen is released. Also, CO_2 produced during respiration is not used by the plants. So, it is released in the air.

Q. 38 How do the guard cells regulate opening and closing of stomatal pores?

Ans. Opening and Closing of Stomata Opening and closing of stomata is an important event in the process of photosynthesis. Along with gaseous exchange, larger amount of water can also be lost through stomata. So, plants close stomata when they does not need carbon dioxide. Each stoma is bounded by two kidney-shaped guard cells.

The opening and closing of stoma is resulted due to the change in the turgidity of these guard cells. When the guard cells take up water and become turgid, the stoma opens and when guard cells lose water and shrink, it closes.



The ends of the two guard cells are joined and thin outer wall bends more readily than the thick inner one. These play an important part in opening and closing of stomatal pore.

- Q. 39** Two green plants are kept separately in oxygen free containers, one in the dark and the other in continuous light. Which one will live longer? Give reasons.

💡 Thinking Process

The plant which is kept in the dark is unable to carry out photosynthesis and hence the container would be filled with CO_2 within a very short span of time. Lack of oxygen in the container would kill the plant.

- Ans.** On the other hand, the plant kept in light would be able to carry out photosynthesis and thus convert CO_2 into oxygen. Hence, this plant would live for a longer duration.

- Q. 40** If a plant is releasing carbon dioxide and taking in oxygen during the day, does it mean that there is no photosynthesis occurring? Justify your answer.

- Ans.** Release of CO_2 and intake of O_2 suggests that either photosynthesis is not taking place or its rate is too low.

Normally during the day, CO_2 generated during respiration is used up in photosynthesis, hence, there is no CO_2 liberates into the atmosphere. *i.e.*, rate of photosynthesis is much more than the rate of respiration. Instead, oxygen release is the major event at this time.

- Q. 41** Why do fishes die when taken out of water?

- Ans.** Fishes take oxygen through their gills. Gills are richly supplied with blood capillaries and can readily absorb oxygen dissolved in water. Gills of fishes are not equipped to take in gaseous oxygen (oxygen from air). Hence, a fish when taken out of water is unable to inhale oxygen rich air and so, dies quickly.

- Q. 42** Differentiate between an autotroph and a heterotroph.

- Ans.** The difference between an autotroph and a heterotroph are as below

Autotroph	Heterotroph
Organisms that prepare their own food.	Organisms that are dependent on other organisms for food.
They have chlorophyll to trap light energy.	They lack chlorophyll.
They are called as producers.	They are called as consumers.
They do not show locomotion.	Most of them show locomotion because they need to move in search of food.

- Q. 43** Is 'nutrition' a necessity for an organism? Discuss.

- Ans.** All the living organisms need energy to perform various life processes. They get this energy from food. Thus, food is the basic requirement of all living organisms.

It is needed for the following purposes

- it provides energy for various metabolic processes in the body.
- it is essential for the growth of new cells and repair or replacement of workout cells.
- it is needed to develop resistance against various diseases.

Thus nutrition is a necessity for an organism.

Q. 44 What would happen if green plants disappear from earth?

Ans. It mean a total disaster of the ecosystem. Green plants are the sources of energy for all organisms. All other organisms directly or indirectly depend on them for food. So, if all the green plants disappear from the earth, all the herbivores will die due to starvation and so will the carnivore, *i.e.*, it would result in extinction of life from the earth.

Q. 45 Leaves of a healthy potted plant were coated with vaseline. Will this plant remain healthy for long? Give reasons for your answer.

Ans. The plant will not remain healthy as layer of vaseline would make an impervious layer on the leaves. *This will adversely effect the plant because*
 (a) Plant will not get oxygen for respiration.
 (b) It will not get carbon dioxide for photosynthesis.
 (c) Upward movement of water and minerals would be hampered due to lack of transpiration.

Q. 46 How does aerobic respiration differ from anaerobic respiration?

Ans. *The aerobic respiration differ from anaerobic respiration is as*

Aerobic Respiration	Anaerobic Respiration
Oxygen is utilised for the breakdown of respiratory substrate.	Oxygen is not required.
Carbon dioxide and water are the end products.	Ethanol and carbon dioxide or lactic acid are the end products.
It takes place in cytoplasm (glycolysis) and inside mitochondria (Krebs cycle).	It takes place in cytoplasm only.
More energy is released due to complete oxidation of glucose.	Less energy is released due to incomplete oxidation of glucose.

Q. 47 Match the words of column I with that of column II.

Column I	Column II
A. Phloem	Excretion
B. Nephron	Translocation of food
C. Veins	Clotting of blood
D. Platelets	Deoxygenated blood

Ans. *The matching of column I with column II is as*

Column I	Column II
A. Phloem	Translocation of food
B. Nephron	Excretion
C. Veins	Deoxygenated blood
D. Platelets	Clotting of blood

Phloem are the transporting tissue which translocates food from leaves to other parts of the plants.

Nephrons are the unit of kidney which helps in excretion.

Veins are the tubes or blood vessels which contains deoxygenated blood.

Platelets are the component of blood which helps in clastting of blood.

Q. 48 Differentiate between an artery and a vein.

Ans. The differentiate between an artery and a vein are as below

Artery	Vein
Have thick elastic, muscular walls. Lumen is narrow.	Have thin, non-elastic walls. Lumen is wide.
Carry oxygenated blood from heart to other body parts (except pulmonary artery).	Carry deoxygenated blood from all body parts to heart (except pulmonary veins).
Valves are absent.	Valves are present to prevent the backflow of blood.

Q. 49 What are the adaptations of leaf for photosynthesis?

Ans. Leaves shows following adaptations for photosynthesis

- Leaves provide large surface area for maximum light absorption.
- The extensive network of veins enable quick transport of substances to and from the mesophyll cells.
- The chloroplasts are present, which are the site for photosynthesis. They are more in number on the upper surface.
- Presence of numerous stomata on leaf's surface for gaseous exchange and transpiration.

Q. 50 Why is small intestine in herbivores longer than in carnivores?

Ans. Cellulose forms the largest part of an herbivore's food. Digestion of cellulose takes a longer time, because the enzymes are produced by the ruminant bacteria that live in the gut of the herbivore. Longer small intestine ensures that the food stays for a longer duration and proper digestion is possible. In the case of carnivores, cellulose is not present in the diet thus the length of small intestine is less.

Q. 51 What will happen if mucus is not secreted by the gastric glands?

💡 Thinking Process

Gastric glands in stomach release hydrochloric acid, enzyme pepsin and mucus. HCl will act on the stomach walls.

Ans. Mucus plays an important role by protecting the inner lining of stomach from hydrochloric acid produced in the stomach. If mucus is not released, it will lead to corrosion of inner lining of stomach causing acidity and ulcers.

Q. 52 What is the significance of emulsification of fats?

💡 Thinking Process

The enzymes can act on smaller molecules effectively.

Ans. Fats are complex molecules which are present in food in the form of large globules that makes it difficult for enzymes to act on them.

Bile salts present in bile (secreted by liver) break down fat molecules mechanically into smaller globules, i.e., emulsify the fats, thus, increasing the efficiency and making it easy for the fat digesting enzymes to act upon them.

Q. 53 What causes movement of food inside the alimentary canal?

Ans. The wall of alimentary canal contains muscle layers. Rhythmic contraction and relaxation of these muscles pushes the food forward. This movement is called **peristalsis**, which occurs all along the gut.

Q. 54 Why does absorption of digested food occur mainly in the small intestine?

Ans. *Maximum absorption occurs in small intestine due to following reasons*

- (a) digestion is completed in small intestine
- (b) inner lining of small intestine is provided with villi which increases the surface area to ensure better absorption
- (c) wall of intestine is richly supplied with blood vessels (which take the absorbed food to each and every cell of the body).

Q. 55 Match the following columns.

Column I	Column II
A. Autotrophic nutrition	Leech
B. Heterotrophic nutrition	<i>Paramecium</i>
C. Parasitic nutrition	Deer
D. Digestion in food vacuoles	Green plant

Ans. *The matching of column I with column II is as*

Column I	Column II
A. Autotrophic nutrition	Green plant
B. Heterotrophic nutrition	Deer
C. Parasitic nutrition	Leech
D. Digestion in food vacuoles	<i>Paramecium</i>

Autotrophic nutrition Plant can synthesise their own food by the process of photosynthesis, thus one called autotrophs and this type of nutrition is called autotrophic nutrition.

Heterotrophic nutrition When organism depends on other plant or animal for their food they are called heterotrophs and nutrition is heterotrophic nutrition. Deer depends on plants for their food thus then heterotrophic nutrition.

Parasitic nutrition When any organism derives or is dependent on other organism for their food causing less to that organism are called parasites and nutrition made is parasitic nutrition. Leech depends on animals for blood as their food causing blood loss to the animals; thus it is a parasite.

Digestion in food vacuoles *Paramecium* digests food by the help of vacuoles. *Paramecium* are prokaryotic cells which do not show cellular arrangement in them.

Q. 56 Why is the rate of breathing in aquatic organisms much faster than in terrestrial organisms?

Ans. Aquatic organisms like fishes obtain oxygen from water present in dissolved state through their gills. Since the amount of dissolved oxygen is fairly low as compared to the amount of oxygen in the air, the rate of breathing in aquatic organisms is much faster than that seen in terrestrial organisms.

Terrestrial organisms use the oxygen present in the atmosphere for respiration. This oxygen is absorbed by different organs in different animals.

Q. 57 Why is blood circulation in human heart called double circulation?

Ans. The blood circulation in human heart is called **double circulation** because the blood passes through the heart twice during one complete cardiac cycle of the body.
Once through the right ventricle to the left auricle of the heart, *i.e.*, **pulmonary circulation**.
Second through the left ventricle to the right auricle of the heart *viz.*, body tissues, *i.e.*, **systemic circulation**.

Q. 58 What is the advantage of having four chambered heart?

Ans. In four chambered heart, left half is completely separated from right half by septa.

It serves following two purposes

- (a) ensures complete segregation of oxygenated and deoxygenated blood (*i.e.*, no mixing).
- (b) allows a highly efficient supply of oxygenated blood to all parts of the body.

This is useful in warm blooded animals (like mammals and birds) which have high energy needs, because they constantly require energy to maintain their body temperature.

Q. 59 Mention the major events during photosynthesis.

Ans. *The major events during photosynthesis are*

- (a) Absorption of sunlight energy by chlorophyll.
- (b) Conversion of light energy into chemical energy, and splitting of water molecule into hydrogen, and oxygen.
- (c) Reduction of CO₂ to produce carbohydrates.

Q. 60 In each of the following situations what happens to the rate of photosynthesis?

- (a) Cloudy days
- (b) No rainfall in the area
- (c) Good manuring in the area
- (d) Stomata get blocked due to dust

Ans.

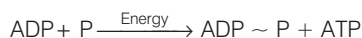
- (a) **Cloudy days** the rate of photosynthesis will decrease because sunlight is necessary for photosynthesis. On a cloudy day, there will be less sunlight.
- (b) **No rainfall in the area** the rate of photosynthesis will decrease because water is one of the raw material needed by plant for photosynthesis. If there is no rainfall in an area, there will be less water available to plant.
- (c) **Good manuring in the area** the rate will increase because plants need raw minerals such as N, P, Fe, Mg, etc., for building their body and take these minerals from the soil. Good manuring in the area will increase the amount of these minerals in the soil thus increasing photosynthesis.
- (d) **Stomata get blocked due to dust** the rate will decrease because CO₂ gas required for photosynthesis enters the leaves through the stomata. If the stomata gets blocked due to dust, less CO₂ or no CO₂ would enter the leaf and the rate of photosynthesis will drop.

Q. 61 Name the energy currency in the living organisms. When and where is it produced?

Ans. ATP (Adenosine Triphosphate) is the energy currency in the living organisms.

It is produced at the end of respiration in mitochondria.

The energy released during respiration is used to make an ATP molecule from ADP and inorganic phosphate.



Q. 62 What is common for *Cuscuta*, ticks and leeches?

Ans. *Cuscuta*, ticks and leeches, all are parasites, i.e., organism which feeds on another living organism called its host.

A parasite receives its food from the host but gives no benefit to host in return. It usually harms the host.

Q. 63 Explain the role of mouth in digestion of food.

Ans. In human beings, food is ingested through the mouth, where the digestion of food begins. *The mouth plays following roles in digestion of food*

- The **teeth** cut the food into small pieces, chew and grind it, i.e., help in physical digestion.
- The **salivary glands** in mouth secrete **saliva** which contain an enzyme called salivary amylase. This enzyme digest the starch present in food into sugar, i.e., help in chemical digestion.
- Tongue helps in thorough mixing of food with saliva.

Q. 64 What are the functions of gastric glands present in the wall of the stomach?

Ans. The stomach wall contains three tubular glands (gastric glands) that secrete gastric juice. *The gastric juice contains*

- Hydrochloric acid** makes the medium acidic so that enzyme pepsin become active. HCl kills bacteria that may enter the stomach with the food.
- Mucus** helps to protect stomach wall from its own secretions HCl.
- Enzyme pepsin** begins the digestion of proteins present in food to form smaller molecules.

Q. 65 Match the terms in column I with those in column II.

Column I	Column II
A. Trypsin	Pancreas
B. Amylase	Liver
C. Bile	Gastric glands
D. Pepsin	Saliva

Ans. The matching of column I with column II is as

Column I	Column I
A. Trypsin	Pancreas
B. Amylase	Saliva
C. Bile	Liver
D. Pepsin	Gastric glands

Trypsin is secreted by pancreas which digests proteins. Amylase is secreted by salivary gland with saliva and digests carbohydrate/starch bile is produced by Liver helps in the emulsification of fat. Pepsin is secreted by gastric glands present in stomach and digests protein.

Q. 66 Name the correct substrates for the following enzymes

(a) Trypsin

(b) Amylase

(c) Pepsin

(d) Lipase

Ans. The correct substrates for the following enzymes are

Enzyme	Substrate
A. Trypsin	Proteins (peptides)
B. Amylase	Starch
C. Pepsin	Proteins
D. Lipase	Emulsified fats

(Also, refer to Q. 65)

Lipases acts an emulsified fat emulsification is brought about by bile from liver.

Q. 67 Why do veins have thin walls as compared to arteries?

Ans. Arteries are thick walled blood vessels which carry blood from heart to all parts of body. They have thick walls because blood emerges from the heart under high pressure.

In comparison, veins are thin walled blood vessels which carry deoxygenated blood from all parts of the body back to the heart. They do not need thick walls because blood flowing through them is no longer under high pressure. Instead, they have valves which prevent the backflow of blood.

Q. 68 What will happen if platelets were absent in the blood?

💡 Thinking Process

Platelets play an important role in coagulation of blood.

Ans. In the absence of platelets, the process of clotting will be affected. This could be a dangerous situation, in case of an injury. No blood coagulation would result in excessive blood loss and can prove fatal for the person.

Q. 69 Plants have low energy needs as compared to animals. Explain.

💡 Thinking Process

Energy is required for movement and other complex activities.

Ans. Plants do not move. Movements in a plant are at the cellular level. In a large plant body, there are many dead cells like schlerenchyma which are not involved in cellular function but provides strength to the plot. As a result it requires less energy.

Animals, on the other hand need more energy to move from one place to another, in search of food and other activities, e.g., mating, predation, shelter, etc. So, the energy need is higher in them.

Q. 70 Why and how does water enter continuously into the root xylem?

Ans. Plants require water to carry out photosynthesis and other activities. The continuous flow of water in root xylem is due to transpiration pull. The water which is lost through the stomata is replaced by xylem vessel. Evaporation of water molecules cells of leaf creates a suction which pulls water from the xylem of roots to leaves.

Thus, transpiration helps in absorption and upward movement of water and minerals dissolved in it from roots to learn.

Cells of roots are in close contact with soil and so actively take up ions by diffusion. The ion-concentration increases inside the root and hence osmotic pressure increases the movement of water from soil into root which occurs continuously.

Q. 71 Why is transpiration important for plants?**💡 Thinking Process**

Transpiration is the evaporation of water from the leaves of a plant. The continuous evaporation of water from the cells of a leaf creates a kind of suction which pulls up water through the xylem vessels.

Ans. Transpiration is important because

- (i) it helps in the absorption and upward movement of water and minerals from roots to leaves.
- (ii) it prevents the plant parts from heating up. i.e., it helps in temperature regulation.

Q. 72 How do leaves of plants help in excretion?

Ans. The leaves help in excretion as given below

- (a) Gaseous waste of respiration and photosynthesis (CO_2 water vapour) are removed through the stomata in leaves.
- (b) Many plants store waste materials in the vacuoles of mesophyll cells and epidermal cells. When older leaves fall, the waste materials are excreted along with the leaves.

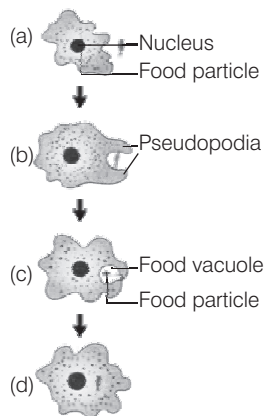
Long Answer Type Questions

Q. 73 Explain the process of nutrition in *Amoeba*.

Ans. *Amoeba* is a unicellular animal which follows holozoic mode of nutrition. The process of obtaining food by *Amoeba* is called phagocytosis.

The various steps involved in phagocytosis are

- (i) **Ingestion** *Amoeba* ingests food particle by forming temporary finger like projections called pseudopodia around it. The food is engulfed forming a food vacuole.
- (ii) **Digestion** Inside the food vacuole, complex substances are broken down into simpler ones by digestive enzymes.
- (iii) **Absorption** The digested food present in the food vacuole diffuse into the cytoplasm. After this, the food vacuole disappears.
- (iv) **Assimilation** Food absorbed is used by *Amoeba* for various processes.
- (v) **Egestion** The remaining undigested material moves to the surface of the cell, the cell membrane ruptures at any place and undigested food is thrown out of the body.



Nutrition in *Amoeba*

Q. 74 Describe the alimentary canal of man.

Ans. **Alimentary canal of man** It measures about 8 to 10 metres in length and extends from mouth to anus. It has following parts

Mouth It consists of the oral cavity, through which food is ingested. It contains teeth, tongue and three pairs of salivary glands.

Pharynx The oral cavity opens into the pharynx. The swallowing mechanism guides the masticated food through the pharynx into a tube, called **oesophagus**.

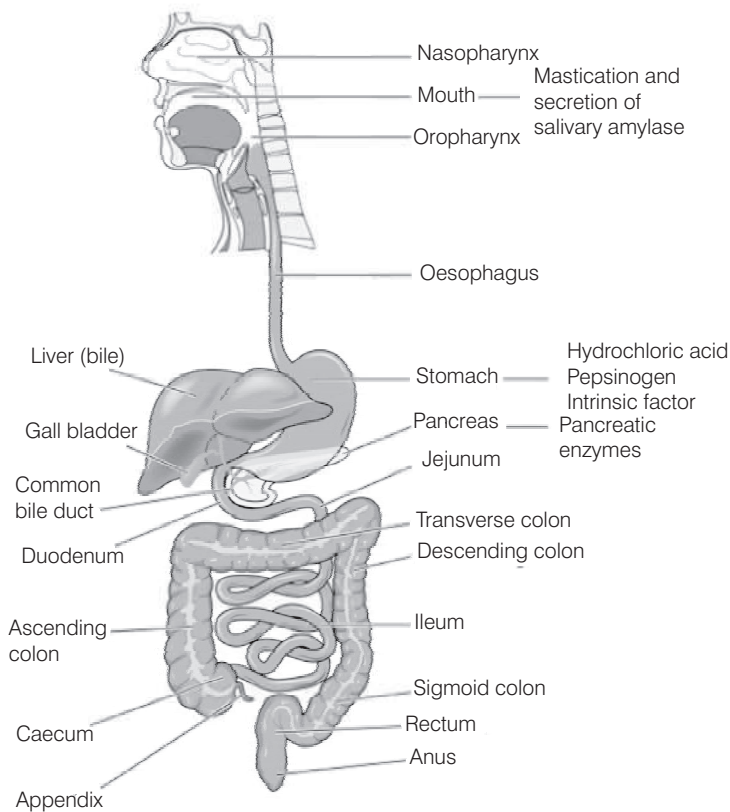
Oesophagus It is a muscular, tubular part of the alimentary canal. The muscular movement called peristalsis carries the food down to the stomach.

Stomach It is a J-shaped sac-like muscular structure that serves as a storehouse of food. It is a large organ which expands when food enters it. The muscular walls helps in mixing food thoroughly with more digestive juices.

Small intestine It is about 6 metres in length and has three divisions, *i.e.*, duodenum (first part as C-shaped, begins from the pyloric stomach), jejunum and ileum.

Large intestine The ileum passes into large intestine, which can be divided into two parts as anterior colon and posterior rectum.

The terminal part of rectum is called **anal canal**. It opens through the **anus**, guarded by sphincter muscles. It allows the faecal matter to be egested out.



Human alimentary canal

Q. 75 Explain the process of breathing in man.

Ans. Breathing in humans involve following steps

- (i) **Inspiration** When we breathe in, the muscles between the ribs contract causing the rib cage to move upward and outward, and the diaphragm contracts and moves downward. As the chest cavity becomes large, air is sucked in from outside into the lungs, the lungs get filled up with air and expand.
- (ii) **Gaseous exchange** Haemoglobin in RBCs binds with oxygen and carries it along the body, with blood. As blood passes through the tissues of the body, the oxygen from blood diffuses into the cell whereas CO_2 produced during respiration diffuse into the blood and is carried to the lungs.
- (iii) **Expiration** When we breathe out, the muscles between the ribs relax causing the rib cage to move downward and inward and the diaphragm relaxes and moves upward. As the chest cavity becomes smaller, air is pushed out from the lungs.

Q. 76 Explain the importance of soil for plant growth.

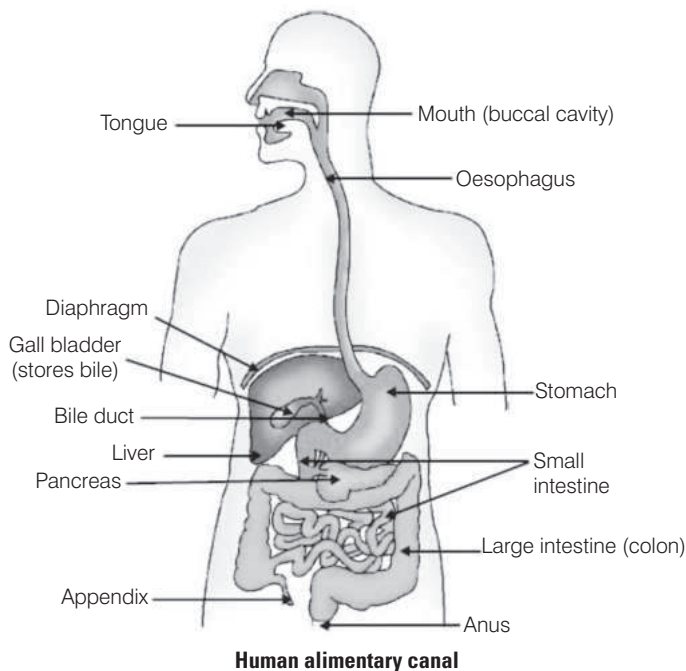
Ans. Soil is important for plant growth due to reasons given below

- (i) Soil provides **anchorage** to plants. It provides the base on which all the terrestrial and some aquatic plants grow.
- (ii) Soil is the **reservoir** of **water and minerals**, essential for all metabolic activities of a plant.
- (iii) The roots of a plant also take the oxygen required for respiration from the soil particles by the process of diffusion.
- (iv) Some plants, (e.g., leguminous plants) also forms symbiotic association with microbes presents in the soil to absorb mineral, like nitrogen.

Q. 77 Draw the diagram of alimentary canal of man and label the following parts.

Mouth, Oesophagus, Stomach, Intestine

Ans. The diagram of alimentary canal of man and its parts are shown as



Q.78 How do carbohydrates, proteins and fats get digested in human beings?

Ans. *Digestion of carbohydrates starts from*

Buccal cavity The food receives saliva in the buccal cavity, which contains salivary amylase. It breaks down starch into simple (glucose) sugar

Starch and glycogen $\xrightarrow{\text{Amylase}}$ Maltose + Isomaltose + Limit dextrins

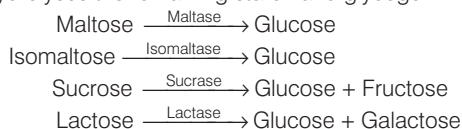
In small intestine

- (i) Pancreatic juice contains pancreatic amylase, which acts in further breakdown the starch.

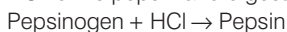
Starch and glycogen $\xrightarrow[\text{Amylase}]{\text{Pancreatic}}$ Maltose + Isomaltose + Limit dextrins

- (ii) Intestinal juice contains the following

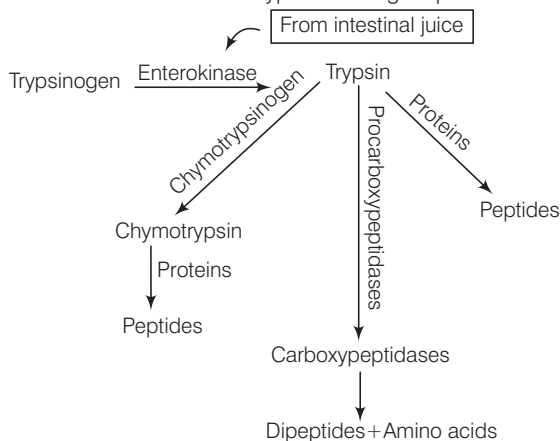
Intestinal amylase hydrolyses the remaining starch and glycogen.

**Digestion of Proteins starts**

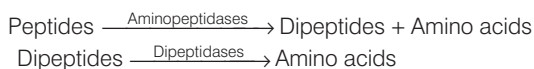
In stomach Inactive form of pepsin is secreted known as pepsinogen which in presence of HCl forms pepsin and digests protein.

**In small intestine**

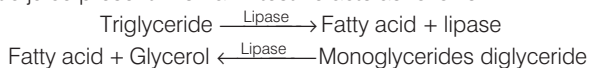
- (i) Pancreatic juice secretes Trypsinogen, chymotrypsinogen (inactive form of trypsin) which on activation forms trypsin and digest protein.



- (ii) Intestinal juice

**Digestion of Fats begins****In small intestine**

- (i) Bile emulsifies fats into smaller droplets
 (ii) Pancreatic juice present in small intestine acts as follows

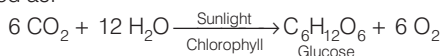


- (iii) Intestinal juice acts as

**Q. 79 Explain the mechanism of photosynthesis.**

Ans. The process by which green plants make their own food in the form of glucose from CO_2 and H_2O by using sunlight energy in the presence of chlorophyll, is called **photosynthesis**. Oxygen gas is released during photosynthesis.

It can be represented as.



The process takes place in the green leaves of a plant glucose in the following steps

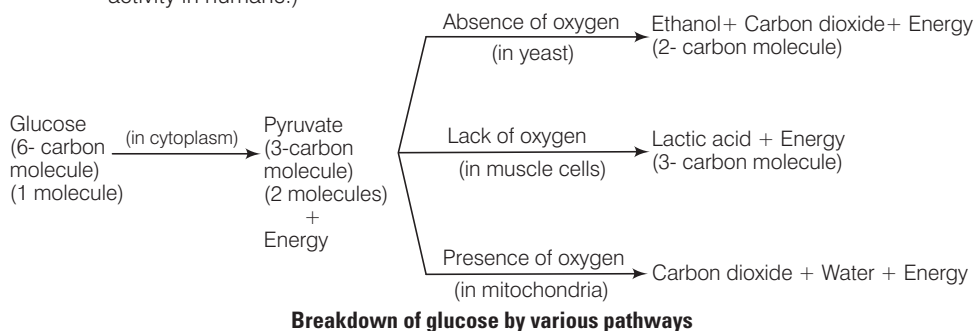
- (i) Sunlight energy is absorbed by chlorophyll (green pigment present in leaves).
- (ii) This light energy is converted into chemical energy.
- (iii) Water molecule is splitted into hydrogen and oxygen. Water required is taken from the soil. It is transported to the leaves from the soil through the roots and stem.
The oxygen gas produced as a by product goes into the air.
- (iv) The hydrogen released during the above reaction is utilised for reduction of CO_2 to form carbohydrate (glucose) by utilising the chemical energy. The CO_2 enters the leaves from the air through tiny pores called stomata.
- (v) The extra glucose is stored as starch in the leaves of the plant.

Q. 80 Explain the three pathways of breakdown in living organisms.

Ans. In the process of respiration, the first step is the breakdown of glucose, a 6-carbon molecule breaks into 2 molecules of 3-carbon molecule called **pyruvate**. This process takes place in the cytoplasm.

The subsequent breakdown pathway depends on the absence or presence of oxygen as follows

- (i) **Aerobic respiration** It takes place in the mitochondria in the presence of oxygen. Pyruvic acid is converted into carbon dioxide, and water. Thus releasing a lot of energy.
- (ii) **Anaerobic respiration** It takes place in the cytoplasm in the absence of oxygen. Pyruvic acid is converted either into CO_2 and ethanol (in case of microbes, *i.e.*, fermentation in yeast) or into lactic acid (in the case of muscle cells during sudden activity in humans.)



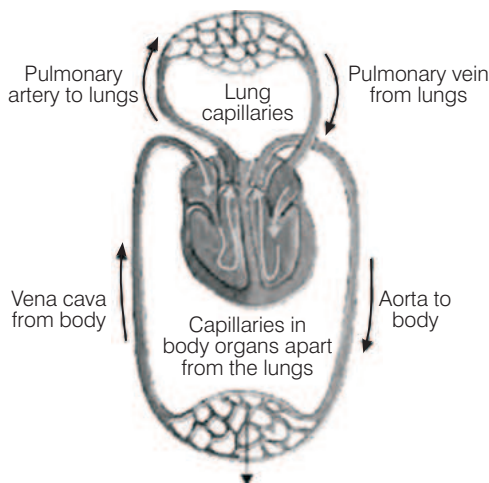
Q. 81 Describe the flow of blood through the heart of human beings.

Ans. Flow of blood through human heart Oxygen rich blood from the lungs comes to the thin-walled upper chamber of the heart on the left, the **left atrium**.

The left atrium relaxes when it is collecting this blood. It then contracts, to enter in the next chamber, the **left ventricle**, which expands, so that the blood is transferred to it. When the muscular left ventricle contracts reaches, the blood is pumped out to the body.

De-oxygenated blood from the body to the upper chamber on the right, the **right atrium**, and it expands. As the right atrium contracts, the corresponding lower chamber, the **right ventricle**, dilates. This transfers blood to the right ventricle, which in turn pumps it to the **lungs** for oxygenation.

Since ventricles have to pump blood into various organs, they have thicker muscular walls than the atria do.



Transport and exchange of oxygen and carbon dioxide

The whole process is repeated continuously. The blood circulatory system in human is an example of **double circulation** since the blood travels twice through the heart in one complete cycle.

- (i) The pathway of blood from heart to the lungs and back to the heart is called **pulmonary circulation**.
- (ii) The pathway of blood from heart to rest of the body and back to the heart is called **systemic circulation**.

These two together make double circulatory system.

Q. 82 Describe the process of urine formation in kidneys.

Ans. Urine is formed in the kidneys in the **nephron**, i.e., the structural and functional unit of kidney. Blood at high pressure travels into these tubules by the tuft of blood capillaries called **glomerulus**.

The following steps are involved in the process

- (i) **Filtration** Blood enters the glomerulus through the afferent arterioles, it passes under pressure that results in filtration of blood. Water and small molecules are forced out of glomerular capillary walls and Bowman's capsule. Large molecules remain in the blood of the glomerulus.
- (ii) **Selective reabsorption** Some molecules are selectively reabsorbed into the blood. The glomerular filtrate flows through the proximal convoluted tubule, the U-shaped Henle's loop and distal convoluted tubule. The useful substances present such as glucose, amino acids and salts are reabsorbed by a process, which requires energy. The filtrate now contains urea, some salts and water. Reabsorption of solutes increases the water concentration of the filtrate. The water is then reabsorbed into blood by osmosis (osmoregulation).
- (iii) **Tubular secretion** Some nitrogenous waste products like creatinine and some other substances like K^+ are removed from blood by DCT (Distal convoluted tubule), and are added to urine.

The urine formed is collected in the **urinary bladder**.

Control and Coordination

Multiple Choice Questions (MCQs)

Q. 1 Which of the following statements is correct about receptors?

- (a) Gustatory receptors detect taste while olfactory receptors detect smell
- (b) Both gustatory and olfactory receptors detect smell
- (c) Auditory receptors detect smell and olfactory receptors detect taste
- (d) Olfactory receptors detect taste and gustatory receptors smell

Ans. (a) A receptor is a cell (or a group of cells) in a sense organ which is sensitive to a particular type of stimulus such as light, sound, etc.

The common type of receptors are

- (i) Photoreceptors (in eyes) detect light
- (ii) Gustatory receptors (in tongue) detect taste
- (iii) Olfactory receptors (in nose) detect smell
- (iv) Thermoreceptors (in skin) detect heat or cold.

Q. 2 Electrical impulse travels in a neuron from

- (a) Dendrite → Axon → Axonal end → Cell body
- (b) Cell body → Dendrite → Axon → Axonal end
- (c) Dendrite → Cell body → Axon → Axonal end
- (d) Axonal end → Axon → Cell body → Dendrite

Ans. (c) The neurons carry messages in the form of electrical signals called electrical impulses or nerve impulses. The dendrites pick up the impulses from receptor and pass them to the cell body, and then along the axon to its end.

At the axonal end, electrical impulse sets off release of some chemicals that crosses the gap or synapse and start a similar impulse in dendrite of the next neuron.

Q. 3 In a synapse, chemical signal is transmitted from

- (a) dendritic end of one neuron to axonal end of another neuron
- (b) axon to cell body of the same neuron
- (c) cell body to axonal end of the same neuron
- (d) axonal end of one neuron to dendritic end of another neuron

Ans. (d) A microscopic gap between a pair of adjacent neurons over which nerve impulses pass when going from one neuron to the next is called **synapse**.
(Also, refer to Q. 2)

Q. 4 In a neuron, conversion of electrical signal to a chemical signal occurs at/in

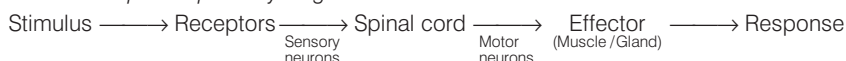
- (a) cell body (b) axonal end (c) dendritic end (d) axon

Ans. (b) At the **axonal** end, the electrical impulse releases small amount of chemical substance (i.e., acetylcholine) into the synapse. This chemical substance crosses the gap and starts a electrical impulse in the dendrite of next neuron.

Q. 5 Which is the correct sequence of the components of a reflex arc

- (a) Receptors → Muscles → Sensory neuron → Motor neuron → Spinal cord
(b) Receptors → Motor neuron → Spinal cord → Sensory neuron → Muscle
(c) Receptors → Spinal cord → Sensory neuron → Motor neuron → Muscle
(d) Receptors → Sensory neuron → Spinal cord → Motor neuron → Muscle

Ans. (d) The pathway taken by nerve impulse in a reflex action is called the reflex arc. A reflex action is involuntary action in response to a stimulus. e.g., coughing, sneezing etc.
It follows a specific pathway as given below



Q. 6 Which of the following statements are true?

- (i) Sudden action in response to something in the environment is called reflex action
(ii) Sensory neurons carry signals from spinal cord to muscles.
(iii) Motor neurons carry signals from receptors to spinal cord.
(iv) The path through which signals are transmitted from a receptor to a muscle or a gland is called reflex arc.

- (a) (i) and (ii) (b) (i) and (iii) (c) (i) and (iv) (d) (i) (ii) and (iii)

Ans. (c) The (ii) statement is false as sensory neurons carry signals from receptors to spinal cord.

The (iii) statement is false as motor neurons carry signals from spinal cord to effectors.
(Also, refer to Q. 5)

Q. 7 Which of the following statements are true about the brain?

- (i) The main thinking part of brain is hind brain.
(ii) Centres of hearing, smell, memory, sight, etc., are located in fore brain.
(iii) Involuntary actions like salivation, vomiting, blood pressure are controlled by the medulla in the hind brain.

(iv) Cerebellum does not control posture and balance of the body.

- (a) (i) and (ii) (b) (i) (ii) and (iii)
(c) (ii) and (iii) (d) (iii) and (iv)

Ans. (c) The **cerebrum** (part of forebrain) is the main thinking part of the brain. It coordinates the voluntary actions of the body. It has different areas for performing different functions such as centres of hearing, smell, memory, etc.

The hindbrain consists of pons (takes part in regulating respiration), cerebellum (helps in maintaining posture and balance of body) and medulla that controls involuntary actions such as breathing etc. The midbrain controls reflex movements.

Q. 8 Posture and balance of the body is controlled by

- (a) cerebrum (b) cerebellum (c) medulla (d) pons

Ans. (b) Cerebellum controls the voluntary actions, e.g., posture. (Also, refer to Q.7)

Q. 9 Spinal cord originates from

- (a) cerebrum (b) medulla (c) pons (d) cerebellum

Ans. (b) Spinal cord is a cylindrical structure enclosed in a bony cage called vertebral column. It begins in continuation with medulla and extends downwards.

Q. 10 The movement of shoot towards light is

- (a) geotropism (b) hydrotropism (c) chemotropism (d) phototropism

Ans. (d) A growth movement of a plant part in response to an external stimuli, towards or away from it is called tropism.

Common stimuli in environment are as given in table below

Stimulus	Type of Tropism
Light	Phototropism
Gravity	Geotropism
Touch	Thigmotropism
Chemical	Chemotropism
Water	Hydrotropism

Q. 11 The main function of abscisic acid in plants is to

- (a) increase the length of cells (b) promote cell division
(c) inhibit growth (d) promote growth of stem

Ans. (c) The main function of abscisic acid in plants is to inhibit growth. The other plant hormones with their functions are given in the table below

Phytohormone	Function
Auxins	Promote cell elongation, root formation, cell division and other physiological processes
Gibberellins	Stimulate stem elongation, stem germination and flowering (opposite effect of ABA)
Cytokinins	Promotes cytokinesis, also helps in breaking dormancy and regulating phloem transport.
Absciscic acid	Growth inhibitor, causes dormancy of seeds, etc., promotes leaf and fruit fall, helps in stomata closing to prevent loss of H_2O .
Ethylene	Promotes transverse growth, senescence, abscission of leaves, etc.

Q. 12 Which of the following is not associated with growth of plant?

- (a) Auxin (b) Gibberellins (c) Cytokinins (d) Abscissic acid

Ans. (d) Abscissic acid is a growth inhibitor hormone. It reverses the growth promoting effects of auxins and gibberellins. It causes dormancy of seeds, wilting of leaves, closing of stomata, etc.

Q. 13 Iodine is necessary for the synthesis of which hormone?

- (a) Adrenaline (b) Thyroxine (c) Auxin (d) Insulin

Ans. (b) Iodine is necessary for the synthesis of thyroxine hormone, deficiency of it in the diet of a person produces less thyroxine hormone and causes a disease known as goitre.

The main symptom of goitre is that the neck of the person appears to be swollen (due to the enlargement of thyroid gland situated in the neck).

Q. 14 Choose the incorrect statement about insulin

- (a) It is produced from pancreas
(b) It regulates growth and development of the body
(c) It regulates blood sugar level
(d) Insufficient secretion of insulin will cause diabetes

Ans. (b) **Pancreas** secrete the hormone insulin. Its function is to regulate blood sugar level. Deficiency of insulin causes a disease known as diabetes that is characterised by large quantities of sugar in blood and even urine.

Growth hormone secreted by the pituitary gland regulates growth and development of body.

Q. 15 Select the mis-matched pair

- (a) Adrenaline–Pituitary gland (b) Testosterone–Testes
(c) Estrogen–Ovary (d) Thyroxine–Thyroid gland

Ans. (a) Adrenaline is secreted by the adrenal glands, located on top of kidneys. Growth hormone is secreted by the pituitary gland

Q. 16 The shape of guard cells changes due to change in the

- (a) protein composition of cells (b) temperature of cells
(c) amount of water in cells (d) position of nucleus in the cells

Ans. (c) The shape of guard cells changes due to amount of water in them. The guard cell swell when water flows into them, causing the stomatal pore to open. Similarly, the pore closes if the guard cells shrink, due to the loss of water in them.

Q. 17 The growth of tendril in pea plants is due to

- (a) effect of light
(b) effect of gravity
(c) rapid cell divisions in tendrillar cells that are away from the support
(d) rapid cell divisions in tendrillar cells in contact with the support

Ans. (c) Tendrils are the thin thread-like growths on the stem or leaves of climbing plants. These are sensitive to touch of other objects (*i.e.*, thigmotropic). When a tendril touches an object, then the side in contact with object grows slowly than its other side, (*i.e.*, rapid cell divisions in cells that are away from the support).

This causes the tendril to bend towards the object by growing towards it, wind around object and cling to it.

Q. 18 The growth of pollen tubes towards ovules is due to

- (a) hydrotropism (b) chemotropism (c) geotropism (d) phototropism

Ans. (b) The growth of a pollen tube towards ovule is induced by a chemical (sugar) substance secreted by the ripened stigma of flower into the style towards ovary.

It is an example of chemotropism, pollen grain responds to stimulus by growing a pollen tube in downward direction, towards ovule for fertilisation.

Q. 19 The movement of sunflower in accordance with the path of sun is due to

- (a) phototropism (b) geotropism (c) chemotropism (d) hydrotropism

Ans. (a) The movement of a plant part in response to light is called phototropism

(Also, refer to Q. 10)

Q. 20 The substance that triggers the fall of mature leaves and fruits from plants is due to

- (a) auxin (b) gibberellin (c) abscisic acid (d) cytokinin

Ans. (c) Absciscic acid is responsible for the formation of layer of abscission. This layer disconnect the living tissue of leaf from the other parts.

(Also, refer to Q. 11)

Q. 21 Which of the following statements about transmission of nerve impulse is incorrect?

- (a) Nerve impulse travels from dendritic end towards axonal end
(b) At the dendritic end electrical impulses bring about the release of some chemicals which generate an electrical impulse at the axonal end of another neuron
(c) The chemicals released from the axonal end of one neuron cross the synapse and generate a similar electrical impulse in a dendrite of another neuron
(d) A neuron transmits electrical impulses not only to another neuron but also to muscle and gland cells

Ans. (b) The information acquired at the end of the dendritic tip of a neuron causes a chemical reaction that produces an electrical impulse. This impulse travels from dendrite to cell body and then along axon to its end.

Q. 22 Involuntary actions in the the body are controlled by

- (a) medulla in fore brain (b) medulla in mid brain
(c) medulla in hind brain (d) medulla in spinal cord

Ans. (c) Medulla is part of hind brain. It controls the involuntary action and regulate reflex responses. The other parts of hind brain are cerebellum and pons.

Q. 23 Which of the following is not an involuntary action?

- (a) Vomiting (b) Salivation (c) Heart beat (d) Chewing

Ans. (d) Involuntary actions are those actions that are not under our direct control such as heart beat, breathing, peristaltic movements, etc. These are controlled by medulla.

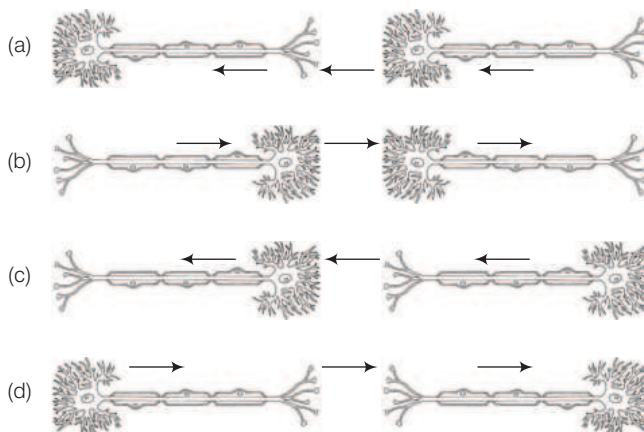
Chewing is under voluntary control and controlled by lobes present in cerebral cortex.

Q. 24 When a person is suffering from severe cold, he or she cannot

- (a) differentiate the taste of an apple from that of an ice cream
- (b) differentiate the smell of a perfume from that of an agarbatti
- (c) differentiate red light from green light
- (d) differentiate a hot object from a cold object.

Ans. (b) During cold, mucus in nasal passages becomes too thick, so that odour molecules can't reach the olfactory receptor cells. Thus, brain receives no signal identifying the odour, and so the smell of a perfume cannot be differentiated from that of an agarbatti.

Q. 25 What is the correct direction of flow of electrical impulses?



Ans. (c) Direction of flow of electrical impulse.
 Impulse → Dendrite → Cell body → Axon → Release of chemicals that cross synapse
 → Dendrite of next neuron

Q. 26 Which statement is not true about thyroxine?

- (a) Iron is essential for the synthesis of thyroxine
- (b) It regulates carbohydrates, protein and fat metabolism in the body
- (c) Thyroid gland requires iodine to synthesise thyroxine
- (d) Thyroxine is also called thyroid hormone

Ans. (a) Iodine is essential for synthesis of thyroxine.

(Also, refer to Q. 13)

Q. 27 Dwarfism results due to

- (a) Excess secretion of thyroxine
- (b) Less secretion of growth hormone
- (c) Less secretion of adrenaline
- (d) Excess secretion of growth hormone

Ans. (b) Deficiency of growth hormone in childhood causes dwarfism while excessive secretion causes gigantism.

Deficiency of thyroxine causes goitre.

Adrenaline is secreted in small amounts all the time but is secreted in large amounts to prepare our body for action during flight or fight.

Q. 28 Dramatic changes of body features associated with puberty are mainly because of secretion of

- (a) oestrogen from testes and testosterone from ovary
- (b) estrogen from adrenal gland and testosterone from pituitary gland
- (c) testosterone from testes and estrogen from ovary
- (d) testosterone from thyroid gland and estrogen from pituitary gland

Ans. (c)

Gland	Hormone	Function
Testes (only in males)	Testosterone	To control the development of male sex organs and male features as deep voice, etc., changes associated with puberty.
Ovaries (only in females)	Oestrogen	To control the development of female sex organs and female features as soft skin, etc.
	Progesterone	To control uterus changes during menstrual cycle and helps in maintenance of pregnancy.

Q. 29 A doctor advised a person to take an injection of insulin because

- (a) his blood pressure was low
- (b) his heart was beating slowly
- (c) he was suffering from goitre
- (d) his sugar level in blood was high

Ans. (d) The person having high sugar level in blood is called a diabetic. Such persons are advised to take less sugar in diet, reduce weight, exercise regularly. Persons with severe diabetes (high sugar level in blood) are treated by giving injections of insulin.
(Also, Refer to Q. 14)

Q. 30 The hormone which increase the fertility in males is called

- (a) oestrogen
- (b) testosterone
- (c) insulin
- (d) growth hormone

Ans. (b) Testosterone is the male sex hormone. It regulates male accessory sex organs. It is responsible for secondary sexual characters like moustache, beard, etc., thus, increasing fertility in males.

Q. 31 Which of the following endocrine glands is unpaired?

- (a) Adrenal
- (b) Testes
- (c) Pituitary
- (d) Ovary

Ans. (c) There are two adrenal glands, one on top of each kidney that make adrenaline hormone. Testes are paired glands presents in males, secretes male sex hormone. Pituitary gland is present just below the brain and is unpaired. It is also called master gland as it secretes a number of hormones. Ovaries are paired glands present in females, secretes female sex hormones.

Q. 32 Junction between two neurons is called

- (a) cell junction
- (b) neuro muscular junction
- (c) neural joint
- (d) synapse

Ans. (d) Junction between two neurons is called synapse.

Cell junction - junction between neighbouring cells

Neuromuscular junction-connects nervous system to muscular system

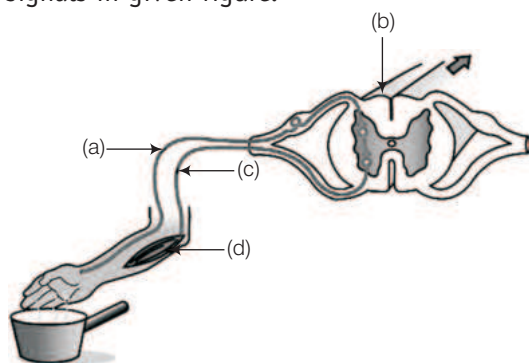
Q. 33 In humans, the life processes are controlled and regulated by

- (a) reproductive and endocrine systems (b) respiratory and nervous systems
(c) endocrine and digestive systems (d) nervous and endocrine systems

Ans. (d) In human beings, nervous system and endocrine system work together to control and coordinate all the activities. Both these systems consist of a number of organs that work together in a systematic way.

Short Answer Type Questions

Q. 34 Label the parts (a), (b), (c) and (d) and show the direction of flow of electrical signals in given figure.

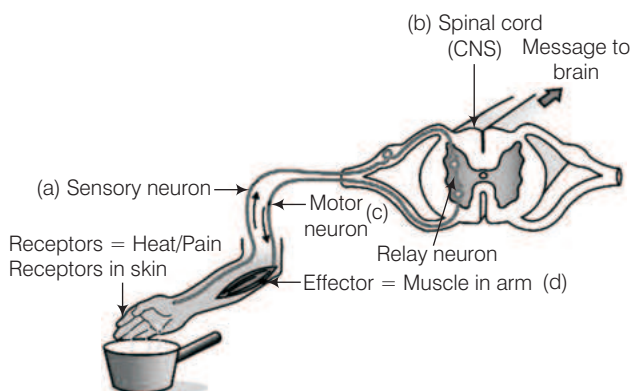


Ans. (a) Sensory neuron

(b) Spinal cord (CNS)

(c) Motor neuron

(d) Effector (muscle in arm)



Q. 35 Name the plant hormones responsible for the following

- (a) elongation of cells (b) growth of stem
(c) promotion of cell division (d) falling of senescent leaves

Ans. (a) Auxin-elongation of cells

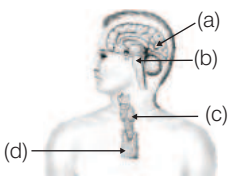
(b) Gibberellin-growth of stem

(c) Cytokinin-promotion of cell division

(d) Absciscic acid-falling of senescent leaves

(Also, refer to Q. 11)

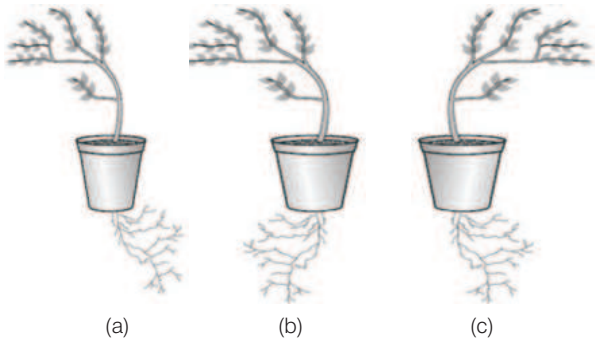
Q. 36 Label the endocrine gland in figure



Ans. The endocrine glands and its location are as follow

Glands	Location
(a) Pineal gland	Attached to dorsal side of brain
(b) Pituitary gland	Just below the brain
(c) Thyroid gland	Attached to the windpipe
(d) Thymus	Lower part of the neck and upper part of chest

Q. 37 In figure (a), (b) and (c), which appears more accurate and why?

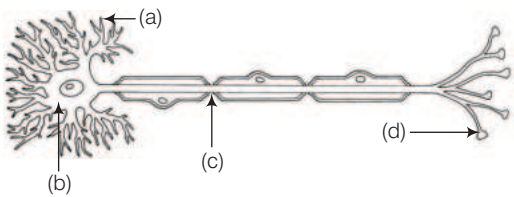


Thinking Process

This is a case of geotropism, i.e., response of plants to gravity.

Ans. Figure (a) is more accurate. The roots of a plant always grow downwards in response the gravity to make sure they find soil and water. On the other hand, the stem always grows up away from gravity pull to make sure they will get light. Both these conditions are shown in figure (a).

Q. 38 Label the part of a neuron in Figure below.



Ans. (a) Dendrite (b) Cell body
(c) Axon (d) Nerve ending

Q. 39 Match the terms of column I with those of column II.

Column I	Column II
A. Olfactory receptors	1. Tongue
B. Thermo receptors (temperature receptors)	2. Eye
C. Gustatoreceptors	3. Nose
D. Photoreceptors	4. Skin

Ans. The matching of the column I with column II

Column I	Column II
A. Olfactory receptors	Nose
B. Thermoreceptors	Skin
C. Gustatoreceptors	Tongue
D. Photoreceptors	Eye

(Also refer to Q. 1)

Q. 40 What is a tropic movement? Explain with an example.

Ans. The movement of a plant in the direction of a stimulus or away from it is called **tropic movement** or **tropism**. It is said to be positive if it is directed towards the source of stimulus and negative if directed away from the source stimulus. e.g., geotropism, the tropic response towards gravity. When a growing portion of a plant is placed horizontally, the stem tip grows away from the pull of gravity. While the root tip grows towards it. Thus, the stem is said to be **negatively geotropic** and the root **positively geotropic**.

Q. 41 What will happen if intake of iodine in our diet is low?

💡 Thinking Process

Iodine is necessary for the thyroid gland to make thyroxine hormone.

Ans. When iodine intake is low, following may happen

- Release of thyroxine from thyroid gland will be less. This will affect the metabolism of carbohydrate, protein and fat and thus slow down the growth of body.
- A person might suffer from goitre. One of the symptoms of this disease is a swollen neck.

Q. 42 What happens at the synapse between two neurons?

💡 Thinking Process

The gap between the ending of an axon (synaptic terminals) and the dendrite of another neuron is called a synapse. A chemical substance is released at axonal end that passes through it.

Ans. When an electrical impulse reaches the end of the axon, it causes the axon bulb to release a chemical which diffuses across the synapse and stimulates the dendrites of adjacent neuron. These dendrites in turn send electrical signals to their cell body to be carried along the axon.

Q. 43 Answer the following

- (a) Which hormone is responsible for the changes noticed in females at puberty?
- (b) Dwarfism results due to deficiency of which hormone?
- (c) Blood sugar level rises due to deficiency of which hormone?
- (d) Iodine is necessary for the synthesis of which hormone?

Ans. (a) Oestrogen At the onset of puberty in females, the ovaries begin to secrete this hormone. It produces secondary sexual characters in females and prepares the body for pregnancy.

- (b) **Growth Hormone** It is secreted by the anterior lobe of pituitary, it promotes growth and development of body.
- (c) **Insulin** It is secreted by pancreas. It controls the rate of oxidation of glucose, helps liver and muscle (cells to absorb glucose from the blood).
- (d) **Thyroxine** It is secreted by the thyroid gland. It regulates the carbohydrate, protein and fat metabolism in the body.

Q. 44 Answer the following

- (a) Name the endocrine gland associated with brain?
- (b) Which gland secretes digestive enzymes as well as hormones?
- (c) Name the endocrine gland associated with kidneys?
- (d) Which endocrine gland is present in males but not in females?

Ans. (a) Pituitary Hypothalamus gland present in brain releases hormones that regulate the secretion of pituitary glands.

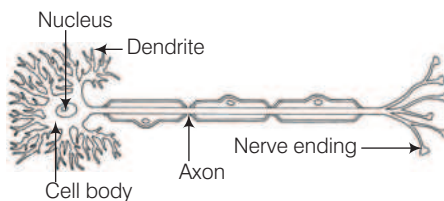
- (b) **Pancreas** It is a digestive gland as well as it secretes the pancreatic juice for digestion of food. It also secretes hormones as insulin.
- (c) **Adrenal** Adrenal glands are located on the top of two kidneys.
- (d) **Testes** These are the glands which are present only in males and secretes male sex hormone, testosterone.

Long Answer Type Questions

Q. 45 Draw the structure of a neuron and explain its function.

Ans. Neurons or nerve cells are the units which make up the nervous system. These are the structural and functional unit of the nervous system. It is the largest cell in the body. It carry messages over large distance in the body quickly. The messages are in the form of electrical signals called electrical impulses or nerve impulses.

A neuron is shown in figure below



A neuron (or nerve cell) has three components

- (i) **Cell body** It contains cytoplasm and a nucleus
- (ii) **Dendrites** These are shorter fibres on the body of a neuron.
- (iii) **Axon** The longest fibre on cell body of a neuron.

The axon has an insulating and protective sheath of myelin around it. A number of long and thin fibres stretch out from the cell body of a neuron called nerve fibres.

Q. 46 What are the major parts of the brain? Mention the function of different parts.

Ans. Brain is the most important coordinating centre in the body. It has three major parts or regions namely the **fore-brain**, **mid-brain** and **hind-brain**.

Parts	Functions
Fore-brain	
Cerebrum	Main thinking part of the brain
Cerebral hemispheres	Intelligence and voluntary actions
Olfactory lobes	Centres of smell
Diencephalon	Has centres of hunger, thirst, etc
Motor areas	Instructs muscles to do various types of jobs
Mid-brain	
Tectum	Controls reflex movements of the neck, head and trunk in response to visual and auditory stimuli
Cerebral peduncle	Also controls the reflex movements of the eye muscles, changes in pupil size and shape of the eye lens
Hind-brain	
Pons	Regulate respiration Relays information between the cerebellum and the cerebrum
Cerebellum	Maintaining posture and balance of the body Enables us to make precise and accurate movements
Medulla	Controls involuntary actions as breathing, etc Controlling centre for reflexes such as swallowing, coughing, vomiting, etc

Q. 47 What constitutes the central and peripheral nervous systems? How are the components of central nervous system protected?

Ans. The central nervous system consists of the brain and the spinal cord

- (i) **Brain** It is the main coordinating centre in the body. It is lodged in the **brain box** or **cranium** which protects it. It is covered by membranes called **meninges**, which also help to protect it. The space between the membranes and the brain is filled with a **cerebrospinal fluid** which protect the brain from mechanical shock. The brain is broadly divided into three regions fore-brain, mid-brain and hind-brain.

- (ii) **Spinal cord** It is a long cylindrical structure. It begins from the medulla oblongata and extends downwards. It is enclosed in a bony cage called **Verebral column** that protects it. It is also surrounded by meninges. It is concerned with spinal reflex actions and conduction of nerve impulses to and from the brain.

Peripheral Nervous System (PNS)

It includes the following

- (i) **Cranial nerves** 12 pairs that emerge from the brain.
- (ii) **Spinal nerves** 31 pairs that arise from the spinal cord and spread throughout the body.
- (iii) **Visceral nerves** They are special kind of nerves mostly arise from the spinal cord and are connected to the internal organs of the body. They carry both sensory and motor neurons. The PNS works according to our will, i.e., it is related to voluntary acts.

Q. 48 Mention one function for each of these hormones

- (a) Thyroxin (b) Insulin (c) Adrenaline (d) Growth hormone
(e) Testosterone

Ans.

Hormone	Function
(a) Thyroxin	It regulates carbohydrate, protein and fat metabolism in the body so as to provide the best balance for growth.
(b) Insulin	It helps in regulating blood sugar levels.
(c) Adrenaline	Increases heart rate and supply of blood to various organs.
(d) Growth hormone	It regulates growth and development in the body.
(e) Testosterone	Controls the changes of body features associated with puberty in male.

Q. 49 Name various plant hormones. Also give their physiological effects on plant growth and development.

Ans.

Plant hormone	Physiological effect
A. Auxin	- Synthesised in the young tip of roots and shoots. It diffuses towards the shady side of plant which stimulates the cells to grow longer, resulting in bending of shoot toward light - Promotes cell elongation and division - Play important role in formation of root and seedless fruit
B. Gibberellin	- Help in growth of stem and flower - Help in germination of seed
C. Cytokinins	- Promote cell division and delay leaf ageing - Also stimulate leaf expansion
D. Absciscic Acid	- Growth inhibitor - Reverses the growth promoting effects of auxins and gibberellins
E. Ethylene	- Promotes transverse growth - Essential for fruit ripening, promotes senescence and abscission of leaves

Q. 50 What are reflex actions? Give two examples. Explain a reflex arc.

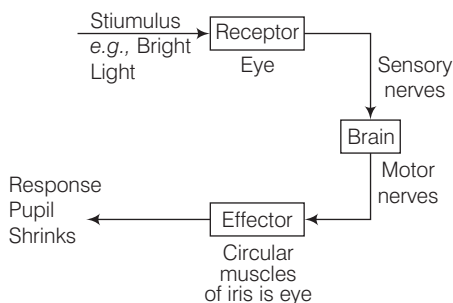
Ans. **Reflex action** is a rapid, automatic response to a stimulus which is not under the voluntary control of the brain *i.e.*, it is an involuntary action. It is a simple form of behaviour in which the same stimulus produces the same response every time. *e.g.*,

(i) If we unknowingly touch a hot plate, we immediately move our hand away from it.

(ii) Moving our foot away on stepping something sharp

Other examples are knee jerk, coughing, yawning, sneezing etc.

The pathway taken by nerve impulse in a reflex action is called the reflex arc



Reflex arcs have evolved in animals because the thinking process of brain is not fast enough. However, even after complex neuron networks have come into existence, reflex arc continue to be more efficient for quick responses.

Q. 51 'Nervous and hormonal systems together perform the function of control and coordination in human beings.' Justify the statement.

💡 Thinking Process

Receptors → Nervous system → Effector (Glands/muscles) Release of Hormone

Ans. The working together of various organs of human being in a systematic, controlled and efficient way to produce a proper response to various stimuli is known as **coordination**. In humans, the nervous and hormonal system together perform this control and coordination.

Nervous System consists of **receptors** that receives the **stimulus** from surrounding environment and send the message conveyed by them to the **spinal cord** and **brain** in form of electrical impulses through the **sensory nerves**. The **motor nerves** then transmit the response to the '**effector**'. The effector are mainly the muscles and glands of our body.

Thus, **endocrine glands** secreting **hormones** are directly or indirectly controlled by the nervous system.

Hence, control and coordination in humans (or animals) depends on two things for transmitting information *i.e.*, chemical signals of hormones and nerve impulses.

If they depended only on nerve impulses through nerve cells, a limited range of tissues would be stimulated. Since, they get additional chemical signals as well, a large number of tissues are stimulated. This is why animals can show a wide - range of response to stimulus.

Q. 52 How does chemical coordination take place in animals?

Ans. Chemical coordination in animals is performed by **hormones** that are **chemical messengers**. Different endocrine glands secrete different hormones to regulate various processes.

These hormones are released into the blood which carry them to specific tissues or organs called **target tissues/organs**. In the target tissue, hormone triggers a particular biochemical or physiological activity.

The hormones are of different types and perform different function. Their timing of release and their amount is gulated by **feedback mechanism**.

Q. 53 Why is the flow of signals in a synapse from axonal end of one neuron to dendritic end of another neuron but not the reverse?

💡 Thinking Process

When an electrical signal reaches the axonal end of a neuron, it releases a chemical substance.

Ans. The **synapse** actually act like a one-way valve because the chemical substance is present only on one side of the gap. This chemical diffuses towards the dendrite end of next neuron where it generates an electrical signal.

Since, the chemicals are absent at the dendritic end of neuron, the nerve impulse can go across only from one side (which contains the chemical substance). In this way, it is ensured that nerve impulses travel in only one direction (through of particular set of neurons).

How do Organisms Reproduce

Multiple Choice Questions (MCQs)

Q. 1 In the list of organisms given below, those that reproduce by the asexual method are

- | | |
|-------------------|---------------------------|
| (i) banana | (ii) dog |
| (iii) yeast | (iv) <i>Amoeba</i> |
| (a) (ii) and (iv) | (b) (i), (iii) and (iv) |
| (c) (i) and (iv) | (d) (ii), (iii) and (iv) |

Ans. (b) Asexual reproduction takes place without the process of gamete formation and only one parent is needed examples include *Amoeba* reproduce by binary fission (division of cell into two similar cells).

Yeast, reproduce by budding (small buds develop from body wall of parent that separate and grow further).

Banana, reproduce by vegetative propagation (vegetative parts of a plant such as root, stem, etc., can produce new plants).

In dog, sexual reproduction takes place, (i.e., it involves two sexes-male and female whose gametes fuse constituting sexual reproduction).

Q. 2 In a flower, the parts that produce male and female gametes (germ cells) are

- | | |
|-----------------------|-------------------------|
| (a) stamen and anther | (b) filament and stigma |
| (c) anther and ovary | (d) stamen and style |

Ans. (c) In a flower, the parts that produce male and female gametes are anther and ovary respectively. Stamen is the male reproductive unit of the flower. It contains a bilobed anther at the top which produces male gametes. Filament is the stalk of the stamen that supports anther.

Pistil (carpel) is the female reproductive part of the flower. It consists of ovary (forms ovules-female gametes), stigma (receives pollen) and style (elongated tube).

Q. 3 Which of the following is the correct sequence of events of sexual reproduction in a flower?

- (a) Pollination, fertilisation, seedling, embryo
- (b) Seedling, embryo, fertilisation, pollination
- (c) Pollination, fertilisation, embryo, seedling
- (d) Embryo, seedling, pollination, fertilisation

Ans. (c) The correct sequence of events are
Pollination (transfer of pollen from stamen to stigma)

↓

Fertilisation (fusion of germ cells to form zygote)

↓

Embryo formation (zygote divides several times to form an embryo within the ovule)

↓

Seedling (ovule develops a tough coat and converts into a seed)

Q. 4 Offspring formed by asexual method of reproduction have greater similarity among themselves because

- (i) Asexual reproduction involves only one parent
- (ii) Asexual reproduction does not involve gametes
- (iii) Asexual reproduction occurs before sexual reproduction
- (iv) Asexual reproduction occurs after sexual reproduction

(a) (i) and (ii) (b) (i) and (iii) (c) (ii) and (iv) (d) (iii) and (iv)

Ans. (a) Offsprings have greater similarity as only one parent is involved in asexual reproduction thus no gametes are formed. The basis of asexual reproduction is mitosis (division of a nucleus into two identical daughter nuclei). Each daughter nucleus has same genetic make up because of replication of parental DNA. The new offsprings produced are called clones.

Q. 5 Characters transmitted from parents to offspring are present in

- (a) cytoplasm (b) ribosome (c) golgi bodies (d) genes

Ans. (d) Characters are transmitted from parents to offsprings through genes. Genes are the hereditary units of the body in living organisms. Chromosomes in the nucleus of a cell contain information for the inheritance of features from parents in the form of DNA (Deoxyribonucleic acid). This DNA contains genes.

Q. 6 Characters that are transmitted from parents to offspring during reproduction show

- (a) only similarities with parents
- (b) only variations with parents
- (c) both similarities and variations with parents
- (d) neither similarities nor variations

Ans. (c) In sexual reproduction, the offsprings are not identical to the parents or to one another. This is because the offsprings receive some genes from mother and some from father.

Because of mixing of genes on re-establishment of number of chromosome in various different combinations, the offsprings show both similarities and variations with characters of parents.

Q. 7 A feature of reproduction that is common to *Amoeba*, *Spirogyra* and yeast is that

- (a) they reproduce asexually
- (b) they are all unicellular
- (c) they reproduce only sexually
- (d) they are all multicellular

Ans. (a) *Amoeba* and yeast are unicellular while *Spirogyra* is multicellular. But all the three reproduce asexually.

Q. 8 In *Spirogyra*, asexual reproduction takes place by

- (a) breaking up of filaments into smaller bits
- (b) division of a cell into two cells
- (c) division of a cell into many cells
- (d) formation of young cells from older cells

Ans. (a) In *Spirogyra*, asexual reproduction takes place by fragmentation, i.e., organism simply breaks up into smaller pieces upon maturation. Each piece grows into new individuals without forming any gametes.

Q. 9 The ability of a cell to divide into several cells during reproduction in *Plasmodium* is called

- (a) budding
- (b) reduction division
- (c) binary fission
- (d) multiple fission

Ans. (d) Multiple fission—organisms divide into many daughter cells simultaneously. e.g., *Plasmodium*

Budding—small bud develops on the cell wall of organism that separates and grows further, e.g., yeast.

Reduction division—meiosis is described as a process of reduction division. The number of chromosomes are reduced to half.

Binary fission—organisms divide into two daughter cells. e.g., *Amoeba*.

Q. 10 The correct sequence of reproductive stages seen in flowering plants is

- (a) gametes, zygote, embryo, seedling
- (b) zygote, gametes, embryo, seedling
- (c) seedling, embryo, zygote, gametes
- (d) gametes, embryo, zygote, seedling

Ans. (a) Correct sequence of reproductive stages in flowering plants is → formation of gametes → fusion of gametes to form zygote → zygote develops into embryo in the ovary → ovule develops a tough coat and converts into a seed.

Q. 11 The number of chromosomes in parents and offsprings of a particular species remains constant due to

- (a) doubling of chromosomes after zygote formation
- (b) halving of chromosomes during gamete formation
- (c) doubling of chromosomes after gamete formation
- (d) halving of chromosomes after gamete formation

Ans. (b) The number of chromosomes in parents and offsprings of a particular species remains constant due to halving of chromosome during gamete formation.

The gametes are special type of cells which contain only half the amount of DNA as compared to normal cells of an organism. So, when a male gamete combines with a female gamete during sexual reproduction, then the new cell 'zygote' will have normal amount of DNA.

Q. 12 In *Rhizopus*, tubular thread-like structures bearing sporangia at their tips are called

- (a) filaments (b) hyphae (c) rhizoids (d) roots

Ans. (b) The thread-like structures of the bread mould, *Rhizopus* are called hyphae. The tiny blobs present on them which contains cells or spores are sporangia. The spores eventually develop into new *Rhizopus* individuals, when they come in the contactment an other most surface.

Q. 13 Vegetative propagation refers to formation of new plants from

- (a) stem, roots and flowers (b) stem, roots and leaves
(c) stem, flowers and fruits (d) stem, leaves and flowers

Ans. (b) In vegetative propagation, vegetative parts of a plant such as root, stem, leaf, etc., can produce new plants. Vegetative reproduction is seen in plants like orchids, ornamental plants, grasses, banana, rose, jasmine, etc.

Q. 14 Factors responsible for the rapid spread of bread mould on slices of bread are

- (i) large number of spores
(ii) availability of moisture and nutrients in bread
(iii) presence of tubular branched hyphae
(iv) formation of round shaped sporangia
(a) (i) and (iii) (b) (ii) and (iv) (c) (i) and (ii) (d) (iii) and (iv)

Ans. (c) Under favourable conditions (like damp and warm conditions, nutrients), the air-borne spores land on food, germinate and produce new plants. These spores spread into air when spore case of plant containing hundreds of spores bursts.

Q. 15 Length of pollen tube depends on the distance between

- (a) pollen grain and upper surface of stigma
(b) pollen grain on upper surface of stigma and ovule
(c) pollen grain in anther and upper surface of stigma
(d) upper surface of stigma and lower part of style

Ans. (b) Length of pollen tube depends on the distance between pollen grain on upper surface of stigma and ovule. A pollen grain falls on the stigma of the carpel, bursts, open and develops a pollen tube downwards through the style towards the ovule in the ovary.

Q. 16 Which of the following statements are true for flowers?

- (i) Flowers are always bisexual
(ii) They are the sexual reproductive organs
(iii) They are produced in all groups of plants
(iv) After fertilisation they give rise to fruits
(a) (i) and (iv) (b) (ii) and (iii) (c) (i) and (iii) (d) (ii) and (iv)

Ans. (d) Flowers are the sexual reproductive organs of a plant. They are produced in angiosperms. Majority of flowers are bisexual. After fertilisation, they give rise to fruits.

Q. 17 Which among the following statements are true for unisexual flowers?

- (i) They possess both stamen and pistil.
- (ii) They possess either stamen or pistil.
- (iii) They exhibit cross pollination.
- (iv) Unisexual flowers possessing only stamens cannot produce fruits.

- (a) (i) and (iv) (b) (ii), (iii) and (iv)
- (c) (iii) and (iv) (d) (i), (iii) and (iv)

Ans. (b) The flowers are unisexual (papaya, watermelon) contains either stamens or carpels. Since only one reproductive organ is present in them, they depend on cross pollination to form zygote after fertilisation. Both stamen and carpels are required for fertilisation, so only one of them cannot produce fruits.

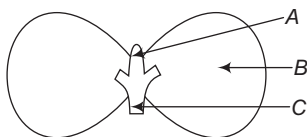
Q. 18 Which among the following statements are true for sexual reproduction in flowering plants?

- (i) It requires two types of gametes.
- (ii) Fertilisation is a compulsory event.
- (iii) It always results in formation of zygote.
- (iv) Offspring formed are clones.

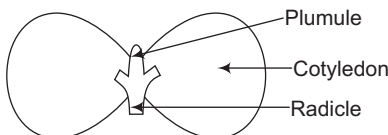
- (a) (i) and (iv) (b) (i), (ii) and (iv)
- (c) (i), (ii) and (iii) (d) (i), (ii) and (iv)

Ans. (c) Sexual reproduction creates variation in organisms. So, clones cannot be produced through it. Clones are identical copy of parent organism. Sexual reproduction needs two type of gametes, i.e., male and female to form zygote after fertilisation.

Q. 19 In figure the parts A, B and C are sequentially



- (a) cotyledon, plumule and radicle (b) plumule, radicle and cotyledon
- (c) plumule, cotyledon and radicle (d) radicle, cotyledon and plumule



Ans. (c) Plumule 'A' grows into shoots, cotyledon 'B' stores food and radicle 'C' grows into roots.

Q. 20 Offspring formed as a result of sexual reproduction exhibit more variations because

- (a) sexual reproduction is a lengthy process
- (b) genetic material comes from two parents of the same species
- (c) genetic material comes from two parents of different species
- (d) genetic material comes from many parents

How do Organisms Reproduce

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Ans. (b) Sexual reproduction occurs between two parents of the same species, not different species. (Also, refer to Q. 6) thus creates new combinations of variants and each combination is novel.

Q. 21 Reproduction is essential for living organisms to order to

- (a) keep the individual organism alive
- (b) fulfill their energy requirement
- (c) maintain growth
- (d) continue the species generation after generation

Ans. (d) Reproduction is not essential for the survival of an individual, but it is an important function of a living being as it helps an organism to perpetuate its own kind.

Through this process, new individuals are produced, that grow and reproduce again, so as to continue the species generation after generation. Reproduction is a process to maintain the progeny of an organism.

Q. 22 During adolescence, several changes occur in the human body. Mark one change associated with sexual maturation in boys

- (a) loss of milk teeth
- (b) increase in height
- (c) cracking of voice
- (d) weight gain

Ans. (c) Cracking of voice is associated with second maturation in boys which occurs during adolescence. During teenage a new set of changes occurs in boys and girls. In boys these changes include growth of hairs on face, cracking of voice etc.

Loss of milk teeth, increase in height and weight gain are the normal process of growth in both boys and girls as the body grows older.

Q. 23 In human females, an event that reflects onset of reproductive phase is

- (a) growth of body
- (b) changes in hair pattern
- (c) change in voice
- (d) menstruation

Ans. (d) In early teenage girls should, breast size increased, with darkening of the skin of nipples at the tip of breasts. Also, girls begin to menstruate at around this time. In human females, this reflects onset of reproductive phase.

Q. 24 In human males, the testes lie in the scrotum, because it helps in the

- (a) process of mating
- (b) formation of sperm
- (c) easy transfer of gametes
- (d) All of these

Ans. (b) Formation of germ cells or sperms takes place in the testes and it lies in the scrotum. In the scrotum (outside the abdominal cavity), the temperature is about 3°C lower than the temperature of the body. Testes lie in scrotum because the sperm formation require a lower temperature than the normal body temperature.

Q. 25 Which among the following is not the function of testes at puberty?

- (i) Formation of germ cells
 - (ii) Secretion of testosterone
 - (iii) Development of placenta
 - (iv) Secretion of estrogen
- (a) (i) and (ii) (b) (ii) and (iii) (c) (iii) and (iv) (d) (i) and (iv)

Ans. (c) Development of placenta and secretion of estrogen are related to female reproductive system, hence are not the function of testes at puberty.

Q. 26 The correct sequence of organs in the male reproductive system for transport of sperms is

- (a) testis → vas deferens → urethra (b) testis → ureter → urethra
(c) testis → urethra → ureter (d) testis → vas deferens → ureter

Ans. (a) Sperms formed in testis are delivered through the vas deferens which joins with another tube called urethra coming from the urinary bladder.

Q. 27 Which among the following diseases is not sexually transmitted?

- (a) Syphilis (b) Hepatitis
(c) HIV-AIDS (d) Gonorrhoea

Ans. (b) The diseases which are spread by sexual contact with an infected person are called Sexually Transmitted Diseases or STD. e.g., gonorrhoea, syphilis and AIDS. Hepatitis is a water borne viral disease which affects liver.

Short Answer Type Questions

Q. 28 In a bisexual flower inspite of the young stamens being removed artificially, the flower produces fruit. Provide a suitable explanation for the above situation.

💡 Thinking Process

Cross pollination can occur leading to fertilisation and formation of fruit.

Ans. A bisexual flower has the male as well as female reproductive organs. If the young stamens, (i.e., male unit) is removed artificially, the flower still has its pistil, (i.e., female unit) intact.

Thus, cross pollination can occur. When the pollen grains from the anther of another flower are transferred to the stigma of this flower with the help of pollinating agents as insects, bees, wind and water, it causes cross-pollination.

After the pollen grain fall on stigma, the next step is fertilisation, formation of fruits and seeds.

Q. 29 Can you consider cell division as a type of reproduction in unicellular organism? Give one reason.

💡 Thinking Process

Two daughter cells are formed by cell division, i.e., it results in production of more individuals of the organism

In unicellular organisms, cell division or fission leads to creation of new individuals that is reproduction.

Ans. Many different patterns of fission have been observed. Many bacteria and Protozoa simply split into two equal halves during cell division. Each half develops into new individual that can lead an independent life.

Thus, cell division can be considered as a type of reproduction in unicellular organism such as *Amoeba*.

Q. 30 What is a clone? Why do offsprings formed by asexual reproduction exhibit remarkable similarity?

Ans. Clones are the offsprings produced by one parent through asexual reproduction. These are genetically identical to the parent. The clones possess exact copies of the DNA of their parent and hence show remarkable similarity to the parent and to one another.

Q. 31 Explain how, offspring and parents of organisms reproducing sexually have the same number of chromosomes?

💡 Thinking Process

Sexual reproduction takes place by the combination of reproductive cells which contain only half the amount of DNA. So $n + n = 2n$ (normal number).

Ans. In sexual reproduction, though the genetic material from a male and female gamete combines together to form a zygote, the amount of DNA in zygote does not get doubled. This is because the gametes are reproductive cell which contain only half the amount of DNA as compared to the parent cell. So, when a male gamete combines with a female gamete during sexual reproduction, the zygote formed will have the normal amount of DNA (i.e., same number of chromosomes as parent).

e.g., if the male gamete has ' n ' chromosomes and female gamete also has ' n ' chromosome. So, during fertilisation, the zygote formed will have $n + n = 2n$ chromosomes, which is the normal number of chromosomes in parents.

Q. 32 Colonies of yeast fail to multiply in water, but multiply in sugar solution. Give one reason for this.

💡 Thinking Process

Energy is needed to carry out all life activities in yeast.

Yeasts are living organisms that require energy for activities like respiration, to multiply, etc.

Ans. When the colony of yeast is in water, it does not get nutrition. Sugar solution, on the contrary provides nutrition. As the yeast gets nutrition and thus energy, it grows and begin to produce buds. This is why colonies of yeast fail to multiply in water, but multiply in sugar solution.

Q. 33 Why does bread mould grow profusely on a moist slice of bread rather than on a dry slice of bread?

💡 Thinking Process

The spores of bread mould need favourable conditions like moist surface to germinate.

Moisture is an important factor for the growth of hyphae of bread mould that contain spores.

Ans. These spores can develop into new *Rhizopus* individuals. The spores are covered by thick walls that protect them until they come into contact with another moist surface and begin to grow.

Moistened bread slice offers both moisture and nutrients to the bread mould, hence it grows profusely.

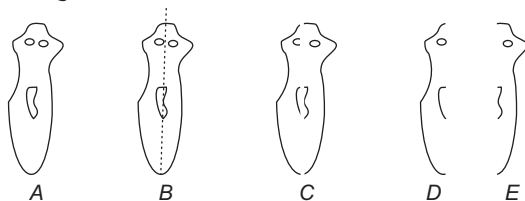
Dry slice of bread offers nutrients but not moisture hence hyphae fail to grow.

Q. 34 Give two reasons for the appearance of variations among the progeny formed by sexual reproduction.

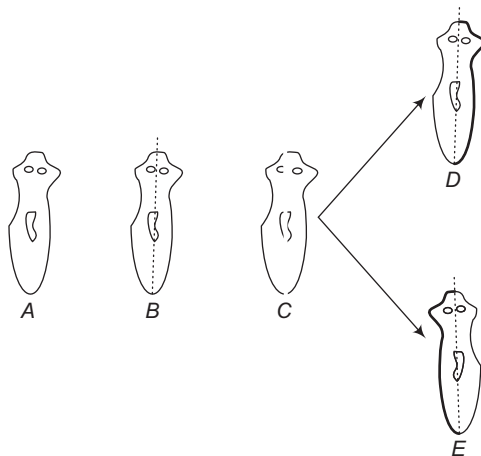
Ans. Variations appear among the progeny formed by sexual reproduction due to following reasons

- (i) Sexual reproduction results in new combinations of gene that are brought together during formation of gametes. This reshuffling of genes in the gametes increase the chance of variation in offsprings.
- (ii) The combination of two sets of chromosomes, one set from each parent during zygote formation, leads to variation within a species.

Q. 35 Would a *Planaria* cut vertically into two halves regenerate into two individuals? Complete the given figure D and E by indicating the regenerated regions.



Ans. *Planaria* possesses great power of regeneration. Thus if the body is cut into two vertical halves, each piece of the body of *Planaria* grows into a complete individual.



Regeneration in *Planaria*

Q. 36 From the internet, gather information about the chromosome numbers of five animals and five plants. Correlate the number with the size of organism and answer the following questions.

- (a) Do larger organisms have more number of chromosomes/cells?
- (b) Can organism with fewer chromosomes reproduce more easily than organisms with more number of chromosomes?
- (c) More the number of chromosomes/cells greater is the DNA content. Justify.

Ans. *Number of chromosomes in somatic cells of some animals and plants*

Animals	Number of Chromosomes
Cat	38
Lion	38
Dog	78
Elephant	58
Kingfisher	132
Plants	
Potato	48
Oats	42
Wheat	42
Mango	40
Rice	24

- (a) No, there is no relationship between size of organism and its chromosome number. A smaller organism may have more chromosomes than a larger organism.
- (b) No, process of reproduction follows a common pattern and is not dependent on the chromosome number.
Ability to reproduce depends on a variety of factors like availability of nutrients, water, suitable mate, etc.
- (c) Yes, since the major component of chromosome is DNA, if there are more chromosomes in a cell, the quantity of DNA will also be more.

Q. 37 In tobacco plant, the male gametes have twenty four chromosomes. What is the number of chromosomes in the female gamete? What is the number of chromosomes in the zygote?

Ans. The number of chromosome in the female gamete would be same as that in the male gamete, *i.e.*, it will have 24 chromosomes.
The number of chromosome in the zygote would be double the number present in the gamete and hence, it would be 48.

Q. 38 Why cannot fertilisation take place in flowers if pollination does not occur?

Ans. In a flower, fertilisation requires both male and female gametes. So, it is necessary that the male gamete reaches the female gamete. This can happen when the pollen grains are transferred to the stigma through any means of pollination.
Hence, fertilisation cannot take place in flowers if pollination does not occur due to absence of pollen, (*i.e.*, the male gamete).

Q. 39 Is the chromosome number of zygote, embryonal cells and adult of a particular organism always constant? How is the constancy maintained in these three stages?

Ans. Yes, the chromosome number of zygote, embryonal cell and adult of a particular organism is always constant.
This constancy is maintained as the cells in all the three structures undergoes only mitotic divisions (non-reductional division).

Q. 40 Where is the zygote located in the flower after fertilisation?

- Ans.** In a flower, fertilisation occurs when the male gamete present in pollen grain fuses with the female gamete (egg) present in ovule. This happens as follows: the pollen grain falls on the stigma of the carpel, bursts open and grows a pollen tube downwards through the style towards the female gamete in ovary. On reaching the ovule, the male gamete comes out of the pollen tube and combines with the nucleus of the egg present in the ovule to form a zygote. So, the zygote is located inside the **ovule** which is present in the **ovary**.

Q. 41 Reproduction is linked to stability of population of a species. Justify the statement.

- Ans.** Every species constantly struggles for its survival. Natural predators and vagaries continue to remove a large section of the population of a particular species. In addition, the natural cycle of life and death also removes a section of the population. Reproduction is a way to replenish the lost section.
- In reproduction, DNA passes from one generation to the next. Copying of DNA takes place with consistency. The consistency of DNA copying during reproduction is important for the maintenance of body design features that allow the organism to use that particular niche. Reproduction is therefore linked to the stability of population of a species.

Q. 42 How are general growth and sexual maturation different from each other?

- Ans.** General growth refers to different types of developmental process in the body like increase in height, weight gain, changes in shape and size of the body. During this phase, the reproductive organs develop at a slower rate.
- During sexual maturation, the changes that occur prepare the body for sexual reproduction. These are specific changes reflected at puberty like cracking of voice, new hair patterns, development of breast in female, etc.

Q. 43 Trace the path of sperm during ejaculation and mention the gland and their functions associated with the male reproductive system.

- Ans. Path of sperm during ejaculation** Sperms come out from testis into the vas deferens which unites with another tube called urethra coming from the urinary bladder. Along the path of vas deferens, glands like the prostate and the seminal vesicle add their secretion so that sperms are in fluid medium to make their transport easier. This fluid also provides nutrition.

Glands associated with male reproductive system are

- (a) **Testis** Secrete the hormone testosterone.
- (b) **Prostate gland** Makes the semen medium alkaline.
- (c) **Cowper's gland** Secretion of this gland lubricates the urethra before ejaculation.
- (d) **Seminal vesicle** Adds fluid content to semen.

Q. 44 What changes are observed in the uterus if fertilisation does not occur?

- Ans.** If the egg is not fertilised, it lives for about one day. Since, the ovary releases one egg every month, the uterus also prepares itself every month to receive a fertilised egg. Its lining becomes thick and spongy. This would be required for nourishing the embryo if fertilisation had taken place. Now, however, this lining is not needed any longer. So, the lining slowly breaks and comes out through the vagina as blood and mucus. This cycle takes place roughly every month and is known as **menstruation**. It usually lasts for about two to eight days.

Q. 45 What changes are observed in the uterus subsequent to implantation of young embryo?

Ans. The uterus prepares itself every month to receive and nurture the growing embryo. The lining thickens and is richly supplied with blood to nourish the growing embryo.

The embryo gets nutrition from the mother's blood with the help of a special tissue called **placenta**. This is a disc which is embedded in the uterine wall. It contains villi on the embryo's side of the tissue. On the mother's side are blood spaces, which surround the villi. This provides a large surface area for glucose and oxygen to pass from the mother to the embryo.

The developing embryo will also generate waste substances which can be removed by transferring them into the mother's blood through the placenta.

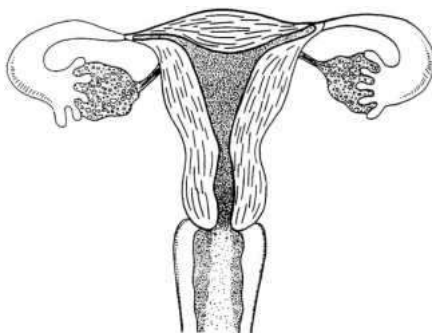
Q. 46 What are the benefits of using mechanical barriers during sexual act?

Ans.

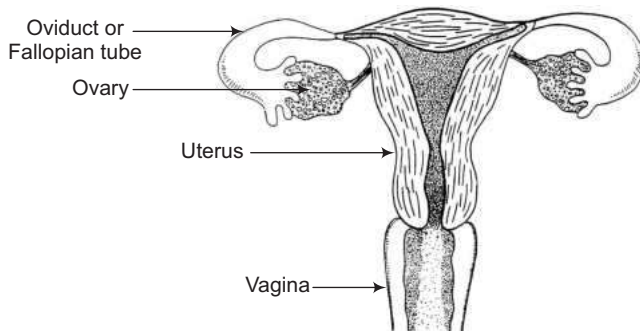
- (i) It creates a barrier so that sperm does not reach the egg, thus it avoids unwanted pregnancy.
- (ii) Use of condoms also protect a person from sexually transmitted diseases as AIDS.

Q. 47 In the given figure label the parts and mention their functions

- | | |
|--------------------------|---------------------------|
| (a) Production of egg | (b) Site of fertilisation |
| (c) Site of implantation | (d) Entry of the sperms |



Ans. (a) Ovary (production of egg) (b) Oviduct (site of fertilisation)
(c) Uterus (site of implantation) (d) Vagina (entry of the sperms)



Q. 48 What would be the ratio of chromosome number between an egg and its zygote? How is the sperm genetically different from the egg?

Thinking Process

A gamete has haploid number of chromosomes.

Ans. The ratio of chromosome number between egg and its zygote is 1 : 2. An egg is a female gamete, it has haploid number of chromosomes. During fertilisation, it fuses with male gamete (also having haploid number of chromosomes) to form a zygote that has diploid number of chromosomes.

Sperms and eggs are genetically different in terms of nature of sex chromosome. The sperm contain either X or Y-chromosome whereas an egg will always have an X-chromosome.

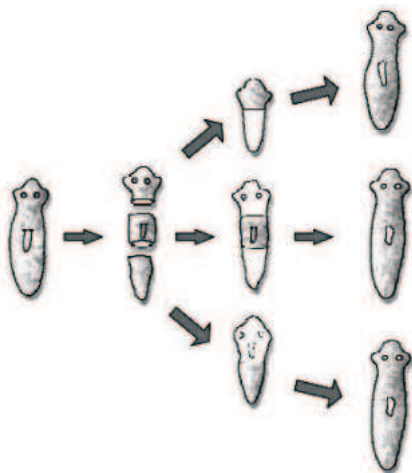
Long Answer Type Questions

Q. 49 Why are budding, fragmentation and regeneration all considered as asexual types of reproduction? With neat diagrams explain the process of regeneration in *Planaria*.

Ans. Budding, fragmentation and regeneration are considered as asexual types of reproduction because all of them involve only one parent and gametes are not involved in reproduction.

Regeneration in *Planaria* It can be cut into any number of pieces and each piece grows into a complete organism. This is known as **regeneration** (see fig.)

Regeneration is carried out by specialised cells. These cells proliferate and make large numbers of cells. From this mass of cells, different cells undergo changes to become various cell types and tissues. These changes take place in an organised sequence referred to as development.



Regeneration in *Planaria*

Q. 50 Write two points of difference between asexual and sexual types of reproduction. Describe why variations are observed in the offspring formed by sexual reproduction.

Ans.

Asexual Reproduction	Sexual Reproduction
It involves only one parent.	It often involves two parents.
Gametes are not produced.	Gametes are produced.
No fertilisation and zygote formation.	Fertilisation and zygote formation is observed.
Meiosis does not occur at anytime during reproduction.	Meiosis occurs at the time of gamete formation.

The reason why variations are observed in offspring formed by sexual reproduction is during sexual reproduction, two types of gametes fuses. Although the gametes have same number of chromosomes, their DNA is not identical. This bring different sets of characters in the offspring.

Q. 51 Distinguish between pollination and fertilisation. Mention the site and product, of fertilisation in a flower. Draw a neat, labelled diagram of a pistil showing pollen tube growth and its entry into the ovule.

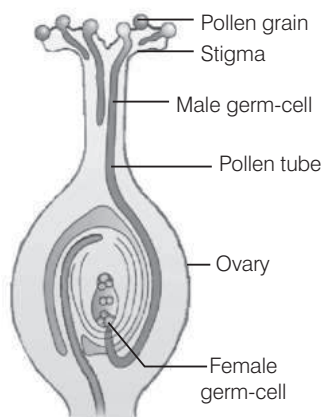
Ans.

Distinguish between Pollination and Fertilisation

Pollination	Fertilisation
It is the transfer of pollen grains from anther to the stigma.	It is the fusion of male and female gametes
It is a physical process.	It is a biological process.

The site of fertilisation is ovule in the ovary.

The product of fertilisation is a zygote.



Pistil showing pollen tube growth and its entry into Ovule

Q. 52 Distinguish between a gamete and zygote. Explain their roles in sexual reproduction.

Ans. **Distinguish between a Gamete and Zygote**

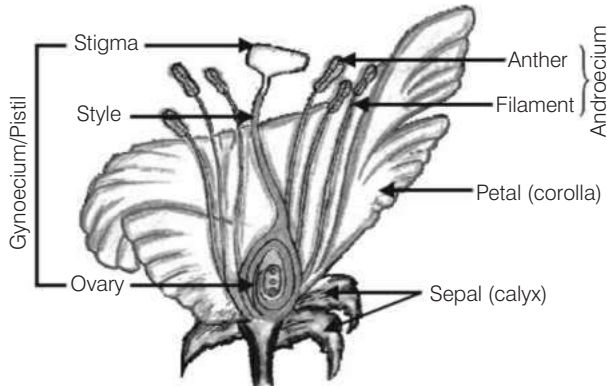
Gamete	Zygote
The cells involved in sexual reproduction are called gametes, e.g., sperm (male) and ova (female). Gametes are unfertilised reproductive cells. The fusion of sperm and egg forms a fertilised ovum or zygote	The fusion of male gamete and female gamete forms zygote during sexual reproduction. Zygote is fertilised egg or fertilised ovum. Zygote undergoes development and forms a newly formed baby.

Importance of gametes and zygote in sexual reproduction Gametes are prerequisite for the sexual reproduction. Both sperm and ova unit to form a zygote. Zygote further develops and forms an embryo which becomes a baby.

Q. 53 Draw the diagram of a flower and label the four whorls. Write the names of gamete producing organs in the flower.

Ans. *The four whorls of a flower are*

- (i) Calyx (comprising of sepals)
- (ii) Corolla (consists of petals)
- (iii) Androecium (male reproductive unit)
- (iv) Gynoecium (female reproductive unit)



Male gamete producing organ is **anther** while

Female gamete producing organ is **ovary** in the flower.

Q. 54 What is placenta? Mention its role during pregnancy.

Ans. Placenta is a disc-like structure in the lining of the uterine wall. It contains villi on the embryo's side of the tissues and blood spaces on the mother's side surround the villi. The embryo is connected to the placenta by a tube called the **umbilical cord**.

Role during pregnancy It is through the placenta that all the requirements of the developing foetus are met.

The villi provide a large surface area that facilitate passage of nutrition and oxygen to embryo from mother through blood.

Waste substances produced by embryo are removed through placenta into mother's blood.

Q. 55 What are various ways to avoid pregnancy? Elaborate any one method.

Ans. Ways to avoid pregnancy are called **contraceptive methods**. It includes a number of ways such as

- (i) Mechanical barrier e.g., condom (so that sperm does not reach the egg)
- (ii) Drugs (or pills)- change the hormonal balance so that eggs are not released.
- (iii) Loop or copper-T prevent pregnancy.
- (iv) Surgical method permanent contraception.

Mechanical barrier There are a number of methods that create barrier between sperm and egg. Some of them are as follows

Condoms It is a fine rubber worn over the penis during sexual intercourse. Semen is collected in it and not discharged into the vagina. This method also prevents the spread of STDS as AIDS.

Diaphragms or caps It can be fitted in the cervix of a woman to prevent semen from reaching the Fallopian tube.

Q. 56 How does fertilisation take place? Fertilisation occurs once in a month.
Comment.

Ans. Fertilisation In human beings, the sperms made in the testis of males are introduced into the vagina of the woman through penis during copulation or mating. The sperms are highly active and mobile and thus move up through cervix into the uterus. From uterus, sperms pass into the oviducts. The oviduct contains an ovum released by the ovary during ovulation. Millions of sperms are released into the vagina at one time, but only one sperm fuses with the ovum in the oviduct to form a zygote. This is called **fertilisation**.

Fertilisation can occur only once in a month because the ovary releases just one mature egg every month during a menstrual cycle (around 14th day).

Q. 57 Reproduction is essentially a phenomenon that is not for survival of an individual but for the stability of a species. Justify.

Ans. All the living organisms need energy for their survival and growth. This energy is obtained from various life processes such as nutrition and respiration. Thus, these phenomenon are essential for the living of an individual.

Compared to these life processes, reproduction may appear to be a waste of energy as it is not essential for survival of an individual. But it is an important function of a living being as it helps in production of new individuals of its own kind. Continuity of life has been possible through reproduction.

Genetic material is transferred from one generation to the next by DNA copying as a result of reproduction. DNA copying takes place with high constancy and considerable variations.

This is advantageous for maintaining features that allow the organism to survive in the changing environment. Therefore, reproduction is related to the stability of a species and is not essential for the survival of an individual.

Q. 58 Describe sexually transmitted diseases and mention the ways to prevent them.

Ans. Sexually Transmitted Diseases (or STDs) are infectious diseases that are transmitted during sexual intercourse.

These include bacterial infections such as gonorrhoea and syphilis, and viral infections such as warts and HIV-AIDS. To prevent STDs, the following precautions can be taken.

- (i) Use a protective covering called condom, over the penis during sex.
- (ii) Avoid multiple sex partners and maintain personal hygiene.

Heredity and Evolution

Multiple Choice Questions (MCQs)

Q. 1 Exchange of genetic material takes place in

- (a) vegetative reproduction
- (b) asexual reproduction
- (c) sexual reproduction
- (d) budding

Ans. (c) Exchange of genetic material takes place in **sexual reproduction** as genes of two parents are involved.

In **asexual reproduction**, no exchange of genetic material takes place as a single parent is involved.

Budding and **vegetative reproduction** are types of asexual reproduction. So there is no exchange of genetic material.

Q. 2 Two pink coloured flowers on crossing resulted in 1 red, 2 pink and 1 white flower progeny. The nature of the cross will be

- (a) double fertilisation
- (b) self pollination
- (c) cross fertilisation
- (d) no fertilisation

Ans. (c) The nature of the cross will be cross fertilisation. Fertilising a plant using pollen from another plant of the same species is cross-fertilisation.

Double fertilisation is a complex fertilisation mechanism of flowering plants. It involves fusion of a male and female gamete. Self-pollination is a form of pollination that occurs in a flower that has both stamen and carpel.

Q. 3 A cross between a tall plant (TT) and short pea plant (tt) resulted in progeny that were all tall plants because

- (a) tallness is the dominant trait
- (b) shortness is the dominant trait
- (c) tallness is the recessive trait
- (d) height of pea plant is not governed by gene 'T' or 't'

Ans. (a) In F_1 generation, the cross between TT and tt will result into all tall plants. Thus, tallness is the dominant trait.

Whenever two traits of a character are crossed, the F_1 plants show only one of the traits. This trait is called the **dominant trait**. The other trait (dwarfness in this case) that does not express itself in F_1 generation is called the **recessive trait**.

Q. 4 Which of the following statement is incorrect?

- (a) For every hormone there is a gene
- (b) For every protein there is a gene
- (c) For production of every enzyme there is a gene
- (d) For every molecule of fat there is a gene

Ans. (d) The incorrect statement is (d) as fat molecules are not related to genes.

On the other hand, genes contain information for making proteins in a cell. The proteins synthesised may be enzymes that catalyse biochemical reactions. The enzymes also controls the process of making hormones. This explains why a, b and c are correct.

Q. 5 If a round, green seeded pea plant (RR yy) is crossed with wrinkled, yellow seeded pea plant, (rr YY) the seeds production in F_1 generation are

- (a) round and yellow
- (b) round and green
- (c) wrinkled and green
- (d) wrinkled and yellow

Ans. (a) The cross between RRyy and rrYY seeds will produce RrYy (round and yellow) seeds in F_1 generation, because round and yellow are dominant traits.

Q. 6 In human males all the chromosomes are paired perfectly except one. This/these unpaired chromosome is/are

- (i) large chromosome
 - (ii) small chromosome
 - (iii) Y-chromosome
 - (iv) X-chromosome
- (a) (i) and (ii) (b) (iii) only (c) (iii) and (iv) (d) (ii) and (iv)

Ans. (c) In human males, one pair called the **sex chromosomes** are unpaired. Here, one is a normal-sized X-chromosome while other is a short Y-chromosome. Women have a perfect pair of sex chromosomes, both called X.

Q. 7 The maleness of a child is determined by

- (a) the X-chromosome in the zygote
- (b) the Y-chromosome in zygote
- (c) the cytoplasm of germ cell which determines the sex
- (d) sex is determined by chance

Ans. (b) The maleness of a child is determined by the Y-chromosome in zygote inherited from the father. If X-chromosome is inherited from the father, the zygote will develop into girl.

Q. 8 A zygote which has an X-chromosome inherited from the father will develop into a

- (a) boy
- (b) girl
- (c) X-chromosome does not determine the sex of a child
- (d) either boy or girl

Ans. (b) A zygote that has an X-chromosome inherited from father will develop into a girl.

All children whether boys or girls inherit an X-chromosome from their mother.

Thus, sex is determined by what they inherit from their father. A child who inherits X-chromosome from her father will be a girl, and one who inherits a Y-chromosome from him will be a boy.

Q. 9 Select the incorrect statement

- (a) Frequency of certain genes in a population change over several generations resulting in evolution
- (b) Reduction in weight of the organism due to starvation is genetically controlled
- (c) Low weight parents can have heavy weight progeny
- (d) Traits which are not inherited over generations do not cause evolution

Ans. (b) The weight reduction due to starvation will not change the DNA of the germ cells, because low weight is not a trait that is genetically controlled or inherited. Thus, low weight parents may have heavy weight progeny.

Traits which are not inherited, i.e., the acquired traits are simply changes in non-reproductive tissues, so, cannot be passed on to DNA of germ cells.

Evolution results from change in frequency of genes in a population over several generations.

Q. 10 New species may be formed if

- (i) DNA undergoes significant changes in germ cells.
- (ii) chromosome number changes in the gamete.
- (iii) there is no change in the genetic material
- (iv) mating does not take place

- (a) (i) and (ii) (b) (i) and (iii) (c) (ii), (iii) and (iv) (d) (i), (ii) and (iii)

Ans. (a) New species may be formed if the DNA changes are severe enough, such as a change in the number of chromosome. This leads to new variations.

Q. 11 Two pea plants one with round green seeds (RRyy) and another with wrinkled yellow (rrYY) seeds produce F_1 progeny that have round, yellow (RrYy) seeds. When F_1 plants are selfed, the F_2 progeny will have new combination of characters. Choose the new combination from the following.

- (i) Round, yellow (ii) Round, green
- (iii) Wrinkled, yellow (iv) Wrinkled, green

- (a) (i) and (ii) (b) (i) and (iv) (c) (ii), and (iii) (d) (i) and (iii)

Ans. (b) The new combination in F_2 progeny will be round yellow and wrinkled green.

The phenotypic ratio of 9 : 3 : 3 : 1 is obtained.

Round Yellow : Round green : Wrinkled yellow : Wrinkled green = 9 : 3 : 3 : 1

Q. 12 A basket of vegetables contains carrot, potato, radish and tomato. Which of them represent the correct homologous structures?

- (a) Carrot and potato (b) Carrot and tomato
- (c) Radish and carrot (d) Radish and potato

Ans. (c) Radish and carrot represent homologous structures as they have the same structure (or basic design) though they are different species.

Q. 13 Select the correct statement

- (a) Tendril of a pea plant and phylloclade of *Opuntia* are homologous
- (b) Tendril of a pea plant and phylloclade of *Opuntia* are analogous
- (c) Wings of birds and limbs of lizards are analogous
- (d) Wings of bird and wings of bat are homologous

Ans. (d) Wings of birds and wings of bat are homologous. Since they have same basic design, however their origin is different.

Q. 14 If the fossil of an organism is found in the deeper layers of earth, then we can predict that

- (a) the extinction of organism has occurred recently
- (b) the extinction of organism has occurred thousands of years ago
- (c) the fossil position in the layers of earth is not related to its time of extinction
- (d) time of extinction cannot be determined

Ans. (b) Fossils refer to the petrified remains of organisms that lived in the past and get preserved in rocks. The fossil position in the layer of earth relates to the time of extinction of organisms.

If the fossil of an organism is found in the deeper layers of earth, then it is predicted that extinction of organism has occurred thousands of years ago. Whereas the fossils found closer to the surface are more recent.

Q. 15 Which of the following statements is not true with respect to variation?

- (a) All variations in a species have equal chance of survival
- (b) Change in genetic composition results in variation
- (c) Selection of variants by environmental factors forms the basis of evolutionary processes
- (d) Variation is minimum in asexual reproduction

Ans. (a) All variations in a species do not have equal chances of survival. Some of the variations may be so drastic that the new DNA copy cannot work with the cellular apparatus it inherits. Such a newborn cell dies soon.

Depending on the nature of variations, different individual will have different kinds of advantage. Variations results from change in genetic composition. Advantageous variations are selected by environmental factors. This leads to evolution and speciation. In asexual reproduction, there is minimum variation that is due to small errors in DNA copy.

Q. 16 A trait in an organism is influenced by

- (a) paternal DNA only
- (b) maternal DNA only
- (c) both maternal and paternal DNA
- (d) neither by paternal nor by maternal DNA

Ans. (c) A trait of an organism is influenced by both maternal and paternal DNA. It is passed down from one generation to another through genes found on the chromosomes. During sexual reproduction, both mother and father pass their genes to their children, thus determining their traits, or characteristic features.

Q. 17 Select the group which shares maximum number of common characters

- (a) two individuals of a species (b) two species of a genus
- (c) two genera of a family (d) two genera of two families

Ans. (a) Two individuals of a species share maximum number of common characters. A species is a population of organisms consisting of similar individuals which can breed together and produce fertile offspring.

Q. 18 According to the evolutionary theory, formation of a new species is generally due to

- (a) sudden creation by nature
- (b) accumulation of variations over several generations
- (c) clones formed during asexual reproduction
- (d) movement of individuals from one habitat to another

Ans. (b) Accumulation of variations over several generations forms new species. Genetic drift accumulate different changes in sub-populations of a species. Also, natural selection may also operate differently in the different geographic locations. Eventually, different groups of new species will be formed.

Q. 19 From the list given below, select the character which can be acquired but not inherited

- (a) colour of eye (b) colour of skin (c) size of body (d) nature of hair

Ans. (c) Acquired traits develop in response to the environment. The size of the body is an acquired trait because it can be changed on the availability of less or more food. The other three colour of eye, and skin and nature of hair are inherited characters from the parents.

Q. 20 The two versions of a trait (character) which are brought in by the male and female gametes are situated on

- (a) copies of the same chromosome (b) two different chromosomes
- (c) sex chromosomes (d) any chromosome

Ans. (a) The two versions of a trait are situated on copies of same chromosome. Each parent contributes one copy of the gene for a particular character. In the gamete however, only one copy is present due to reduction division. At the time of fusion, the diploid condition is maintained. Sex chromosomes carry genes for sexual characters.

Q. 21 Select the statements that describe characteristics of genes

- (i) genes are specific sequence of bases in a DNA molecule
- (ii) a gene does not code for proteins
- (iii) in individuals of a given species, a specific gene is located on a particular chromosome
- (iv) each chromosome has only one gene

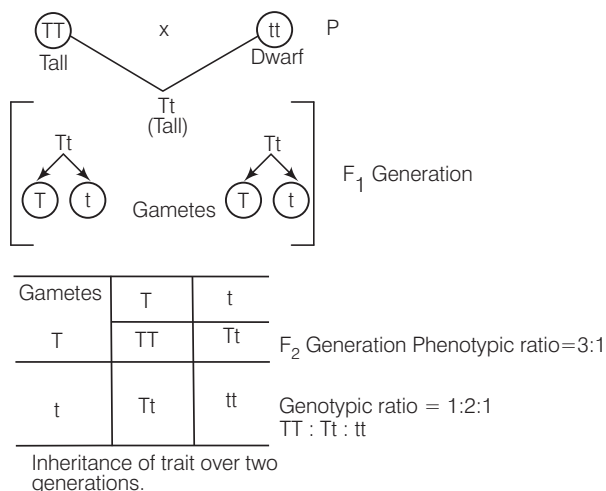
- (a) (i) and (ii) (b) (i) and (iii) (c) (i) and (iv) (d) (iii) and (iv)

Ans. (b) Genes are stretches of DNA found on chromosomes of a cell. A gene contains information for making proteins in a cell. A specific gene is located on a particular chromosome in individuals of a given species.

Q. 22 In peas, a pure tall plant (TT) is crossed with a short plant (tt). The ratio of pure tall plants to short plants in F_2 is

- (a) 1 : 3 (b) 3 : 1 (c) 1 : 1 (d) 2 : 1

Ans. (b) In F_2 generation, the phenotypic ratio is 3 : 1, in a monohybrid cross.



Q. 23 The number of pair(s) of sex chromosomes in the zygote of humans is

- (a) one (b) two (c) three (d) four

Ans. (a) The number of sex chromosomes in the zygote of humans is one pair. It is not always a perfect pair. In females, it is perfect with two X-chromosomes. However in males, there is one X-chromosome and one Y-chromosome.

Q. 24 The theory of evolution of species by natural selection was given by

- (a) Mendel (b) Darwin (c) Morgan (d) Lamarck

Ans. (b) The theory of evolution of species by natural selection was given by **Charles Darwin**.

Mendel gave the laws of inheritance of traits. **Morgan** worked on *Drosophila melanogaster* and gave concept of linked genes, **Lamarck** gave the theory of inheritance of acquired characteristics.

Q. 25 Some dinosaurs had feathers although they could not fly but birds have feathers that help them to fly. In the context of evolution this means that

- (a) reptiles have evolved from birds
(b) there is no evolutionary connection between reptiles and birds
(c) feathers are homologous structures in both the organisms
(d) birds have evolved from reptiles

Ans. (d) In the context of evolution, the use of feathers by birds for flying means that birds have evolved from reptiles. Dinosaurs had feathers but could not fly using them. Birds, later adapted the feathers for flight. Since, dinosaurs were reptiles, this means that birds have evolved from them.

Short Answer Type Questions

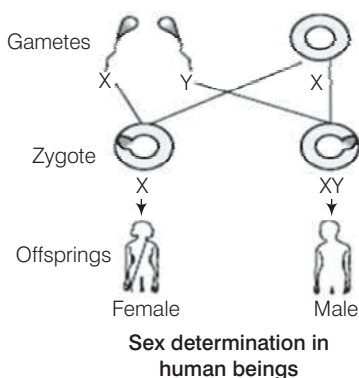
Q. 26 How is the sex of a new born determined in humans?

Ans. In human beings, the sex of the individual is genetically determined. *i.e.*, the genes inherited from parents decide whether it will be boys or girl.

Most human chromosomes have a maternal and a paternal copy, and we have 22 such pairs. But one pair, called the **sex chromosomes**, is odd in not always being a perfect pair.

Women have a perfect pair of sex chromosomes, both called X. But men have a mismatched pair in which one is a normal-sized X while the other is a short one called Y. So women are XX, while men are XY.

All children will inherit an x-chromosome from their mother regardless of whether they are boys or girls. Thus, **the sex of the children will be determined by what they inherit from their father**. A child who inherits an x-chromosome from her father will be a girl, and one who inherits a Y-chromosome from him will be a boy.



Q. 27 Do genetic combination of mothers play a significant role in determining the sex of a new born ?

Ans. The mothers have a pair of X-chromosomes. So they do not play significant role in determining the sex of a new born.

All children will inherit an X-chromosome from their mother regardless whether they are boys or girls. Therefore, sex is always determined by the other sex chromosome that they inherit from the father. One who inherits the X-chromosome of father will be girl while who inherits Y-chromosome of the father will be boy.

Q. 28 Mention three important features of fossils which help in the study of evolution.

Ans. *Three important features of fossils which help in the study of evolution are*

- (i) Fossils represents mode of preservation of ancient species.
- (ii) Fossils help in establishing evolutionary relationship between organisms and their ancestors.
- (iii) Fossils help in establishing the time period in which organisms lived.

Q. 29 Why do all the gametes formed in human females have an X-chromosome?

Ans. Human females have two X-chromosomes called sex chromosome. During meiosis at the time of gamete formation, one X-chromosome enters each gamete. Hence, all the gametes formed in human females possess an X-chromosome.

Q. 30 In human beings, the statistical probability of getting either a male or female child is 50 : 50. Give a suitable explanation.

💡 Thinking Process

In human beings, the sex is determined by the type of sperm as it can either have a X-chromosome or Y-chromosome.

Ans. The sex of an infant is determined by the type of sex chromosome contributed by the male gamete. A male produces two types of sperms-one type bears 22 + X composition and the other, 22 + Y. Therefore, a male have 50% sperms with X-chromosomes and other 50% with Y-chromosome.

Any one of the two types of sperms can fertilise the egg. If a Y-bearing sperm fertilises the egg, the zygote will be XY (male) and when an X-bearing sperm fertilises the egg, the resulting zygote will be XX (female). Since the ratio of X-chromosome and Y-chromosome in male gamete is 50 : 50. The statistical probability of male or a female infant is also 50 : 50.

Q. 31 A very small population of a species faces a greater threat of extinction than a larger population. Provide a suitable genetic explanation.

💡 Thinking Process

Small population gets reproductively isolated so no exchange of genes, that would lead to less variations.

Ans. Fewer individuals of a species in a population will face reproductive isolation. So, they would impose extensive inbreeding among them, due to which no exchange of genes take place. This limits the appearance of variations and put the species at a disadvantage. If there are changes in the environment, the individuals would fail to cope up with the environmental changes, and thus face a greater threat of extinction.

Q. 32 What are homologous structures? Given an example. Is it necessary that homologous structures always have a common ancestor?

Ans. Structures which have the same basic structure (or same basic design) but perform different functions are called **homologous structures**. e.g., forelimbs of reptiles, amphibians, birds and mammals.

Yes, homologous structures are inherited from a common ancestor. As in the above example, the mammals as well as birds, reptiles and amphibians all have four limbs. The basic structure of limbs is similar though it has been modified to perform different functions in various vertebrates. Thus, showing they have evolved from a common ancestor.

Q. 33 Does the occurrence of diversity of animals on earth suggest their diverse ancestry also? Discuss this point in the light of evolution.

Ans. Since animals shows a vast diversity in their structures, they probably have diverse ancestry. Because common ancestry may greatly limit the extent of diversity, e.g., all birds are closely related, they have common ancestors but birds and reptiles are also related.

We can thus build up small groups of species with recent common ancestors, then super-groups of these groups with more distant common ancestors, and so on. In theory, we can keep going backwards like this until we come to the notion of a single species at the very beginning of evolutionary time.

As many of these diverse animals are inhabiting the same habitat, their evolution by geographical isolation and speciation is also not likely to occur. Thus, occurrence of diversity of animals on earth suggest their diverse ancestry also.

Q. 34 Give the pair of contrasting traits of the following characters in pea plant and mention which is dominant and recessive

(i) yellow seed

(ii) rounds seed

Ans.

Character	Contrasting traits	
	Dominant	Recessive
(i) seed colour	yellow	green
(ii) seed shape	round	wrinkled

Q. 35 Why did Mendel choose pea plant for his experiments?

Ans. Mendel choose pea plant for his experiment due to following reasons

- Pea plants were self pollinating (which enabled them to produce next generation of plants easily).
- Pea plants were easy to cultivate and had short life span.
- They had sharply defined and contrasting traits such as height (tall, dwarf), seed colour (yellow-green), etc.

Q. 36 A woman has only daughters. Analyse the situation genetically and provide a suitable explanation.

💡 Thinking Process

A daughter is born when a sperm carrying X-chromosome fertilises the ovum.

Ans. The sex of a child depends on the type of male gamete (sperm) as the man produces sperms with either X or Y-chromosome. If a sperm carrying X-chromosome fertilises the ovum (female gamete with X-chromosome), then the child born will be a girl.

This is because the child will have XX pair of sex chromosome. A woman has only daughters. This indicates that in every fusion, the sperm carrying X-chromosome fertilised the ovum.

Long Answer Type Questions

Q. 37 Does geographical isolation of individuals of a species lead to formation of a new species? Provide a suitable explanation.

💡 **Thinking Process**

Geographical isolation interrupts the flow of genes between isolated populations, thus leading to speciation.

Ans. Yes, geographical isolation of individuals of a species lead to formation of a new species. If we consider the population of same species splitting into two separate groups and getting isolated from each other geographically by various barriers as sea, mountain, etc.

The geographical isolation of the two groups lead to their **reproductive isolation** due to which **no genes** are **exchanged** between them. However, breeding continues within the isolated populations producing more and more generations. Over the generations, **genetic drift** and **natural selection** is operated in the isolated group and make them more different from other.

After many years, the individuals of isolated groups will become so different that they will be incapable of reproducing with each other (if they meet again) and new species will originate evolved.

Q. 38 Bacteria have a simpler body plan when compared with human beings. Does it mean that human beings are more evolved than bacteria? Provide a suitable explanation.

Ans. Taking the totality of life characteristics into account, it is hard to label either bacteria or human being as evolved. **Evolution** is simply the generation of diversity and the shaping of the diversity by environmental selection.

Also, it is not as if the newly generated species are in any way better than the older one. It is just that **natural selection** and **genetic drift** have together led to the formation of a population that cannot reproduce with the original one.

The only progressive trend in evolution seems to be that more and more complex body designs have emerged over time. However, again, it is not as if the older designs are inefficient. In fact, one of the simplest life forms—bacteria—inhabit the most inhospitable habitats like hot springs, deep-sea thermal vents and the ice in Antarctica. In other words, human beings are not the crest of evolution, but simply yet another species in the abundant spectrum of evolving life.

Q. 39 All the human races like Africans, Asians, Europeans, Americans and others might have evolved from a common ancestor. Provide a few evidences in support of this view.

Ans. There is no biological basis to the notion of human races. All humans are a single species irrespective of their race. All of them have a common body plan, structure, physiology and metabolism. All of them have a constant chromosome number, i.e., 46. Their genetic make up is also similar, almost 99.9% DNA is same in all humans. And also all the humans, can freely inter-breed to produce offsprings.

Not only this, study of evolution of human beings indicates that all of us come from Africa. The earliest members of the human species, *Homo sapiens* can be traced there.

All these evidences clearly indicates that all of us whether Africans, Asians Europeans or Americans, etc; have evolved from a common ancestor.

Q. 40 Differentiate between inherited and acquired characters. Give one example for each type.

Ans. **Differences between inherited and acquired characters**

Inherited characters	Acquired characters
Characters that are passed from parents to offsprings are inherited characters	Characters that do not pass from parents to offsprings and develops in response to the environment are acquired characters.
They cause changes in the reproductive cells (gametes) of an organism. <i>e. g., eye colour, seed colour.</i>	They do not cause changes in the reproductive cells of an organism. <i>e.g., obese body, loss of finger in an accident</i>

Q. 41 Give reasons why acquired characters are not inherited.

Ans. Acquired characters do not produce change in the genes (or DNA) of germ cells, so they cannot be inherited.

e.g., If tails of mice are cut in first generation, and then breeding is done, the new mice born will have full tails. The cut tail of mice is an acquired trait which is never passed on to their progeny. This is because cutting of mice tails does not change the genes of their reproductive cells.

Q. 42 Evolution has exhibited a greater stability of molecular structure when compared with morphological structures. Comment on the statement and justify your opinion.

💡 Thinking Process

There is immense diversity in form, structure, size, etc., but basic molecules have the same basic structure.

Ans. Evolution's progressive trend seems to be more and more complex in body designs (or morphological structures), emerged overtime. *e.g.,* immense diversity and increasing complexity is seen in body size, form and structure from non-chordates to chordates.

But at the molecular level, these diverse type of organisms exhibit unbelievable similarity, *e.g.,* the chemical composition of DNA is basically same in all living organisms except for differences in the sequence of nitrogenous bases. Other biomolecules like RNA, proteins, etc., also show remarkable similarity in all organisms.

Thus, it shows that evolution has exhibited a greater stability of molecular structure as compared to morphological structures.

Q. 43 In the following crosses write the characteristics of the progeny

Cross

(a) RR YY × RR YY

Round, yellow, Round, yellow

(b) Rr Yy × Rr Yy

Round, yellow, Round, yellow

(c) rr yy × rr yy

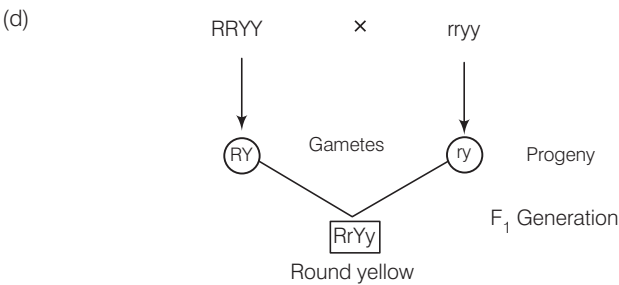
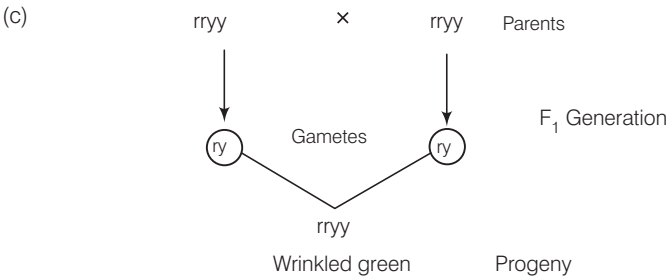
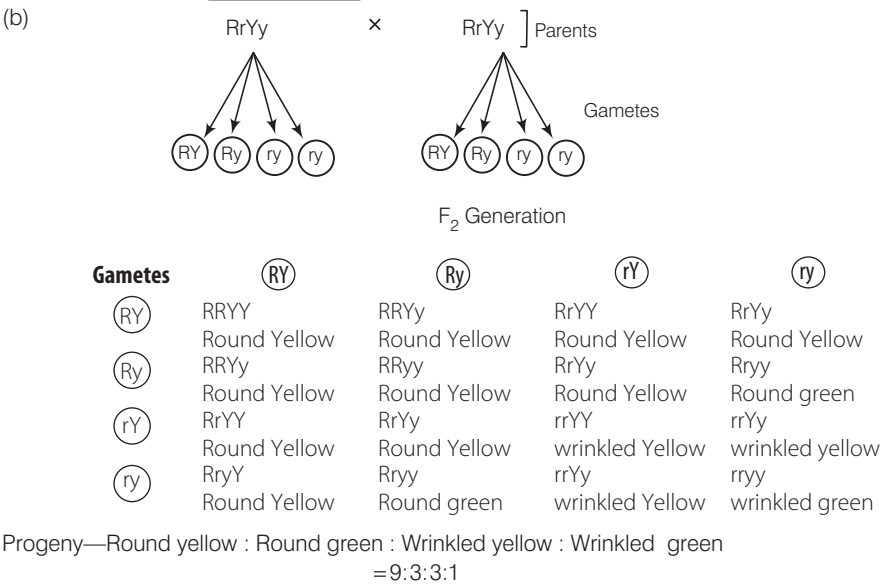
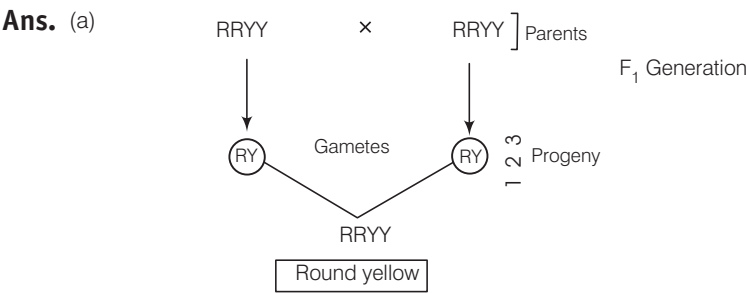
wrinkled, green, wrinkled, green

(d) RR YY × rr yy

Round, yellow, wrinkled, green

💡 Thinking Process

Round and yellow are the dominant characters while wrinkled and green are recessive



Q. 44 Study the following cross showing self pollination in F_1 . Fill in the blank

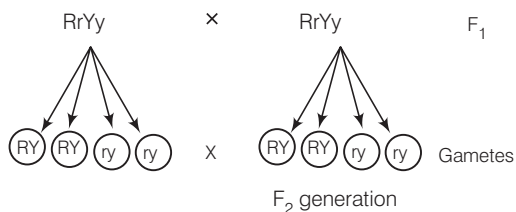
Parents RRYY x rryy
 Round, yellow Wrinkled, green
 F_1 — Rr Yy x ?
 Round, yellow

Ans. Self cross in F_1 shows dominant character (how of dominance) in F_1 generation.
i.e.,

 RrYy x RrYy
 Round yellow Round yellow

Q. 45 In Question 44, what are the combination of characters in the F_2 progeny? What are their ratios?

Ans. When F_1 generation dihybrid parts one crossed with each other the characters segregates to produce new combination the cross can be shown as



Gametes	RY	Ry	rY	ry
RY	RRYY Round Yellow	RRyy Round Yellow	RrYY Round Yellow	RrYy Round Yellow
Ry	RRYy Round Yellow	RRyy Round green	RrYy Round Yellow	Rryy Round green
rY	RrYY Round Yellow	RrYy Round Yellow	rrYY wrinkled Yellow	rrYy wrinkled yellow
ry	RrYy Round Yellow	Rryy Round green	rrYy wrinkled Yellow	rryy wrinkled green

Combination of characters

Round yellow (9), Round green (3), Wrinkled yellow (3), Wrinkled green (1)

in the ratio 9 : 3 : 3 : 1

Q. 46 Give the basic features of the mechanism of inheritance.

Ans. Inheritance is the transmission of genetically controlled characteristic (traits) from one generation to the next. *The basic features of mechanism of inheritance are following*

- (i) Characters are controlled by genes. Each gene control called allele one character.
- (ii) There may be two or more forms of a gene. One form may be dominant while other is recessive.
- (iii) Genes are present on chromosomes.
- (iv) Each parent possesses a pair of genes for each characteristic on a chromosomes.
- (v) When a male gamete fuses with the female gamete during fertilisation, they make a new cell called zygote with a full set of genes. This zygote carries characteristics of both the parents which are inherited through the genes.
- (vi) The offsprings inherit two genes (or a pair of genes) for each trait from its parents characters. The trait shown by the offsprings in first of enation dominant.

Q. 47 Give reasons for the appearance of new combination of characters in the F_2 progeny in question 45.

Ans. All the F_1 plants had round and yellow seeds. Cron between of generation, they, they give rise to new combinations in F_2 generation with round-yellow, round-green, wrinkled yellow and wrinkled green in the ratio of 9 : 3 : 3 : 1.

It indicates that the chances for the pea seeds to be round or wrinkled do not depend on their chances to be yellow or green. Each pair of alleles is independent of the other pairs. This is called the principle of **independent assortment** (3rd law of Mendelian gametes).

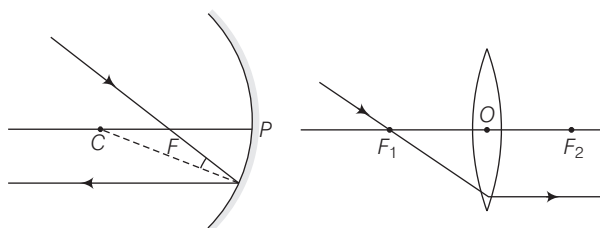
Light : Reflection and Refraction

Multiple Choice Questions (MCQs)

Q. 1 Which of the following can make a parallel beam of light when light from a point source is incident on it?

- (a) Concave mirror as well as convex lens
- (b) Convex mirror as well as concave lens
- (c) Two plane mirrors placed at 90° to each other
- (d) Concave mirror as well as concave lens

Ans. (a) A ray passing through the principal focus of a concave mirror or convex lens, after reflection/refraction, will emerge parallel to the principal axis.



Q. 2 A 10 mm long awl pin is placed vertically in front of a concave mirror. A 5 mm long image of the awl pin is formed at 30 cm in front of the mirror. The focal length of this mirror is

- (a) –30 cm
- (b) –20 cm
- (c) –40 cm
- (d) –60 cm

Ans. (b) Given, object size, $h = +10.0\text{ mm}$ ($\because 1\text{ cm} = 10\text{ mm}$)
 $= +1.0\text{ cm}$

Image size, $h' = 5.0 \text{ mm} = 0.5 \text{ cm}$

Image distance, $v = -30 \text{ cm}$

(For real image)

Focal length, $f = ?$

As, magnification, $m = \frac{h'}{h}$ (image size)
(object size)

Also, magnification, $m = \frac{-v}{u} \Rightarrow \frac{h'}{h} = \frac{-v}{u}$

$$\frac{0.5}{1} = \frac{-30}{u} \Rightarrow u = -60 \text{ cm}$$

Using mirror formula, $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$$\Rightarrow \frac{1}{f} = \frac{1}{-30} - \frac{1}{60} = \frac{-2-1}{60} = \frac{-3}{60}$$

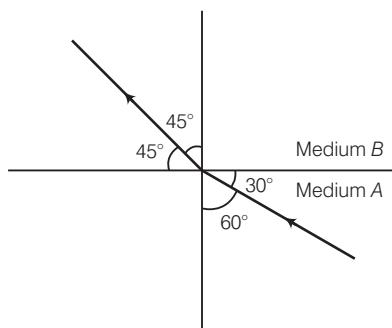
$$\Rightarrow f = -20 \text{ cm}$$

Q. 3 Under which of the following conditions, a concave mirror can form an image larger than the actual object?

- (a) When the object is kept at a distance equal to its radius of curvature
- (b) When object is kept at a distance less than its focal length
- (c) When object is placed between the focus and centre of curvature
- (d) When object is kept at a distance greater than its radius of curvature

Ans. (c) A concave mirror can form an image enlarged, real and inverted than the actual object, beyond centre of curvature (C) when object is placed between the focus (F) and centre of curvature.

Q. 4 Figure shows a ray of light as it travels from medium **A** to medium **B**. Refractive index of the medium **B** relative to medium **A** is



(a) $\frac{\sqrt{3}}{2}$

(b) $\frac{\sqrt{2}}{\sqrt{3}}$

(c) $\frac{1}{\sqrt{2}}$

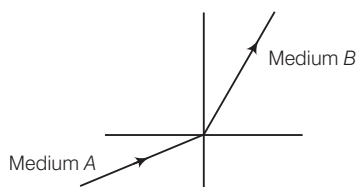
(d) $\sqrt{2}$

Ans. (a) Given, angle of incidence, $i = 60^\circ$, angle of refraction, $r = 45^\circ$

Refractive index of the medium **B** relative to medium **A**,

$$\mu_{BA} = \frac{\sin i}{\sin r} = \frac{\sin 60^\circ}{\sin 45^\circ} = \frac{\left(\frac{\sqrt{3}}{2}\right)}{\left(\frac{1}{\sqrt{2}}\right)} = \frac{\sqrt{3}}{\sqrt{2}}$$

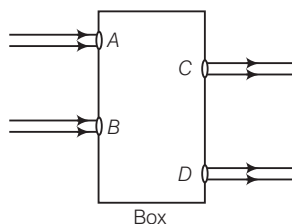
Q. 5 A light ray enters from medium A to medium B as shown in the figure. The refractive index of medium B relative to A will be



- (a) greater than unity (b) less than unity
(c) equal to unity (d) zero

Ans. (a) Since light rays in the medium B goes towards normal. So it has greater refractive index and lesser velocity of light w.r.t. medium A. So refractive index of medium B w.r.t. medium A is greater than unity.

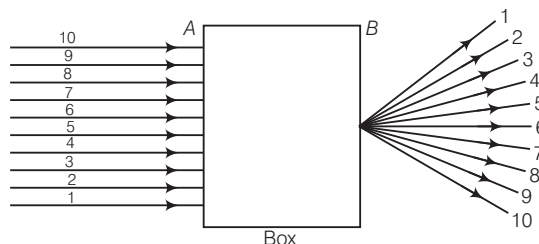
Q. 6 Beams of light are incident through the holes A and B and emerge out of box through the holes C and D respectively as shown in the figure. Which of the following could be inside the box?



- (a) A rectangular glass slab (b) A convex lens
(c) A concave lens (d) A prism

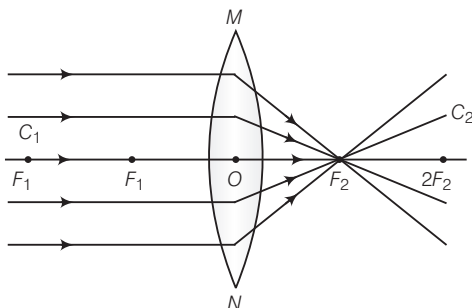
Ans. (a) Here, the emergent rays are parallel to the direction of the incident ray. Therefore, a rectangular glass slab could be inside the box as the extent of bending of light ray at the opposite parallel faces AB (air-glass interface) and CD (glass-air interface) of the rectangular glass slab are equal and opposite. This is why the ray emerges parallel to the incident ray.

Q. 7 A beam of light is incident through the holes on side A and emerges out of the holes on the other face of the box as shown in the figure. Which of the following could be inside the box?



- (a) Concave lens (b) Rectangular glass slab
(c) Prism (d) Convex lens

Ans. (d) Since in the diagram all the parallel rays converge at a point. So inside the box there will be a convex lens.



Q. 8 Which of the following statements is true?

- (a) A convex lens has 4 dioptre power having a focal length 0.25 m
- (b) A convex lens has 4 dioptre power having a focal length -0.25 m
- (c) A concave lens has 4 dioptre power having a focal length 0.25 m
- (d) A concave lens has 4 dioptre power having a focal length -0.25 m

Ans. (a) The power P of a lens of focal length f is given by $P = \frac{1}{f}$, where f is the focal length in metre and power in dioptre.

$$P = \frac{1}{f} \text{ or } f = \frac{1}{P} = \frac{1}{4} = 0.25 \text{ m}$$

Q. 9 Magnification produced by a rear view mirror fitted in vehicles

- (a) is less than one
- (b) is more than one
- (c) is equal to one
- (d) can be more than or less than one depending upon the position of the object in front of it.

Ans. (a) The convex mirror forms virtual, erect and diminished image of the object. And rear view mirror also form some type of image. Therefore, magnification (m) produced by a rear view mirror fitted in vehicles is less than one *i.e.*, $m < 1$.

Q. 10 Rays from sun converge at a point 15 cm in front of a concave mirror. Where should an object be placed so that size of its image is equal to the size of the object?

- (a) 15 cm in front of the mirror
- (b) 30 cm in front of the mirror
- (c) between 15 cm and 30 cm in front of the mirror
- (d) more than 30 cm in front of the mirror

Ans. (b) The rays from sun *i.e.*, from infinity, are parallel to principal axis after reflection converge at a point, known as focus. Therefore, focal length (F) of concave mirror is 15 cm. And we know that, same size, real and inverted image is formed by concave mirror when object is placed at focus $2F$ or centre of curvature so to form same size image object will be placed at $15 \times 2 = 30$ cm.

Q. 11 A full length image of a distant tall building can definitely be seen by using

- (a) a concave mirror (b) a convex mirror
(c) a plane mirror (d) both concave as well as plane mirror

Ans. (b) The convex mirror forms virtual, erect and diminished image of the objects. So, it can form full length image of a distant tall building.

Q. 12 In torches, search lights and headlights of vehicles, the bulb is placed

- (a) between the pole and the focus of the reflector
(b) very near to the focus of the reflector
(c) between the focus and centre of curvature of the reflector
(d) at the centre of curvature of the reflector

Ans. (b) Concave mirrors are commonly used in torches, search lights and vehicles headlights to get powerful parallel beams of light.

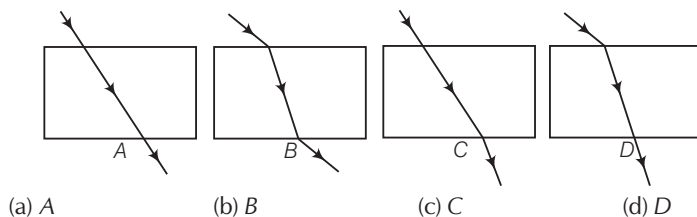
Here the bulb is placed very near to the focus of the reflector as the incident rays, which passes through the focus of concave mirror, after reflection become parallel to the principal axis of the mirror.

Q. 13 The laws of reflection hold good for

- (a) plane mirror only (b) concave mirror only
(c) convex mirror only (d) all mirrors irrespective of their shape

Ans. (d) The laws of reflection holds good for light reflected from any smooth surface *i.e.*, all mirrors regardless of its shape.

Q. 14 The path of a ray of light coming from air passing through a rectangular glass slab traced by four students shown as **A**, **B**, **C** and **D** in the figure. Which one of them is correct?



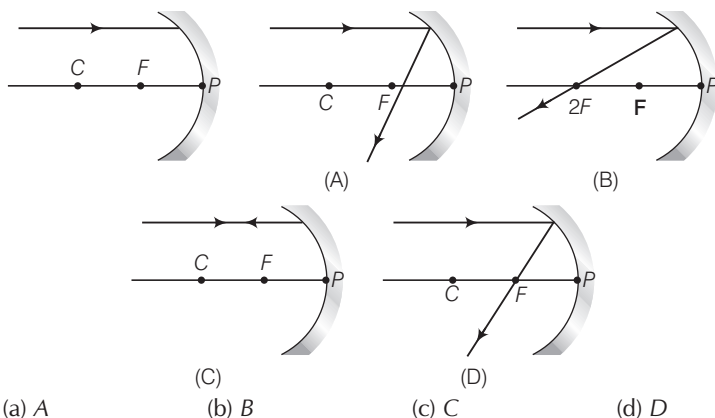
Ans. (b) In a rectangular glass slab, the emergent rays are parallel to the direction of the incident ray, because the lateral deviation of bending of the ray of light at the opposite parallel faces (air-glass interface) and (glass-air interface) of the rectangular glass slab are equal and opposite. This is why the ray emerges are parallel to the incident ray.

Q. 15 You are given water, mustard oil, glycerine and kerosene. In which of these media, a ray of light incident obliquely at same angle would bend the most?

- (a) Kerosene (b) Water (c) Mustard oil (d) Glycerine

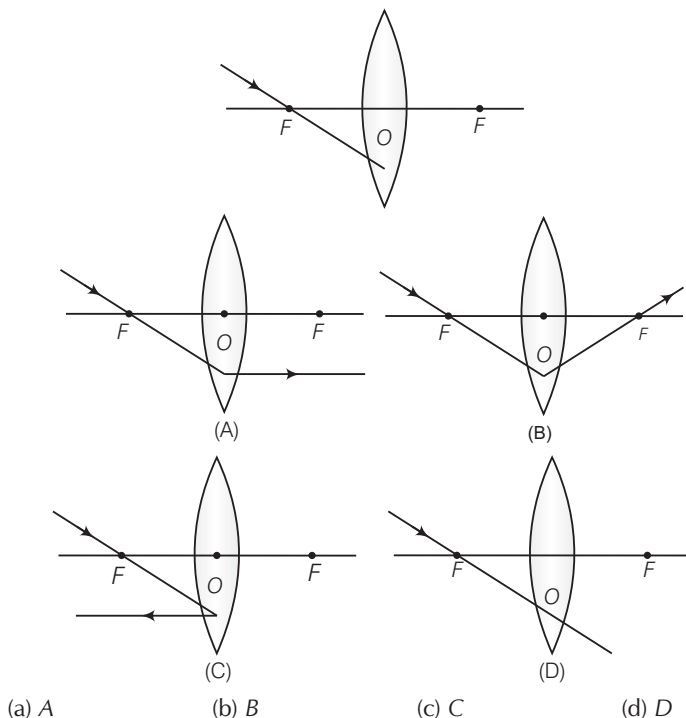
Ans. (d) The given material having their refractive index as kerosene is 1.44, water is 1.33, mustard oil is 1.46 and glycerine is 1.74. Thus, glycerine is most optically denser and hence have the largest refractive index. Therefore, ray of light bend most in glycerine.

Q. 16 Which of the following ray diagrams is correct for the ray of light incident on a concave mirror as shown in figure?



Ans. (d) A ray parallel to the principal axis, after reflection will pass through the principal focus in case of a concave mirror.

Q. 17 Which of the following ray diagrams is correct for the ray of light incident on a lens shown in figure?



Ans. (a) A ray of light passing through the principal focus of a convex lens after refraction from a convex lens, will emerge parallel to the principal axis.

Q. 18 A child is standing in front of a magic mirror. She finds the image of her head bigger, the middle portion of her body of the same size and that of the legs smaller. The following is the order of combinations for the magic mirror from the top.

- (a) Plane, convex and concave (b) Convex, concave and plane
(c) Concave, plane and convex (d) Convex, plane and concave

Ans. (c) Concave mirrors (of large focal length) can be used to see a larger image of the head, the plane mirror for middle portion to see her body of the same size and convex mirror to see the diminished image of leg. Hence, the combinations for magic mirror from the top is concave mirror, plane mirror and convex mirror.

Q. 19 In which of the following, the image of an object placed at infinity will be highly diminished and point sized?

- (a) Concave mirror only
(b) Convex mirror only
(c) Convex lens only
(d) Concave mirror, convex mirror, concave lens and convex lens

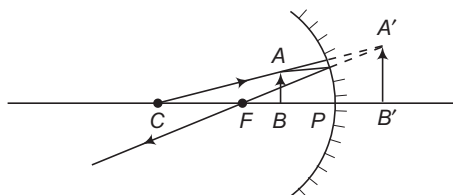
Ans. (d) The incident rays which comes from an object placed at infinity will be parallel and the rays parallel to the principal axis, after reflection/refraction by concave mirror, convex mirror, concave lens and convex lens, will pass or appear to pass through the principal focus. Hence, image will be highly diminished and point sized.

Short Answer Type Questions

Q. 20 Identify the device used as a spherical mirror or lens in following cases, when the image formed is virtual and erect in each case.

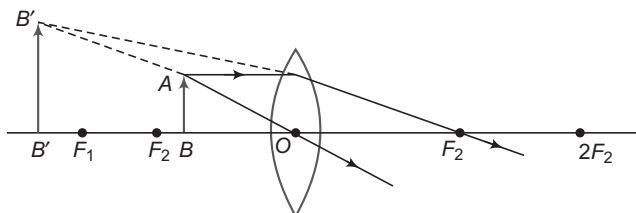
- (a) Object is placed between device and its focus, image formed is enlarged and behind it
(b) Object is placed between the focus and device, image formed is enlarged and on the same side as that of the object
(c) Object is placed between infinity and device, image formed is diminished and between focus and optical centre on the same side as that of the object
(d) Object is placed between infinity and device, image formed is diminished and between pole and focus, behind it

Ans. (a) The spherical mirror is used as concave mirror.



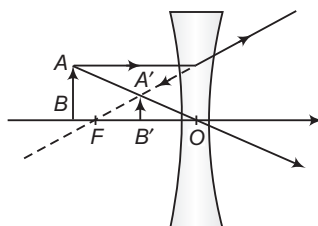
Object between F and P

(b) The spherical lens is used as convex lens.



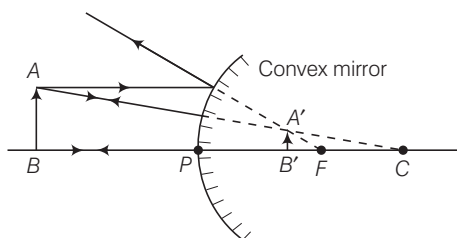
Object between focus (F) and device (O)

(c) The spherical lens is used as concave lens.



Object between infinity and device (O)

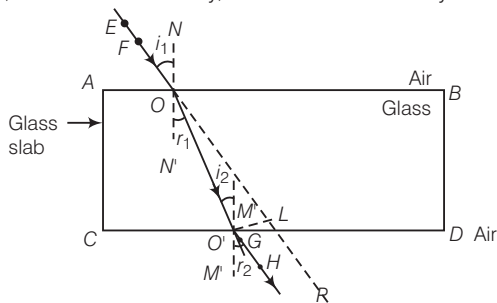
(d) The spherical mirror is used as convex mirror.



Object between infinity and device (P)

Q. 21 Why does a light ray incident on a rectangular glass slab immersed in any medium emerges parallel to itself? Explain using a diagram.

Ans. In the given figure, EO is the incident ray, OO' is the refracted ray and $O'H$ is the emergent ray.



The extent of bending of the ray of light at the opposite parallel faces AB (air-glass interface) and CD (glass-air interface) of the rectangular glass slab is equal and opposite. This is why the ray emerges parallel to the incident ray on a rectangular glass slab. However, the light ray is shifted sideward slightly. When glass slab is immersed in any medium the interface AB (medium-glass) and CD glass medium are equal and opposite so, the emergent ray will always be parallel to the incident ray.

Q. 22 A pencil when dipped in water in a glass tumbler appears to be bent at the interface of air and water. Will the pencil appear to be bent to the same extent, if instead of water we use liquids like, kerosene or turpentine? Support your answer with reason.

Thinking Process

As it is based on refraction of light.

Ans. A pencil partly immersed in water in a glass tumbler, it appears to be displaced at the interface of air and water. The light reaching out from the portion of the pencil inside water seems to come from a different direction, compared to the part above water due to refraction of light.

The pencil appear to be bent to the different extent, if instead of water, liquids like, kerosene or turpentine are used as their refractive indices are different which in turn produces deviation from incident ray by different extent.

Q. 23 How is the refractive index of a medium related to the speed of light? Obtain an expression for refractive index of a medium with respect to another in terms of speed of light in these two media?

Ans. The refractive index of the medium is defined as the ratio of speed of light in vacuum to the speed of light in the medium. It is expressed as

$$\text{Refractive index, } \mu = \frac{\text{speed of light in vacuum, } c}{\text{speed of light in medium, } v}$$

μ_1 = refractive index of first medium

μ_2 = refractive index of second medium

v_1 = velocity in first medium

v_2 = velocity in second medium

For medium 1, we have

$$\mu_1 = \frac{c}{v_1}$$

For medium 2, we have

$$\mu_2 = \frac{c}{v_2}$$

$$\therefore \mu_{21} = \frac{\mu_2}{\mu_1} = \frac{\frac{c}{v_2}}{\frac{c}{v_1}} = \frac{v_1}{v_2}$$

Q. 24 Refractive index of diamond with respect to glass is 1.6 and absolute refractive index of glass is 1.5. Find out the absolute refractive index of diamond.

Ans. Given, refractive index of diamond with respect to glass, ${}_g\mu_d = 1.6 = \frac{\mu_d}{\mu_g}$

Absolute refractive index of glass, ${}_a\mu_g = 1.5, = \mu_g / \mu_a$

Absolute refractive index of diamond, ${}_a\mu_d = \mu_d / \mu_a = ?$

$${}_g\mu_d = \frac{{}_a\mu_d}{{}_a\mu_g}$$

$\Rightarrow$

$${}_a\mu_d = {}_g\mu_d \times {}_a\mu_g = 1.6 \times 1.5 = 2.4$$

Q. 25 A convex lens of focal length 20 cm can produce a magnified virtual as well as real image. Is this a correct statement? If yes, where shall the object be placed in each case for obtaining these images?

Ans. Yes, the statement is correct. The convex lens of focal length 20 cm can produce a magnified, virtual as well as real image. *The object should be placed*

- (i) Between focus F_1 and optical centre O (i.e., a distance less than 20 cm from the lens) for magnified, virtual and erect image.
- (ii) Between F_1 and $2F_1$ (i.e., at a distance between 20 cm to 40 cm) for real, inverted and enlarged image.

Q. 26 Sudha finds out that the sharp image of the window pane of her science laboratory is formed at a distance of 15 cm from the lens. She now tries to focus the building visible to her outside the window instead of the window pane without disturbing the lens. In which direction will she move the screen to obtain a sharp image of the building? What is the approximate focal length of this lens?

Ans. Sudha should move the screen towards the lens to obtain a sharp image of the building because window pane was lying beyond focus $2F$ or centre of curvature and convex lens forms its image, on the other side between F and $2F$ and when she tries to focus the building visible to her outside the window (i.e., at distant object at infinity), then lens forms the image of building at a distance of focal length. The approximate focal length of this lens is 15 cm.

Q. 27 How are power and focal length of a lens related? You are provided with two lenses of focal length 20 cm and 40 cm, respectively. Which lens will you use to obtain more convergent light?

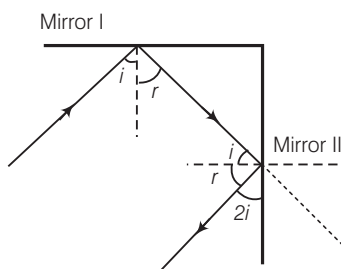
Ans. The power of a lens is related to its focal length as

$$P = \frac{1}{f \text{ (in m)}}$$

For greater convergent light, lens of higher power and smaller focal length is needed i.e., the lens of focal length 20 cm is needed for the same.

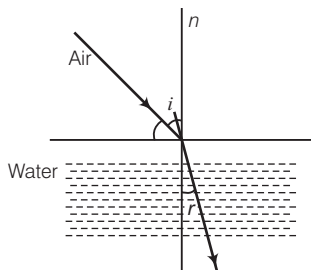
Q. 28 Under what condition in an arrangement of two plane mirrors, incident ray and reflected ray will always be parallel to each other, whatever may be angle of incidence. Show the same with the help of diagram.

Ans. When two plane mirrors are placed at right angle with each other, then incident ray and reflected ray will always be parallel to each other, whatever may be angle of incidence.

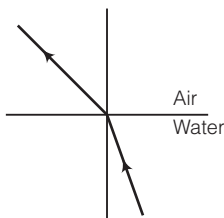


Q. 29 Draw a ray diagram showing the path of rays of light when it enters with oblique incidence (i) from air into water; (ii) from water into air.

Ans. (i) When ray of light enters with oblique incidence from air into water, then it goes from optical rarer medium to optical denser medium and velocity of light decreases which in turn bends the incident light towards the normal. Also, $i > r$



(ii) When ray of light enters with oblique incidence from water into air, then it goes from optical denser medium to optical rarer medium and velocity of light increases which in turn bends the incident light away from the normal. Also, $i < r$.

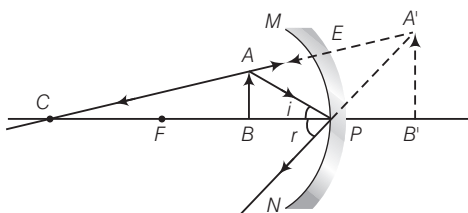


Long Answer Type Questions

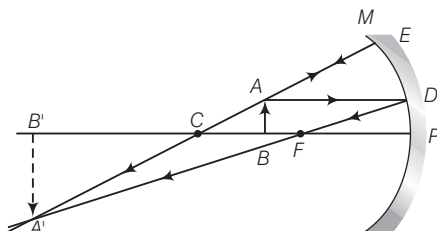
Q. 30 Draw ray diagrams showing the image formation by a concave mirror when an object is placed

- between pole and focus of the mirror
- between focus and centre of curvature of the mirror
- at centre of curvature of the mirror
- a little beyond centre of curvature of the mirror
- at infinity

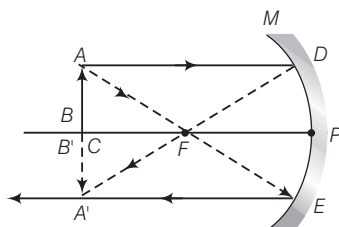
Ans. (a) The enlarged, virtual and erect image forms behind the mirror when the object is placed between pole and focus of the mirror.



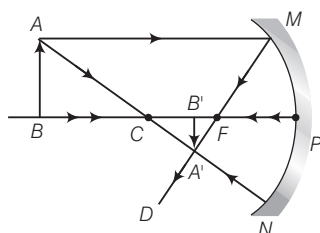
- (b) The enlarged, real and inverted image forms beyond centre of curvature when the object is placed between focus and centre of curvature of the mirror.



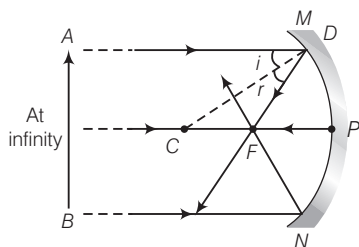
- (c) The real and inverted image equal to the size of object forms at centre of curvature when the object is placed at the centre of curvature of the mirror.



- (d) The diminished, real and inverted image forms between centre of curvature and focus when the object is placed a little beyond centre of curvature of the mirror.



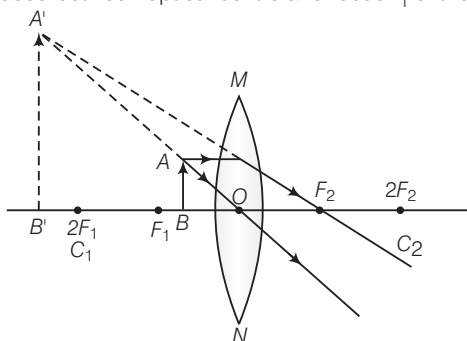
- (e) The real, inverted and highly reduced image forms at focus F when the object is placed at infinity.



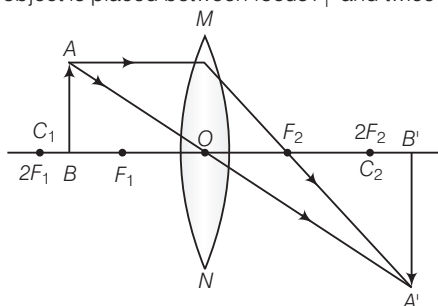
31. Draw ray diagrams showing the image formation by a convex lens when an object is placed

- between optical centre and focus of the lens
- between focus and twice the focal length of the lens
- at twice the focal length of the lens
- at infinity
- at the focus of the lens

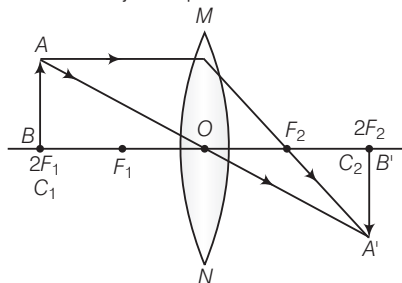
- Ans.** (a) The enlarged, virtual and erect image forms beyond $2F_1$ in the same side of object when the object is placed between optical centre and focus F_1 of the lens.



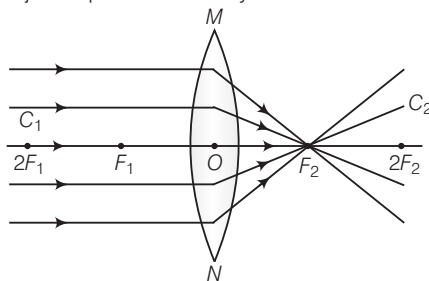
- (b) The enlarged, real and inverted image forms beyond focus $2F_2$ on the other side of the object when the object is placed between focus F_1 and twice the focal length of the lens.



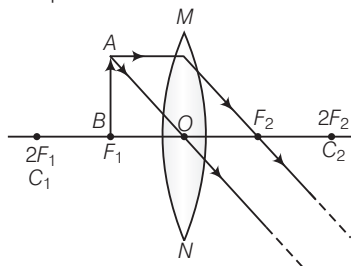
- (c) The real and inverted image of equal to the size of object forms at focus $2F_2$ on the other side of the object when the object is placed at twice the focal length of the lens.



- (d) The real, inverted and highly reduced image forms at focus F_2 on the other side of the object when the object is placed at infinity.



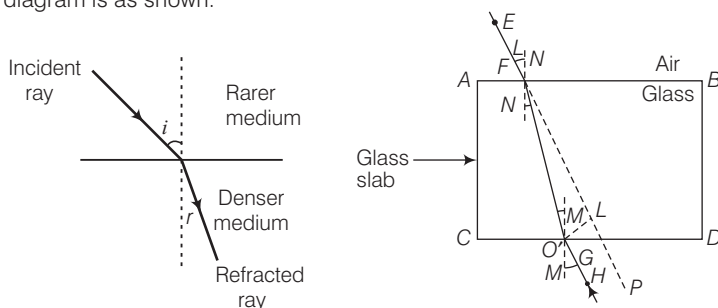
- (e) The real, inverted and highly magnified image forms at infinity on the other side of the object when the object is placed at the focus of the lens.



Q. 32 Write laws of refraction. Explain the same with the help of ray diagram, when a ray of light passes through a rectangular glass slab.

Ans. The following are the laws of refraction of light

- The incident ray, the refracted ray and the normal to the interface of two transparent media at the point of incidence, all lie in the same plane.
- The ratio of sine of angle of incidence to the sine of angle of refraction is constant, for the light of a given colour and for the given pair of media. This law is also known as Snell's law of refraction. And the constant is called refractive index. The ray diagram is as shown.

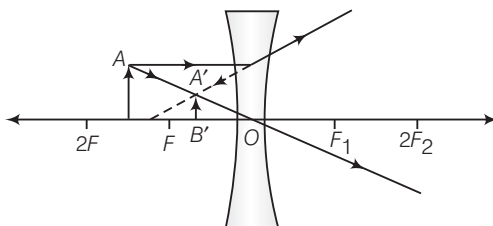


In a rectangular or glass slab, the emergent rays are parallel to the incident ray because the extent of bending of the ray of light at the opposite parallel faced of rectangular glass slab are equal and opposite, so that emergent ray is parallel to incident ray.

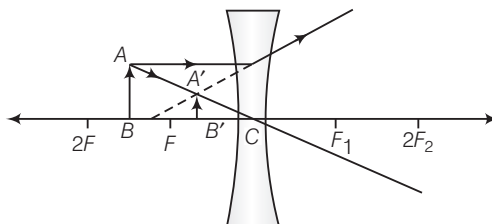
Q. 33 Draw ray diagrams showing the image formation by a concave lens when an object is placed

- at the focus of the lens
- between focus and twice the focal length of the lens
- beyond twice the focal length of the lens

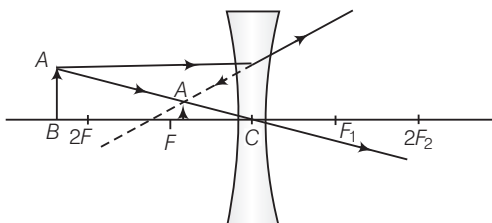
Ans. (a) The image formed is virtual, erect, diminished in size and between F and O when the object is placed at focus



- (b) The image formed is virtual, erect, diminished in size and between optical centre and focus F when the object is placed between focus and twice the focal length of the lens.



- (c) The image formed is virtual, erect, diminished in size and between optical centre and focus F when the object is placed beyond twice the focal length of the lens.

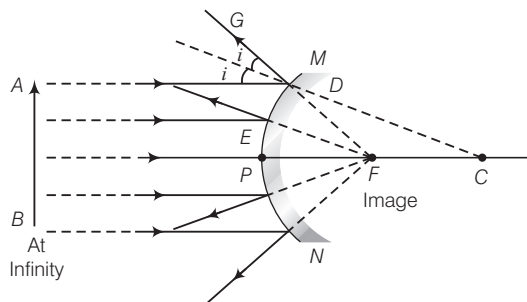


Q. 34 Draw ray diagrams showing the image formation by a convex mirror when an object is placed

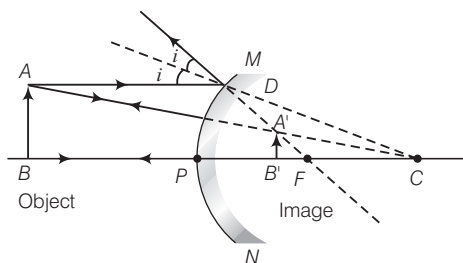
(a) at infinity

(b) at finite distance from the mirror

Ans. (a) The virtual, erect and highly diminished image of the object forms at focus F behind the mirror when the object is placed at infinity.



- (b) The virtual, erect and diminished image forms between focus F and pole P behind the mirror when the object is placed at finite distance from the mirror.



- Q. 35** The image of a candle flame formed by a lens is obtained on a screen placed on the other side of the lens. If the image is three times the size of the flame and the distance between lens and image is 80 cm, at what distance should the candle be placed from the lens? What is the nature of the image at a distance of 80 cm and the lens?

Thinking Process

Using magnification find the object distance. Then, using lens formula, calculate the value of focal length. Using sign of convention nature of lens can be identified.

Ans. The image is real as only the real image can be taken on the screen.

Here, image distance $v = +80$ cm

Magnification, $m = -3$

Object distance, $u = ?$

Since, magnification,

$$m = \frac{v}{u} \Rightarrow -3 = \frac{80}{u}$$

$$u = \frac{-80}{3} \text{ cm}$$

Nature of image

→ Real, inverted

→ magnified

→ formed beyond $2F$

Using lens formula, we have

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{f} = \frac{1}{80} - \frac{3}{-80} = \frac{4}{80} = \frac{1}{20}$$

$$\Rightarrow f = 20 \text{ cm}$$

Positive focal length denotes that lens is convex.

- Q. 36** Size of image of an object by a mirror having a focal length of 20 cm is observed to be reduced to $1/3$ rd to its size. At what distance, the object has been placed from the mirror? What is the nature of the image and the mirror?

Thinking Process

Solve object distance for both concave mirror and convex mirror using the relation of magnification and mirror formula.

Ans. Here, considering (the case for both type of possible spherical mirrors)
For concave mirror

Focal length, $f = -20$ cm

Magnification, $m = -\frac{1}{3}$

Since, magnification, $m = -\frac{v}{u}$

$$\therefore \text{Magnification, } m = -\frac{1}{3} = -\frac{v}{u}$$

$$v = \frac{u}{3}$$

Using mirror formula, we have

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{f} = \frac{1}{u} + \frac{3}{u} = \frac{4}{u}$$

$$\Rightarrow u = 4f = 4(-20) = -80 \text{ cm}$$

The object should be placed at a distance 80 cm from the concave mirror.

For convex mirror,

Focal length, $f = +20$ cm

$$\text{Magnification, } m = +\frac{1}{3}$$

$$\text{Since, magnification, } m = -\frac{v}{u}$$

$$\therefore \text{Magnification, } m = \frac{1}{3} = -\frac{v}{u}$$

$$v = -\frac{u}{3}$$

Using mirror formula, we have

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{f} = \frac{-3}{u} + \frac{1}{u} = \frac{-2}{u}$$

$$u = -2f = -2(20) = -40 \text{ cm}$$

The object should be placed at a distance 40 cm from the convex mirror to form virtual, erect and diminished image.

Q. 37 Define power of a lens. What is its unit? One student uses a lens of focal length 50 cm and another of -50 cm. What is the nature of the lens and its power used by each of them?

Ans. Power of lens is defined as the ability of a lens to bend the rays of light. It is given by the reciprocal of focal length in metre. Its unit is dioptre.

$$\text{If focal length, } (f) = 50 \text{ cm, then } P = \frac{100}{f} = \frac{100}{50} = 2 \text{ D, lens is convex.}$$

$$\text{If focal length, } (f) = -50 \text{ cm, then } P = \frac{100}{f} = \frac{100}{-50} = -2 \text{ D, lens is concave.}$$

Q. 38 A student focussed the image of a candle flame on a white screen using a convex lens. He noted down the position of the candle, screen and the lens as under

position of candle = 12.0 cm

position of convex lens = 50.0 cm

position of the screen = 88.0 cm

- What is the focal length of the convex lens?
- Where will the image be formed, if he shifts the candle towards the lens at a position of 31.0 cm?
- What will be the nature of the image formed, if he further shifts the candle towards the lens?
- Draw a ray diagram to show the formation of the image in case as said above.

Ans. Object distance u = Position of the convex lens – Position of candle
 $= 50 - 12 = 38 \text{ cm}$

By sign convention, $u = -38 \text{ cm}$

Similarly, image distance, v = position of the screen – position of convex lens
 $= 88 - 50 = 38 \text{ cm}$

By sign convention, $v = +38 \text{ cm}$

(i) Using lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$\frac{1}{f} = \frac{1}{38} - \frac{1}{-38} = \frac{2}{38} = \frac{1}{19}$$

$$f = 19 \text{ cm}$$

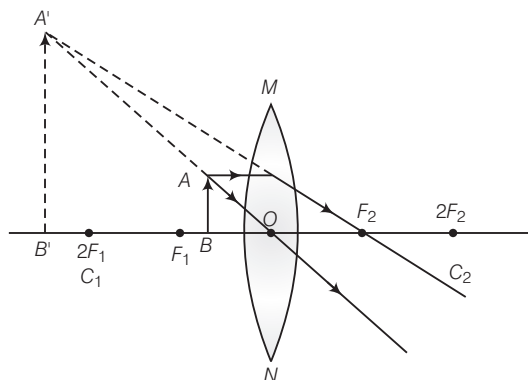
The focal length of the convex lens is 19 cm.

(ii) After shifting the candle towards the lens at a position of 31.0 cm, then
 object distance u = position of convex lens – position of candle
 $= 50 - 31 = 19$

By sign convention, $u = -19 \text{ cm}$, but here focal length of the convex lens is also 19 cm.
 So, the candle lies at the focus of lens, hence its image forms at infinity.

(iii) When student further shifts the candle towards the lens *i.e.*, candle lies between optical centre and focus of convex lens, then lens forms enlarged, virtual and erect image of the candle.

(iv) The ray diagram showing the formation of image



The Human Eye and the Colourful World

Multiple Choice Questions (MCQs)

Q. 1 A person cannot see distinctly objects kept beyond 2 m. This defect can be corrected by using a lens of power

- (a) +0.5 D (b) -0.5 D (c) +0.2 D (d) -0.2 D

💡 Thinking Process

If a person cannot see distantly, objects kept beyond 2 m then he is suffering from myopia.

Ans. (b) As the person has the eye defect, myopia, therefore, a concave lens has to be used whose focal length will be, $f = -2$ m (using sign convention). Thus,

$$\begin{aligned} \text{Power, } P &= \frac{1}{f} \text{ (where, } f \text{ is focal length in metre)} \\ &= \frac{1}{-2} = -0.5 \text{ D} \end{aligned}$$

Q. 2 A student sitting on the last bench can read the letters written on the blackboard but is not able to read the letters written in his text book. Which of the following statements is correct?

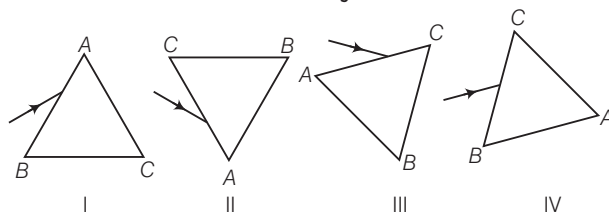
- (a) The near point of his eyes has receded away
(b) The near point of his eyes has come closer to him
(c) The far point of his eyes has come closer to him
(d) The far point of his eyes has receded away

💡 Thinking Process

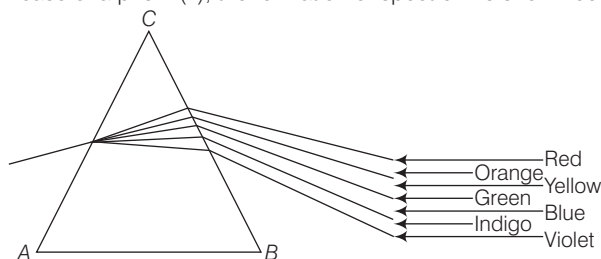
Hypermetropia may have blurred vision to a person when looking at objects close to them and clearer vision when looking at objects in the distance.

Ans. (a) The student sitting on the last bench can read the letters written on the blackboard but is not able to read the letters written in his text book because he is suffering from hypermetropia or far sightedness. He can see distant objects clearly but cannot see nearby objects distinctly.

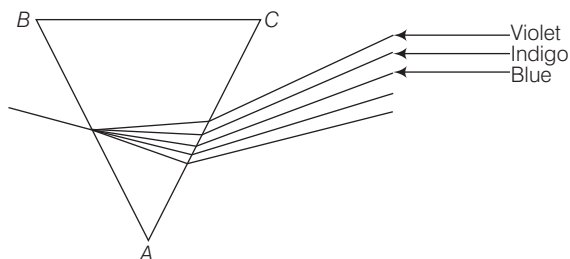
Q. 3 A prism ABC (with BC as base) is placed in different orientations. A narrow beam of white light is incident on the prism as shown in figure. In which of the following cases, after dispersion, the third colour from the top corresponds to the colour of the sky?



Ans. Generally, in case of a prism (II), the formation of spectrum is shown below



In the above figure, from top the third colour is yellow. But we can see that from bottom the third colour is blue which is the colour of sky. So, we can obtain the correct situation by inverting the prism. Thus the required orientations can be bound in case II.



Q. 4 At noon the sun appears white as

- (a) light is least scattered
- (b) all the colours of the white light are scattered away
- (c) blue colour is scattered the most
- (d) red colour is scattered the most

Ans. (a) At noon the sun appears white because the light from the sun is directly over head and travel relatively shorter distance. The sun appears white as only a little of the blue and violet colours are scattered.

Q. 5 Which of the following phenomena of light are involved in the formation of a rainbow?

- (a) Reflection, refraction and dispersion
- (b) Refraction, dispersion and total internal reflection
- (c) Refraction, dispersion and internal reflection
- (d) Dispersion, scattering and total internal reflection

Ans. (c) A rainbow is caused by dispersion, refraction and internal reflection of sunlight by tiny water droplets, present in the atmosphere and always formed in a direction opposite to that of the sun. The water droplets act like small prisms. They refract and disperse the incident sunlight, then reflect it internally and finally refract it again when it comes out of the raindrop.

Q. 6 Twinkling of stars is due to atmospheric

- (a) dispersion of light by water droplets
- (b) refraction of light by different layers of varying refractive indices
- (c) scattering of light by dust particles
- (d) internal reflection of light by clouds

Ans. (b) The twinkling of a star is due to atmospheric refraction of lights of stars. These lights, on entering the earth's atmosphere, undergoes refraction continuously before it reaches the earth.

The path of rays of light coming from the distant star goes on varying slightly, the apparent position of the star fluctuates and the amount of starlight entering the eye flickers. The star sometimes appears brighter and at some other time, fainter gives us the twinkling effect.

Q. 7 The clear sky appears blue, because

- (a) blue light gets absorbed in the atmosphere
- (b) ultraviolet radiations are absorbed in the atmosphere
- (c) violet and blue lights get scattered more than lights of all other colours by the atmosphere
- (d) light of all other colours is scattered more than the violet and blue colour lights by the atmosphere

Ans. (c) The clear sky appears blue is due to Rayleigh scattering of sunlight. The molecules in the air scatter blue light from the sun more than they scatter red light.

Q. 8 Which of the following statements is correct regarding the propagation of light of different colours of white light in air?

- (a) Red light moves fastest
- (b) Blue light moves faster than green light
- (c) All the colours of the white light move with the same speed
- (d) Yellow light moves with the mean speed as that of the red and the violet light

Ans. (c) The propagation of light of different colours of white light in air or vacuum move with the same speed but different wavelengths and frequencies.

Q. 9 The danger signals installed at the top of tall buildings are red in colour. These can be easily seen from a distance because among all other colours, the red light

- (a) is scattered the most by smoke or fog (b) is scattered the least by smoke or fog
- (c) is absorbed the most by smoke or fog (d) moves fastest in air

Ans. (b) The danger signals installed at the top of tall buildings are red in colour because among all other colours, red colour is scattered the least by smoke or fog. This is primarily because wavelength of red colour is the largest. Thus it can be easily seen from a distance.

Q. 10 Which of the following phenomena contributes significantly to the reddish appearance of the sun at sunrise or sunset?

- (a) Dispersion of light (b) Scattering of light
- (c) Total internal reflection of light (d) Reflection of light from the earth

Ans. (b) The reddish appearance of the sun at sunrise or sunset is due to least scattering of red light. The red light have the maximum wavelength. The rays of light have to travel a larger part of the atmosphere.

Q. 11 The bluish colour of water in deep sea is due to

- (a) the presence of algae and other plants found in water
- (b) reflection of sky in water
- (c) scattering of light
- (d) absorption of light by the sea

Ans. (c) The bluish colour of water in deep sea is due to scattering of light as the very fine particles in water scatter mainly blue light.

The red, green, orange and yellow having longer wavelengths are absorbed more strongly by water than blue which has shorter wavelength. So, when white light from the sun enters the sea, it is mostly the blue that gets returned.

Q. 12 When light rays enter the eye, most of the refraction occurs at the

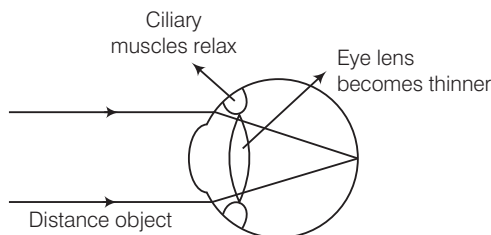
- (a) crystalline lens (b) outer surface of the cornea
- (c) iris (d) pupil

Ans. (b) When the light rays enters the eye through a thin membrane, forms the transparent bulge on the front surface of the eyeball, called the cornea. Most of the refraction for the light rays entering the eye occurs at this outer surface of the cornea.

Q. 13 The focal length of the eye lens increases when eye muscles

- (a) are relaxed and lens becomes thinner
- (b) contract and lens becomes thicker
- (c) are relaxed and lens becomes thicker
- (d) contract and lens becomes thinner

Ans. (a) The focal length of the eye lens increases when eye muscles relaxed and becomes thinner. The sharp image of the distant object is formed at the retina. This enables us to focus accurately on distant objects.



Q. 14 Which of the following statements is correct?

- (a) A person with myopia can see distant objects clearly
- (b) A person with hypermetropia can see nearby objects clearly
- (c) A person with myopia can see nearby objects clearly
- (d) A person with hypermetropia cannot see distant objects clearly

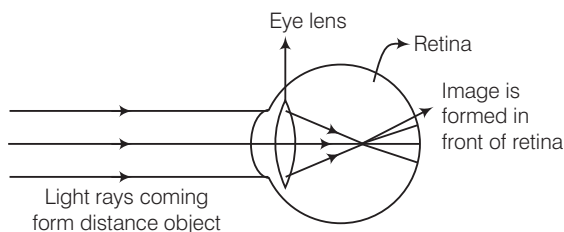
Ans. (c) A person with myopia can see nearby objects clearly. While a person with hypermetropia can see distant objects clearly.

Short Answer Type Questions

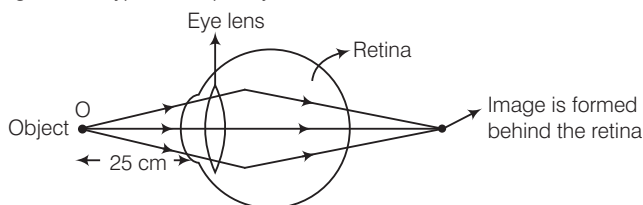
Q. 15 Draw ray diagrams each showing

- (i) myopic eye and
- (ii) hypermetropic eye.

Ans. (i) The ray diagram for myopic eye is as shown below



(ii) The ray diagram for hypermetropic eye is as shown below

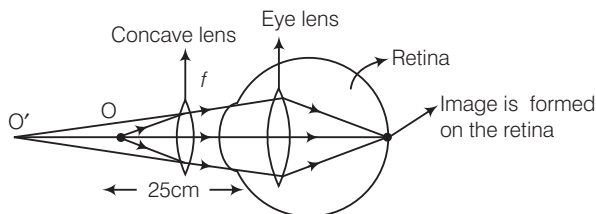


Q. 16 A student sitting at the back of the classroom cannot read clearly the letters written on the blackboard. What advice will a doctor give to her? Draw ray diagram for the correction of this defect.

Ans. The student is suffering from myopia or near-sightedness. In myopia, one can see nearby objects clearly but cannot see distant objects distinctly.

The doctor advises a concave lens of suitable power to bring the image back onto the retina.

The ray diagram for the correction of this defect is shown below



Q. 17 How are we able to see nearby and also the distant objects clearly?

Ans. We are able to see nearby and also the distant objects clearly due to the ability of the eye lens to adjust its focal length which is known as power of accommodation.

When the muscles are relaxed, the lens becomes thin and its focal length increases. This enables us to see the distant objects clearly. When the ciliary muscles are contract, the curvature of the eye lens increases and the eye lens becomes thicker. Consequently, the focal length of the eye lens decreases. This enables us to see nearby objects clearly.

Q. 18 A person needs a lens of power -4.5 D for correction of her vision.

(a) What kind of defect in vision is she suffering from?

(b) What is the focal length of the corrective lens?

(c) What is the nature of the corrective lens?

Thinking Process

Calculate the focal length. If focal length is positive, then lens used is convex and if negative, then lens used is concave lens.

Ans. (a) As the power of lens is negative, she must be suffering from myopia.

(b) Power, $P = -4.5$ D, focal length, $f = ?$

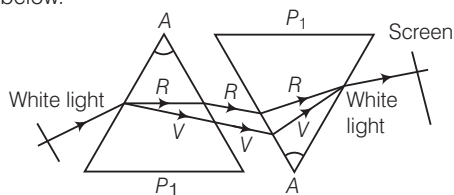
$$P = \frac{1}{f}$$

$$\Rightarrow f = \frac{1}{P} = \frac{1}{-4.5} = -0.222 \text{ m} = -22.2 \text{ cm}$$

(c) The nature of the corrective lens is concave or divergent.

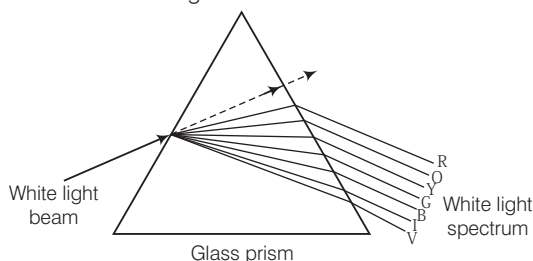
Q. 19 How will you use two identical prisms so that a narrow beam of white light incident on one prism emerges out of the second prism as white light? Draw the diagram.

Ans A narrow beam of white light incident on one prism emerges out of the identical prism placed in an inverted position with respect to the first prism close to the first prism as shown in figure given below.



Q. 20 Draw a ray diagram showing the dispersion through a prism when a narrow beam of white light is incident on one of its refracting surfaces. Also indicate the order of the colours of the spectrum obtained.

Ans. The ray diagram showing the dispersion through a prism when a narrow beam of white light is incident on one of its refracting surfaces are shown below

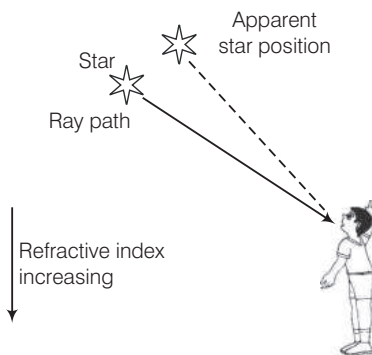


The order of the colour is Violet, Indigo, Blue, Green, Yellow, Orange and Red starting from the base of the prism to upwards.

Q. 21 Is the position of a star as seen by us its true position? Justify your answer.

Ans. No, what is seen is not true. The starlight on entering the earth's atmosphere, undergoes refraction continuously before it reaches the earth. The atmospheric refraction occurs in a medium of gradually changing refractive index.

Since, the atmosphere bends starlight towards the normal, the apparent position of the star is slightly different from its actual position. Thus, star appears slightly higher (above) than its actual position.



Q. 22 Why do we see a rainbow in the sky only after rainfall?

Ans. We see a rainbow in the sky only after rainfall due to dispersion of sunlight by tiny water droplets, present in the atmosphere due to rainfall.

The water droplets act like small prisms. They refract and disperse the incident sunlight, then reflect it internally and finally refract it again when it comes out of the raindrop and hence due to dispersion of light and internal reflection, rainbow formation takes place during rainfall.

Q. 23 Why is the colour of the clear sky blue?

Ans. The blue colour of clear sky is due to Rayleigh scattering of sunlight. The molecules of air and other fine particles in the atmosphere have size smaller than the wavelength of visible light. These are more effective in scattering light of shorter wavelengths at the blue end than light of longer wavelengths at the red end.

The red light has a wavelength about 1.8 times greater than blue light. Thus, when sunlight passes through the atmosphere, the fine particles in air scatter the blue colour (shorter wavelengths) more strongly than red. The scattered blue light enters our eyes and we see, colour of sky is blue.

Q. 24 What is the difference in colours of the sun observed during sunrise/sunset and noon? Give explanation for each.

Ans. During sunrise/sunset, the sun looks reddish because at this stage, rays from the sun have to travel a much larger part of atmosphere. The red colour having largest wavelength is scattered the least.

At noon, the sun is nearly overhead. The sunlight has to pass through much smaller, portion of earth's atmosphere. The scattering is much less and hence the sun looks white.

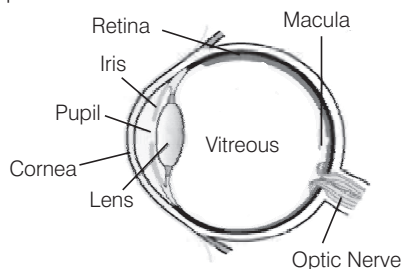
Long Answer Type Questions

Q. 25 Explain the structure and functioning of human eye. How are we able to see nearby as well as distant objects?

Ans. The structure and functioning of human eye are as given below

Cornea Cornea is transparent. It is responsible for, most of the refraction of light rays towards the retina.

Iris Iris is a dark, circular muscular diaphragm behind the cornea. It gives the eye its colour. It also controls the pupil size and thus controls the amount of light entering the eye.



Pupil Pupil is surrounded by the iris. It regulates and controls the amount of light entering the eye.

Retina The retina is the innermost layer of the eye. It contains an outer pigmented layer and an inner nervous layer which contains the photo receptors (rods and cones). Cone cells are coned shaped and rod cells are rod shaped.

Optic nerve The optic nerve connects the eye to the brain. The nerve fibres carry impulses from the retina to the visual cortex.

Eye lens The eye lens is made up of a fibrous, jelly-like material. It is transparent and has biconvex structure. It forms an inverted real image of the object on the retina and also separates the aqueous and the vitreous humours.

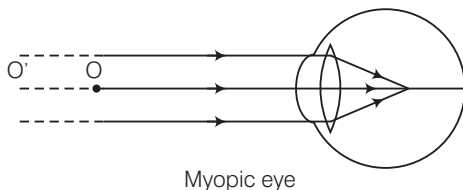
Vitreous humour Vitreous humour is clear, semi-solid supporting the eye ball.

The eye lens forms an inverted real image of the object on the retina which sends 'image-electric signals to brain via the optic nerves. The brain reconstruct erect image of objects and make us see as they are. For second part, refer to Q. 17.

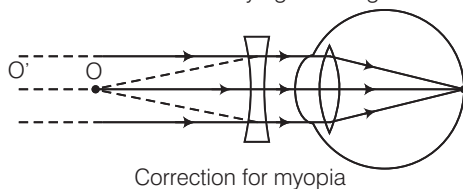
Q. 26 When do we consider a person to be myopic and hypermetropic? Explain using diagrams how the defects associated with myopic and hypermetropic eye can be corrected?

Ans. Myopic This term is used to define short sightedness. Light from a distant object forms an image before it reaches the retina. This could be because the eye is too long or the cornea or crystalline lens is too strong.

A myopic person has clear vision when looking at objects close to them but distant objects will appear blurred.

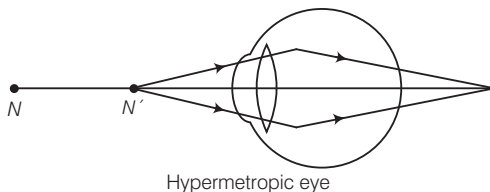


To correct this defects a concave lens (minus powered) is placed in front of a myopic eye, moving the image back to the retina and clarifying the image.

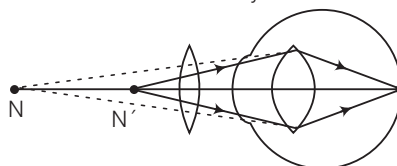


Hypermetropic It means long sightedness, and, in which the image of a nearby object is formed behind the retina. This could be because the eye is too short or the cornea or crystalline lens does not refract the light enough.

A hypermetropic person may have blurred vision when looking at objects close to them and clearer vision when looking at objects in the distance.



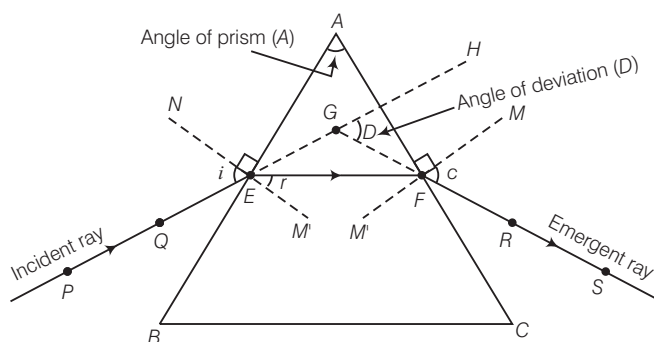
To correct this defects a convex lens (plus powered) in front of a hypermetropic eye, the image is moved forward and focuses correctly on the retina.



Correction for hypermetropic eye

Q. 27 Explain the refraction of light through a triangular glass prism using a labelled ray diagram. Hence, define the angle of deviation.

Ans. The refraction of light through a triangular glass prism is shown below



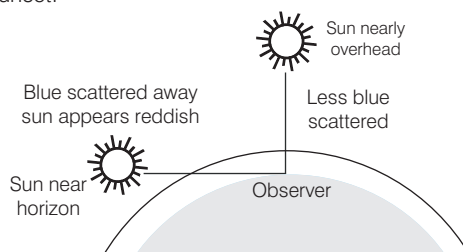
A ray of light PE is entering from air to glass at the first surface AB . The light ray EF on refraction has bent towards the normal. At the second surface AC , the light ray FS has entered from glass to air and bent away from normal.

The angle made by extending incident ray with the emergent ray is called angle of deviation.

Q. 28 How can we explain the reddish appearance of sun at sunrise or sunset? Why does it not appear red at noon?

Ans. The reddish appearance of the sun at sunrise or sunset is due to scattering of light by the molecules of air and other fine particles in the atmosphere have size smaller than the wavelength of visible light from the sun near the horizon. It passes through thicker layers of air and larger distance in the earth's atmosphere before reaching our eyes and most of the blue light and shorter wavelengths are scattered away by the particles.

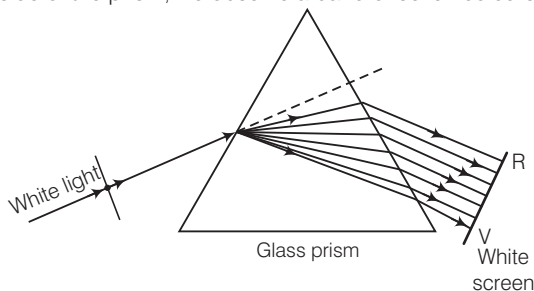
So, only red light, being of higher wavelength reaches us which gives reddish appearance of sun at sunrise or sunset.



At noon, the sun appears white, not red, as only a little of the blue and violet colours are scattered as light from the Sun overhead would travel relatively shorter distance.

Q. 29 Explain the phenomenon of dispersion of white light through a glass prism, using suitable ray diagram.

Ans. When the white light such as sunlight is passed through a pin to reduce it to a narrow beam. This narrow beam is made to fall on a triangular glass prism. When a white screen is held on the other side of the prism, we observe a band of seven colours on the screen.



The colour sequence obtained on the screen from its lower end is given by the famous acronym **VIBGYOR**, where V stands for violet, I stands for Indigo, B stands for Blue, G stands for Green, Y stands for Yellow, O stands for Orange and R stands for Red.

The red colour bends the least on passing through the prism and violet colour bends through maximum angle on passing through the prism. This is why red is at top and violet is at bottom. Thus, the phenomenon of splitting of white light into its constituent seven colours on passing through a glass prism is known as dispersion of light.

Q. 30 How does refraction take place in the atmosphere? Why do stars twinkle but not the planets?

Ans. The refraction of light by the earth's atmosphere is due to the gradually changing refractive index of air layers. The physical conditions of the refracting medium (air) are not stationary and the apparent position of the object as seen through the hot air fluctuates. This wavering is thus an effect of atmospheric refraction on a small scale in our local environment.

Since, the atmosphere bends light from the astronomical bodies viz, stars towards the normal, the apparent position of the astronomical bodies is slightly different from its actual position. The astronomical bodies appear slightly higher (above) than its actual position when viewed near the horizon.

The **twinkling of a star** is due to atmospheric refraction of starlight. The starlight, on entering the earth's atmosphere, undergoes refraction continuously before it reaches the earth. The path of rays of light coming from the distant star goes on varying slightly, the apparent position of the star fluctuate and the amount of starlight entering the eye flickers that the star sometimes appears brighter and at some other time fainter which gives us the twinkling effect.

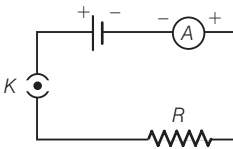
The planets are much closer to the earth and are thus seen as extended sources whereas stars are very distant, the approximate point-sized sources of light. If we consider a plane as a collection of a large number of point-sized sources of light, the total variation in the amount of light entering our eye from all the individual point-sized sources will average out to zero, thereby nullifying the twinkling effect. Hence, the planets do not twinkle.

12

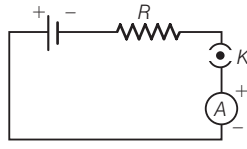
Electricity

Multiple Choice Questions (MCQs)

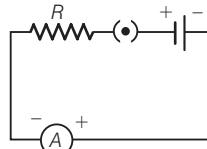
Q. 1 A cell, a resistor, a key and an ammeter are arranged as shown in the circuit diagrams of figure. The current recorded in the ammeter will be



(i)



(ii)

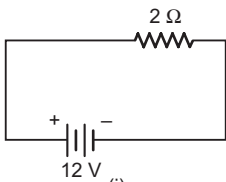


(iii)

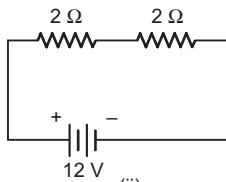
- (a) maximum in (i) (b) maximum in (ii)
(c) maximum in (iii) (d) the same in all the cases

Ans. (d) A cell, a resistor, a key and an ammeter are arranged in series as shown in the circuit diagrams. In series connections the order of elements in the circuit does not matter to the amount of current flows through it.

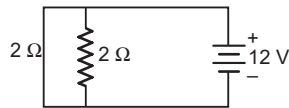
Q. 2 In the following circuits, heat produced in the resistor or combination of resistors connected to a 12 V battery will be



(i)



(ii)



(iii)

- (a) same in all the cases (b) maximum in case (i)
(c) maximum in case (ii) (d) maximum in case (iii)

Ans. (d) In Case (i), $R = 2\Omega$

Case (ii), $R = 2 + 2 = 4\Omega$

Case (iii), $\frac{1}{R} = \frac{1}{2} + \frac{1}{2} = 1 \Rightarrow R = 1\Omega$

Since, $H = \frac{V^2}{R} \times t$, As voltage in the three cases for equivalent resistance is same
 so, $H \propto \frac{1}{R}$. So heat produced is maximum in case (iii).

Q. 3 Electrical resistivity of a given metallic wire depends upon

- (a) its length (b) its thickness
 (c) its shape (d) nature of the material

Ans. (d) Electrical resistivity of a given metallic wire depends on number density of free electrons in the conductor which is the nature of material. i.e., $\rho \propto \frac{1}{n}$; n = number of free electrons per unit volume. It also depends on the temperature of conductor.

Q. 4 A current of 1 A is drawn by a filament of an electric bulb. Number of electron passing through a cross-section of the filament in 16 s would be roughly

- (a) 10^{20} (b) 10^{16} (c) 10^{18} (d) 10^{23}

Ans. (a) Given, current, $I = 1$ A

Time, $t = 16$ s

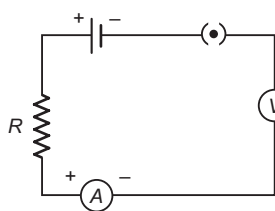
Number of electrons $n = ?$

We know that,

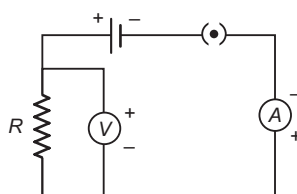
$$\text{Current, } I = \frac{Q}{t} = \frac{ne}{t} \quad [\because e = \text{charge of an electron} = 1.6 \times 10^{-19} \text{ C and } Q = ne]$$

$$\Rightarrow n = \frac{It}{e} = \frac{1 \times 16}{1.6 \times 10^{-19}} = 10^{20}$$

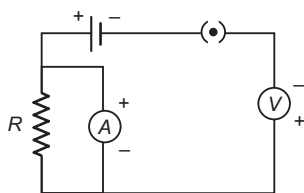
Q. 5 Identify the circuit in which the electrical components have been properly connected.



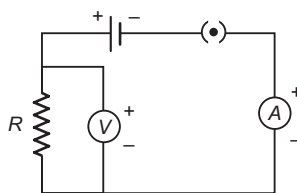
(i)



(ii)



(iii)



(iv)

(a) (i)

(b) (ii)

(c) (iii)

(d) (iv)

Ans. (b) While identifying the circuit, the following conditions must be satisfied.

- (a) An ammeter is always connected in series.
- (b) The voltmeter should be connected in parallel
- (c) The positive terminals of V and A should be joined to positive terminal of the cell and their negative terminals should be joined to the negative terminal of the cell.

Thus, the above conditions are satisfied in case (ii).

Q. 6 What is the maximum resistance which can be made using five resistors each of $1/5\Omega$?

- (a) $1/5\Omega$ (b) 10Ω (c) 5Ω (d) 1Ω

Ans. (d) The maximum resistance is obtained when resistors are connected in series combination. Thus equivalent resistance

$$R_s = n \times R = 5 \times \frac{1}{5} = 1\Omega, R_s = \text{equivalent resistance for series combination.}$$

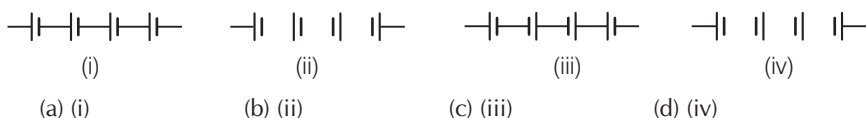
Q. 7 What is the minimum resistance which can be made using five resistors each of $1/5\Omega$?

- (a) $1/5\Omega$ (b) $1/25\Omega$ (c) $1/10\Omega$ (d) 25Ω

Ans. (b) The minimum resistance is obtained when resistors are connected in parallel combination. Thus, equivalent resistance,

$$R_p = \frac{R}{n} = \frac{1/5}{5} = \frac{1}{25}\Omega, R_p = \text{equivalent resistance for parallel combination.}$$

Q. 8 The proper representation of series combination of cells obtaining maximum potential is



Ans. (a) The proper representation of series combination of cells is that the negative terminal of the first cell is connected to the positive terminal of the second cell and the negative terminal of the second cell is connected to the positive terminal of the third cell and so on.

Q. 9 Which of the following represents voltage?

- (a) $\frac{\text{Work done}}{\text{Current} \times \text{Time}}$ (b) Work done $\times$ Charge
- (c) $\frac{\text{Work done} \times \text{Time}}{\text{Current}}$ (d) Work done $\times$ Charge $\times$ Time

Ans. (a) As we know that,

$$\text{Work done, } W = Vq = VIt \quad (q = It)$$

where, $V = \text{Voltage}$

$I = \text{Current flowing}$

$t = \text{Time taken}$

$$\therefore V = \frac{W}{I \times t} = \frac{\text{Work done}}{\text{Current} \times \text{Time}}$$

Q. 10 A cylindrical conductor of length l and uniform area of cross-section A has resistance R . Another conductor of length $2l$ and resistance R of the same material has area of cross-section.

- (a) $A/2$ (b) $3A/2$ (c) $2A$ (d) $3A$

Ans. (c) In first case,

$$\text{Resistivity of the conductor, } \rho = \frac{RA'}{l} \quad \dots(i)$$

In second case,

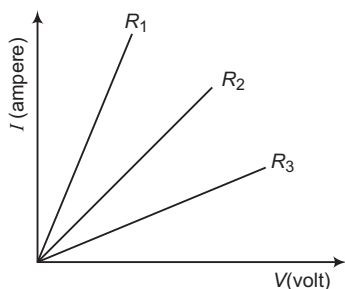
$$\text{Resistivity of the conductor, } \rho = \frac{RA'}{2l} \quad \dots(ii)$$

$\therefore$ For same material, resistivity of the conductor will be same.

So, on substituting the value of ρ in Eq. (ii), we have

$$\frac{RA}{l} = \frac{RA'}{2l} \Rightarrow A' = 2A$$

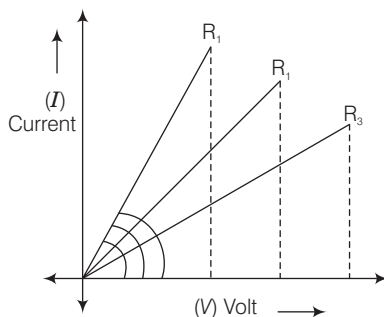
Q. 11 A student carries out an experiment and plots the $V-I$ graph of three samples of nichrome wire with resistances R_1 , R_2 and R_3 respectively as shown in figure. Which of the following is true?



- (a) $R_1 = R_2 = R_3$ (b) $R_1 > R_2 > R_3$ (c) $R_3 > R_2 > R_1$ (d) $R_2 > R_3 > R_1$

Ans. (c) While performing an experiment, a student plot $V-I$ graph of three samples of nichrome wire with resistances R_1 , R_2 and R_3 . And we know that slope of V and I tells about the resistance and (slope of V and I) $\propto \frac{1}{\text{Resistance}}$

i.e.,



So, $R_3 > R_2 > R_1$

Q. 12 If the current I through a resistor is increased by 100% (assume that temperature remains unchanged), the increase in power dissipated will be

- (a) 100% (b) 200% (c) 300% (d) 400%

Ans. (c) Power,

$$P = I^2 R$$

$$P_1 = I^2 R$$

and

$$P_2 = (2I)^2 R = 4I^2 R$$

[$\therefore$ 100% increase in current means that current becomes $2I$]

$$\therefore \text{Increase in power dissipated} = P_2 - P_1$$

$$= 4I^2 R - I^2 R$$

$$= 3I^2 R$$

$$(\because I^2 R = P_1)$$

$$\text{Percentage increase in power dissipated} = \frac{3P_1}{P_1} \times 100 = 300\%$$

Q. 13 The resistivity does not change if

- (a) the material is changed
(b) the temperature is changed
(c) the shape of the resistor is changed
(d) both material and temperature are changed

Ans. (c) Resistivity depends on the nature of the material and the temperature but does not depend on the shape of the resistor.

Q. 14 In an electrical circuit three incandescent bulbs. **A**, **B** and **C** of rating 40W, 60 W and 100 W, respectively are connected in parallel to an electric source. Which of the following is likely to happen regarding their brightness ?

- (a) Brightness of all the bulbs will be the same
(b) Brightness of bulb A will be the maximum
(c) Brightness of bulb B will be more than that of A
(d) Brightness of bulb C will be less than that of B

Ans. (c) In an electrical circuit, three incandescent bulbs A, B and C of rating 40 W, 60 W and 100 W are connected in parallel to an electric source, then the bulb with the highest wattage glows with maximum brightness. Power dissipated $\propto$ Brightness. Thus, the brightness of bulb B (100 W) is maximum.

This brightness different bulbs are in following order

$$B_{100} > B_{60} > B_{40}$$

Q. 15 In an electrical circuit, two resistors of 2Ω and 4Ω respectively are connected in series to a 6V battery. The heat dissipated by the 4Ω resistor in 5s will be

- (a) 5 J (b) 10 J (c) 20 J (d) 30 J

💡 Thinking Process

First of all we find the net current flowing through the resistor. And then use formula

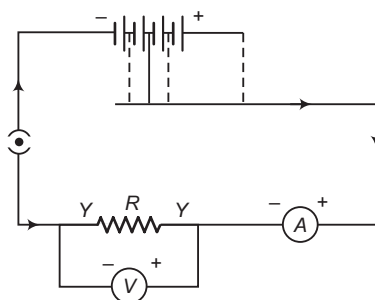
$$(H = I^2 R t)$$

Ans. (c) Given, resistors, $R_1 = 2\ \Omega$ and $R_2 = 4\ \Omega$
 Voltage, $V = 6\text{ V}$ Time, $t = 5\text{ s}$
 Resistance, $R = 4\ \Omega$
 Heat dissipated, $H = ?$
 Resistor, $R_s = R_1 + R_2 = 2 + 4 = 6\ \Omega$
 Current, $I = \frac{V}{R} = \frac{6}{6} = 1\text{ A}$.
 Heat dissipated $H = I^2 R t = 1 \times 4 \times 5 = 20\text{ J}$

Q. 16 An electric kettle consumes 1 kW of electric power when operated at 220 V. A fuse wire of what rating must be used for it?

- (a) 1 A (b) 2 A (c) 4 A (d) 5 A

Ans. (d) Given, power, $P = 1\text{ kW} = 1000\text{ W}$



Voltage,
Current,

$$V = 220\text{ V}$$

$$I = ?$$

$$I = \frac{P}{V} = \frac{1000}{220} = 4.5\text{ A}$$

Thus, the rating of fuse-wire is 5 A which is greater than 4.5 A.

Q. 17 Two resistors of resistance $2\ \Omega$ and $4\ \Omega$ when connected to a battery will have

- (a) same current flowing through them when connected in parallel
 (b) same current flowing through them when connected in series
 (c) same potential difference across them when connected in series
 (d) different potential difference across them when connected in parallel

Ans. (b) In series combination of resistances same current flows through each resistor where as in parallel combination same voltage exist across each resistor. Thus, the two resistors of resistances $2\ \Omega$ and $4\ \Omega$ are connected in series has same current flowing through it.

Q. 18 Unit of electric power may also be expressed as

- (a) volt ampere (b) kilowatt hour (c) watt second (d) joule second

Ans. (a) Electric power is given by $P = \text{voltage} \times \text{current}$

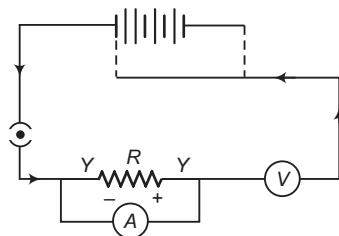
SI Unit of voltage = Volt

SI Unit of current = Ampere

So, its unit is volt ampere.

Short Answer Type Questions

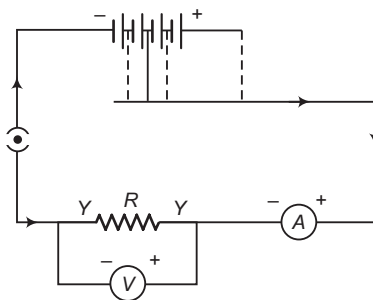
Q. 19 A child has drawn the electric circuit to study Ohm's law as shown in figure. His teacher told that the circuit diagram needs correction. Study the circuit diagram and redraw it after making all corrections.



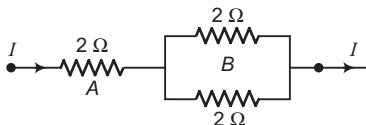
Ans. The circuit diagram is not correct because of following reasons

- Ammeter is connected in parallel with R and voltmeter connected in series with it in contrary of the fact that ammeter should be connected in series and voltmeter in parallel. This is so because same current flows in series and same voltage exist in parallel combination.
- The current is drawn from negative terminal and enter into the battery at possible terminal which is not possible in one battery circuit.
- Cells are not connected in series combination in the battery of the circuit.

The correct diagram is as follow



Q. 20 Three 2Ω resistors, **A**, **B** and **C** are connected as shown in figure. Each of them dissipates energy and can with stand a maximum power of 18 W without melting. Find the maximum current that can flow through the three resistors?



Ans. Resistance, $R = 2\Omega$
 Maximum power, $P_{\max} = 18\text{ W}$
 Maximum current, $I_{\max} = ?$
 $P = I^2 R$

$$\Rightarrow I = \sqrt{\frac{P}{R}} = \sqrt{\frac{18}{2}} = 3\text{ A} = I_{\max}$$

Maximum current that can flow through 2Ω resistor is 3 A . This current divides along B and C because in parallel combination, voltage across B and C remain same and hence, $I \propto \frac{1}{R}$.

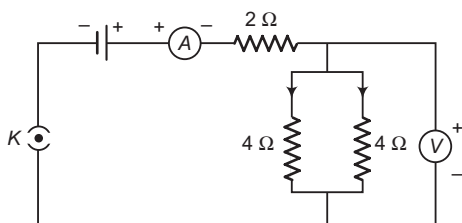
Since, B and C have same resistance 2Ω each, therefore, same current i.e., $\frac{3}{2} = 1.5\text{ A}$ flows through B and C .

Q. 21 Should the resistance of an ammeter be low or high? Give reason.

Ans. The resistance of an ammeter should be low. An ammeter has to be connected in series with the circuit to measure current. In case, its resistance is not very low, its inclusion in the circuit will reduce the current to be measured. In fact, an ideal ammeter is one which has zero resistance.

Q. 22 Draw a circuit diagram of an electric circuit containing a cell, a key, an ammeter, a resistor of 2Ω in series with a combination of two resistors (4Ω each) in parallel and a voltmeter across the parallel combination. Will the potential difference across the 2Ω resistors be the same as that across the parallel combination of 4Ω resistors? Give reason.

Ans. The circuit diagram containing a cell, a key, a resistor of 2Ω in series with a combination of two resistors (4Ω each) in parallel and a voltmeter across the parallel combination is shown below



The resistance of parallel combination of two resistors (4Ω each) is 2Ω .

The same current passes through the 2Ω resistor and 2Ω (equivalent of two parallel resistor of 4Ω). So, voltage drop by them will be same, hence the voltage V is same across 2Ω resistor and the parallel combination of two resistors, cell of 4Ω .

Q. 23 How does use of a fuse wire protect electrical appliances?

Ans. A fuse is a circuit connection that will be destroyed by excessive current flow, creating an open circuit, to protect electrical appliances from receiving too high current. It is designed to be the weakest link in the circuit, which will fail before the other components are damaged.

The simplest fuses contain thin wire strips that melts when heated by excessive current flow. A fuse wire consists of low melting point and high resistivity and it is always connected in live wire.

Q. 24 What is electrical resistivity? In a series electrical circuit comprising a resistor made up of a metallic wire, the ammeter reads 5A. The reading of the ammeter decreases to half when the length of the wire is doubled. Why?

Ans. The electrical resistivity of a material is defined as the resistance of a conductor made of that material of unit length and unit cross-sectional area.

The resistance of an uniform conductor is given by

$$R = \rho \frac{l}{A}$$

where, l is the length of the conductor

ρ is the electrical resistivity of the material

A is the cross-sectional area.

If l is doubled, then R is also doubled.

So, in this case,

$$R' = \rho \frac{2l}{A} = 2 \rho \frac{l}{A} = 2R$$

Since,

$$I = \frac{V}{R}$$

So,

$$I' = \frac{V}{2R} = \frac{I}{2}$$

when, R is doubled, I becomes $\frac{I}{2}$.

Q. 25 What is the commercial unit of electrical energy? Represent it in terms of joules.

Ans. The commercial unit of electrical energy is kilowatt hour. It is written as kWh

$$1 \text{ kWh} = 1 \text{ kW} \times \text{h}$$

$$= 1000 \text{ W} \times 3600 \text{ s}$$

$$= 3.6 \times 10^6 \text{ J}$$

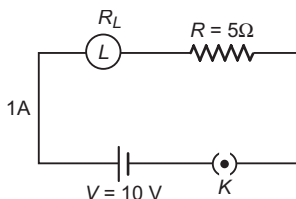
$$(\because 1 \text{ kW} = 1000 \text{ W}, 1 \text{ h} = 3600 \text{ s})$$

Q. 26 A current of 1 A flows in a series circuit containing an electric lamp and a conductor of 5Ω when connected to a 10V battery. Calculate the resistance of the electric lamp.

Now, if a resistance of 10Ω is connected in parallel with this series combination, what change (if any) in current flowing through 5Ω conductor and potential difference across the lamp will take place? Give reason.

Ans. Given current, $I = 1 \text{ A}$

Resistance of conductor, $R = 5\Omega$



Voltage, $V = 10 \text{ V}$

Resistance of lamp, $R_L = ?$

Total resistance in the circuit,

$$R_T = \frac{V}{I} = \frac{10}{1} = 10\Omega$$

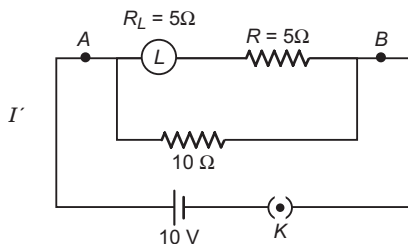
$\therefore$

$$R_L = R_T - R = 10 - 5 = 5\Omega$$

Potential difference across the lamp

$$= IR_L = 1 \times 5 = 5V$$

When 10Ω resistance is connected in parallel with total resistance R_T ($R_L + R = 10\Omega$), then total resistance R' in the circuit is given by



$$\frac{1}{R'} = \frac{1}{10} + \frac{1}{R_T} = \frac{1}{10} + \frac{1}{10} = \frac{2}{10} = \frac{1}{5}$$

$$R' = 5\Omega$$

Current through the circuit,

$$I = \frac{V}{R'} = \frac{10}{5} = 2A$$

Since, 10Ω and R_T (10Ω) are in parallel, current through R_T is $\frac{I}{2} = \frac{2}{2} = 1A$

Thus, current through lamp and conductor of 5Ω in series is $1A$. Also PD across lamp

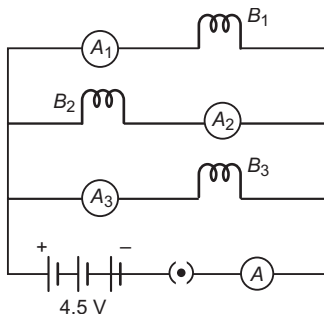
$$= \frac{I}{2} \times 5 = 1 \times 5 = 5V$$

Q. 27 Why is parallel arrangement used in domestic wiring?

Ans. Parallel arrangement used in domestic wiring because of the following

- All the appliances work at the same voltage as that of the electric supply.
- If one of the appliances is out of order, e.g. if a bulb gets fused, all other appliances keep on working as the circuit is not broken in parallel arrangement of devices.

Q. 28 B_1 , B_2 and B_3 are three identical bulbs connected as shown in figure. When all the three bulbs glow, a current of $3A$ is recorded by the ammeter A.



- (i) What happens to the glow of the other two bulbs when the bulb B_1 gets fused?
- (ii) What happens to the reading of A_1 , A_2 , A_3 and A when the bulb B_2 gets fused?
- (iii) How much power is dissipated in the circuit when all the three bulbs glow together?

Ans. Resistance of combination of three bulbs in parallel

$$R_{\text{eq}} = \frac{V}{I} = \frac{4.5}{3} = 1.5 \Omega$$

If R is the resistance of each wire, then

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} \text{ or } \frac{1}{R_{\text{eq}}} = \frac{3}{R}$$

or

$$R = 3R_{\text{eq}} = 3 \times 1.5 = 4.5 \Omega$$

Current in each bulb,

$$I = \frac{V}{R} = \frac{4.5 \Omega}{4.5 \Omega} = 1 \text{ A}$$

- (i) When bulb B_1 gets fused, the currents in B_2 and B_3 remain same $I_2 = I_3 = 1 \text{ A}$, because voltage across the B_2 and B_3 bulb remains same, so their glow remains unaffected.
- (ii) When bulb B_2 gets fused, the current in B_2 becomes zero and current in B_1 and B_3 remains 1 A. Because voltage across B_1 and B_3 bulb remaining same

$$\text{Total current } I = I_1 + I_2 + I_3 = 1 + 0 + 1 = 2 \text{ A}$$

Current in ammeter, $A_2 = 0$

Current in ammeter, $A_3 = 1 \text{ A}$

Current in ammeter, $A_1 = 1 \text{ A}$

Current in ammeter, $A = 2 \text{ A}$

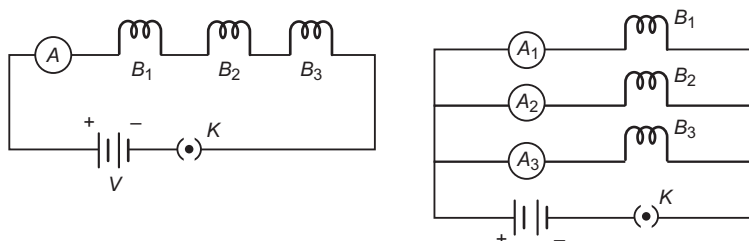
- (iii) When all the three bulbs are connected.

$$\text{Power dissipated, } P = \frac{V^2}{R_{\text{eq}}} = \frac{(4.5)^2}{1.5} = 13.5 \text{ W}$$

Q. 29 There incandescent bulbs of 100 W each are connected in series in an electric circuit. In another set of three bulbs of the same wattage are connected in parallel to the same source.

- (a) Will the bulb in the two circuits glow with the same brightness? Justify your answer
- (b) Now let one bulb in both the circuits get fused. Will the rest of the bulbs continue to glow in each circuit? Give reason

Ans. (a) Let us assume that the resistance of each bulb be R . The circuit diagram in two cases may be drawn as given below



Equivalent resistance in series combination $R_3 = R + R + R = 3R$, voltage = V

Let current through each bulb in series combination be I_1 . By Ohm's law,

$$V = I_1 \times 3R \Rightarrow I_1 = \frac{V}{3R}$$

So, power consumption of each bulb in series combination.

$$\begin{aligned} P_1 &= I_1^2 (3R) = \left(\frac{V}{3R}\right)^2 \times 3R = \frac{V^2}{9R^2} \times 3R \\ &= \frac{V^2}{3R} = \frac{V^2}{3R} \end{aligned} \quad \dots(i)$$

For parallel circuit, (1)

The resistance of each bulb = R

Voltage across each bulb = V [$\because$ same voltage in parallel combination]

$\therefore$ Power consumption of each bulb in parallel combination

$$P_2 = \frac{V^2}{R} \quad \dots(ii)$$

$\therefore$ From Eqs. (i) and (ii), we have

$$\begin{aligned} \frac{P_2}{P_1} &= \frac{V^2 / R}{V^2 / 3R} \\ \frac{V^2}{R} \times \frac{3R}{V^2} &= 3 \Rightarrow P_2 = 3P_1 \end{aligned}$$

Therefore, each bulb in parallel combination glows 3 times brighter to that of each bulb in series combination.

- (b) When one bulb gets fused in both the circuit, then in series combination, circuit gets broken and current stops flowing and remaining bulb don't glow. Whereas in parallel combination, same voltage continue to act on the remaining voltage and hence, other bulbs continue to glow with same brightness.

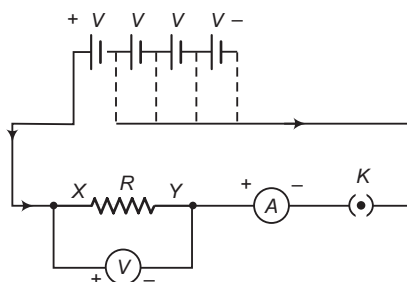
Q. 30 State Ohm's law? How can it be verified experimentally? Does it hold good under all conditions? Comment.

Ans. Ohm's law states that the electric current flowing through a conductor is directly proportional to the potential difference applied across its ends, provided the physical conditions such as temperature remains unchanged.

If V is the potential difference applied across the ends of conductor through which current I flows, then according to Ohm's law.

$$\begin{aligned} V &\propto I \Rightarrow V = IR \\ I &= \frac{V}{R} \end{aligned}$$

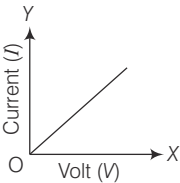
where, R is the constant of proportionality called resistance of conductor at a given temperature. The experimental set up of Ohm's law is made as the circuit shown below in the figure, consisting of a nichrome wire XY , an ammeter (A). A voltmeter (V) and four cells of 1.5 V each.



The reading in the ammeter A for the current and reading of the voltmeter V for the potential difference across the nichrome wire XY is measure and recorded in the table given below, first by using only one cell and then by the two, three and four cells in successive readings.

S.No.	Number of cells used in the circuit	Current through the nichrome wire, I (ampere)	Potential difference across the nichrome wire, V (volt)	V / I (Volt/Ampere)
1.	1			
2.	2			
3.	3			
4.	4			

Calculating the ratio of V to I for each pair of potential difference V and current I . A graph is plotted between V and I and the nature of the graph is observed as shown here. The same value for V / I is obtained corresponding to one, two, three and four cells in successive readings. Also, the $V - I$ graph is obtained as straight line that passes through the origin. This law is only valid for ohmic conductor e.g. metals.



Thus, V / I is a constant, i.e., $V \propto I$

This verifies Ohm' law.

Ohm's law does not hold good under all conditions as it is not a fundamental law of nature like Newton's law. It is obeyed by metallic conductors only when physical conditions like temperature etc. are kept unchanged. It is not obeyed by a lamp filament, junction diode, thermistor etc. These are called non-ohmic conductor.

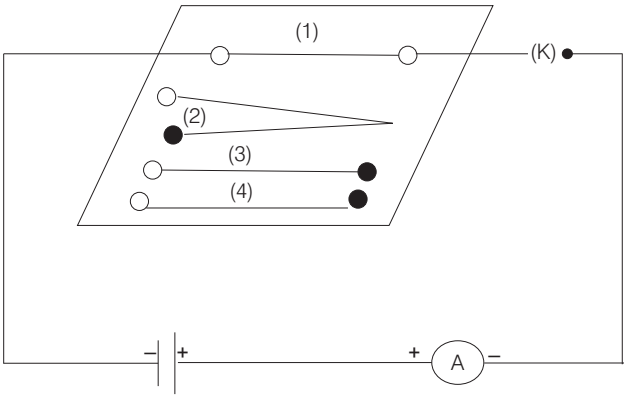
Q. 31 What is electrical resistivity of a material? What is its unit? Describe an experiment to study the factors on which the resistance of conducting wire depends.

Ans. The electrical resistivity of a material is defined as the resistance of a conductor made of that material of unit length and unit cross-sectional area. Its SI unit is Ohm metre (Ωm).

$$R = \rho \frac{l}{A} \Rightarrow \rho = \frac{RA}{l}$$

(where, $A = 2\text{m}^2$
 $l = 1\text{m}$)

Consider an electric circuit consisting of a cell, an ammeter, a nichrome wire of length L (marked 1 to 4) and a plug key as shown below



At point plug the key and note the current in ammeter. Replace the nichrome wire by another nichrome wire of same thickness but twice the length that is $2l$ at point 2. Again note the reading. Now, replace the wire by a thicker nichrome wire of same length l (marked 3). A thicker wire has a larger cross-sectional area.

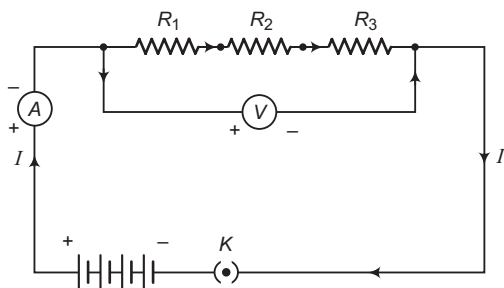
Again note down the current through the circuit. Replace nichrome wire with copper wire of same length and same area of cross-section at point 4. Note the value of current. Notice the difference in the current in all cases. When the length of the wire is doubled, then the ammeter reading decreases to half its previous value *i.e.*, current through the wire is halved. Since, resistance of the wire $R = \frac{V}{I}$, then R is doubled which implies $R \propto L$.

When the nichrome wire is replaced by a thicker one of same material and length, the current in the wire increases which means that the resistance of the thicker wire (3) is less than that of the thinner wire (1). This implies $R \propto \frac{1}{A}$.

When the nichrome wire is replaced by a copper wire (4) of the same length and same cross-sectional area, then the current recorded by the ammeter is more. This means that the resistance of copper wire is less than that of the nichrome wire of the same dimensions *i.e.*, the resistance of the wire depends on the nature of its material.

Q. 32 How will you infer with the help of an experiment that the same current flows through every part of the circuit containing three resistances in series connected to a battery?

Ans. Let the experimental set up comprises of three resistors R_1 , R_2 and R_3 of three different values such as 1Ω , 2Ω and 3Ω which are connected in series. Connect them with a battery of 6 V, an ammeter and plug a key, as shown in figure,

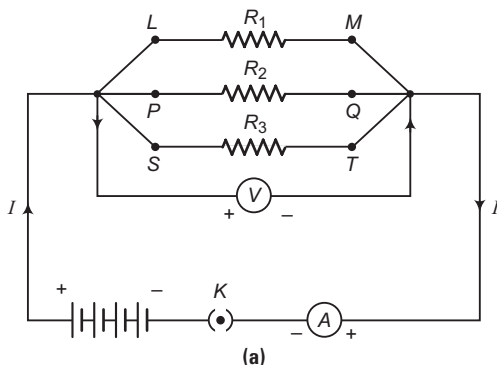


The key K is closed and the ammeter reading is recorded. Now, the position of ammeter is changed to anywhere in between the resistors again, the ammeter reading is recorded each time. It's observed that there was identical reading each time, which shows that same current flows through every part of the circuit containing three resistances in series connected to a battery.

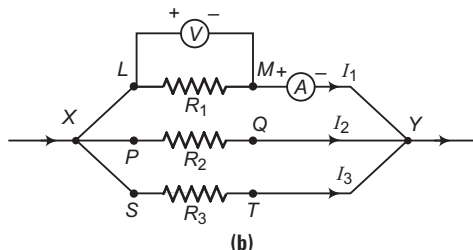
Q. 33 How will you conclude that the same potential difference (voltage) exists across three resistors connected in a parallel arrangement to a battery?

Ans. The experimental set-up comprises of three resistors R_1 , R_2 and R_3 which are joined in parallel combination and connecting them with a battery, an ammeter (A), a voltmeter (V) and a plug key K , as shown in figure (a).

The key K is closed and the voltmeter and ammeter readings are recorded.



The key K is open and removing the ammeter and voltmeter from the circuit and insert the voltmeter V in parallel with R_1 and ammeter in series with the resistor R_1 , as shown in fig. (b). Again, the voltmeter and ammeter readings are recorded.



Similarly, measuring the potential differences across resistances R_2 and R_3 . It is found that voltmeter gives identical reading which leads to conclude that the voltage or potential difference across each resistors is same and equal to the potential difference across the combination.

Q. 34 What is Joule's heating effect? How can it be demonstrated experimentally? List its four applications in daily life.

Ans. The heating effect of current is defined by Joules law of heating. It states that the heat H produced by a resistor of resistance R due to current flowing through it for time t is $H = I^2 R t$. It is also called ohmic heating and resistive heating.

In a conductor when an electric field is applied across its ends, the free electrons available in it start drifting opposite to the direction of the electric field. These electrons collide with the atoms which have lost the electrons.

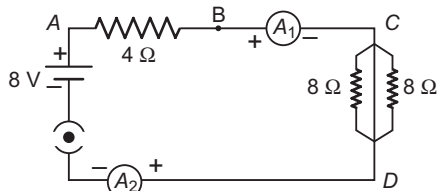
As a result of these collisions some energy of the electrons is transferred to the atoms which vibrate violently as they gain energy. Thus, heat is developed in the conductor. Greater the current, greater will be the rate of collision and so greater will be the heat produced.

The four applications using heating effect of current are as follows

- | | |
|---------------------|--------------------|
| (i) room heater | (ii) electric bulb |
| (iii) electric iron | (iv) electric fuse |

Q. 35 Find out the following in the electric circuit given in figure

- effective resistance of two $8\ \Omega$ resistors in the combination
- current flowing through $4\ \Omega$ resistor
- potential difference across $4\ \Omega$ resistance
- power dissipated in $4\ \Omega$ resistor
- difference in ammeter readings, if any



Ans. (a) Since, two $8\ \Omega$ resistors are in parallel, then their effective resistance R_p is given by

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_p} = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}\ \Omega$$

- (b) Total resistance in the circuit $R = 4\ \Omega + R_p$
 $= 4\ \Omega + 4\ \Omega$
 $= 8\ \Omega$

$$\text{Current through the circuit } I = \frac{V}{R} = \frac{8}{8} = 1\text{ A}$$

(where, V = Voltage of battery and
 R = Net equivalent resistance of circuit)

Thus, current through $4\ \Omega$ resistor is 1 A as $4\ \Omega$ and R_p are in series and same current flows through them.

- (c) Potential difference across $4\ \Omega$ resistor is potential drop by the $4\ \Omega$ resistor

$$\text{i.e., } V = IR = 1 \times 4 = 4\text{ V}$$

- (d) Power dissipated in $4\ \Omega$ resistor $P = I^2 R = 1^2 \times 4 = 4\text{ W}$

- (e) There is no difference in the readings of ammeters A_1 and A_2 as same current flows through all elements in a series current.

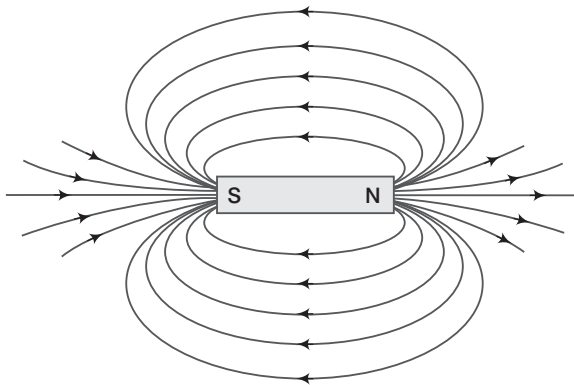
Magnetic Effects *of* Electric Current

Multiple Choice Questions (MCQs)

Q. 1 Choose the incorrect statement from the following regarding magnetic lines of field

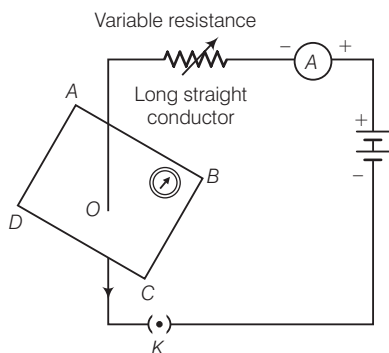
- (a) The direction of magnetic field at a point is taken to be the direction in which the North pole of a magnetic compass needle points
- (b) Magnetic field lines are closed curves
- (c) If magnetic field lines are parallel and equidistant, they represent zero field strength
- (d) Relative strength of magnetic field is shown by the degree of closeness of the field lines

Ans. (c) Let us consider the diagram given below to represent the magnetic field lines around a bar magnet.



From this diagram and our basic knowledge we can infer that options (a), (b) and (d) are correct but option (c) is incorrect as, the parallel lines of magnetic field represent the uniform magnetic field.

Q. 2 If the key in the arrangement as shown below is taken out (the circuit is made open) and magnetic field lines are drawn over the horizontal plane $ABCD$, the lines are

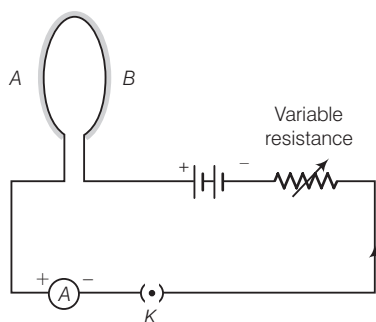


- (a) Concentric circles
- (b) Elliptical in shape
- (c) Straight lines parallel to each other
- (d) Concentric circles near the point O but of elliptical shapes as we go away from it

Ans. (c) When the key is taken out (the circuit is made open), no current flows through the wire hence, no magnetic field exists due to the conductor.

The only magnetic field is due to Earth's magnetic field and is straight lines parallel to each other. Whose horizontal component is directed from geographical South to geographical North on the horizontal plane $ABCD$.

Q. 3 A circular loop placed in a plane perpendicular to the plane of paper carries a current when the key is ON. The current as seen from points A and B (in the plane of paper and on the axis of the coil) is anti-clockwise and clockwise respectively. The magnetic field lines point from B to A . The N-pole of the resultant magnet is on the face close to



- (a) A
- (b) B
- (c) A if the current is small and B if the current is large
- (d) B if the current is small and A if the current is large

Ans. (a) The N-pole of the resultant magnet is on the face close to A because, the magnetic field lines enter in loop from B and come out from A. Also, as a matter of fact magnetic lines come out of the N-pole of magnet. Therefore, face close to A represents N-pole. The currents in A and B are same.

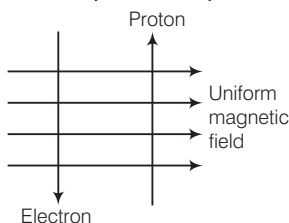
Q. 4 For a current in a long straight solenoid N and S-poles are created at the two ends. Among the following statements, the incorrect statement is

- (a) The field lines inside the solenoid are in the form of straight lines, which indicates that the magnetic field is the same at all points inside the solenoid
- (b) The strong magnetic field produced inside the solenoid can be used to magnetise a piece of magnetic material like soft iron, when placed inside the coil
- (c) The pattern of the magnetic field associated with the solenoid is different from the pattern of the magnetic field around a bar magnet
- (d) The N and S-poles exchange position when the direction of current through the solenoid is reversed

Ans. (c) Here, options (a), (b) and (d) are correct and knowledge based but option (c) is incorrect as the pattern of magnetic field inside a solenoid is uniform and similar to that of bar magnet.

Also the one end of the solenoid behave as magnetic north pole, while the other behaves as the south pole.

Q. 5 A uniform magnetic field exists in the plane of paper pointing from left to right as shown in figure. In the field an electron and a proton move as shown. The electron and the proton experience



- (a) Forces both pointing into the plane of paper
- (b) Forces both pointing out of the plane of paper
- (c) Forces pointing into the plane of paper and out of the plane of paper, respectively
- (d) Force pointing opposite and along the direction of the uniform magnetic field respectively

Ans. (a) In the given figure, the proton and electron are moving in opposite direction to each other and in perpendicular to the direction of magnetic field. Now, we know that, the direction of current is taken opposite to the direction of motion of electron.

So, both electron and proton has current in same direction. Therefore, the forces acting on it is given by Fleming's left hand rule and they are pointing into the plane of the paper.

Q. 6 Commercial electric motors do not use

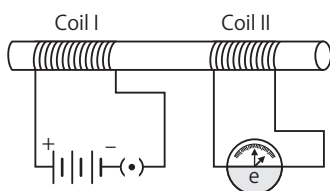
- (a) An electromagnet to rotate the armature
- (b) Effectively large number of turns of conducting wire in the current carrying coil
- (c) A permanent magnet to rotate the armature
- (d) A soft iron core on which the coil is wound

Ans. (c) A commercial electric motor is one which uses the following

- (i) An electromagnet in place of permanent magnet.
- (ii) Large number of turns of the conducting wire in the current-carrying coil.
- (iii) A soft iron core on which the coil is wound. The combination of soft iron core and coil is called an armature. It enhances the power of motor.

Thus, commercial electric motors do not use permanent magnet to rotate the armature because permanent magnets are weak and do not produce strong magnetic field in the region.

Q. 7 In the arrangement shown in figure, there are two coils wound on a non-conducting cylindrical rod. Initially, the key is not inserted. Then the key is inserted and later removed. Then



- (a) The deflection in the galvanometer remains zero throughout
- (b) There is a momentary deflection in the galvanometer but it dies out shortly and there is no effect when the key is removed.
- (c) There are momentary galvanometer deflections that die out shortly, the deflections are in the same direction.
- (d) There are momentary galvanometer deflections that die out shortly, the deflections are in opposite directions. Thus the galvanometer shows momentary deflections in opposite directions.

Ans. (d) In the above arrangement, whenever an electric current through the first coil is changed, an emf is induced in the coil due to change in magnetic field lines which pass through the neighbouring second coil.

When key is inserted and removed, then the magnetic field lines pass through second coil increases and decreases in two cases respectively. Therefore, the direction of current in two cases is in opposite directions. Thus the galvanometer shows momentary deflections in opposite directions.

Q. 8 Choose the incorrect statement.

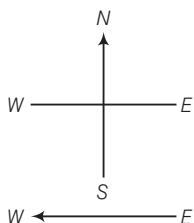
- (a) Fleming's right hand rule is a simple rule to know the direction of induced current.
- (b) The right hand thumb rule is used to find the direction of magnetic fields due to current carrying conductors.
- (c) The difference between the direct and alternating currents is that the direct current always flows in one direction, whereas the alternating current reverses its direction periodically.
- (d) In India, the AC changes direction after every $\frac{1}{50}$ second.

Ans. (d)

- (a) According to Fleming's right hand rule, if the direction of magnetic field and the direction of motion of conductor are given then we can find the direction of the induced current. So option (a) is correct.

- (b) In the right-hand thumb rule, we find the direction of magnetic field due to a straight current-carrying conductor. Thus option (b) is correct.
- (c) The direction and magnitude of AC changes periodically whereas in case of DC these remain constant. So option (c) is also correct.
- (d) The frequency of 'AC' in India is 50 cycle/s. Therefore, time period of AC is $1/50$ s. But AC changes its direction after every half time period, i.e., $1/(2 \times 50)$ i.e., $1/100$ s. Thus in India, the AC changes direction after every $1/100$ s. Hence option (d) is incorrect.

Q. 9 A constant current flows in a horizontal wire in the plane of the paper from east to west as shown in figure. The direction of magnetic field at a point will be north to south.



- (a) Directly above the wire
 (b) Directly below the wire
 (c) At a point located in the plane of the paper on the north side of the wire
 (d) At a point located in the plane of the paper on the south side of the wire

Ans. (b) According to right hand thumb rule, when conductor is held in right hand, keeping thumb from east to west as shown in the figure, the curve of the finger will be from north to south at a point lying directly below the wire.

Q. 10 The strength of magnetic field inside a long current carrying straight solenoid is

- (a) more at the ends than at the centre
 (b) minimum in the middle
 (c) same at all points
 (d) found to increase from one end to the other

Ans. (c) The strength of magnetic field lines inside a long current carrying straight solenoid same at all points because the magnetic field lines are straight, equi-spaced and parallel to the axis of solenoid and hence uniform magnetic field exist inside the solenoid.

Q. 11 To convert an AC generator into DC generator

- (a) split ring type commutator must be used
 (b) slip rings and brushes must be used
 (c) a stronger magnetic field has to be used
 (d) a rectangular wire loop has to be used

Ans. (a) To convert an AC generator into DC generator the split ring type commutator are used which converts the direction of flow of induced current, in the external circuit, in the same direction throughout the complete rotation of the armature.

Q. 12 The most important safety method used for protecting home appliances from short circuiting or overloading is

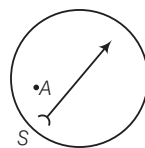
- (a) earthing (b) use of fuse
(c) use of stabilizers (d) use of electric meter

Ans. (b) The most important safety method used for protecting home appliances from short circuiting or overloading is the electric fuse. This is a safety device having thin wire of short length made of tin (25%) and lead (75%) alloy having low melting point around 200°C .

The fuse wire is of chosen thickness, so as to fix its resistance and hence amount of heating on passage of a particular amount of current. Whenever current through the fuse exceeds the set limit, the fuse wire melts and breaks the circuit. This saves the main circuit components from damage.

Short Answer Type Questions

Q. 13 A magnetic compass needle is placed in the plane of paper near point **A** as shown in figure. In which plane should a straight current carrying conductor be placed so that it passes through **A** and there is no change in the deflection of the compass? Under what condition is the deflection maximum and why?



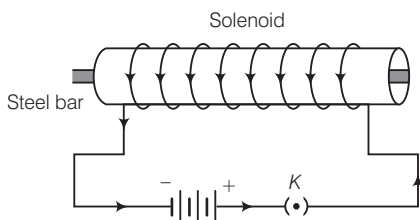
Ans. The straight current carrying conductor should be placed in the plane of paper so that it passes through **A**. This produces magnetic field in a plane perpendicular to plane of paper. The compass needle remains undeflected due to vertical magnetic field produced by wire since compass needle are free to rotate only in horizontal plane.

The deflection in compass needle is maximum when the conductor through **A** is perpendicular to the plane of paper and the magnetic field due to straight current conductor lies in the plane of paper.

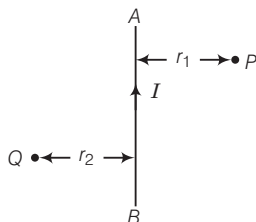
Q. 14 Under what conditions permanent electromagnet is obtained if a current carrying solenoid is used? Support your answer with the help of a labelled circuit diagram.

Ans. The conditions to obtain permanent electromagnet as following if a current carrying solenoid is used are

- The current through the solenoid should be direct current.
- The rod inside is made of a magnetic material such as steel.
- The magnetic material should be of high retentivity and should not lose it easily.



- Q. 15** AB is a current carrying conductor in the plane of the paper as shown in figure. What are the directions of magnetic fields produced by it at points P and Q ? Given $r_1 > r_2$, where will the strength of the magnetic field be larger?



Thinking Process

The direction of magnetic field can be found by using Maxwell's Right Hand Thumb rule.

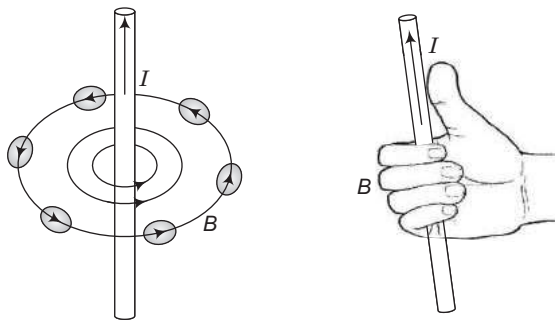
- Ans.** In the given figure, the current carrying conductor AB is in the plane of the paper. Thus, the direction of magnetic field is into the plane of the paper at P and out of it at the point Q . The magnitude of magnetic field B is inversely proportional to distance r

$$\text{i.e.,} \quad B \propto \frac{1}{r}$$

Magnetic field B is stronger near the conduct at Q than at P as $r_1 > r_2$.

- Q. 16** A magnetic compass shows a deflection, when placed near a current carrying wire. How will the deflection of the compass get affected, if the current in the wire is increased? Support your answer with a reason.

- Ans.** A magnetic compass shows a deflection when placed near a current carrying wire as shown in the figure



This deflection of the compass increases because the magnetic field B due to current carrying straight conductor is directly proportional to the strength of the current I passing through it. It is expressed as

$$B \propto I$$

If the current is increased then the magnetic field produced is stronger and vice-versa.

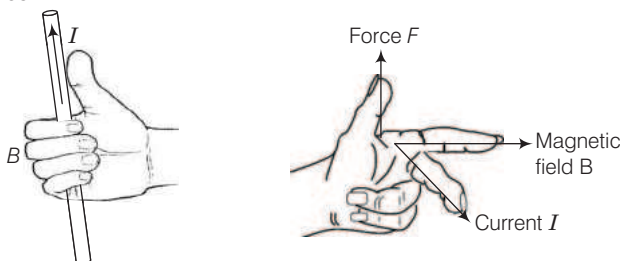
Q. 17 It is established that an electric current through a metallic conductor produces a magnetic field around it. Is there a similar magnetic field produced around a thin beam of moving (i) alpha particles, (ii) neutrons? Justify your answer.

Ans. In case of movement of a charged particle the magnetic field is created around the path on which the charged particle moves.

- (i) Yes, a thin beam of alpha particles (which are positively charged) is like straight conductor carrying current in the direction of motion.
- (ii) No, as neutrons carry no charge so no magnetic field would be created around its path.

Q. 18 What does the direction of thumb indicate in the right-hand thumb rule? In what way this rule is different from Fleming's Left-hand rule.

Ans. According to Maxwell's right-hand thumb rule, the thumb shows the direction of electric current as shown in the figure. The curled finger holding the conductor give the direction of magnetic lines.



Whereas, Fleming's left-hand rule gives the direction of force experienced by a current carrying straight conductor placed in an external magnetic field.

Q. 19 Meena draws magnetic field lines of field close to the axis of a current carrying circular loop. As she moves away from the centre of the circular loop she observes that the lines keep on diverging. How will you explain her observation?

Ans. We know that the magnetic field is stronger near the current carrying conductor and decreases as we move away from the conductor. In case of a current carrying circular loop, the magnetic field is stronger near the periphery, but weaker near the centre of the loop. Due to this, the magnetic field lines appear as straight lines near the centre.

As we move towards the periphery of the circular loop, the magnetic field lines appear to be diverging so that they can be circular around the wire of the loop.

Q. 20 What does the divergence of magnetic field lines near the ends of a current carrying straight solenoid indicate?

Ans. The divergence (degree of closeness) of magnetic field lines near the ends of a current carrying straight solenoid indicates a decrease in the strength of the magnetic field near the ends of the solenoid.

Q. 21 Name four appliances wherein an electric motor, a rotating device that converts electrical energy to mechanical energy, is used as an important component. In what respect motors are different from generators?

Ans. The four appliances such as electric fans, washing machines, mixers grinders, electric drills, tape recorders, etc, uses an electric motor as a rotating device that converts electrical energy to mechanical energy. The motor is a machine which converts the electric energy into mechanical energy.

The generator is the device which converts mechanical energy into electrical energy. It is used to convert the mechanical power to electric power which varies in the magnitude with a specific frequency.

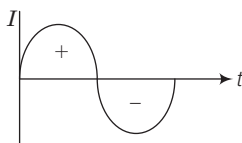
Q. 22 What is the role of the two conducting stationary brushes in a simple electric motor?

Ans. The two conducting stationary brushes draw current from the battery and supply it to the armature of the motor because as they touch the outer sides of the two halves of the split rings. The inner sides of split rings are insulated and attached to the axle of the motor.

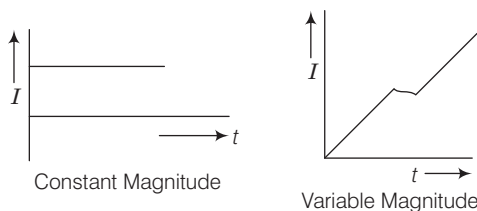
Q. 23 What is the difference between a direct current and an alternating current? How many times does AC used in India change direction in one second?

Ans. The basic difference between direct current and an alternating current is as follows

Alternating Current is an electric current which changes its direction (i.e., polarity after a certain fixed interval of time. Graphically it can be represented as following



Direct Current is an electric current whose magnitude is either constant or variable, but the direction of flow in a conductor remains the same. Sources of DC are voltaic cell, a dry cell, battery, DC generator etc and graphically it is shown as



The frequency of AC in India is $f = 50$ cycle/s

$\therefore$ Time period of AC is, $T = \frac{1}{f} = \frac{1}{50}$ s

but AC changes its direction after every half time period, $T = \frac{1}{50 \times 2} = \frac{1}{100}$ s

Thus, in India AC change its direction after every $\frac{1}{100}$ s

So, in one second AC changes its direction 100 times.

Q. 24 What is the role of fuse, used in series with any electrical appliance? Why should a fuse with defined rating not be replaced by one with a larger rating?

Ans. The main role of fuse used in series with any electrical appliance limits the current through the appliance. It protects the appliance due to short-circuiting or over / loading as when current exceeds the limit for which the fuse is rated, it blows off and the appliance is disconnected from the electric supply.

When a fuse with defined rating for a particular appliance is replaced by one with larger rating, it does not blow off and the appliance is damaged due to larger current passing through it.

Long Answer Type Questions

Q. 25 Why does a magnetic compass needle pointing north and south in the absence of a nearby magnet get deflected when a bar magnet or a current carrying loop is brought near it? Describe some salient features of magnetic lines of field concept.

Ans. When a magnetic compass needle pointing north and south in the absence of a nearby magnet or a current loop, it is acting upon by the earth's magnetic field only. But in the presence of a magnet or a current loop (which also has a magnetic field) the earth's magnetic field near the compass is modified and the needle is deflected from north and south directions.

The salient features of magnetic field lines are

- (i) A magnetic field line is directed from north-pole to south-pole outside the magnet.
- (ii) A magnetic field line is a closed and continuous curve
- (iii) The magnetic field lines never intersect each other, because if two lines meet, it means that force is acting in two directions at that point which is not possible.
- (iv) Closer the field lines, stronger is the magnetic field and *vice-versa* is also true.
- (v) Magnetic field lines are parallel and equidistant, those represent a uniform magnetic field.

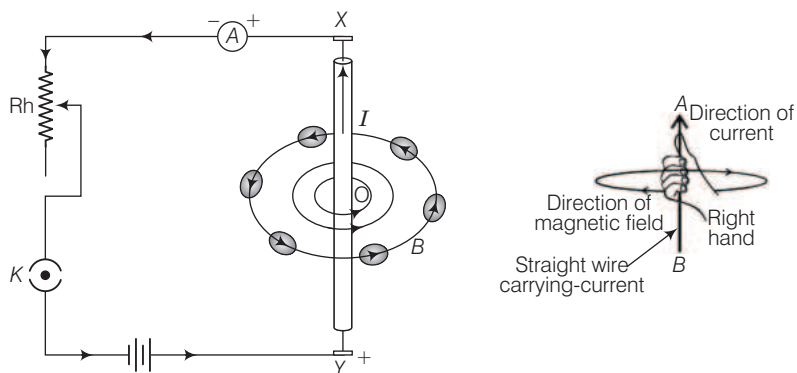
Q. 26 With the help of a labelled circuit diagram illustrate the pattern of field lines of the magnetic field around a current-carrying straight long conducting wire. How is the right hand thumb rule useful to direction of magnetic field associated with a current-carrying conductor?

Ans. Take a thick straight conductor XY and pass it through the centre of a thick cardboard. Connect the ends of the conductor to the terminals of a battery through a Rheostat Rh, a key K and an ammeter (A) so that the current flows from Y to X. when we sprinkle some iron fillings on the cardboard, these average themselves in concentric circles around O when cardboard is gently tapped.

Thus, the pattern of concentric circles indicating the field lines of a magnetic field is observed. These concentric circles represent the magnetic field around a current carrying straight wire become larger and larger as we move away from it.

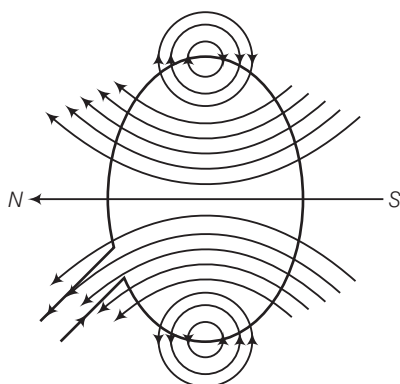
We use right hand thumb rule to find the direction of magnetic field. It follows as imagine the straight conductor in your right hand such that the thumb points in the direction of current.

Then, the direction of curling of fingers of the right hand gives the direction of magnetic field lines.



Q. 27 Explain with the help of a labelled diagram the distribution of magnetic field due to a current through a circular loop. Why is it that if a current carrying coil has n turns the field produced at any point is n times as large as that produced by a single turn?

Ans. The magnetic field pattern due to a circular coil is shown in the given figure. At every point of current carrying circular loop the concentric circles representing the magnetic field around it becomes larger and larger as we move away from the wire. At the centre of the loop the field appears as straight line.



Magnetic lines of force due to a current through a circular loop

The magnetic field produced by current carrying circular wire at a given point, depend on

- (i) **Amount of current flowing through the wire** The strength of magnetic field (B) due to current carrying circular wire is directly proportional to the amount of current (I) flowing through it.
i.e., $B \propto I$... (i)
- (ii) **Number of turns of the circular wire** The strength of the magnetic field due to current carrying wire is directly proportional to the number of turns (N) of the wire.
i.e., $B \propto N$ (ii)

If there is a circular coil having n turns, the field produced is n times as large as that produced by a single turn. It happens because the current in each circular turn has same direction and the field due to each turn just adds up.

Q. 28 Describe the activity that shows that a current-carrying conductor experiences a force perpendicular to its length and the external magnetic field. How does Fleming's left-hand rule help us to find the direction of the force acting on the current-carrying conductor?

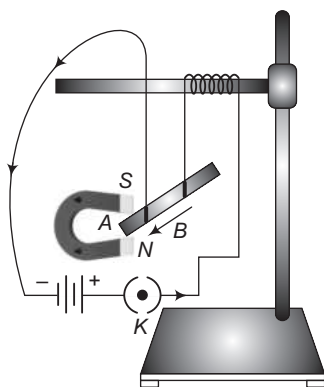
Ans. Aim To show that force is exerted on a current carrying conductor placed in a strong magnetic field.

Materials Required

A conducting rod (AB), connecting wires, horse shoe magnet, battery, switch and clamp stand.

Procedure

- (i) Connect the conducting rod AB to the battery and key.
- (ii) Place a strong horse shoe magnet in such a way that the rod AB lies between the poles with magnetic field directed upwards, i.e., N-pole is vertically above the rod.
- (iii) Plug in the key.



Current carrying conductor placed in a strong magnetic

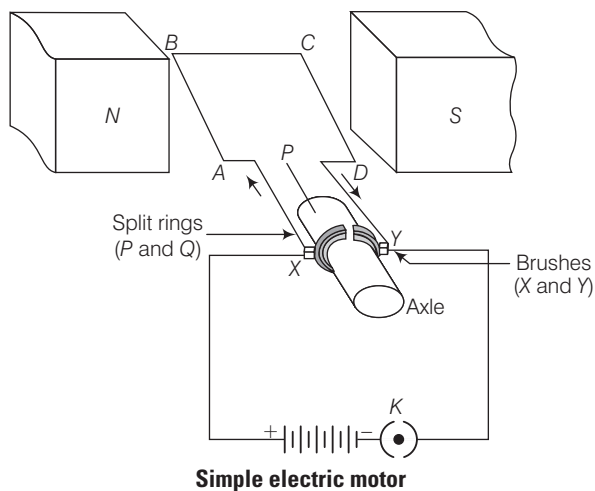
Observation

A current-carrying rod AB experiences a force perpendicular to its length and the magnetic field.

Based on above observation, it can be said that when a current carrying conductor is placed within a magnetic field, the conductor experiences deflection. Fleming's left hand rule explains the direction of displacement in this case. Let us assume that the current is moving in anticlockwise direction in the loops. In that case, the magnetic field would be in clockwise direction at the top of the loop. Moreover, it would be in anticlockwise direction at the bottom of the loop

Q. 29 Draw a labelled circuit diagram of a simple electric motor and explain its working. In what way these simple electric motors are different from commercial motors?

Ans. The circuit diagram of a simple electric motor is shown below



Working

- (i) Let coil $ABCD$ be in horizontal position. When the key is switched on, the current flows in direction $DCBA$ and then leaves through the brushes B_2 via ring R_2 .
- (ii) Applying Fleming's left-hand rule, no force acts on arm BC and AD as they are parallel to magnetic field, arm AB experiences a force in downward direction and arm CD experiences an equal force in upward direction. A torque acts on the coil and it rotates in anti-clockwise direction.
- (iii) While rotating, the coil reaches the vertical position, brushes lose contact with the rings and current stops flowing. But the coil does not stop due to inertia of motion.
- (iv) When the coil passes the vertical position, the rings automatically change their positions and come in contact with opposite brushes.
- (v) This reverses the direction of current through the coil but the direction of current on right-hand side remains same.
- (vi) So, the force on right hand side is always upward and a force on left-hand side is always in downward direction. Thus, the coil continues to anti-clockwise direction.

A commercial electric motor is one which uses the following

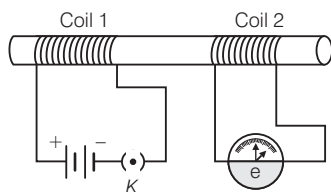
- (i) An electromagnet in place of permanent magnet.
- (ii) Large number of turns conducting wire in current carrying coil.
- (iii) A soft iron core on which the coil is wound. The combination of soft iron core and coil is an **armature**. It enhances the power of motor.

Thus, commercial electric motors do not use permanent magnet to rotate the armature because permanent magnets are weak and do not produce strong magnetic field in the region.

Q. 30 Explain the phenomenon of electromagnetic induction. Describe an experiment to show that a current is set up in a closed loop when an external magnetic field passing through the loop increases or decreases.

Ans. This phenomenon was discovered by Michael Faraday and Joseph Henry in 1831. The phenomenon of generation of an electric current in a closed circuit from magnetic effects *i.e.*, by changing the magnetic field is called electromagnetic induction. Electric current produced due to this phenomenon is called induced current. The potential difference set in the circuit due to induced current is called induced emf.

Experiment Let two different coils of copper wire, namely coil 1 and coil 2, having large number of turns are arranged as per the circuit diagram as shown in figure.

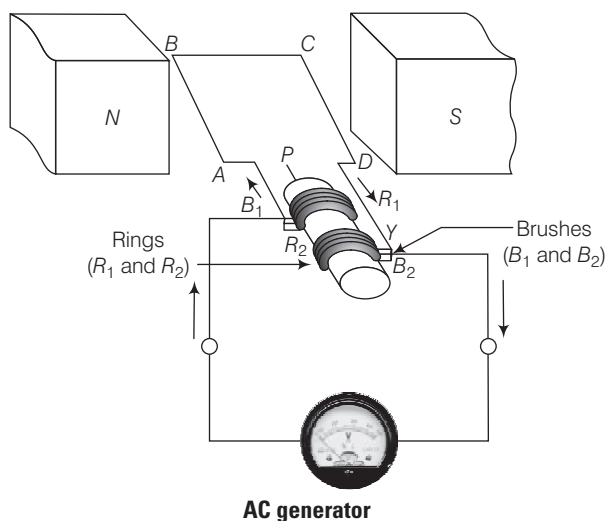


The coil 1, having larger number of turns, is connected in series with a battery and a plug key whereas the other coil 2 is connected with a galvanometer as shown in circuit diagram. When key K is closed, the galvanometer shows a deflection in its needle instantly jumps to one side and just as quickly returns to zero, indicating a momentary current in coil 2.

This happens due to an external magnetic field passing through the loop increases. When key K is open, the coil 1 is disconnected from the battery. The needle momentarily moves, but to the opposite side. It means that now the current flows in the opposite direction in coil 2. This happens due to an external magnetic field passing through the loop decreases.

Q. 31 Describe the working of an AC generator with the help of a labelled circuit diagram. What changes must be made in the arrangement to convert into a DC generator?

Ans The working of an AC generator is as follows



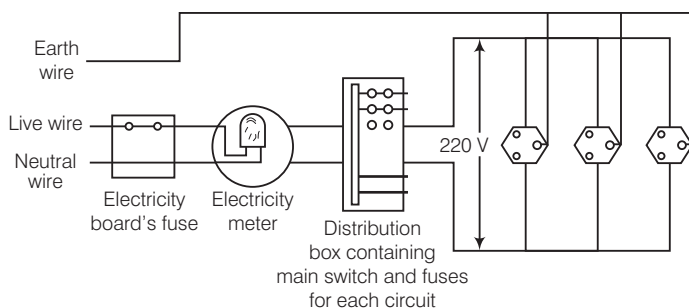
As the coil rotates, the two rings also rotate with the coil. The brushes P and Q keep contact with the rotating ring. When the coil starts rotating with arm AB moving up and CD moving down (clockwise) cutting the magnetic lines. According to Fleming's right-hand rule, the current is induced in these arms in the direction $ABCD$.

After half rotation, arm CD moves down and AB moves up. As a result, the direction of current in each segment changes, giving rise to net induced current in the direction $DCBA$. Thus, after every rotation, polarity of current in the respective arm changes, thereby generating an alternating current.

To convert an AC generator into a DC generator, a split ring commutator is used. This helps in producing direct current.

Q. 32 Draw an appropriate schematic diagram showing common domestic circuits and discuss the importance of fuse. Why is it that a burnt out fuse should be replaced by another fuse of identical rating?

Ans The schematic diagram of common domestic circuit is shown below



Fuse is a safety device having short length of thin wire made of tin (25%) and lead (75%) alloy having low melting point around 200°C . It is an important component of all domestic circuits. A fuse in a circuit prevents damage to the appliances and the circuit due to overloading and short-circuiting. The use of an electric fuse prevents the electric circuit and the appliance from a possible damage by stopping the flow of unduly high electric current. The fuse is placed in series with the device.

A fuse wire works because of its lower melting point which is possible. If a fuse with larger rating is used with an appliance, the fuse wire shall not melt and hence would fail to serve the required purpose. Due to this, a fuse with defined rating should not be replaced by one with a larger rating.

Sources of Energy

Multiple Choice Questions (MCQs)

Q. 1 Which of the following is a non-renewable source of energy?

- (a) Wood (b) Sun (c) Fossil fuels (d) Wind

Ans. (c) Non-renewable sources of energy are those which are exhaustible and cannot be replaced, once they have been used.

Non-renewable sources of energy also known as conventional sources of energy. The fossil fuels are non-renewable sources of energy whereas wood, the sun and wind are renewable sources of energy.

Q. 2 Acid rain happens because

- (a) sun leads to heating of upper layer of atmosphere
(b) burning of fossil fuels release oxides of carbon, nitrogen and sulphur in the atmosphere
(c) electrical charges are produced due to friction amongst clouds
(d) the earth atmosphere contains acids

Ans. (b) Acid rain happens because of several human activities of polluting the atmosphere. The emission of carbon, sulphur and nitrogen from the industries, burning of fossil fuels, etc., and also by the natural phenomena such as emission from volcanoes make the water vapour in the cloud more acidic thereby causing acid rain.

Q. 3 Fuel used in thermal power plants is

- (a) Water (b) Uranium (c) Biomass (d) Fossil fuels

Ans. (d) The thermal power plant generates electric power from heat produced by burning fossil fuels *i.e.*, coal and petroleum. Everyday we burn large amount of fossil fuels to heat up water to produce steam. The steam so produced runs turbines to generate electricity.

Q. 4 In a hydro power plant

- (a) potential energy possessed by stored water is converted into electricity
(b) kinetic energy possessed by stored water is converted into potential energy
(c) electricity is extracted from water
(d) water is converted into steam to produce electricity

Ans. (a) In a hydro power plant, water from the top of the dam is allowed to fall through pipelines over the blades of turbine at the bottom of the dam. In this process, the potential energy of water changes into its kinetic energy which is transferred to the turbine.

The moving turbine rotates the armature of a generator to produce electricity *i.e.*, turbine changes the kinetic energy into electricity. Thus, potential energy possessed by stored water is converted into electricity.

Q. 5 Which is the ultimate source of energy?

- (a) Water (b) Sun (c) Uranium (d) Fossil fuels

Ans. (b) The sun is considered as the ultimate source of energy for several reasons. The main reason is that all form of energy, directly or indirectly is derived from sun. Without the sun, no food (meat or plant based) would grow and thus, we would not be able to gain the energy needed to live.

Q. 6 Which one of the following forms of energy leads to least environmental pollution in the process of its harnessing and utilisation?

- (a) Nuclear energy (b) Thermal energy
(c) Solar energy (d) Geothermal energy

Ans. (c) Solar energy leads to least environmental pollution in the process of its harnessing and utilisation. In nuclear energy, emission of radiation is a high risk for environmental contamination.

In thermal energy, heat produced by burning fossil fuels causes huge environmental pollution. In geothermal energy, noise pollution is created by drilling operations at geothermal sites.

Q. 7 Ocean thermal energy is due to

- (a) energy stored by waves in the ocean
(b) temperature difference at different levels in the ocean
(c) pressure difference at different levels in the ocean
(d) tides arising out in the ocean

Ans. (b) The water at the surface of the sea or ocean is heated by the sun while the water in deeper sections is relatively cold. This difference in temperature between these layers ranges from 10°C to 30°C is exploited to obtain energy. Thus, ocean thermal energy is due to temperature difference at different levels in the ocean.

Q. 8 The major problem in harnessing nuclear energy is how to

- (a) split nuclei
(b) sustain the reaction
(c) dispose of spent fuel safely
(d) convert nuclear energy into electrical energy

Ans. (c) The major hazard of nuclear power generation is the storage and disposal of spent or used fuels. Improper nuclear-waste storage and disposal result in environmental contamination, as well as, risk of accidental leakage of nuclear radiation.

It happened in Chernobyl disaster 1986, Fukushima Nuclear disaster 2011 caused great damage to the living beings and habitats.

Q. 9 Which part of the solar cooker is responsible for greenhouse effect?

- (a) Coating with black colour inside the box
- (b) Mirror
- (c) Glass sheet
- (d) Outer cover of the solar cooker

Ans. (c) Glass sheet present in the solar cooker easily passes the radiation into the solar cooker and these radiation get absorbed and reflected back by the black coating is of longer wavelength and can't pass back out through the glass. Thus glass sheet produces greenhouse effect in solar cooker.

Q. 10 The main constituent of biogas is

- (a) methane
- (b) carbon dioxide
- (c) hydrogen
- (d) hydrogen sulphide

Ans. (a) The composition of biogas is

- (i) methane (CH_4), 65-75% (combustible).

And other constituent of biogas are as follows

- (ii) carbon dioxide (CO_2), 20-30% (non-combustible)
- (iii) hydrogen (H_2), 5-10% (combustible)
- (iv) hydrogen sulphide (H_2S), traces (combustible)
- (v) nitrogen (N_2), 2-6% (non-combustible)

Thus, the main constituent of biogas is methane.

Q. 11 The power generated in a windmill

- (a) is more in rainy season since, damp air would mean more air mass hitting the blades
- (b) depends on the height of the tower
- (c) depends on wind velocity
- (d) can be increased by planting tall trees close to the tower

Ans. (c) Wind energy farms can be located only in vast open areas located in favourable wind conditions as the minimum velocity for a windmill to function is 11 km/h to 16 km/h and is called as cut-in speed. Thus, the power generated in a windmill depends on wind velocity.

Q. 12 Choose the correct statement

- (a) Sun can be taken as an inexhaustible source of energy
- (b) There is infinite storage of fossil fuel inside the earth
- (c) Hydro and wind energy plants are non-polluting sources of energy
- (d) Waste from a nuclear power plant can be easily disposed off

Ans. (a) The sun has been radiating an enormous amount of energy at the present rate for nearly 5 billion years and will continue radiating at that rate for about 5 billion years more, so, the sun can be taken as an inexhaustible source of energy.

Q. 13 In a hydroelectric power plant more electrical power can be generated if water falls from a greater height because

- (a) its temperature increases
- (b) larger amount of potential energy is converted into kinetic energy
- (c) the electricity content of water increases with height
- (d) more water molecules dissociate into ions

Ans. (b) In a hydroelectric power plant more electrical power can be generated if water falls from a greater height because the rise in water level, causes the increase in potential energy of water.

Thus when it flows from higher position more amount of kinetic energy is formed by the conversion of higher potential energy and this kinetic energy in the form of moving water can produce more electrical power.

Q. 14 Choose the incorrect statement regarding wind power.

- (a) It is expected to harness wind power to minimum in open space
- (b) The potential energy content of wind blowing at high altitudes is the source of wind power
- (c) Wind hitting at the blades of a windmill causes them to rotate. The rotation thus achieved can be utilised further
- (d) One possible method of utilising the energy of rotational motion of the blades of a windmill is to run the turbine of an electric generator

Ans. (b) To generate wind power, we require wind at a very high speed. Due to this motion it possess kinetic energy and as such they are capable of doing mechanical work by virtue of its motion.

The energy possessed by the wind is due to its high speed. When the blowing wind strikes across the blades of a windmill, it exerts a force on them due to which the blades of the windmill start rotating. This rotational motion of the blades is used to run the turbine of an electric generator. Thus, the option (b) is the incorrect statement.

Q. 15 Choose the incorrect statement.

- (a) We are encouraged to plant more trees so as to ensure clean environment and also provide biomass fuel
- (b) Gobar-gas is produced when crops, vegetable wastes etc., decompose in the absence of oxygen
- (c) The main ingredient of biogas is ethane and it gives a lot of smoke and also produces a lot of residual ash
- (d) Bio-mass is a renewable source of energy

Ans. (c) Encouraging to plant more trees we ensure clean and pollution free environment and it also provide bio-mass fuel. Gobar-gas is made from the decomposition of cow dung; crops, vegetable wastes, etc., decompose in the absence of oxygen.

The main ingredient of bio-gas is methane. It burns without smoke, leaves no residue like ash in wood, charcoal and coal burning. Bio-mass is living matter or its residues and is a renewable source of energy. Thus, part (c) is the incorrect statement.

Short Answer Type Questions

Q. 16 Why is there a need to harness non-conventional sources of energy? Give two main reasons.

Ans. *There is a need to harness non-conventional sources of energy due of the following reasons*

- (i) The energy demands are increasing rapidly because of population explosion and our efforts to improve the quality of life by adopting faster means of transportation, rapid industrialisation and extensive use of energy-fed appliances.
- (ii) The sources of energy which are available are mainly fossil fuels which are non-renewable sources of energy and are limited. It will get exhausted after a time being. So, we need to harness non-conventional sources of energy.

Q. 17 Write two different ways of harnessing energy from ocean.

Ans. *The two different ways of harnessing energy from ocean are as follows*

- (i) **Tidal energy** The energy derived from rising and falling ocean tides is called tidal energy. The tidal energy can be harnessed by constructing a tidal barrage or tidal dam across a narrow opening to the sea.
- (ii) **Ocean thermal energy** Solar energy stored in the oceans in the form of heat is known as ocean thermal energy. Temperature differences between the deep ocean water and upper level is used to produce ocean thermal energy. The process of harnessing the thermal energy of the sea is called ocean thermal energy conversion.

Q. 18 What steps would you suggest to minimise environmental pollution caused by burning of fossil fuels?

Ans. *The following steps can be would suggested to minimise environmental pollution caused by burning of fossil fuels*

- (i) Using smokeless appliances and using various techniques to reduce the escape of harmful gases.
- (ii) Converting land into forest by planting trees.
- (iii) Using clean fuels like CNG, LPG, etc.
- (iv) By using public transport instead of private vehicles.
- (v) By increasing the efficiency of combustion process.

Q. 19 What is the role of a plane mirror and a glass sheet in a solar cooker?

Ans. *The role of a plane mirror and a glass sheet in a solar cooker are as follows*

- (i) **Plane mirror** They are used as reflector in solar cookers to focus the maximum rays of the sun into the cooker to achieve a higher temperature.
- (ii) **Glass sheet** The transparent glass sheet kept over the open end of the heating devices allows the infrared rays and visible rays to enter the device but does not allow the infrared radiations to move out of the heating device due to increase in the wavelength of radiation inside the cooker and the temperature of heating device rises appreciably.

Q. 20 Mention three advantages of a solar cell.

Ans. *The three advantages of a solar cell are as follows*

- (i) Solar cell is pollution-free during use.
- (ii) Its maintenance cost is very low and work quite satisfactorily without the use of any focusing device. It is also a ultimate source of energy.
- (iii) It can be set up in remote and inaccessible hamlets or very sparsely inhabited areas in which laying of a power transmission line may be expensive and not commercially viable.

Q. 21 What is bio-mass? What can be done to obtain bio-energy using biomass?

Ans. The waste material of living things (cattle dung) and dead parts of plants and animals are called bio-mass. It has been a traditional source of energy, e.g., wood, crop residue, bagasse (residue of sugarcane after extracting juice) cow-dung cake are used as fuels for domestic as well as industrial uses.

Bioenergy in the form of biogas can be produced from Biomass by the decomposition in the absence of air. Biomass such as wood, cow-dung etc, are directly used as fuel.

Q. 22 What are the limitations in obtaining energy from wind?

Ans. *The limitations in obtaining energy from wind are as follows*

- (i) It can be established only at those places where wind blows whole of the year.
- (ii) The minimum wind speed necessary for satisfactory working of a wind generator is 15 km/h, which is not gained continuously.
- (iii) Wind energy farm requires quite large area of land nearly 2 hectares land is needed for 1 MW generator.
- (iv) The setting up of wind energy farm is very expensive.
- (v) As the blades are exposed to vagaries of nature like rain, the sun, storm and cyclone, they need high level of maintenance.
- (vi) The wind energy farms disturb rainfall pattern.

Long Answer Type Questions

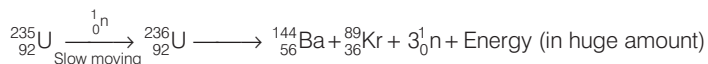
Q. 23 Which is the process used to harness nuclear energy these days? Explain it briefly.

Ans. Nuclear reactor is the process used to harness nuclear energy these days. Nuclear fission reaction takes place in the nuclear reactor. The reaction in which a heavy nucleus splits into two or more smaller nuclei with the evolution of large amount of energy, when it is bombarded with slow moving neutron is called nuclear fission.

The nucleus of a heavy atom (such as uranium, plutonium or thorium), when bombarded with low energy neutrons, can be split apart into lighter nuclei along with the release of tremendous amount of energy. *U-235 nucleus splits up broadly into two groups of nuclei*

- (i) A heavy group of nuclei, with mass number in the range $A = 130$ to $A = 149$.
- (ii) A light group of nuclei, with mass number in the range $A = 85$ to $A = 104$.

In the form of a nuclear reaction, we represent nuclear fission of U-235 as follows



The major hazards of nuclear power generation are as follows

- (i) The improper nuclear-waste storage and disposal may result into environmental contamination.
- (ii) There is a risk of accidental leakage of nuclear radiation and its leakage causes huge loss to living things.
- (iii) The high cost of installation of a nuclear power plant and high risk of environmental contamination and limited availability.

Q. 24 How can solar energy be harnessed? Mention any two limitations in using solar energy. How are these limitations overcome?

Ans. *Solar energy can be harnessed directly as well as indirectly which are as follows*

- (i) **Direct utilisation** The direct utilisation of solar energy can be done either by collecting it as heat (solar cooker, solar heater) or by converting it to electricity (solar cells).
- (ii) **Indirect utilisation** It can be done by converting solar energy into chemical energy like bio-mass of plants, etc.

The limitations in using solar energy are as follows

- (i) Energy reaching the surface is very much diffused so, direct utility is limited.
- (ii) It is not available uniformly all the time and at all the places.
- (iii) It is not available in night.
- (iv) It is not available on a cloudy day.

These limitations of using solar energy are overcome by using solar cells that convert solar energy into electricity.

Q. 25 Make a list of conventional and non-conventional sources of energy. Give a brief description of harnessing one non-conventional source of energy.

Ans. *The following are the list of conventional and non-conventional sources of energy*

- (i) **Conventional sources of energy** They are those which are used extensively and meet a major portion of our energy requirement. These are fossil fuels (wood, coal and petrol), hydro energy, biomass energy and wind energy.
- (ii) **Non-conventional sources of energy** They are those which are not used as extensively as conventional ones and meet our energy requirement only on a limited scale. These are solar energy, ocean energy (tidal energy, wave energy and ocean thermal energy), geothermal energy and nuclear energy.

The harnessing of one non-conventional source of energy is as follows

Nuclear energy Nuclear energy is produced by the release of heat from unstable elements such as uranium. The energy is harnessed by using the energy to heat water.

The radioactive water is then pumped through a heat exchanger where the dirty water is used to heat clean water. The clean water can then be used to drive turbines and other forms of engine.

Q. 26 Why is there a need for harnessing non-conventional sources of energy? How can energy be harnessed from the sea in different ways?

Ans. *There is a need for harnessing non-conventional sources of energy because of following reasons*

- (i) The demand for energy is increasing day-by-day to meet out the basic requirement of our changed life-styles, growing use of machines and industrialisation in order to improve our living standards.
- (ii) The fossil fuels are non-renewable sources of energy and were formed over millions of years ago and there are only limited reserves of fossil fuels.

The energy from the sea can be harnessed in the different forms are as follows

- (i) **Tidal energy** It is harnessed by constructing a dam across a narrow opening to the sea. A turbine fixed at the opening of the dam converts tidal energy to electricity.
- (ii) **Wave energy** A wide variety of devices have been developed to trap huge waves near the seashore for rotation of turbine and production of electricity.

- (iii) **Ocean thermal energy** The water at the surface of the sea or ocean is heated by the sun while the water in deeper sections is relatively cold. This difference in temperature is exploited to obtain energy in ocean thermal energy conversion plants. These plants can operate if the temperature difference between the surface water and water at depths up to 2 km is 293 K (or 20°C) or more.

The warm surface water is used to boil a volatile liquid like ammonia. The vapours of the liquid are then used to run the turbine of the generator. The cold water from the depths of the oceans is used to condense vapour again to liquid.

The devices used to harness this form of ocean energy are known as Ocean Thermal Energy Conversion [OTEC] power plants.

Q. 27 What are the environmental consequences of using fossil fuels? Suggest the steps to minimise the pollution caused by various sources of energy including non-conventional sources of energy causes global warming.

Ans. *The environmental consequences of using fossil fuels are as follows*

- (i) The air pollution caused by burning of coal or petroleum products.
- (ii) The oxides of carbon, nitrogen and sulphur that are released on burning fossil fuels are acidic oxides. These lead to acid rain which affects our water and soil resources.
- (iii) The generation of greenhouse effect of gases like carbon dioxide leading to global warming.

The following are the steps to minimise the pollution caused by various sources of energy including non-conventional sources of energy

- (i) The pollution caused by burning fossil fuels can be reduced by increasing the efficiency of the combustion process and using various techniques to produce the smokeless appliances.
- (ii) The air pollution caused by burning of coal or petroleum products can be reduced by afforestation.
- (iii) The planned and judicious use of energy can minimise the pollution e.g., use of Liquid Petroleum Gas (LPG) and Compressed Natural Gas (CNG) as domestic fuel and transportation vehicles.
- (iv) Proper and safe disposal of nuclear wastes.

Q. 28 Energy from various sources is considered to have been derived from the sun. Do you agree? Justify your answer.

Ans. Yes, Sun is the ultimate source of energy directly or indirectly, all the forms of energy are derived from solar energy. *Because of the following reasons*

- (i) **Non-renewable sources of energy** Fossil fuels like coal, petroleum and natural gas are formed due to burial of large plants and ancient creatures whose ultimate source of energy is the sun.
- (ii) **Renewable sources of energy** *They are indirectly derived from solar energy such as*
 - (a) **Energy from flowing water** Clouds are formed when water in lakes, rivers, seas, etc., evaporate due to solar energy. They bring rainfall and snowfall. The rain and melting snow feed rivers, streams, etc. This flowing water can be used for getting hydroelectricity.
 - (b) **Wind energy** Wind energy arises due to uneven heating of the earth's surface by the sun rays at two different adjoining places. Due to this, a pressure difference is created and wind possesses kinetic energy.

- (c) **Bio-energy** Plants in the process of photosynthesis convert the solar energy into food (chemical energy). This food is consumed by animals. Thus, the animal wastes and remains of the plants constitute bio-mass which can be utilised as a source of energy.
- (d) **Wave energy** The waves are generated by strong winds (due to solar energy) blowing across the sea.
- (e) **Ocean thermal energy** Sun is responsible for the temperature difference between the water at the surface and water at depth in seas and oceans.
- (iii) **Solar heating devices** They derive their energy directly from solar energy and convert it into other usable forms of energy. Thus, the energy from various sources are considered to have been derived from the sun.

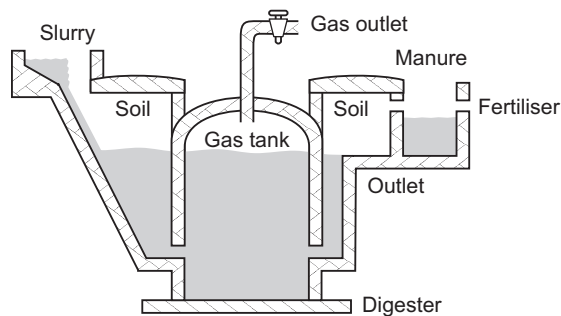
Q. 29 What is bio-mass? Explain the principle and working of a biogas plant using a labelled schematic diagram.

Ans. Bio-mass refers to the organic fuel obtained from plants and animal wastes like wood, cow-dung, residue after harvesting the crop, vegetable waste and sewage, etc. It has been used as a fuel for a long time. These fuels, however, do not produce much heat on burning and a lot of smoke is given out when they are burnt.

Bio-energy can be produced in a plant, known as 'bio-gas plant or gobargas plant' using bio-mass like cow-dung, various plant materials like the residue after harvesting the crops, vegetable waste and sewage are decomposed in the absence of oxygen to give bio-gas. *The labelled diagram of 'gobar gas' plant is given below*

Principle

It is based on the principle that the anaerobic micro-organisms decompose or breakdown complex compounds of the cow-dung slurry, in absence of oxygen, in a few days and generate gases like methane, carbon dioxide, hydrogen and hydrogen sulphide which burn without smoke and leaves no residue like ash.



Working The plant has a dome-like structure built with bricks. A slurry of cow-dung, various plant materials like the residue after harvesting the crops, vegetable waste, sewage and water is made in the mixing tank from where it is fed into the digester. The digester is a sealed chamber which is free from oxygen.

Anaerobic micro-organisms decompose or break down complex compounds of the cow-dung slurry in a few days and generate gases like methane, carbon dioxide, hydrogen and hydrogen sulphide. The bio-gas is stored in the gas tank above the digester from which they are drawn through pipes for use.

Our Environment

Multiple Choice Questions (MCQs)

Q. 1 Which one of the following is an artificial ecosystem?

- (a) Pond (b) Crop field (c) Lake (d) Forest

Ans. (b) Crop field is an artificial ecosystem. It is an agricultural land created by man. Pond, lake and forests are natural ecosystem, as they are self-sustainable and do not need human interference for their maintenance.

Q. 2 In a food chain, the third trophic level is always occupied by

- (a) carnivores (b) herbivores (c) decomposers (d) producers

Ans. (a) In a food chain, the third trophic level is always occupied by carnivores that feed on herbivores.

The various steps in a food chain at which the transfer of food (or energy) takes place are called **trophic levels**.

Plants (producers or autotrophs) occupy the first trophic level, herbivores constitute the second trophic level, carnivores constitute the third trophic level and higher carnivores occupy the fourth trophic level.

Q. 3 An ecosystem includes

- (a) all living organisms
(b) non-living objects
(c) both living organisms and non-living objects
(d) sometimes living organisms and sometimes non-living objects

Ans. (c) All the interacting organisms in an area together with the non-living constituents of the environment form an **ecosystem**.

Thus, an ecosystem consists of biotic components comprising living organisms and a biotic components comprising physical factors like temperature, rainfall, wind, soil and minerals.

Q. 4 In the given food chain, suppose the amount of energy at fourth trophic level is 5 kJ, what will be the energy available at the producer level?

Grass → Grasshopper → Frog → Snake → Hawk

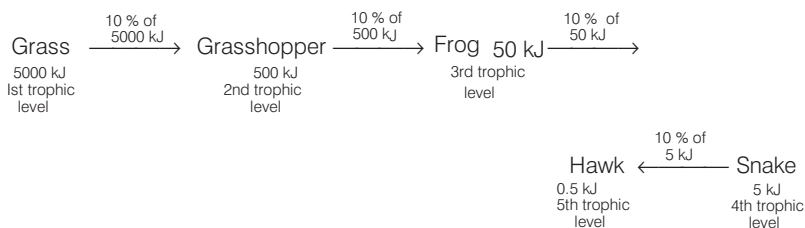
(a) 5 kJ

(b) 50 kJ

(c) 500 kJ

(d) 5000 kJ

Ans. (d) According to 10% law, only 10% of the energy entering a particular trophic level of organisms is available for transfer to the next higher trophic level. In this food chain, at the 4th trophic level, only 5 kJ energy is available to the snake. So, the energy available at the producer level will be 5000 kJ. *It can be shown as*



Q. 5 Accumulation of non-biodegradable pesticides in the food chain in increasing amount at each higher trophic level is known as

(a) eutrophication

(b) pollution

(c) bio-magnification

(d) accumulation

Ans. (c) **Bio-magnification** is the accumulation of non-biodegradable pesticides in the food chain in increasing amount at each higher trophic level. The maximum concentration of these chemicals are accumulated in human beings as they occupy top level in any food chain.

Eutrophication is the process by which a water body becomes enriched in dissolved nutrients that stimulate the growth of aquatic plant life usually resulting in depletion of dissolved oxygen.

Pollution is the introduction of contaminants into the natural environment that cause adverse change.

Q. 6 Depletion of ozone is mainly due to

(a) chlorofluorocarbon compounds

(b) carbon monoxide

(c) methane

(d) pesticides

Ans. (a) **Depletion** of ozone is mainly due to chlorofluorocarbons (CFCs). These are synthetic chemicals which are used as refrigerants and in fire extinguishers.

Carbon monoxide leads to air pollution. It causes harm to humans by combining with haemoglobin, thus affecting O_2 carrying capacity of blood.

Methane is a greenhouse gas, released from power stations, chemical industries etc.

Pesticides are not degradable, thus causes biological magnification.

Q. 7 Organisms which synthesise carbohydrates from inorganic compounds using radiant energy are called

- (a) decomposers
- (b) producers
- (c) herbivores
- (d) carnivores

Ans. (b) Organisms which synthesise carbohydrates from inorganic compounds using radiant energy are called **producers** e.g., all green plants, blue-green algae.

Organisms which consume the food produced, either directly from producers or indirectly by feeding on other consumers are the **consumers**. They can be classified as herbivores, carnivores and omnivores. Organisms which feed on dead plants and animals are called **decomposers** or **saprophytes**.

Q. 8 In an ecosystem, the 10% of energy available for transfer from one trophic level to the next is in the form of

- (a) heat energy
- (b) light energy
- (c) chemical energy
- (d) mechanical energy

Ans. (c) The 10% of energy available for transfer from one trophic level to the next in an ecosystem is in the form of **chemical energy**.

The autotrophs (or producers) capture the energy present in sunlight and convert it into chemical energy. This energy is passed onto other trophic levels to support all the activities of the living world.

Q. 9 Organisms of a higher trophic level which feed on several types of organisms belonging to a lower trophic level constitute the

- (a) food web
- (b) ecological pyramid
- (c) ecosystem
- (d) food chain

Ans. (a) Organisms of a higher trophic level feeding on several types of organisms belonging to a lower trophic level constitute the **food web**.

A series of organisms through which food energy flows in an ecosystem is called a **food chain**.

An **ecosystem** consists of all the living beings of an area and non-living components of their environment.

The graphic summary of the trophic structure and energy transfer in an ecosystem is called **ecological pyramids**.

Q. 10 Flow of energy in an ecosystem is always

- (a) unidirectional
- (b) bidirectional
- (c) multidirectional
- (d) no specific direction

Ans. (a) **Flow of energy** in an ecosystem is always unidirectional. This is because the energy that is captured by the autotrophs cannot revert back to the solar input and the energy which passes to herbivores cannot come back to autotrophs.

The energy moves progressively through various trophic levels so, it may not be available to previous level.

Q. 11 Excessive exposure of humans to UV-rays results in

- (i) damage to immune system (ii) damage to lungs
- (iii) skin cancer (iv) peptic ulcers
- (a) (i) and (ii) (b) (ii) and (iv) (c) (i) and (iii) (d) (iii) and (iv)

Ans. (c) The UV-rays have extremely harmful effects on human beings, other animals as well as plants.

It can cause skin cancer, damage the eyes by causing cataract, also damage immune system by lowering the body's resistance to diseases.

Q. 12 In the following groups of materials, which group(s) contains only non-biodegradable items?

- (i) Wood, paper, leather (ii) Polythene, detergent, PVC
- (iii) Plastic, detergent, grass (iv) Plastic, bakelite, DDT
- (a) (iii) (b) (iv) (c) (i) and (iii) (d) (ii) and (iv)

Ans. (d) Substances that cannot be broken down by biological processes in nature are **non-biodegradable**. e.g., polythene, detergent, PVC, plastics, bakelite, DDT etc.

On the other hand, substances that are broken down by biological processes are said to be **biodegradable**. e.g., wood, paper, leather, grass, animal bones etc.

Q. 13 Which of the following limits the number of trophic levels in a food chain?

- (a) Decrease in energy at higher trophic levels
- (b) Defficient food supply
- (c) Polluted air
- (d) Water

Ans. (a) Decrease in energy at higher trophic levels limits the number of trophic levels in a food chain. At each trophic level, a large portion of energy is utilised for the maintenance of organisms that occur at that trophic level.

So, organism at higher level gets less and less energy at successive levels. The number of trophic levels are limited to 3-4 because after that, the energy available for the next level will be too small i.e., it will be insufficient to sustain life of the organisms.

Q. 14 Which of the statement is incorrect?

- (a) All green plants and blue-green algae are producers
- (b) Green plants get their food from organic compounds
- (c) Producers prepare their own food from inorganic compounds
- (d) Plants convert solar energy into chemical energy

Ans. (b) Green plants get their food from inorganic compounds using radiant energy of the sun in the presence of chlorophyll.

All green plants and blue-green algae are called **producers** as they can prepare food from inorganic substances by photosynthesis.

Producers capture the solar energy and convert it into chemical energy.

Q. 15 Which group of organisms are not constituents of a food chain?

- (i) Grass, lion, rabbit, wolf
 - (ii) Plankton, man, fish, grasshopper
 - (iii) Wolf, grass, snake, tiger
 - (iv) Frog, snake, eagle, grass, grasshopper
- (a) (i) and (iii) (b) (iii) and (iv) (c) (ii) and (iii) (d) (i) and (iv)

Ans. (c) The sequence of living organisms in a community in which one organism consumes another organism to transfer energy is called a **food chain**.

Food chain (ii) is an aquatic food chain in which grasshopper cannot take part. The correct sequence is

Plankton → Zooplankton → Fishes → Man

In food chain (iii) wolf, snake and tiger are carnivores, there is no herbivore to eat grass.

(i) is forest food chain

Grass → Rabbit → Wolf → Lion

Similarly (iv), is also food chain in a grassland.

Grass → Grasshopper → Frog → Snake → Eagle

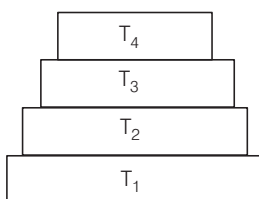
Q. 16 The percentage of solar radiation absorbed by all the green plants for the process of photosynthesis is about

- (a) 1% (b) 5% (c) 8% (d) 10%

Ans. (a) The green plants capture about 1% of the energy of sunlight that falls on their leaves for the process of photosynthesis.

This energy is converted into chemical energy as food. At the subsequent level, 10% energy is available.

Q. 17 In the given figure, the various trophic levels are shown in a pyramid. At which trophic level is maximum energy available?



- (a) T_4 (b) T_2 (c) T_1 (d) T_3

Ans. (c) The maximum energy is available at T_1 . There is progressive decline in the amount of energy available from producer to higher trophic levels, i.e., $T_1 > T_2 > T_3 > T_4$ (energy).

This is because at each trophic level, a large portion of energy is utilised for maintenance of organisms at that trophic level and lost as heat.

Q. 18 What will happen if deer is missing in the food chain given below?

Grass → Deer → Tiger

- (a) The population of tiger increases
- (b) The population of grass decreases
- (c) Tiger will start eating grass
- (d) The population of tiger decreases and the population of grass increases

Ans. (d) If deer is missing in the given food chain, there will not be sufficient food for the tigers. Some of the tigers will die because of starvation and hence, the population of tigers will decrease. Since grass is eaten by deer, the population of grass will also increase when deer is missing.

Q. 19 The decomposers in an ecosystem

- (a) convert inorganic material to simpler forms
- (b) convert organic material to inorganic forms
- (c) convert inorganic materials into organic compounds
- (d) do not breakdown organic compounds

Ans. (b) The decomposers in an ecosystem convert organic material to inorganic forms, that go into the soil and are once again used up by the plants. Decomposers feed on dead remains and waste products of organisms. Producers convert inorganic materials into organic compounds like sugar and starch using sunlight in presence of chlorophyll.

Q. 20 If a grasshopper is eaten by a frog, then the energy transfer will be from

- (a) producer to decomposer
- (b) producer to primary consumer
- (c) primary consumer to secondary consumer
- (d) secondary consumer to primary consumer

Ans. (c) In a food chain, if a grasshopper is eaten by a frog, then the energy transfer will be from primary consumer to secondary consumer. Grasshopper feeds on producers *i.e.*, the grass/plants. So, it occupies the level of primary consumer. Frogs, eating grasshopper thus become the secondary consumer.

Q. 21 Disposable plastic plates should not be used because

- (a) they are made of materials with light weight
- (b) they are made of toxic materials
- (c) they are made of biodegradable materials
- (d) they are made of non-biodegradable materials

Ans. (d) Disposable plastic plates should not be used because they are made of non-biodegradable materials like plastics etc. The non-biodegradable materials cannot be broken down into harmless substances in nature, so they are major pollutants of the environment.

Short Answer Type Questions

Q. 22 Why is improper disposal of waste a curse to environment?

Ans. Disposal of waste means 'to get rid of waste'. Improper disposal of waste is a curse to environment because it can cause substantial harm to human health and safety and pollute our environment (air, water and soil).

Improper waste storage or disposal frequently contaminates surface and groundwater supplies.

People living in homes built near old and abandoned waste disposal sites may be in a particular vulnerable position.

Q. 23 Write the common food chain of a pond ecosystem.

Ans. *In a freshwater aquatic ecosystem like a pond, the following organisms are included*

Producers Rooted or floating plants usually algae, called phytoplankton.

Consumers Animals of the ecosystem which depend on the producers for food directly.

Herbivores The algae are eaten up by Protozoa (zooplankton).

Carnivores The Protozoa are eaten up by small fish.

Higher carnivores big fishes and fish-eating bird or animal eats the small fish.

Algae (Producer) → Protozoa (Herbivore) → Small fish (Carnivore) → Large fish/bird/animal (Large carnivore)

Pond ecosystem of food chain

Q. 24 What are the advantages of cloth bags over plastic bags during shopping?

💡 Thinking Process

Cloth bags are biodegradable, while plastic bags are non-biodegradable

Ans. *Cloth bags have following advantages over plastics bags*

- (i) Cloth bags are biodegradable so they do not pollute the environment.
- (ii) Cloth bags are capable of carrying more things.
- (iii) They can be reused over and over again.

Q. 25 Why are crop fields known as artificial ecosystem?

💡 Thinking Process

All the components of a crop field are taken care of by humans

Ans. Artificial ecosystems are those ecosystems which are modified and managed by human beings.

Crop fields are man-made, plants do not grow naturally rather most of the plants are grown by humans according to the season, type of soil etc. Crop fields are not like wild forest area which is left to the care of nature and can sustain itself.

In crop fields, the land is managed, soil is prepared for sowing seeds, then irrigated and further progress is also kept under observation for getting good yield.

Thus, crop field are known as artificial ecosystems.

Q. 26 Differentiate between biodegradable and non-biodegradable substances. Give examples.

Ans. *Differentiation between biodegradable and non-biodegradable substances are*

Biodegradable substances	Non-biodegradable substances
Substances that are broken down into simpler substances by biological processes	Substances that cannot be broken down into simpler ones by biological processes.
They are environment friendly as they usually do not pollute the environment	They are not environment friendly as they are major pollutants of the environment
<i>e.g.</i> , Wool, paper, cotton, fruit and vegetable peels, leaves etc.	<i>e.g.</i> , DDT, polythene bags, plastics, metal articles like iron nails, synthetic fibres etc.

Q. 27 Suggest one word for each of the following statement/definitions.

- The physical and biological world where we live in
- Each level of food chain where transfer of energy takes place
- The physical factors like temperature, rainfall, wind and soil of an ecosystem
- Organisms which depend on the producers either directly or indirectly for food

Ans. (a) Environment
(b) Trophic level
(c) Abiotic factors
(d) Consumers/heterotrophs

Q. 28 Explain the role of decomposers in the environment.

Ans. Organisms that feed on dead plants and animals are called decomposers, *e.g.*, bacteria, fungi etc.

They breakdown the complex organic compounds present in the dead remains into simpler substances and obtain nutrition from them. These substances are released into the soil and the atmosphere. *Thus, they play following roles*

- Help in recycling of materials, replenishment of the soil's nutrients etc.
- They also clean up our surroundings by decomposing organisms and organic wastes.

Q. 29 Select the mis-matched pair in the following and correct it.

(a) Bio-magnification	Accumulation of chemicals at the successive trophic levels of a food chain
(b) Ecosystem	Biotic components of environment
(c) Aquarium	A man-made ecosystem
(d) Parasites	Organisms which obtain food from other living organisms

Ans. (b) An ecosystem consists of biotic components (comprising living organisms) and abiotic components (comprising physical factors like temperature etc.) of environment.

Q. 30 We do not clean ponds or lakes, but an aquarium needs to be cleaned. Why?

💡 Thinking Process

Ponds or lakes are natural ecosystem while aquarium is artificial ecosystem.

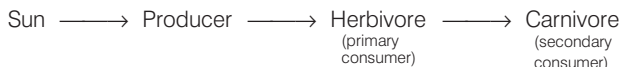
Ans. A pond or lake is a natural ecosystem. They are self-sustaining and complete, in them, all the organisms of food chain are available. If any organism dies, there are decomposers to decompose their bodies into simpler substances.

An aquarium, on the contrary is an artificial and incomplete ecosystem. The abiotic components are not supplied naturally to it. It may not have all the biotic components in it. If a fish die in an aquarium in the absence of decomposer, it will lie there as a rotten body, polluting the water of aquarium. Thus, an aquarium needs regular cleaning.

Long Answer Type Questions

Q. 31 Indicate the flow of energy in an ecosystem. Why is it unidirectional? Justify.

Ans. *The flow of energy in an ecosystem occurs in following sequence*



The flow of energy is unidirectional because of reasons given below

- (i) Energy flows progressively from one trophic level to another and cannot revert back. Energy given out as heat is lost in the environment, it does not return to be used again
- (ii) Also, the available energy decreases at higher trophic level. Out of the total energy available, at a particular trophic level, only 10% is passed on to the next trophic level, making it impossible for energy to flow in the reverse direction.

Q. 32 What are decomposer? What will be the consequence of their absence in an ecosystem?

Ans. Organisms which breakdown the complex organic compounds present in dead and decaying matter are called **decomposers**. e.g., certain bacteria and fungi.

Decomposers acts as cleaning agents of environment by decomposing the dead bodies of plants and animals. They also help in recycling of materials, replenishment of soil's nutrients etc.

The consequence of their absence in an ecosystem can be disastrous as discussed below

- (i) The dead bodies would keep lying, it would lead to their accumulation and thus, polluting the environment.
- (ii) The elements by which bodies are made would never be returned to the environments. As a result, all the nutrients present in soil, air and water would soon be exhausted and the whole life cycle of organisms will be disrupted.

Q. 33 Suggest any four activities in daily life which are eco-friendly.

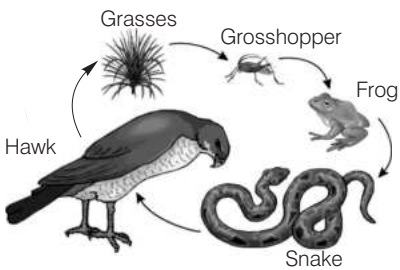
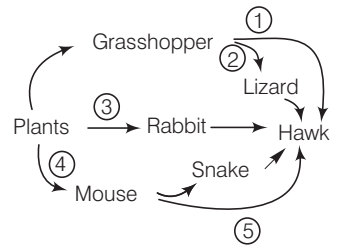
Ans. **Eco-friendly activities** are those activities that are beneficial for the environment.

Four activities in daily life which are eco-friendly are as follows

- (i) Separation of bio-degradable and non-biodegradable wastes.
- (ii) Use of cloth/paper bags in place of polythene/plastic bags.
- (iii) Using public transport for commuting, walking or using cycle for short distances.
- (iv) Harvesting rain water.
- (v) Switching off light/fans in unoccupied rooms.
- (vi) Use of compost and vermicompost in place of fertilisers.

Q. 34 Give two differences between food chain and food web.

Ans. The two differences between food chain and food web

Food chain	Food web
It is a series of organisms feeding on one another	It consists of a number of interconnected food chains, involving many organisms.
Members of higher trophic level feeds upon a single type of organism of the lower trophic level.	Member of higher trophic level can feed upon organisms of the lower trophic levels of other food chains.
e.g., food chain in grassland ecosystem 	e.g., food web of a grassland ecosystem. 

Q. 35 Name the wastes which are generated in your house daily. What measures would you take their disposal?

Ans. The waste generated in our house and measure for its disposal are given in the table below

Household wastes	Measures for disposal
Kitchen waste like bottles, plastics, food etc	Prepare a compost pit
Paper wastes like newspaper, envelopes etc	Should be recycled
Plastic bags	Should be safely dumped in garbage bins for non-biodegradable wastes
Vegetable/fruit peels/rind	Can be placed near trees/plants, so that on decomposition enrich the soil with nutrients
All other wastes	Segregation into biodegradable and non-biodegradable wastes

Q. 36 Suggest suitable mechanism(s) for waste management in fertiliser industries.

Ans. Effluents and harmful gases are the main wastes which are produced in a fertiliser factory.

Suitable mechanism for waste management are

- (i) For control of gaseous pollutants **combustion equipments** are used for pollutants which can be oxidised. The pollutants are exposed to a high temperature (1650°) in the process.
Air pollutants such as certain gases and vapour and inflammable compounds are controlled through the use of **adsorption equipments**. Adsorption is a surface phenomenon and it needs the presence of large solid surface area. Toxic and odoriferous compounds are efficiently removed.
- (ii) *Three options are available for controlling the effluents*
 - (a) Control can take place at the point of generation within the factory.
 - (b) Waste water can be pre-treated for discharge to municipal treatment systems.
 - (c) Waste water can be treated completely at the factory and either reused or discharged directly into receiving waters.

Q. 37 What are the by-products of fertiliser industries? How do they affect the environment?

Ans. The by-products of fertiliser industries are pesticides and some chemical fertilisers. They are non-biodegradable, so they get accumulated at each trophic level. These mix up with soil and water. From there, these pesticides are absorbed by the growing plants along with water and minerals.

When herbivores eat these plants, these poisonous chemicals pass into their bodies through the food chain. When carnivores eat herbivores, these pesticides transfer to their bodies. Man being omnivore, eats both plants and herbivores and these get accumulated into their body.

Thus, pesticides enter the food chain at the producer level and in the process of transfer of food through food chains, these harmful chemicals get concentrated at each trophic level. This is called **bio-magnification**.

Q. 38 Explain some harmful effects of agricultural practices on the environment.

Ans. *Some harmful effects of agricultural practices on the environment are as follows*

- (i) **Soil degradation** Extensive cropping causes loss of soil fertility. Also, over the time it can lead to soil erosion and finally to desertification.
- (ii) **Pollution** Use of synthetic fertilisers and pesticides leads to soil, water and air pollution.
- (iii) **Water shortage** Excess use of ground water for agriculture lowers the water table. This results in acute water shortage at many places.
- (iv) **Bio-magnification** The chemical pesticides, being non-biodegradable accumulates in organisms in increasing amounts at each trophic level.
- (v) **Deforestation** Indiscriminate felling of trees for agriculture has resulted in loss of habitat for wildlife. Thus, it also causes damage to natural ecosystem.

16

Management *of* Natural Resources

Multiple Choice Questions (MCQs)

Q. 1 From the list given below pick the item that is not a natural resource

- (a) soil (b) water (c) electricity (d) air

Ans. (c) Natural resources occur naturally within environment that exists relatively undisturbed by human, in a natural form. Electricity is not a natural resource. It is developed by man. While soil, water and air are natural resources found in environment.

Q. 2 The most rapidly dwindling natural resource in the world is

- (a) water (b) forests
(c) wind (d) sunlight

Ans. (b) Forests are the most rapidly dwindling natural resource in the world. They are destroyed indiscriminately as they provide raw material for various industries as timber industry, paper industry, sports industry, etc.

Q. 3 The most appropriate definition of a natural resource is that it is a substance/commodity that is

- (a) present only on land
(b) a gift of nature which is very useful to mankind
(c) a man-made substance placed in nature
(d) available only in the forest

Ans. (b) A 'natural resource' is a substance/commodity that is a gift of nature which is very useful to mankind. These are air, water, soil etc.

Q. 4 The main cause for abundant coliform bacteria in the river Ganga is

- (a) disposal of unburnt corpses into water
(b) discharge of effluents from electroplating industries
(c) washing of clothes
(d) immersion of ashes

Ans. (a) The main cause for abundant coliform bacteria in the river Ganga is disposal of unburnt corpses into water. Coliform are the group of bacteria found in human intestines whose presence in water indicates contamination by disease-causing microorganisms. The disposal of unburnt corpses contaminate water of the river and release these bacteria in water.

Q. 5 The pH of water sample collected from a river was found to be acidic in the range of 3.5-4.5, on the banks of the river were several factories that were discharging effluents into the river. The effluents of which one of the following factories is the most likely cause for lowering the pH of river water?

- (a) Soap and detergent factory (b) Lead battery manufacturing factory
(c) Plastic cup manufacturing factory (d) Alcohol distillery

Ans. (b) Lead is a major constituent of lead-acid battery used extensively in car batteries. The discharge of lead from battery manufacturing factory can lower the pH of river water by making it acidic.

Q. 6 The pH range most conducive for life of freshwater plants and animals is

- (a) 6.5-7.5 (b) 2.0-3.5 (c) 3.5-5.0 (d) 9.0-10.5

Ans. (a) The pH of water body should be in the range of 6.5-7.5 for the survival of freshwater plants and animals. Below and above this range is harmful for the aquatic life.

Q. 7 The three R's that will help us to conserve natural resources for long term use are

- (a) recycle, regenerate, reuse (b) reduce, regenerate, resuse
(c) reduce, reuse, redistribute (d) reduce, recycle, reuse

Ans. (d) *The three R's that help us to conserve natural resources for long term use are*

Reduce This means less use of natural resources, e.g., save water by repairing leaky taps.

Recycle This means collection of materials like plastic, paper, glass etc., to make required things instead of synthesising or extracting fresh plastic, paper, glass etc.

Reuse Instead of throwing articles, (e.g., used envelopes) use it again. This is better than recycling because recycling process uses some energy.

Q. 8 Given below are a few statements related to biodiversity. Pick those that correctly describe the concept of biodiversity

- (i) Biodiversity refers to the different species of flora and fauna present in an area.
(ii) Biodiversity refers to only the flora of a given area.
(iii) Biodiversity is greater in a forest.
(iv) Biodiversity refers to the total number of individuals of a particular species living in an area.

- (a) (i) and (ii) (b) (ii) and (iv) (c) (i) and (iii) (d) (ii) and (iii)

Ans. (c) Biodiversity refers to the different species of flora and fauna present in an area. It is greater in a forest due to availability of plenty of resources.

Q. 9 Among the statements given below select the ones that correctly describe the concept of sustainable development

- (i) planned growth with minimum damage to the environment.
- (ii) growth irrespective of the extent of damage caused to the environment.
- (iii) stopping all development work to conserve the environment.
- (iv) growth that is acceptable to all the stakeholders.

(a) (i) and (iv) (b) (ii) and (iii) (c) (ii) and (iv) (d) only (iii)

Ans. (a) Sustainable development means planned growth with minimum damage to the environment. The growth of environment should not be harmful to the stakeholders as well. It should be acceptable to them, e.g., if some trees have to be cut for various purposes, then damage to the environment can be minimised by planting new samplings in place of cut trees.

Q. 10 In our country, vast tracts of forests are cleared and a single species of plant is cultivated. This practice promotes

- (a) biodiversity in the area
- (b) monoculture in the area
- (c) growth of natural forest
- (d) preserves the natural ecosystem in the area

Ans. (b) Monoculture in the area is promoted by the practice of clearing vast tracts of forests and cultivating a single species of plant. This destroys the biodiversity in the area as the number of species lower. It also disturbs the natural ecosystem in the area. It deaccelerates the growth of natural forest.

Q. 11 A successful forest conservation strategy should involve

- (a) protection of animals at the highest trophic level
- (b) protection of only consumers
- (c) protection of only herbivores
- (d) comprehensive programme to protect all the physical and biological components

Ans. (d) Forest conservation strategy should involve comprehensive programme to protect all the physical and biological components, (i.e., the ecosystem). If organisms only at a particular trophic levels are protected, the ecosystem would be disturbed, as the organisms at the lower trophic level would vanish fast. This will lead to situation where the organisms at all the upper trophic levels would die due to starvation.

Q. 12 The important message conveyed by the 'Chipko Movement' is

- (a) to involve the community in forest conservation efforts
- (b) to ignore the community in forest conservation efforts
- (c) to cut down forest trees for developmental activities
- (d) government agencies have the unquestionable right to order destruction of trees in forests

Ans. (a) The important message conveyed by the 'Chipko Movement' is to involve the community in forest conservation efforts.
People's participation can help in increasing forest as well as help in their conservation.

Q. 13 In our country, there are attempts to increase the height of several existing dams like Tehri and Almati, dams across Narmada. Choose the correct statements among the following that are a consequence of raising the height of dams

- (i) Terrestrial flora and fauna of the area is destroyed completely.
- (ii) Dislocation of people and domestic animals living in the area.
- (iii) Valuable agricultural land may be permanently lost.
- (iv) It will generate permanent employment for people.

- (a) (i) and (ii) (b) (i), (ii) and (iii)
- (c) (ii) and (iv) (d) (i), (ii) and (iv)

Ans. (b) *The consequences of raising the height of dams are as follows*

- (i) Terrestrial flora and fauna of area is destroyed completely. It leads to deforestation and loss of biological diversity.
- (ii) Dislocation of people and domestic animals living in an area causes various social problems.
- (iii) Valuable agricultural land may be permanently lost.
- (iv) The people are not even given adequate compensation for their losses.
- (v) This leads to economic problems.
- (vi) It generates temporary employment for people.

Q. 14 Expand the abbreviation GAP

- (a) Governmental Agency for Pollution Control
- (b) Gross Assimilation by Photosynthesis
- (c) Ganga Action Plan
- (d) Governmental Agency for Animal Protection

Ans. (c) GAP is the abbreviation for Ganga Action Plan. This multi-crore project came about in 1985 because the quality of water in the Ganga was getting poor due to pollution.

Q. 15 Select the incorrect statement.

- (a) Economic development is linked to environmental conservation
- (b) Sustainable development encourages development for current generation and conservation for resources for futures generations
- (c) Sustainable development does not consider the view points of stakeholders
- (d) Sustainable development is a long planned and persistent development

Ans. (c) Sustainable development consider the view points of stakeholders. It is a process in which decentralised economic growth and ecological conservation go hand in hand. It encourages development for current generation and conservation of resources for future generation. It is a long planned and persistent development. Today, economic development is linked to environmental conservation. The environment is not degraded or put to risk in the name of economic development.

Q. 16 Which of the following is not a natural resource?

- (a) Mango tree (b) Snake (c) Wind (d) Wooden house

Ans. (d) Wooden house is not a natural resource. It is man-made. While a mango tree, snake and wind are natural resource. They are present in the environment naturally.

Q. 17 Select the wrong statement.

- (a) Forests provide variety of products (b) Forests have greater plant diversity
(c) Forests do not conserve soil (d) Forests conserve water

Ans. (c) Forests are major site of conservation of soil. The roots of trees firmly hold the soil particles to protect it from flowing away. The humus is more in the soil of forests. Forest provide variety of products, e.g., timber, paper etc. They have a greater plant diversity as different type of plants are naturally present. Forest also helps in conservation of water.

Q. 18 Arabari forests of Bengal is dominated by

- (a) teak (b) sal
(c) bamboo (d) mangroove

Ans. (b) Arabari forests of Bengal is dominated by sal trees. With the active and willing participation of the local community, these forests underwent a remarkable recovery by 1983.

Q. 19 Ground water will not be depleted due to

- (a) afforestation
(b) thermal power plants
(c) loss of forest, and decreased rainfall
(d) cropping of high water demanding crops

Ans. (a) Ground water will not be depleted due to afforestation (i.e., plantation of trees). Most of the groundwater is polluted and depleted by thermal power plants, loss of forest, decreased rainfall and due to use of excess water for cropping of high water demanding crops.

Q. 20 Opposition to the construction of large dams is due to

- (a) social reasons (b) economic reasons
(c) environmental reasons (d) All of these

Ans. (d) Opposition to the construction of large dams by local people is due to social, economic and environmental problems.
(Also, Refer to Q. 13)

Q. 21 Khadins, Bundhis, Ahars and Kattas are ancient structures that are examples for

- (a) grains storage (b) wood storage
(c) water harvesting (d) soil conservation

Ans. (c) Khadins tanks in Rajasthan, Bundhis in Madhya Pradesh and Uttar Pradesh, Ahars in Bihar, and Kattas in Karnataka are some of the ancient water harvesting structures of India. They are still in use today for the conservation of water.

Q. 22 Pick the right combination of terms which has no fossil fuel.

- (a) Wind, ocean and coal (b) Kerosene, wind and tide
(c) Wind, wood, sun (d) Petroleum, wood, sun

Ans. (c) Wind, wood, sun has no fossil fuel. Coal, petroleum and kerosene are fossil fuels in the given list.

Q. 23 Select the eco-friendly activity among the following

- (a) Using car for transportation
- (b) Using polybags for shopping
- (c) Using dyes for colouring clothes
- (d) Using windmills to generate power for irrigation

Ans. (d) Wind is a renewable source of energy. So, using windmills to generate power for irrigation is a eco-friendly activity.

Car uses petrol or diesel that is non-renewable.

Dyes are synthetic chemical compounds and polybags are non-biodegradable. So, these three are not eco-friendly activities.

Q. 24 It is important to make small check dams across the flooded gullies because they

- (i) hold water for irrigation
- (ii) hold water and prevent soil erosion
- (iii) recharge ground water
- (iv) hold water permanently

(a) (i) and (iv)

(b) (ii) and (iii)

(c) (iii) and (iv)

(d) (ii) and (iv)

Ans. (b) It is important to make small check dams across the flooded gullies because they recharge groundwater. These small dams helps to hold water. This provides moisture for the vegetation cover, thus preventing soil erosion.

Short Answer Type Questions

Q. 25 Prepare a list of five items that you use daily in the school. Identify from the list such items that can be recycled.

Ans. Five items that are list daily in school are Paper, pen, plastic box, steel spoon, scale, eraser, compass and divider (metallic), steel lunch box etc.

The items that can be recycled are paper, plastic box, compass, steel lunch box and steel spoon.

Q. 26 List two advantages associated with water harvesting at the community level.

Ans. **Water harvesting** is a method of holding rainwater on the surface of the earth, to make it percolate under the ground so as to recharge groundwater. *Two advantages associated with water harvesting at the community level are*

- (i) Giving people control over their local water resources ensures that mismanagement is reduced.
- (ii) Over-exploitation of these resources is reduced. The water stored in the ground does not evaporate, provides moisture for vegetation over a wide area, and is protected from contamination by human and animal waste.

Q. 27 In a village in Karnataka, people started cultivating crops all around a lake which was always filled with water. They added fertilisers to their field in order to enhance the yield. Soon they discovered that the waterbody was completely covered with green floating plants and fishes started dying in large numbers.

Analyse the situation and give reasons for excessive growth of plants and death of fish in the lake.

💡 Thinking Process

Use of fertilisers adds nutrients to the waterbody leading to eutrophication and thus, results in loss of fishes in lake.

Since people used excessive fertilisers in the fields, these added fertilisers flow away with rain into the water body.

Ans. As many fertilisers contain phosphates and nitrates, the water body became enriched with these chemicals. This enriches the nutrients in water leading to excessive growth of small, green, aquatic plants and the surface of water was completely covered with plants (eutrophication).

These small plants consume most of the dissolved oxygen of the water leading to its deficiency. Due to this, the fishes and the other aquatic life in the lake die because of oxygen starvation and depletion of light.

Q. 28 What measures would you take to conserve electricity in your house?

Ans. The following measures can be taken to conserve electricity in our house

- (i) Put off the fans and lights in unoccupied rooms and when they are not required.
- (ii) Maximise use of solar radiation. It is a pollution free and cost free resource that is easily available.
- (iii) During winters, instead of electric heaters, solar water heating system should be used.
- (iv) Fluorescent tubes or CFL should be used instead of electric bulbs as the former consume less electricity.

Q. 29 Although coal and petroleum are produced by degradation of bio-mass, yet we need to conserve them. Why?

Ans. Coal and petroleum were formed from the degradation of bio-mass millions of years ago. As these resources are being utilised at a much faster rate than their formation, they will be exhausted in the near future. And then we would need to look for alternative sources of energy.

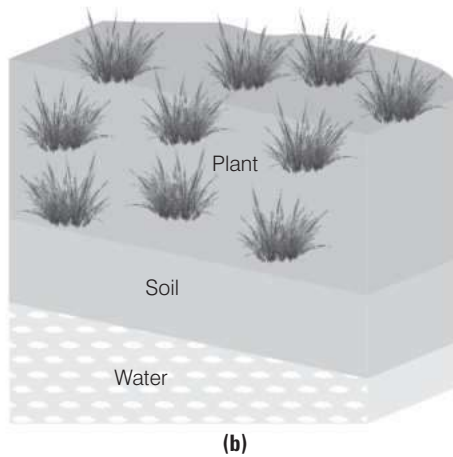
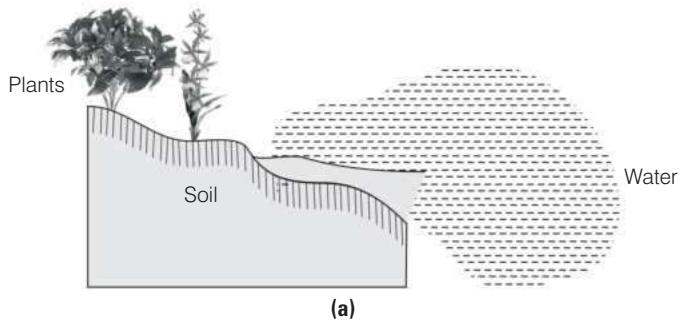
This is the reason why we need to conserve them, although these resources are produced by degradation of bio-mass.

Q. 30 Suggest a few measures for controlling carbon dioxide levels in the atmosphere.

Ans.

- (i) Reduce the consumption of petrol in automobiles. Using car-pools and public transport helps in reducing petrol usage.
- (ii) Use of alternative fuels such as CNG (clean fuel) instead of coal and petroleum.
- (iii) Manure should be prepared out of litter instead of burning it.
- (iv) The smoke coming out of the thermal power stations and other industries should be well treated to remove harmful gases, before discharging it into atmosphere.
- (v) Planting more and more trees.

Q. 31 I. Locate and name the water reservoirs in figures (a) and (b).



II. Which has an advantage over the other and why?

- Ans.** I. The water reservoir is a pond in fig. (a) and underground water body in fig. (b)
i.e., underground water body has more advantage over fig. (a) and water reservoir.
- II. There are many advantages of water stored in the ground. *Some of these are*
- (i) It does not evaporate.
 - (ii) It spreads out to recharge wells and provides moisture for crops over a wide area.
 - (iii) It does not promote breeding of mosquitoes (unlike stagnant water collected in ponds etc).
 - (iv) It is free from contamination by human and animal wastes.
 - (v) It is utilised for the benefit of local population.

Long Answer Type Questions

Q. 32 In the context of conservation of natural resources, explain the terms reduce, recycle and reuse. From among the materials that we use in daily life, identify two materials for each category.

Ans. *In the context of natural resources, the following terms refer to*

Reduce This means to use a material/commodity in lesser quantity.

Recycle Means a material that is used once, is collected and sent back to a manufacturer so that they can make some other useful materials from it.

Reuse It means using a thing over and over again instead of throwing it away. This is actually even better than recycling because the process of recycling uses some energy.

From among the materials we use in daily life, we can **reduce** electricity use by switching off unnecessary lights and fans and save water by repairing leaky taps.

Recycle used paper, plastic bottles, metal objects can be recycled.

Reuse The plastic bottles in which we buy various food-items like jam or pickle can be reused for storing things in kitchen, the used envelopes can be reused to store things, reversed inside out to be used again.

Q. 33 Prepare a list of five activities that you perform daily in which natural resource can be conserved or energy utilisation can be minimised.

Ans. *Activities that we perform daily in which natural resources can be conserved or energy utilisation can be minimised are*

- (i) Use fan and light only when required. Switch them off in unoccupied rooms.
- (ii) Use CFL in place of conventional bulbs/tubes.
- (iii) Avoid using a hose pipe for watering plants as a lot of water is wasted through it.
- (iv) Use solar energy whenever possible.
- (v) Unused water in the water bottles may be used for watering plants.
- (vi) Close taps when not in use.
- (vii) Use public transport or car pools to save fuel.
- (viii) Use CNG as in alternative fuel.
- (ix) Pressure cookers should be used for cooking food to save fuels like LPG.
- (x) Use stairs to climb atleast upto three floors of building instead of taking lift. This will save electricity (any five).

Q. 34 Is water conservation necessary? Give reasons.

Ans. *Conservation of water is necessary due to following reasons*

- (i) **Uneven distribution** Some parts of our country have high resources of water whereas other parts suffer from chronic water shortage.
- (ii) Wide seasonal as well as yearly fluctuation in rainfalls.
- (iii) Water in rivers and ponds is polluted by dumping of untreated sewage and industrial wastes into it.
- (iv) The changing life style of people, especially in urban areas, is consuming more water.
- (v) High yielding varieties of crop uses much more water for irrigation. Also insufficient vegetation cover on ground allow little water to seep in and gets stored as ground water.

Q. 35 Suggest a few useful ways of utilising waste water.

Ans. *Some usefull ways of utilising waste water are as follows*

- (i) For recharging the ground water.
- (ii) For irrigation.
- (iii) Treated municipal water can be used for washing cars, watering gardens etc.
- (iv) Waste water from the kitchen can be collected and reused in toilet flushing.
- (v) Certain pollutants in municipal water can become fertiliser for various crops.

Q. 36 What is the importance of forest as a resource?

Ans. *Forest are renewable natural resource. They are important as a resource due to following reasons*

- (i) Provide habitat, food and protection to wildlife.
- (ii) Help in balancing CO₂ and O₂ of atmosphere.
- (iii) Improves water holding capacity of soil.
- (iv) Regulates water cycle.
- (v) For human beings, they are the source of various essential commodities like fuel, wood, timber, fruits, resins, etc.
- (vi) Forest helps in conservation of biodiversity as a large number of species (of plants and animals) live inside them.

Q. 37 Why are the Arabari forests of Bengal known to be a good example of conserved forest?

Ans. The Arabari forest of Bengal were badly degraded. Then in 1972, the West Bengal Forest Department planned to revive the degraded forest by involving the local people. The villagers of the surrounding area were involved in the protection of 1272 hectares of this forest.

With the active participation of local people living around the forest, the degraded sal forest of Arabari became thick and green within ten years. In return, the villagers were given employment in both silviculture and harvesting operations of the forest and allowed to collect firewood and fodder from the forest area on a nominal payment. Thus, the Arabari forest provides a good example of conserved forest.